

FINAL REPORT OF THE MACAYA BIOSPHERE RESERVE PROJECT

by

Florence E. Sergile

Charles A. Woods

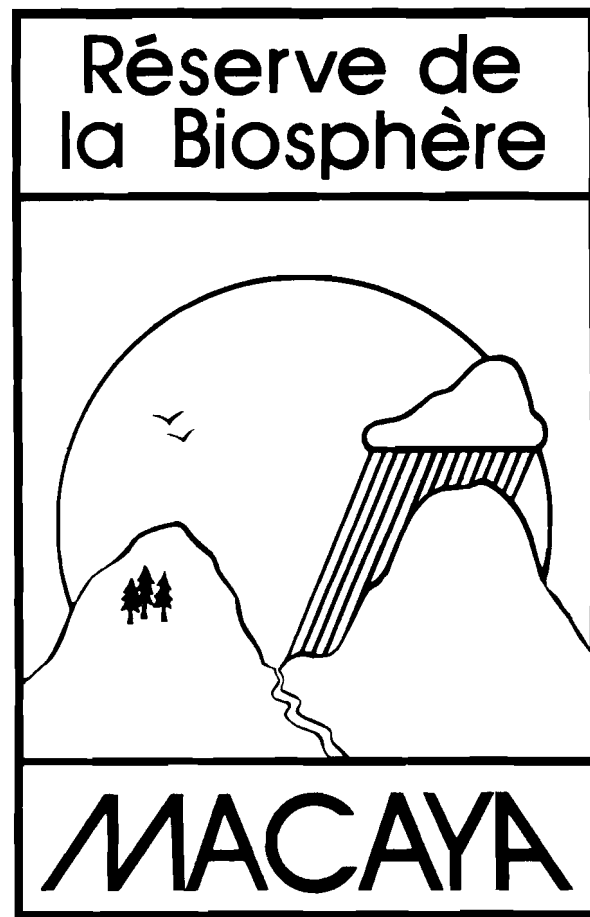
Paul E. Paryski



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PROJECT

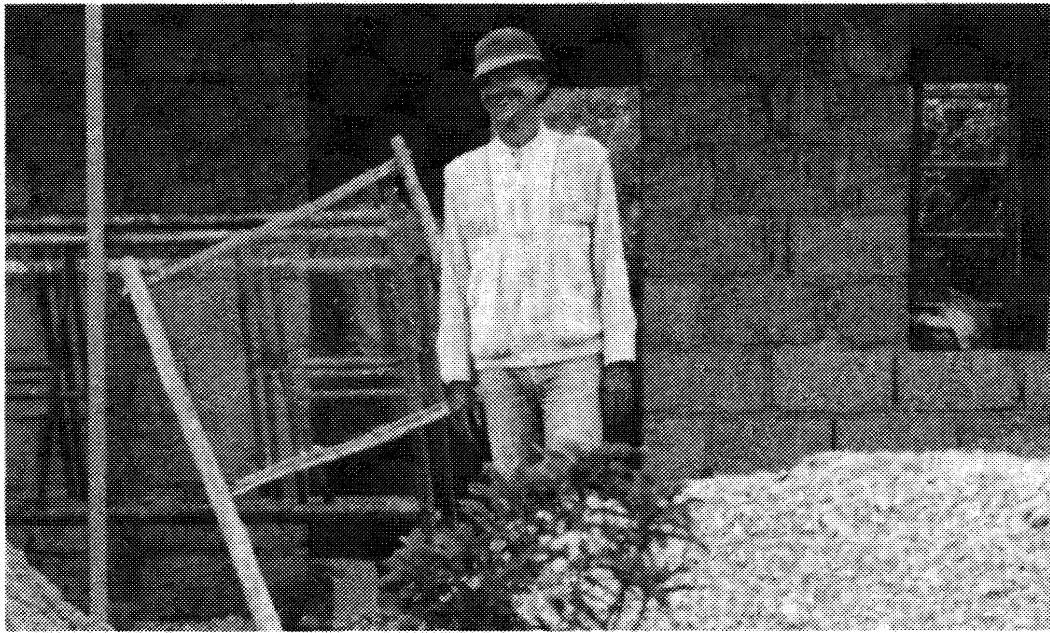
By
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Florida Museum of Natural History

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1992



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In Memory of Michel Aubry

This volume is dedicated to Michel Aubry who gave his life for his, and our, dream of establishing a conservation program in Haiti. Michel was born in Port-au-Prince, Haiti, June 19, 1943, and died on October 30, 1990, at the village of Canot near Parc Macaya in a tragic accident that occurred when the truck he was driving turned over while crossing the flooded l'Acul River. He was returning from a hard week of field work.

Michel was responsible for organizing logistics, supervising all construction, and obtaining supplies, materials and equipment for the project. He was able to develop wonderful strategies due to his exceptional gift of communication and his ability to make people happy, while bringing them together to work productively. He was especially sensitive to needs of others and was very bright and perceptive--talents that served both him and the project well, under the very trying work conditions of a project based in the high, wild mountains of Haiti. He radiated exuberance and inspired all those who knew him.

Michel was a true actor, and at every appropriate moment used this talent to entertain and amuse. He was one of the pillars of the Haitian theater and participated in numerous theatrical production in Haiti and other countries. He was devoted to the conservation of nature and Haiti's natural heritage.

Michel Aubry was truly loved by all, even people who barely knew him. He was not only gifted but a gift to those who did know and love him. His dancing, his sense of drama, his laughter and, at times, his tears sang songs of life itself. His passage was a loss not only to the project, but to Haiti and its fantastic and unique natural heritage. I hope that his life, which was given in the fight to save Macaya, and all it represents was not in vain. He was a great friend and comrade whom I shall never, never forget.

Paul Paryski

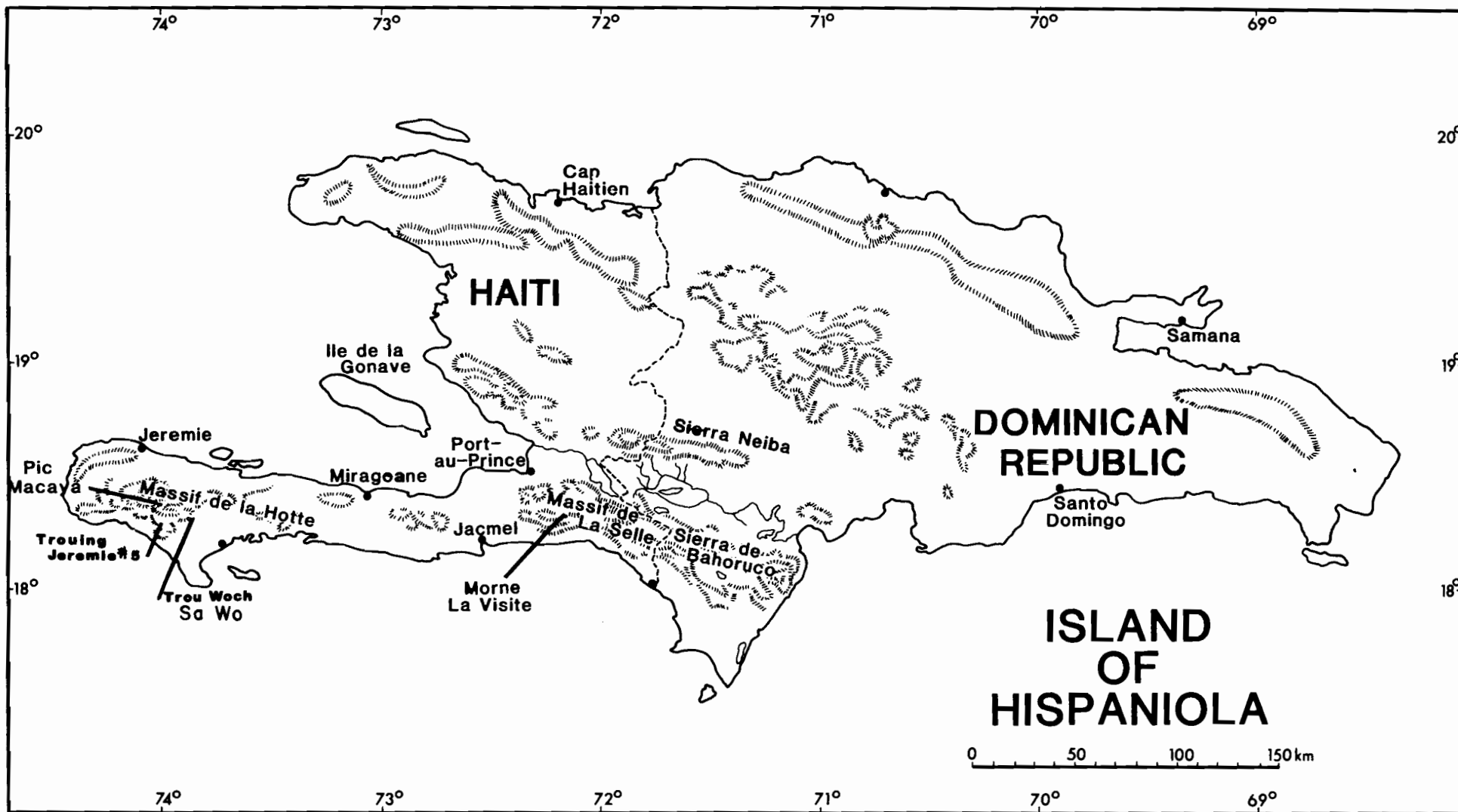
I met Michel three times in my life, but it was as if I had known him a long time. The Macaya Biosphere Reserve population and technicians will never forget Michel. There is not a day on the field that someone will not talk about his laughter, his drama play, his colorful Creole and his dedication.

We all miss you, Petro!

Florence Sergile

Acknowledgements

We would like to thank all of our co-workers of the University of Florida Macaya Biosphere Reserve Project who participated in various aspects of this project, and who contributed so much under very difficult conditions. Some of our colleagues who contributed valuable ideas and much-needed assistance were Paul Monaghan, Micki Swishser, Jenness McBride, Paul Paryski and John Hermanson. We acknowledge and appreciate the support of Edmond Magny and Louis Buteau of MARNDR and Albert Mangones of ISPAN. We also acknowledge with appreciation our associates at USAID, particularly Michelet Fontaine, Larry Harms, Catherine McIntyre, Kevin Mullally, and Charles-Emile Philoctete. We are very grateful to those many Haitians who assisted us as we did field work. And, we are grateful for the assistance of the following people in producing this volume: Beth Ramey for editing and layout, and Laurie Walz and Linda Chandler for artwork. Of course, none of the work would have been possible without the support of Missy Woods.



ACRONYMS

Many organizations, individuals and terms are referred to in this volume. We are providing a list of the major terms used. When a name for an institution or organization is used for the first time in the text it is spelled out completely, but thereafter usually the acronym is used.

COSAR	Cooperative de Saint Andre de Rendel
DMRE	Departement des Mines et des Ressources Energetiques
DRN	Direction des Ressources Naturelles
EMAF	Ecole Moyenne d'Agroforesterie
FLMNH	Florida Museum of Natural History
GOH	Government of Haiti
ISPAN	Institute de Sauvegarde du Patrimoine National
IUCN	International Union for the Conservation of Nature and Natural Resources
MBRP	Macaya Biosphere Reserve Project of the University of Florida
MBR	Macaya Biosphere Reserve
MARNDR	Ministere de l'Agriculture, des Ressources Naturelles et du Developpement Rural; sometimes known as Damien
NGO	Non-Governmental Organization
PVO	Private Voluntary Organization
SPE	Service de Protection de l'Environnement
Parcs Haiti	Parcs Nationaux Naturels d'Haiti
SAHPE	Societe Audubon d'Haiti pour la Protection de l'Environnement
SRF	Service des Ressources Forestieres
USAID	United States Agency for International Development
UNEP	United National Environment Program
UNESCO	United Nations Educational, Scientific and Cultural Organizations
WBP	World Bank Project
WCS	World Conservation Strategy
WWF	World Wildlife Fund

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RESUME

Ce document traite de l'histoire du projet de réserve de la biosphère de Macaya. Il est divisé en trois chapitres. Le premier est celui des sondages de 1988-1990. C'est un document de base qui statue les caractères socio-économiques et biophysiques du parc national Macaya et de ses zones tampons. Il fournit des recommandations sur l'utilisation de l'espace et propose des stratégies à court et long terme. Le deuxième chapitre est le rapport des activités entreprises par le projet du 1er octobre 1988 au 31 mai 1992. Le troisième propose un plan futur de gestion pour le parc national et la réserve de la biosphère de Macaya. En dernier lieu, la synthèse des documents produits à partir des recherches de l'Université de Floride grâce aux fonds de l'USAID y a été inclus.

LE SONDAGE DE LA REGION DE MACAYA

Entre novembre 1988 et janvier 1989, l'université de Floride entreprit une enquête de reconnaissance autour du parc national du Pic Macaya. L'objectif de cette enquête était d'évaluer le statut de l'environnement socio-économique et biophysique de ces régions, en vue d'établir un plan de gestion pour la réserve de la biosphère de Macaya. Deux concepts sont pris en considération: 1) le parc national du Pic Macaya représentant la zone réservée est compris à l'intérieur d'un système global rationnellement géré où les habitants de la zone en sont les gérants; 2) la gestion du parc dépend de celle des terres l'entourant qui devraient être zonées, classifiées, stabilisées et améliorées pour leur utilisation durable au bénéfice des populations avoisinantes. Le zonage comprend: 1) la zone réservée qui abrite les terres les plus fragiles et accuse une diversité biologique importante. 2) La zone d'utilisation spéciale est composée des aires à caractéristiques écologiques et biologiques où une stricte protection est impossible. Cette zone comprend les têtes de cours d'eau et les étangs qui sont d'une importance capitale pour la population et la balance écologique. 3) La zone de forêt qui est caractérisée par de fortes pentes où les sols mériteraient d'être stabilisés mais n'accuse aucune priorité écologique. 4) la zone d'agrosylviculture qui comprend les pentes modérées où l'élevage, la fruiticulture, caféiculture et production d'herbes sont possibles et recommandés. 5) la zone d'agriculture qui inclut les plateaux et les aires à faible pente où la production agricole intensive est possible moyennant d'y apporter une technologie adéquate pour augmenter les revenus de la population (Appendix I. Map of the Macaya Biosphere Reserve: Functional zones). A l'intérieur de ces zones, beaucoup d'espaces doivent être réhabilités parce que leur régénération naturelle est impossible. Ce volet est important dans l'aménagement des zones forestières et d'agrosylviculture et implique un engagement à long terme de la part des agences de financement et de gestion envers la population de la région.

Statut socio-économique et de l'environnement

La sphère d'étude comprend les zones de Formond, Trois Sources et Déglacis où de grands changements biophysiques se sont matérialisés entre 1983 et 1988. L'économie de la région est basée sur la vente des produits récoltés et du bétail vers les marchés de Ducis, Camp Perrin, Chantal, Rendel et Port-à-Piment. Les cultures les plus "rentables" de la région sont les pois noirs, l'igname, la patate, le café et la banane. Le travail agricole est effectué par des équipes ou "esquades" ou par des employés agricoles rémunérés à la journée. D'autres emplois existent et comprennent les professeurs, les petits commerçants, les représentants de l'état, et les artisans. La saison de plantation débute entre novembre et décembre avec la préparation de sol. Les semis sont effectués en janvier et les femmes entreprennent le sarclage vers le mois de mars. En mai elles récoltent les pois noirs et du séchage et de la commercialisation. Entre mai et juin, le maïs est récolté, les sols sont encore préparés pour la grande saison des pois, en juillet les semis sont effectués. En octobre la patate et les pois sont récoltés et plus tard le manioc et les ignames le sont à leur tour.

L'élevage est extrêmement important dans la région et comprend le gros et menu bétail. Les paysans les plus aisés possèdent des animaux tandis que les plus pauvres servent de gardien. Très peu de familles possèdent des mules ou chevaux. Les parasites particulièrement tiques et taons constituent un gros problème.

La région du parc constitue une source de bois de construction. Les efforts des agents de parcs n'ont pu arrêter cette activité que dans la région de Formond. On compte deux ou trois charbonniers. Le bois est utilisé pour les clôtures, les tuteurs, et la fabrication de la chaux qui en consomme beaucoup.

La zone de Formond comprend les localités de Portal Formond ou Formond, Cavalier, Durand, Casse Cou, Dalest, Sous Bois, Kay Tilus, Kay Itil, Kay Chale. Cette zone est habitée par environ 785 âmes qui exploitent les terres de l'état comme fermiers, en deux moitiés et/ou vendent leur main d'oeuvre. Les fermiers les plus riches de la région travaillent leurs propres terres et celles du domaine de l'état. La région comprend un conseil communautaire dominé par les plus riches familles, deux églises et une petite école primaire.

Les mornes du domaine de l'état exploités par les plus riches familles de la région, se régénèrent naturellement grâce à la présence des agents de parc qui interdisaient l'agriculture dans ces zones. Le maquis de Sous Bois souffre de dommages considérables en faveur de jardins. De gros problèmes d'érosion sont visibles à Casse Cou et à Dalest. Le haut Morne Cavalier se régénère naturellement et les Remparts sont restés intacts. Dans la région de Kay Tilus/Kay Itil (1100-1700 m) les pentes sont très dégradées et onze éboulements ont été rapportés. Le brûlis est coutume à chaque saison de plantation sur ces avalanches, et menace la forêt de pins en amont. Cette zone représente un important bassin versant de la rivière de l'Acul.

La région de Trois Sources est habitée par 125 âmes et comprend les localités de Grand'Plaine, Graine Mouton, Plaine Mou, Tombe Kriye, Tremper et Sivette. Son système agricole est différent de celui de Formond à cause de l'altitude, du relief et de la précipitation. Les pois noirs et la banane sont les denrées principales. La majorité des cultivateurs de la zone ne sont pas de la région mais viennent de Rendel. Les terres sont exploitées en fermes de l'état et en deux moitiés particulièrement sur les propriétés d'absentéismes. La COSAR occupe 596 hectares. L'impact de la COSAR et les liens unissant les coopérateurs et les cultivateurs de Trois Sources sont très importants. Le fragile bassin versant de ces sources est complètement dégradé par la coupe abusive des pins.

Déglacis est habité par 400 âmes environ et comprend les localités de Palencont et Déglacis. Cette région a été exploitée vers 1965 par Ti Yoyo Valerius. Elle est caractérisée par un canyon et des éboulements où se pratique une "dangereuse" agriculture. Les versants des mornes sont trop escarpés pour supporter une production rentable. Une forte immigration s'opère dans

cette région lors de la saison de plantation des pois noirs. Le brûlis s'étend sur des pentes fragile. La dégradation de cette zone est extensive et menace les mornes Formond et Macaya, une importante part du bassin versant de la Grande Ravine du Sud et les régions en aval.

La préservation du parc Macaya exige une exploitation rationnelle des terres et la collaboration des organisations communautaires. L'interdiction de cultiver dans les zones réservées a défavorisé les paysans les plus pauvres.. L'agrosylviculture représente le système le mieux approprié de l'utilisation des terres. Les programmes suggérés à court terme mettent l'accent sur la rémunération par une amélioration des systèmes cultureux et l'emploi des paysans les plus démunis par le projet dans les programmes de réhabilitation. Les programmes à long termes comprennent l'établissement et exploitation de bosquets d'arbres et de parcelles protégées par des haies vives, la production et utilisation de fourrage, l'installation d'infrastructures, et la stabilisation de zones dégradées.

LE PROJET DE RESERVE DE LA BIOSPHERE DE MACAYA

L'objectif principal de ce projet est la conservation des ressources naturelles et la réhabilitation des écosystèmes uniques de la zone du Pic Macaya en assurant le développement économique durable de la région.

L'université de Floride développa un plan de gestion des ressources naturelles en appliquant dans la zone tampon les principes de réserves de la biosphère. Cette importante région est le bassin versant de 5 importantes rivières (la Grande Ravine du Sud, la rivière de l'Acul, de Port-à-Piment, des Roseaux et la Guinaudée) qui fournissent l'eau potable et d'irrigation aux régions en aval. Les dommages causés par la dégradation de ces mornes sont estimés à plus de 19 million de dollars américains chaque année. Les forêts des mornes Formond et Macaya abritent beaucoup d'espèces endémiques menacées d'extinction à cause de la rapide déforestation des zones avoisinantes.

Le projet fut financé par l'USAID du 1er Octobre 1987 au 31 mai 1992. L'aire d'intervention se situe autour du Morne Macaya et inclue environ 50.000 ha. Le parc national Macaya est unique. Il contient les derniers vestiges des forêts pluvieuses de montagne, accuse 6000 mm de pluies par an et une immense diversité biologique composée de 149 espèces de plantes, 2 mammifères et 20 espèces d'oiseaux endémiques dont plusieurs menacées d'extinction.

Les activités principales du projet n'ont commencé qu'en novembre 1988. Les premiers plans de gestion se concentrant essentiellement sur la gestion du parc national ont été amendés pour y inclure la zone tampon à cause des troubles politiques.

Objectifs et stratégies

Le projet a travaillé en collaboration avec le Service de Protection de l'Environnement, l'agronome du district agricole sud et le service des Forêts du Ministère de l'Agriculture, le Corps de la paix, l'ISPAN, l'ORE et l'UNICORS. Il comprenait un effectif de 25 employés. Les objectifs spécifiques et les stratégies pour atteindre l'objectif principal sont les suivants:

1. Gestion et développement du parc et de ces environs en réserve de la biosphère.

Les réalisations pour atteindre cet objectif comprennent le zonage fonctionnel pour l'utilisation rationnelle des terres, la production de cartes de base, la pose de 4000 mètres de clôture autour de zones critiques de la zone réservée, des photographies aériennes et images satellites qui révèlent une dégradation accrue, le développement d'un plan de rétablissement des espèces menacées Ti Chitte ou Ti Quatre yeux (*Xenolgea montana*), Nez long (*Solenodon paradoxus*) Chat-Ouan ou Diablotin (*Pterodroma hasitata*), la production d'enseignes, de panneaux explicatifs, de posters et autocollants, et la création d'infrastructures physiques et institutionnelles.

2. Assistance à 1750 paysans de la réserve pour augmenter leur revenu en utilisant des techniques compatibles à la conservation des ressources naturelles.

Ce volet comprend le choix des coopérateurs primaires du projet (PPC), la création de l'Association des Guides de Macaya (AGM), l'assistance technique et la vulgarisation agrosylvicole, des essais sur 300 parcelles de démonstration et 30 variétés de légumes, la distribution des semences de légumes, de pois noirs et de porcs rustiques et l'établissement d'environ 1000 mètres de haies vives chez des coopérateurs. Trois études socio-économiques ont révélés les dynamiques de population de la région. Les femmes sont les plus enclines à migrer vers les villes, plus d'enfants ont été scolarisés entre 1989 et 1991, la plupart des gens interviewés étaient nés dans la région et la majorité de la population est jeune. Nous recommandons une diversification dans la production agricole, un programme intensif d'éducation environnementale et de vulgarisation agricole sur un plus long terme. Nous estimons que 650.000 gourdes ont été générées dans la région grâce aux porcs.

3. Réhabilitation des zones critiques et dégradées c'est à dire des pentes, sources, habitats menacées pour l'augmentation de la capacité des bassins versants et la protection de la diversité biologique des zones réservées.

Ce programme s'est effectué dans les régions de Malini/Dupouy, Trois Sources, Kay Tilus et Chambellan/Déglacis. Il comprend l'identification des communautés de plantes et du pourcentage de chaque espèce évoluant dans le système, la collection des semences, leur reproduction en pépinière et leur mise en terre. Près de 40.000 semences et 120.000 plantules ont été mise en terre. Des fiches techniques de pré-mise en terre sont prêtes pour douze espèces. Des structures de stabilisation d'éboulements ont été établies sur 60 avalanches. Cependant, leur succès dépend de l'utilisation des sols et des systèmes de production de la région. Un total de 169 seuils a été érigé dans les région de Ravine Cresson et Fond Bleu.

4. Création d'une base d'information comprenant des données sur les études effectuées dans la région de Macaya.

Une importante bibliographie comptant 3500 titres sur l'histoire et les ressources naturelles d'Haiti et de la Caraïbe est en phase de publication. Les études menées par l'université de Floride ont déjà été distribuée en Haiti.

5. Développement d'un programme éducatif visant à: i) la formation du personnel local; ii) la production de brochures et livrets sur les aires protégées et les oiseaux; iii) production d'un programme d'éducation environnementale; iv) production et pose d'enseignes dans le parc.

Une série de séminaires a été produite pour les techniciens et les villageois des régions de Formond, Cavalier et Déglacis. La direction technique du projet a assisté à deux séminaires sur les parcs nationaux et espaces protégés, à un séminaire sur le diagnostic des facteurs de l'érosion et raisonnement des aménagements. Les techniciens du projet en ont assistés à cinq sur les propositions de projet, premiers soins vétérinaires, tenure de terre et techniques de conservation des ressources naturelles.. Ils en ont produits deux pour les villageois de la réserve.

Un premier vidéo a été produit en Septembre 1989. Un second "The Forest in the Clouds" (La forêt dans les nuages) a été produit pour émission sur PBS (Public Broadcasting System). Plusieurs conférences ont été prononcées pour informer le public de l'importance de la région de Macaya.

Le temps est un facteur limitant dans les programmes de conservation et gestion des ressources naturelles. L'élevage libre reste encore un des plus gros problèmes à résoudre dans la région. Les parcelles emblavées de plantules sont piétinées par le bétail, malgré la clôture qui délimite les zones de production de la zone réservée. Le programme de fourrage mixte (pâture et

foin) a débuté par la multiplication et la distribution de matériel végétal chez des coopérateurs, cependant le temps a fait défaut pour son implantation effective. Les programmes de formation sont nécessaires. La collaboration avec le Ministère de l'Agriculture s'est accrue, et l'entente avec d'autres institutions a été initiée. L'expérience d'une ONG devrait être partagée avec les autres.

LE FUTUR DU PARC NATIONAL MACAYA

Le concept de protection des aires naturelles en Haïti remonte à 1926. Il est étroitement lié à la protection des réserves forestières et des bassins versants et comporte une copieuse législation. Les objectifs du programme de parcs nationaux naturels en Haïti sont: 1) La protection des conditions et processus écologiques naturels; 2) la promotion du patrimoine naturel national et de son unique valeur; 3) le développement de programmes touristiques. Le programme de conservation des parcs nationaux devrait être développé dans les plus brefs délais car la flore et la faune du pays se détériorent à un rythme alarmant.

Les objectifs des parcs nationaux naturels sont:

1. *La protection des conditions et processus écologiques naturels.* Le bénéfice de ces actions est: a) la préservation des bassins versants, donc l'amélioration de la qualité de vie de tous les Haïtiens dans les aires adjacentes aux parcs nationaux; b) la préservation de la diversité des espèces naturelles donc du patrimoine national naturel.

2. *La promotion du patrimoine national naturel.* Ses bénéfices sont: a) l'enrichissement des connaissances des citoyens haïtiens sur les caractéristiques uniques de leur pays; b) la possibilité que de sages décisions soient prises pour le développement et l'utilisation durable des ressources naturelles à long terme.

3. *Le développement de programmes touristiques et récréatifs.* Ce volet prendrait avantage de la situation et de la beauté unique des parcs et des caractéristiques spéciales comme la flore, la faune ou la géologie. Il est certain que les Haïtiens tireraient parti des parcs au niveau régional et national sans abîmer leur qualité, si le plan de gestion est développé et établi.

Administration des parcs nationaux et réserves naturelles

Ce plan propose une unité administrative, "*Parcs Nationaux Naturel d'Haïti*" où Parcs Haïti, pour gérer toutes les divisions fonctionnelles d'un programme global de conservation. Cette organisation aurait à sa tête un directeur et serait sous la tutelle d'un seul ministère. Parcs Haïti serait en charge des sites naturels déjà identifiés par le gouvernement, proposerait d'autres sites uniques (plages, grottes, paysages uniques, jardin botanique etc.) et travaillerait en étroite collaboration avec le MARNDR, l'ISPAN et d'autres institutions concernées.

Parcs Haïti devrait être une institution autonome qui mettrait sur pied un programme de conservation fonctionnel pour Haïti. Elle serait supervisée par un conseil, le Conseil Consultatif des Parcs Nationaux (CCPN) formé des directeurs de l'ISPAN, des Ressources Naturelles du MARNDR, de l'Office National du Tourisme, du président de la Société Audubon Haïti, d'un citoyen haïtien dûment intéressé à la conservation, d'un représentant d'une organisation internationale de conservation et du directeur de Parcs Haïti qui serait le président et coordonnateur de ce conseil. Les membres se réuniraient au moins une fois l'an pour discuter des programmes et de leur développement qui serait effectué avec le personnel proposé dans le tableau suivant.

**PARCS NATIONAUX NATURELS D'HAÏTI
(PARCS HAÏTI)**

Bureau Central

Directeur
Assistant Directeur en Education et Récréation
Assistant Directeur en Conservation et Recherche
Secrétaire-comptable
Secrétaire-bibliothécaire
Chauffeur
Messager
Gardien

Parc La Visite

Superviseur de Parc
Gardien
Cuisinière
Agents de Parcs (7)
Ouvriers (15)

Parc Macaya

Superviseur de Parc
Gardien
Cuisinière
Agents de Parcs (7)
Ouvriers (15)

Deux alternatives sont proposées pour la création immédiate de Parc Haïti. La première prévoit un contrat avec une institution internationale de conservation qui s'occuperait du programme en employant et entraînant des Haïtiens sur une période de cinq ans. La deuxième suggère, comme le projet de la Banque Mondiale, d'utiliser le personnel du Service de Protection de l'Environnement du MARNDR. Cependant, il est recommandé que Parcs Haïti n'ait qu'un seul directeur.

Les responsabilités de Parcs Haïti sont les suivantes: Exécution du plan de gestion, fonctionnement de Parcs Haïti, usage public, gestion de l'environnement des parcs, observation et évaluation des caractéristiques naturelles, et développement des programmes éducatifs. Ces différentes activités seraient entreprises par le directeur et ces assistants pendant 1) la phase d'exécution qui comprendrait la révision du plan de gestion, les appels d'offres et embauchages, et 2) la phase de fonctionnement qui engloberait la coordination entre différentes institutions, les présentations et les relations publiques, la supervision du budget et la coordination sur le terrain. Le budget quinquennal de ces opérations est de 12.573.950 gourdes et comprend l'installation, les constructions, les salaires et les opérations.

Le Parc National Pic Macaya

Il est situé dans le massif de la Hotte, à 36 km au nord-ouest de la ville des Cayes. On y accède difficilement, par la seule route voiturable de Ducis-Le Prêtre-Platons. Il s'étend autour de deux montagnes dominantes, le morne Formond (2219 m) et le Morne Macaya (2347 m) et inclut de profondes ravines, d'autres mornes de formation karstique et volcanique et des forêts de pins et de feuillus riches en espèces endémiques. Son climat accuse des précipitations de plus de 4000mm/an due aux alizés, aux vents d'est et aux nordés. Le parc constitue le château d'eau de la Grande Ravine du Sud, la rivière de l'Acul, de Port-à-Piment, des Roseaux et de la rivière Glace, dont la stabilité est affectée par le déboisement excessif en "faveur" de cultures non-rentables. Ce parc est grandement menacé par une poignée de paysans qui mettent en jeu la vie de milliers de gens en aval.

Le décret du 23 juin propose arbitrairement une surface de 2.000 ha, cependant elle devrait être de 7.500 ha et comprendraient les régions et caractéristiques suivantes (Appendix III. Réserve de la Biosphère de Macaya. Proposed management plan):

1) Le morne Macaya (2.347 m), le morne Formon (2.219 m), le pic le Ciel (2.170 m), Civette (1.533 m) et Grande Plaine (1.900 m) zones de recharge de la nappe phréatique et sources de 5 rivières. Ces zones devraient rester boisées car elles accusent des pentes de plus de 60%, retiennent des milliards de tonnes de sols et protègent les systèmes d'irrigation en aval. La réhabilitation de ces zones sera extrêmement coûteuse à la nation haïtienne, car d'accès difficile.

2) Bois Formond, Bois Durand situé entre la crête du morne Cavalier (1570 m), Sous Bois et les plaines de Durand et Formond. Zone de recharge de la nappe phréatique et d'importance capitale car particulièrement riche en espèces endémiques animales (oiseaux, agouti, reptiles et amphibiens) et végétales (orchidées, sapotilliers, mélastomes, calbassiers).

3) Mare Cochon/Diquillon de formation karstique et riche en espèces endémiques. Habitat unique et essentiel des nez longs (*Solenodon paradoxus*).

Cette surface de 7,500 ha ne représente que 0.27% du territoire national, et le minimum requis pour un programme de conservation de la région qui contient le plus haut taux d'endémisme de l'île et constitue le château d'eau des milliers d'hectares des régions en aval.

Géologie

La géologie de la région de Macaya est complexe et se situe entre le Crétacé inférieur et le Récent. Elle est caractérisée par deux formations, celle de Macaya composée de roches calcaires à topographie karstique et celle de Dumisseau caractérisée par un conglomérat de roches volcaniques basaltiques, de turbidites calcaires, de cherts, de grès siliceux, et par un relief très accidentés avec des gorges étroites. Ce sont des zones à fort ruissellement. De nombreuses failles traversent le parc du nord au sud et d'autres de l'est à l'ouest. La plus importante faille qui sépare le morne de Formond de celui de Macaya date du Miocène et traverse une bonne partie de la presqu'île du sud et le parc via la Grande Ravine du Sud. Des affleurements de basaltes peuvent être observés jusqu'à 1600 mètres sur le morne Formond. La plupart des géologues pensent que la formation Macaya superpose celle de Dumisseau.

Sols

Les formations géologiques régissent la composition des sols de la région qui varient entre des oxisols et des ultisols à pH légèrement acide ou neutre. La plaine de Formond et le "Rak Bois" Formond comporte de profond oxisols à pH neutre, tandis que celle de Durand, les mornes vers Cavalier, la zone de Kay Tilus, du Morne Formond et de Trois Sources ont des ultisols. Ces sols sont très susceptibles à l'érosion et sont déjà menacés dans beaucoup de régions.

Flore

La flore du parc comporte près de 500 espèces de plantes vasculaires dont 102 fougères (3 endémiques), 99 espèces de mousses et 49 espèces d'hépatiques, 1 conifère endémique, 141 orchidées (38 endémiques du massif de la Hotte et 58 de l'île d'Haïti), 367 autres plantes à fleurs (55 endémiques du massif de la Hotte, 18 du Massif de la Hotte et de la Selle, 39 de l'île). Trois espèces sur dix sont endémiques de Macaya. Un bon nombre de ces espèces sont sensibles à la déforestation et risquent de disparaître malgré leur potentiel sylvicole, médicinal et ornemental non encore exploité. Ces plantes sont groupées dans la forêt de pins, des forêts de feuillus, des bosquets denses, et/ou fragmentés, des jardins en jachère à l'intérieur des zones écologiques de Holdridge. La forêt très humide de montagne de basse altitude se situe dans la région de la plaine de Formond et des "rak bois" Formond/Durand, la forêt très humide de la zone sous-tropicale au nord du pic Macaya. Les crêtes des mornes Formond et Macaya sont du type de forêt très humide de montagne. Les régions du haut de la Grande Ravine du Sud et de Diquillon sont des forêts pluvieuses de montagne de basse altitude et finalement la forêt pluvieuse de la zone sous tropicale se situe dans la région du plateau de Mare Cochon vers Catiche.

Faune

Le parc national de Macaya compte 11 espèces de papillons dont le *Calisto loxias* connu seulement de cette région. Cinquante sept espèces de mollusques ont été collectés en 1983 dont 2 nouveaux genres, 27 nouvelles espèces, 23 endémiques. L'inventaire herpétologique de la zone compte 18 amphibiens et 14 reptiles (analyses en cours).

La faune aviaire d'Haïti est riche et comporte 73 espèces d'oiseaux dont 24 endémiques. Un grand nombre de migrateurs nichent uniquement dans les forêts montagneuses d'Haïti comme le "Chat-huant", diabolon ou pétrel (*Pterodroma hasitata*). Le parc national Macaya compte 65 espèces dont le Caleçon Rouge (*Temnotrogon roseigaster*), le kat Je Sid (*Phaenicopohilus poliocephalus*), le Ti Chit Kat Je (*Xenoligea montana*) et le colibri (*Todus angustirostris*). Les analyses d'enquêtes confirment qu'onze espèces sont vulnérables à cause de la dégradation de leur habitat et menacés d'extinction. Des programmes de rétablissement et de conservation sont urgents, ils seront difficiles mais pas impossibles. La disparition d'espèces entraîne tout un déséquilibre qui affecte grandement les hommes à court et à long terme.

Les mammifères terrestres endémiques du parc ne sont qu'au nombre de deux. Ce sont l'agouti ou zagouti (*Plagiodontia aedium*) et le nez long ou nen long (*Solenodon paradoxus*). L'agouti est un rongeur nocturne qui vit dans des crevasses de formation karstique boisée ou se réfugie entre les racines de gros arbres particulièrement le bois tremblé (*Didymopanax tremulum*) dans les montagnes aux environs de 2.200 m d'altitude. Il se nourrit d'écorce de jeunes branches, de rameaux, de feuilles et de fruits de bois tremblé, d'avocat marron (*Persea anomala*), de lamandier (*Prunus* sp.). Cependant, il est considéré comme un fléau, puisqu'il s'attaque aux racines de malanga, d'igname, de patates, aux parties végétales des pois noirs, au maïs quand ces cultures sont établies près de ces sites naturels. Il boit très peu d'eau, donc l'obtient probablement de sa nourriture. Les agoutis sont de passables grimpeurs et utilisent leur queue semi-préhensile pour se mouvoir de branches en branches. Ils ont une vie assez longue, mais un taux de reproduction bas. Les femelles donnent naissance à un seul petit par portée après une gestation de plus de quatre mois. La couleuvre endormie (*Epicrates striatus*), les chiens et les chats "marrons" sont ces plus grands prédateurs. Il a toujours été un animal rare, et est encore vivant grâce à son comportement secret, cependant si sa niche écologique est détruite, il a de fortes chances de disparaître.

Le nez long est un insectivore sylvestre nocturne qui vit dans les crevasses des formations karstiques de la région de Catiché et Duchity, entre 500 et 1.000 m d'altitude. Il se nourrit d'escargots, de centipèdes, de scorpions, de petits vertébrés et d'œufs d'anolis, grenouilles, oiseaux. Le taux de fécondité est faible. La femelle donne naissance à 2 petits par an. La gestation est de plus de 90 jours. La couleuvre endormie (*Epicrates striatus*), la chouette ou frisé (*Tyto alba*), les chiens et les chats "marrons", la mangouste sont ses prédateurs. Il est très vulnérable car mauvais grimpeur et pas aussi alerte que l'agouti. Il est très rare dans les limites du parc et disparaîtra dans les 10 ans à venir si de gros efforts de conservation ne sont pas entrepris.

Paléontologie

La faune des fantômes du passé est abondante. Elle comptait 25 espèces, il y a 20.000 ans. Des fossils trouvés dans les dépressions (touings) documentent la présence de 5 insectivores, 13 rongeurs, 1 saï (singe), 4 aïs et 1 mégathérium (paresseux de l'ordre des édentés). Les fouilles ont révélées que les mammifères collectés dans la région de Formond vivaient depuis le Pleistocène supérieur et l'Holocène inférieur.

Gestion et utilisation de l'espace

La gestion du parc national Macaya comprendrait les volets suivants sur des espaces bien spécifiques: 1) la récréation ouvertes aux touristes; 2) l'enrichissement où se déroulent les expositions pour l'information du public sur le parc; 3) la préservation biologique qui représente l'activité la plus importante du programme de conservation; 4) la régénération qui permet la

restauration des systèmes naturels; 5) la recherche qui comprend des stations climatologiques et laboratoires; 6) les guérites de sécurité et d'information; 7) les zones d'utilisation spéciale qui ne peuvent être visitées qu'avec un permis; 8) les zones de service et d'entretien où sont placés les dépôts et où le personnel fait la mise en place pour les différents programmes.

Le plan de gestion du Parc National Pic Macaya est divisé en neuf thèmes: 1) l'Administration dont serait responsable un superviseur de parc; 2) l'entretien qui serait entrepris par une équipe à plein temps; 3) la sécurité assurée par des agents de parcs entraînés; 4) la récréation et le tourisme; 5) l'enrichissement et l'interprétation; 6) les relations publiques; 7) la recherche; 8) le plan quinquennal de conservation qui comprend la préservation des systèmes existants et leur régénération dans les aires critiques, la préservation de la faune et de la flore, la stabilisation des bassins versants, des aires érodées et la régénération des forêts; 9) le plan décennal qui comprend le bornage du parc et des zones réservées, les appels d'offres, l'emploi et la formation du personnel nécessaire, la construction des guérites de sécurité, d'autres centres d'accueil dans les régions de la Guinaudée, Déglacis, Trois Sources, l'établissement des plans de conservation et de préservation, le développement des attractions pour touristes, la pose d'affiches et panneaux, les recherches sur les espèces endémiques, l'amélioration et l'entretien des structures et des routes de pénétration.

Les Réserves de la Biosphère

Une stratégie qui établirait un programme de conservation des ressources naturelles en Haïti doit combiner développement durable et conservation pour satisfaire les besoins d'une population sans endommager les écosystèmes naturels. Ceci est possible grâce au développement de réserves de la biosphère. Ce concept permet de promouvoir et maintenir la balance entre les hommes et le milieu naturel par une utilisation adéquate de l'espace. Il semble convenir à la gestion des aires protégées en Haïti. Une réserve idéale est composée d'une aire réservée entourée de zones fonctionnelles. Dans la région de Macaya, le projet de l'université de Floride a développé ce modèle qui n'est toujours pas officiel auprès de l'UNESCO. Elle consiste en une zone réservée qui est le parc national Macaya, et des zones fonctionnelles qui comprennent l'agriculture, l'agrosylviculture et l'élevage, des zones d'utilisation spéciale et de forêts. Chacune des zones peut être gérée par des organismes différents moyennant une bonne collaboration. Ce serait le rôle de Parcs Haïti de protéger le parc national et de coordonner les activités de production avec d'autres organisations.

La gestion des aires protégées requiert un plan actif et dynamique et son exécution à long terme devra être implantée dans les plus brefs délais, autrement les 2% de forêts et de végétation naturelles qui restent encore au pays seront rapidement détruits. Le projet de l'université de Floride financé par l'USAID a fonctionné pendant cinq ans et aurait encore beaucoup à faire pour protéger le parc. Plus de 12 sites naturels devraient être gérés. Le projet de la Banque Mondiale qui embrasserait un grand volet de conservation et de protection de l'environnement est en suspens. En attendant qu'il soit implanté, l'université de Floride a obtenu de la Fondation MacArthur un don qui supporte des activités de conservation et d'éducation environnementale entre 1991 et 1994. Ce projet est un "pont" entre celui financé par l'USAID entre 1986 et 1992. Son effectif ne comporte que 7 employés. La conservation des ressources, parcs et sites naturels en Haïti est urgente.



Dupouy Nursery



On site seminar with Primary Project Cooperators

INTRODUCTION

The original project strategy concentrated on establishing Parc National Pic Macaya, and conservation activities. By design, the University of Florida (UF) was to provide direct assistance to the Government of Haiti (GOH), specifically the Ministère de l'Agriculture, Ressources Naturelles et Développement Rural (MARNDR), to make Parc Macaya functional. Under modification of the project, the UF/MBRP was to cooperate with the MARNDR with no direct assistance.

The strategy was expanded to involve applying the concept of a biosphere reserve to the park and the buffer zone and to offer the peasant farmers of the area, who were and are the principal destroyers of the forests and ecosystems, new environmentally-appropriate agricultural techniques and other alternate means of generating income. This innovative strategy combining ecosystem protection and rehabilitation with sustainable development had been used in very few other countries in the world, and was on the cutting edge of new techniques for environmental protection. The theme of the highly-publicized June 1992 UN Conference of the Environment and Development in Rio de Janeiro was environmental protection and rehabilitation and sustainable development. This was also one of the recurrent themes of the International Union for the Conservation of Nature conference in Caracas held in February, 1992. In Haiti, as is true for many very poor countries, it is impossible to protect ecosystems, forest cover and natural resources by simply fencing off a park. Intense demographic pressure and increasing oppressive poverty had driven the peasant population surrounding the park area, roughly 7,000 persons, to clear and burn the very steep forested slopes in order to plant such crops as black beans and yams. Although the peasants realize that deforestation means the drying up of springs and rivers, soil erosion and seasonal flooding with loss of life, property and agricultural productivity, they are so poor and de-capitalized, they can think only of day-to-day survival. Thus non-destructive income-generating techniques and activities were provided for the peasant farmers in the region; in addition, the biosphere reserve concept of functional zoning (i.e., designating zones for specific functions such as total protection, forestry, agro-forestry or agriculture based on ecological, geophysical and social considerations) was applied to the region. Parc Macaya, with about 2,000 hectares and its buffer zone of 20,000 hectares were included as the Macaya Biosphere Reserve.

UF/MBRP activities were concentrated in three critical areas within the MBR: 1. Formon/Cavalier on the southern slopes of the Formon ridge, an area containing a relatively large peasant population and some less steep "plains"; this area contains the headwaters of the Acul and Torbeck rivers; 2. Riviere Trois Sources, the area to the west of Pic Formon and Macaya being exploited by non-resident peasant farmers and UNICORS. It includes the headwaters of the Riviere Guinaudee; 3. Deglacié, the area between the Formon and Macaya ridges to the east of Pic Macaya being exploited by pioneer peasant farmers and

containing the headwaters of the Ravine du Sud. All three areas were important as major watershed catchment areas and as important ecosystems and centers of biodiversity. A majority of the land in these target zones was state-owned and considered to be a part of Parc Macaya. All three areas were under great pressure from poor peasant farmers wishing to extend and expand their gardens. Much of the farming in these target zones was carried out using the traditional Haitian "deux moities" system in which a richer peasant would lease or seize state land and let poorer peasants clear and work the land in return for half the harvest. In the Riviere Trois Sources area, UNICORS owned large tracts of land supposedly dedicated to coffee plantations, but in fact, UNICORS often used the "deux moities" technique of exploitation.

The project was funded under USAID's Targeted Watershed Management Project, a large project whose objective was the protection from environmental degradation, primarily soil erosion, and the increase of agricultural productivity of the entire watershed area south of Pic Macaya. The project had two basic components, the Macaya protection component with the UF and Projet Sove Te (PST). PST was composed of four different local NGOs and a contracted umbrella organization, ARD, which was to provide administrative and technical assistance to the four NGOs and act as a intermediary between the NGOs and USAID. The UF/MBRP worked closely with two of the NGOs, the Organization for the Rehabilitation of the Environment (ORE) and the Union des Cooperatives de la Region du Sud (UNICORS), both of which undertook activities in the MBR zone. ORE worked in the Formon area where it promoted the use of alley cropping ("haies vives") and planting of fruit and coffee trees and improved the genetic stock of major crops. COSAR, one of UNICORS's member cooperatives, had purchased large tracts of land to clear cut to plant coffee and black beans and to harvest lumber. The UNICORS program under the TWMP concentrated on the construction of "haies vives", but principally in land lower than the MBR. Under the TWMP two meetings were held every six months to: 1) discuss and coordinate strategies and administer procedures and problems; and 2) to evaluate progress. The four NGOs, ARD/PST, USAID, and the UF/MBRP attended all meetings; the strategy meetings were also attended by MARNDR and other invited local organizations. The NGOs held separate meetings to which the UF/MBRP was not invited to discuss their particular problems.

The original strategy of the UF/MBRP included the selection of primary project cooperators (PPCs), selected peasant farmers representing a cross section of socioeconomic classes and geographic locations. First, the PPCs would help test new environmentally-appropriate agricultural techniques using the UF-developed Farming Systems Research methods. The PPCs received intensive training in the construction of "haies vives", in introducing new high revenue crops such as cabbage and carrots, in raising livestock, particularly the rustic pigs introduced by the UF/MBRP to augment incomes. These PPCs in turn would choose seven additional peasant farmers and teach these farmers what they had learned. These peasant farmers would then teach seven more resulting in a pyramidal effect of spreading technologies and knowledge.

The UF/MBRP was granted a first extension in 1990 to continue its activities until September 30, 1991 and a second extension in 1991 to continue until March 31, 1992. After three years of activities, USAID requested that the UF/MBRP modify its program again to concentrate more on conservation and rehabilitation, leaving most agricultural development activities to ORE. Plans were made for an intensive rehabilitation program.

On September 30, 1991, a bloody coup d'etat overthrew the elected government of Haiti forcing its President, Jean-Bertrand Aristide into exile. In reaction to the coup d'etat, an embargo on trade with Haiti was applied by the OAS and its member states, and USAID suspended all activities of the projects it funded, requesting that all U.S. based employees be repatriated. Therefore, all UF/MBRP activities in Haiti were suspended as of 1 October 1991. The administrative coordinator and the technical coordinator were requested to leave Haiti which they did under USAID's departure and evacuation procedures. The administrative coordinator received permission and return to Haiti on February 22, 1992. Upon his return, the administrative coordinator requested a two-month extension to secure UF/MBRP facilities and to complete in an orderly manner essential activities such as the planting of trees already in the UF/MBRP nurseries, training of community leader in natural resource protection, securing facilities and completing the training of three local veterinary technicians. The extension received almost immediate verbal acceptance but it was not until the beginning of May that the final extension document was signed, leaving very little time to complete all activities. The UF/MBRP officially closed on May 31, 1992.

Chapter I

The 1988-1989 Sondeo of the Macaya Area

Introduction

A special consultation team was formed by the University of Florida for the purpose of conducting a rapid rural appraisal (Sondeo) of the target region of the Macaya Biosphere Reserve (MBR). The team conducted the Sondeo during the period 27 November 1988 to 12 December 1988. Team members included: Charles Woods, M. E. Swisher, John Lichte, Roy Voss, Paul Monaghan, Paul Paryski, Suze Mathieu, and Michel Aubry.

The Sondeo team's efforts were preceded by a study conducted by Mr. Paul Monaghan in the period May to July, 1988. Mr. Monaghan submitted a final report "Three Communities in the Macaya Biosphere Reserve: A Report on Fieldwork in the Southern Mountains of Haiti," on November 1, 1988, to the Office of International Programs, University of Florida. Mr. Monaghan's report, as well as his extensive briefing of Sondeo team members, provided a crucial input for this Sondeo. Many of the conclusions, as well as a great deal of the content, of Mr. Monaghan's report is included here.

The Sondeo team attempted to achieve four major objectives.

- (1) An assessment of the current status and recent changes in the biogeophysical environment in the proposed MBR area. This assessment was viewed as a way to provide the basis for evaluating the impact of future activities within the region, as well as to provide the MBR project team with baseline data to utilize in planning project activities.
- (2) A second major objective was to gain a basic understanding of existent farming systems within the region. This included identifying problems, constraints, and potentials that were common in the region then, as well as today.

- (3) A major purpose of the Sondeo was to develop generalizations and conclusions based on the observations that were made. These generalizations and conclusions permitted the development of hypotheses which served as the basis for future planning and activities in the Macaya area.
- (4) The team also formulated a strategy for the MBR project for 1989-1991. This strategy was not designed to provide a detailed plan of work for the resident project team, but rather to provide an overview or direction for team activities. It provided the basis for development of a detailed plan of work. This plan of work, which was originally planned on being completed by March 1, 1989, was the basis for the entire center part of the project.

The word "Sondeo" comes from the Spanish *Sondear* (to sound, to fathom, to find out, to learn about). A Sondeo is conducted by identifying a well-balanced team with representatives of all of the major disciplines of concern. In this case, the team was composed of representatives with advanced training in agriculture, farming systems, agricultural economics, anthropology, history, Haitian culture, conservation, and biological diversity. The team included three individuals with complete fluency in Haitian Creole. The methodology of this Sondeo was to divide the overall team into three groups, with at least one advanced Creole speaker in each group. The teams interviewed farm families in depth. Each interview followed a predesignated series of questions. However, each interview was allowed to concentrate on areas of special concern of each individual farmer.

The members of the Sondeo team met as a group each night to review the findings of the day, and to discuss techniques and developments. The teams therefore were able to work in a very coordinated manner, and to revise the scope of the questions each day to respond to new information that appeared in the interviews. The composition of the teams changed each day to insure the group worked as an overall Sondeo rather than as three smaller teams. All of the results from one target area were summarized by the Sondeo team in a debriefing before the team moved on to a second target area. The Sondeo team worked at Formon and Trois Sources (see itinerary below). A debriefing was held at Camp Perrin and in Port-au-Prince at the end of the Sondeo.

The Sondeo team traveled to Port-au-Prince on November 27 and on November 28 was briefed by BRED, Robert Cassagnol, and Michelle Oriol at a morning meeting at USAID/Haiti. On November 29 the team traveled to the Plain of Formon and spent November 29-December 2 conducting interviews and undertaking an analysis of local conditions on the Plain of Formon, the Plain of Durand, and the valley of Cavalier. On December 3 the team traveled to Rendel where they spoke with Father Roger Hallee of UNICORS. On December 4 the team went by mule to the "Gran Plenn" (Grande Plaine) on the west side of Pic Macaya where they worked from December 4-6. The team discussed the results of the Sondeo and produced a preliminary draft report on December 7-8. On the morning of December 9 the Sondeo team met with representatives of ARD, ORE, DCCH, and UNICORS to discuss our preliminary analysis, and to seek commentary from the NGOs. The evening of December 9 the team met with Robert Cassagnol in Port-au-Prince. Additional meetings were held on December 10-11, and the document was further refined. A debriefing was held at USAID/Haiti on the morning of December 12, and the team returned to Florida that same day. Between December 13 and January 24 the document was reviewed and critiqued at the University of Florida. The date of completion of the final Sondeo report is 25 January 1989.

In writing this report a number of regional names were encountered. When an official name for a locality existed we followed that spelling. When a locality had only a local name we have reproduced that spelling in Creole as closely as possible based on the usage in the Haitian Creole Dictionary by Albert Valdman. When a Creole name is used for the first time it is usually in quotes, with a French translation following it in parentheses. As an example, the "Gran Plenn" (Grande Plaine) region west of Pic Formon does not have an official designation on maps or in the Gazetteer, and so the Creole name has been used to match local usage. We hope that this is not confusing to readers, and have used Creole names so that the Sondeo document will be of maximum value in Haiti.

CURRENT STATUS AND RECENT CHANGES IN THE BIOGEOPHYSICAL ENVIRONMENT

Since the creation of Parc National Macaya (Macaya National Park) in 1983, there have been significant changes in the biogeophysical environment in the MBR. The following is a summary description of those changes observed during the Sondeo, with comments on their concomitant probable causes.

Formon. The vegetative cover on the slopes above the Plain of Durand ("Diran") and the Plain of Formon is undergoing natural regeneration. Previously, these very steep slopes, largely Terre l'Etat (state land), were controlled by richer families in the area and were intensively sharecropped by farmers of lower socioeconomic status. Active enforcement of interdiction of agriculture by the Agents de Parc (National Park Guards), hired and supervised by the Institut de Sauvegard du Patrimoine National (ISPAN), has played a major role in allowing regeneration to occur. There appear to be no actively cultivated parcels (gardens) above 1300 meters, although regeneration is being somewhat retarded by use of these slopes as pasture for livestock (cattle, goats, and sheep).

In contrast, the region known as Kay Tilus/Kay Itil, the highlands (1100-1700 meters) to the northeast of Formon proper and east of the Ravine Seche (including the beginnings of the Ravine Casse Cou), has suffered extreme environmental degradation. In this area, the major source of water generation for the Riviere l'Acul, eleven very large (20 meters + wide, 50 meters long) landslides were observed. Numerous gardens are present on slopes of more than sixty degrees and vegetative cover is reduced to short grasses. Much of the erosion in this area was probably caused by the very heavy rainfall accompanying Hurricane Gilbert. Continual and progressive destruction (slash and burn clearing) of the montane cloud forest close to the top of the ridge of Formon is occurring. This area is privately owned by wealthier farmers, some of whom are Agents de Parc. Effective land use control has not been attained.

The upper Cavalier area (Pic Cavalier to upper Sous Bois) is also regenerating. Again, these are primarily state lands on which agriculture, other than pasturing, has ceased.

The Rak Bois Sous Bois area has suffered considerable environmental damage, caused by an exponential increase in the number of gardens present. This increase may be due to the interdiction of agriculture on the higher slopes and mountains of the Park. The Rak Bois Sous Bois has the greatest biological diversity of the entire MBR.

The larger state-owned lands of humid forest on the very steep slopes to the south of Plain of Durand, called the Rampes de Cavalier, have generally remained intact, probably as a result of the activities of the Agents de Parc.

In general, Hurricane Gilbert did not significantly impact the vegetative cover of the Formon area, although it caused major erosion damage, particularly on the principal paths and those higher slopes that did not have effective vegetative cover. Destruction of crops and livestock by Gilbert has increased pressure for agricultural land use on the cover that still remains. Approximately 40% of the livestock of the region was destroyed by Hurricane Gilbert.

Riviere Trois Sources. The entire area of the Riviere Trois Sources catchment basin, an environmentally very fragile area, has been heavily impacted by clearing pine forest and virgin montane cloud forest in order to plant gardens. Most of these gardens have been cleared by non-resident farmers from Rendel who plant black beans, which they believe requires complete removal of natural vegetative cover from the land. The entire river valley, bordered on both sides by very steep (60° + slopes) is now scarred by numerous landslides. In fact, some paths are no longer used by the farmers, due to danger from rock slides. The greatest damage occurs on the southern slopes of the Riviere Trois Sources, extending to 1650 meters on the Formon Ridge. In this area, virgin montane cloud forest with a unique ecosystem of rare native and endemic species is being destroyed.

The small farmer community at Sivette has enlarged its gardens to the east along the top of the Macaya Ridge and southeast on the extreme southern slope of the Riviere Trois Sources. New avalanches were observed on these slopes, and the access path has been partially destroyed. Damage extends to less than one kilometer from the saddle connecting the Formon and Macaya Ridges. In this area, a pioneer farmer, Mr. Pierre Edmond Paul, has occupied state land and opened this very critical zone to sharecroppers from Rendel. At least twenty hectares have been cleared and large landslides are present.

Again, the immediate cause of much of the erosion damage is Hurricane Gilbert. This is especially pronounced below the southwest face of Pic Macaya, on which large fires, probably started by farmers, have destroyed the natural vegetative cover. The immense loss of livestock and crops caused by Gilbert has also increased economic pressure on these very fragile lands. Gilbert did not greatly damage the forests of this area.

In this part of the MBR, the Agents de Parc have been totally ineffective. COSAR, however, has played an important part in the deforestation of the zone. COSAR opened land for colonization. COSAR has cleared forest, ostensibly to plant coffee. However, it would appear that approximately 200 hectares of this land is actually utilized to plant beans and only about 56 hectares for coffee. These activities have legitimized the process of slash and burn agriculture, and have encouraged the use of these fragile lands as an income-generating resource for non-resident farmers from Rendel.

The Sondeo team also observed that pine (*Pinus occidentalis*) is being cut for lumber. At least seven "scieries" (logging frames) were seen. The trees are being purchased by lumber merchants in Rendel who hire teams to cut the trees into planks and beams. The lumber is sold in Rendel, Port-a-Piment, Les Cayes, and Port-au-Prince. The team, while ascending to the Trois Sources area, observed twenty-two planks being carried down to Rendel. COSAR's lumbering complex high on the Formon Ridge (1600 meters) has suspended operations.

As in the Formon target zone, fewer birds were sighted than in previous expeditions to the region. Hurricane Gilbert probably is the cause of the reduction in numbers of birds.

Overview of the Target Zones: Physical Factors

The target region consists of at least three separate zones which will be referred to here as Formon, Trois Sources, and Deglacié. These zones are treated separately for most planning purposes because they differ in biogeophysical character, social structure, land tenure, and historical development. A brief description of each is provided below. The Sondeo team did not visit the Deglacié region during this investigation, which was evaluated in January-February, 1989.

The overall Formon area extends from Sous Bois in the southeast, to Cavalier in the valley south of the escarpment of Formon/Durand, to Morne Cavalier in the west. It includes the steep slopes of the Massif Formon to the north and the ravines and rolling hills of the Plain of Formon to the east in the area of Ravine Casse Cou.

Trois Sources includes the region surrounding the COSAR building on the Gran Plenn. A road is currently being built into the region, but at this time access is via a rough trail from the end of the road. The designation Trois Sources is based on the central feature of the zone, the Riviere Trois Sources. This river flows west from the saddle (ridge) connecting Pic Formon and Pic Macaya. The steep slopes of each mountain dominate the region.

Geology. Karst topography predominates east and south of Morne Cavalier and between 1800 and 2000 meters elevations east of Pic Formon and along the ridge between Pic Formon and Pic Macaya. Exposure of karst is associated with the karst hills covering the edge of the plain of Formon and the plain of Durand, the "rak bwa." Solution pipes and sinkholes are common on the plain of Formon and on the plain of Durand as well. Exposure of blocks of limestone stand like monuments on the surface.

The limestone varies in hardness and fracturing and may not be able to be treated as a single formation. Water penetration through cracks and fissures and consequent runoff will be associated with these differences. Below the limestone lies a complex that was formed under the sea in the late Cretaceous

period.

Basalt exists as pillow lavas, dikes and other thin strata intruded into the basement rock. This basement rock is composed of deep water marine limestone, shale and other sedimentary rocks veined with calcite.

When weathered, these rocks are highly erodible. The most notable feature caused by this erosion in the park area can be seen in the undercutting of the rock under the limestone of the Demisseau Formation, causing the great cliffs of the massif. Differences in topography are reflections of the predominate underlying material. Those areas underlain with limestone are usually rounded on the ridges and in the ravines. Soils and rock are usually more stable on these materials. On the other hand, where the sedimentary/basaltic rock is found, ridges are sharp and "V-shaped" ravines are common, as the material is highly erodible.

Based on the predominant geological formations in the area, three general types of topography can be defined. These correspond to some degree to current land use patterns and are also an important element in developing a land use plan for the future.

- 1). The first type consists of very steep, upper slopes. Such lands are in some cases utilized today for agriculture, particularly in the Trois Sources area. They are not, however, generally appropriate for agricultural land use. It is recommended that long term planning move toward a system in which use of these lands is restricted to conservation purposes and/or very limited use of natural resources (e.g., selective exploitation of native vegetation).
- 2). The second type consists of moderately sloping to steep hillsides, and corresponds in large part to the karst outcroppings, known locally as rak bwa. While some farmers are presently almost entirely dependent on these areas for agricultural production, most farmers who utilize the rak bwa also have access to more productive lands at lower elevations and/or with lower slopes. These rak bwa lands are highly variable. In some cases they are appropriate for limited agricultural use (particularly agroforestry), but in other cases extreme degradation has already occurred, limiting their utility in the future. In the latter case, rehabilitation is critical--that is, systems must be devised which can improve the quality of these lands.
- 3). Finally, much of the area consists of formations which are quite appropriate for intensive, sustained agricultural production. These lands include the nearly level plains (such as Durand) and gently sloping hillsides. Most of these lands are in agricultural production today and should continue to be utilized for agriculture in the future. Conservation agricultural practices should be developed which can assist farmers in making the most productive use of such lands.

Soils. In the central portion of the Plain of Formon, the soils are deep oxisols formed on predominantly limestone parent rock. These soils are dark red and deep with moderate fertility levels. Soil pH is nearly neutral. Lower down the slope, the soil changes to a brown ultisol formed on the non-limestone rock. Because slopes are not severe, the soils are relatively deep and somewhat fertile with nitrogen and phosphorus being the most commonly deficient nutrients. The pH of these soils is slightly acid.

On the slopes of the upper area of Durand, the soils are relatively deep and coarse-textured ultisols that may be formed on the colluvium and alluvium that has washed downslope from underneath the Demisseau formation. Some explosive volcanics of a later geologic era may have also contributed to the parent material of these soils. The soils are less weathered and the coarse texture leaves them droughty and dryer than surrounding soils.

Toward Kay Tilus and the ridge of Formon, the steep slopes have been cleared, exposing the ultisols. Severe erosion is in process, but farming continues in spite of landslide and landslip on nearly all of the deforested slopes.

On areas of rak bwa, a large portion of the soil surface is covered by limestone rock formations and strewn rocks, but the soils are nonetheless deep and fertile oxisols. Production is severely limited by the portion of the soil surface not covered with rock.

Across the escarpment toward Cavalier, the soils are predominantly brown ultisols. On the upper slopes below the market of Sous Bois the soils are relatively recently cleared and hence moderately deep. Because

of the nature of the soil, however, they are subject to erosion and may rapidly degrade. Because of the limestone on the rak bwa above them, they retain a neutral pH. Lower on the valley floor, the ultisols have been cultivated much longer and even with less slopes are eroded moderately.

To the west of the Park in the areas near the COSAR complex, the soils follow the same pattern of ultisol formation. Cleared land is showing signs of erosion and in some areas the plinthite subsoil is at the surface. High up the Port-a-Piment river, clearing and fire during the last ten years have caused severe avalanche hazard that should be addressed immediately.

The three predominant soil types described above coincide to some degree with the major topographical features of the region. However, it must be stressed that soils in the entire region are extremely variable and form a mosaic of intermixed soil types in some areas which will require extremely careful attention when agricultural land use is suggested.

In general, the oxisols occupy the nearly flat plains and very lowest slopes in the region. They are much more prevalent in the Formon area than in the Trois Sources area. While they do offer problems of fertility, their structure is excellent and they are highly resistant to erosion, in addition to occupying the least erodible sites. Intensive agricultural production on these soils should be possible.

Deep ultisols occupy many of the intermediate slopes, including large portions of the rak bwa area. These soils are much more susceptible to erosion than the oxisols and must be treated with great care. In many cases, current land use will not be sustainable on these soils, especially when it occurs on steep slopes. These soils should, over the long term, receive the greatest attention regarding appropriate agricultural and/or non-agricultural land use practices. These soils have also been highly degraded in many areas and will require rehabilitation if they are to prove productive in the future.

The upper slopes are occupied in many cases by thinner ultisols, even more susceptible to erosion than those occupying the lower slopes and rak bwa. These soils are very prevalent in the Trois Sources zone, and contribute to the very severe erosion problems that are so visible in that area. In the vast majority of cases, prolonged agricultural production will not be possible on these soils. It should be noted that the upper slopes show extreme variability in soil type and that generalization is difficult. Further study of these soils is critical, particularly in areas of intensive utilization.

Recommended Land Uses. Two concepts are critical to implementing the Macaya Biosphere Reserve, both of which represent a radical departure from land management as it is currently practiced in the region. The first concept is that the core, protected area, the national park itself, is seen as one element in a rational, managed use of a much larger land area. Thus, management of lands outside the core area becomes fully as important as management of park lands themselves. The core area is seen as critical to the successful management of the outlying lands (as a source of water, for example), and, in turn, the outlying lands are seen as part of a whole whose successful management contributes to preservation of the inner core.

Second, the population of the area itself must become the stewards of all the lands in the biosphere reserve. This means, in essence, that individuals are being asked to become stewards for lands which are not in fact their own and that they are being asked to engage in land use practices whose benefits accrue not only to them and their immediate families, but also to other inhabitants of Haiti dependent on the resources provided by MBR.

The implications of these concepts for this project were very important. First, the University of Florida and USAID turned some attention away from strict park management in order to concentrate on management of lands outside the park. Second, land utilization outside the park boundaries must be stabilized and improved in a way that is beneficial to the surrounding population. Only by doing so can the local population become involved in a rational land management scheme, which is necessary for the survival of the protected core itself. Third, given the extreme poverty which characterizes most of the population residing in the MBR, this can be achieved only by increasing the actual monetary well-being of these people. Given the current low standard of living and increasing population pressure, immediate, fairly short-term success must be achieved in developing land use strategies which will permit people in the area to significantly improve their standard of living in a sustainable, upwardly spiraling manner.

Five general types of land use zones are recommended for the MBR. A brief description of each, the goals associated with the particular land use, and types of proposed land uses are given below.

Three general concepts are embodied in these land use recommendations. The first is a traditional concept associated with Pic Macaya National Park--some core lands must be preserved in a relatively untouched state. Preservation is the major goal of land use in the Core Area and Special Use Zones described below. The development of national and local institutions which can effectively oversee the utilization of lands in these zones far into the future, is critical to the successful management of these zones.

Other lands must be rehabilitated. These are lands which have already suffered moderate to extensive degradation. Simple preservation or maintenance of the status quo is insufficient here. These lands must be improved through planned, scientifically-based land use management. In some cases, rehabilitation will involve returning these lands to a status more nearly resembling their former, untouched state. In other cases, however, lands require rehabilitation for more intensive human use. Some parts of the *rak bwa*, for example, should be returned to a more natural state and others should be managed for more profitable, intensive human uses. Rehabilitation is the major goal of land use management in the Forestry and Agroforestry Zones.

Unlike preservation, development of methods for rehabilitation requires extensive scientific input. Only rarely has rehabilitation of lands been attempted anywhere in the world, and little scientific basis for rehabilitation exists in Haiti. Utilization of native species, for example, is an important element in rehabilitation. Yet, for many of the tree species of the region, virtually nothing is known about how to rear seedlings in nurseries, whether they can be coppiced for use in agroforestry systems, and how well they will survive in a highly altered environment. Further, most research in agroforestry on a worldwide basis has been conducted in a lowland environment. Relatively little attention has been devoted to the highland environment. Development of the scientific basis for rehabilitation requires a long-term commitment to the MBR.

Finally, land use on much of the land in the area must be greatly intensified. Current farming practices will not permit farmers in the area to gain a higher standard of living from their lands. Intensification of agriculture is therefore critical to long term success of the MBR. Intensification means, in this context, both more intensive use of the land and higher value production. Technologies which can permit much more intensive production in the area are known. They require testing and adaptation under local conditions. Adapting these practices to local conditions and disseminating them to farmers in the region was a short term goal of this project. Intensification was the major goal of project intervention in the Agricultural Zone.

Success in this area is largely based on increasing cash flow for farmers. The current farming systems employ, for all practical purposes, no outside inputs aside from labor (only one family has ever used fertilizer, for example). It is questionable that further intensification of labor is possible, and even if it is possible, labor intensification alone will not significantly increase the productivity and profitability of the farming systems described below. Therefore, the use of agricultural inputs such as improved seed and fertilizer is critical, which requires that farmers attain cash flow beyond that needed for immediate consumption. The University of Florida's short-term strategy, therefore, concentrated on ways of improving cash flow within the local economy.

Land Use Zones

The following idealized zones are designed to indicate the optimal land use activity for all of the habitats of the Macaya Biosphere Reserve. These determinations were made on the basis of the gradient of the slope and the location within the reserve. During the course of the project, the definitions of these zones were refined, and the concept was expanded to cover all of the micro-habitats of the reserve. The expanded definitions take soil and landscape characteristics as well as socioeconomic factors into consideration in addition to the steepness of the slopes. In this way, the working definitions function at the level of micro-zones, so that the most suitable activity for each location within the Macaya Biosphere Reserve can be designated on a high resolution land use map.

Core zone

Description. These are the most fragile areas where a loss of forest cover and soils would result in extensive degradation of the surrounding areas. The core area is not yet ecologically degraded, and includes endemic plants and animals of unique importance. However, some habitat restoration may be necessary where the process of degradation has progressed to the point where the habitat is in danger of being lost.

Goals. Soil, water, and forest conservation, and preservation of biological diversity are the critical goals.

Uses. These lands should be totally protected from agricultural and forest practices. Suitable land uses include tourism (hiking, camping, photography), and research (with permits).

Special use zones

Description. These are biologically and ecologically valuable zones where total protection is not possible because of the proximity of the areas to high density human activities, the small size of the habitat (i.e., pond or spring), or multi-use requirements (i.e., springs, streams, and rivers). These areas also include corridors connecting disjunct habitats that are necessary to maintain biological diversity in the overall zone of the biosphere reserve.

Goals. The major objective of land use planning is to protect the habitat without excluding the use of the limited resource involved.

Uses. Only specific activities such as gathering water, washing clothes, searching for renewable resources (i.e., poles, honey) should be allowed.

Forestry zone

Description. These are steep areas where soil stabilization is necessary, but where the habitat is not critically important to maintain biological diversity or prevent extreme erosion.

Goals. Important goals include protection of the soil, increasing overall precipitation levels (i.e., comb water as "dew" from the frequent cloud cover), and providing an alternate revenue source for local farmers based on wise use of a renewable natural resource.

Uses. Selective harvesting (no clear-cutting allowed) of trees would be allowed. Where possible, native trees should be utilized because of the proximity of these areas to the Core Zone. Fast-growing native species of economic importance can be supplied from nursery stock as part of the forestry process.

Agroforestry zone

Description. These are areas of moderate slope where intensive forestry, moderate agriculture, and some grazing can occur without exposing the soil to rapid erosion or degradation.

Goals. The major goal is to provide agricultural sites in otherwise fragile habitats for local farmers, as well as the appropriate technology to use these sites without destroying them. A major concern is to raise the productivity of these sites and to increase the income levels of local farmers who utilize them.

Uses. Any of several possible mixes of agroforestry techniques may be utilized, including alley cropping, grazing, production of fruit, intensive forestry, and production of other tree crops.

Agricultural zone

Description. These are relatively flat areas with deep soils (usually oxisols) where well-planned intensive agriculture will not irrevocably damage the environment. Lands on moderately steep slopes where erosion is not a problem can also be included within the agricultural zone.

Goals. These areas should be utilized to maintain a high level of economic return and to raise productivity and incomes of local farmers.

Uses. Intensive agricultural production should be practiced on these sites.

The following table summarizes the relative amount of land that was recommended for inclusion in each land use category as of January, 1989. The final distribution of the land into each category was planned on being modified as the project progressed, and as our understanding of the optimal total size of the Macaya Biosphere Reserve changed.

Area of MBR (January 1989) 16,364 ha

Zones of MBR (January 1989)

Core Zone	4,337 ha (27%)
Forestry Zone	4,070 ha (24%)
Agroforestry Zone	3,555 ha (22%)
Agricultural Zone	3,128 ha (19%)
Regeneration Zone	737 ha (05%)
Special Use Zone	496 ha (03%)

Overall percent of MBR in Conservation related activities 35%

Overall percent of MBR in Production related activities 65%

Specific Recommendations for Formon. The ban on garden activities and the cutting of trees above 1600 meters on the steep hills of Formon should continue.

The rolling hillsides between 1000 and 1600 meters should be zoned for agroforestry and forestry uses. An active program to utilize this zone for firewood production, use of local endemics of economic importance, coffee and plantain production, and production of shrub crops (such as pigeon pea, *Cajanus cajanus*) should be developed. There is a significant problem of soil erosion in the Ravine Casse Cou and the Ravine Dalest. These ravines, and their associated connecting ravines, should be reforested (Agroforestry and Special Use Zones).

The Plain of Formon and Plain of Durand are appropriate locations for intensive agriculture with some attention being paid to soil stabilization in the ravines of the area.

There are several areas of unique value on the Plain of Formon and Plain of Durand, such as the small pond near Madam Robert's house and several karst domes scattered throughout the plains. These should be designated as Special Use Zones. The margins of the small pond, for example, should be reforested with appropriate endemic trees and shrubs (as well as some food trees such as plantains). Continued access to the small pond for washing and watering animals would be available at one location near the main trail.

The rak bwa at Sous Bois, Formon, Durand, and Cavalier are of great significance to the natural patrimony of the country. These areas are also of significance to the daily activities of the local residents of the region. Poles and firewood are collected there. Some pockets are suitable for agriculture.

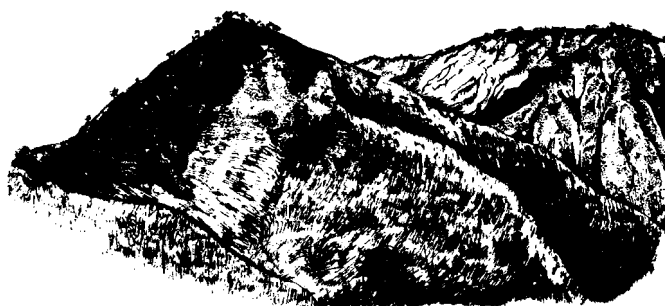
The rak bwa should be maintained as a continuous unit from the Ravine Dalest and Ravine Casse Cou to the Rak Bois Sous Bois, Bwa Formon, Bwa Durand, and the upper ridges of Morne Cavalier. This will provide a continuous buffer zone for the area of the biosphere reserve. Careful zoning will allow pockets in this continuous rak bwa (once reconstructed) to be used to agricultural purposes, and should allow limited exploitation of the forest for poles and firewood, since these resources are renewable.

Specific Recommendations for Trois Sources. All land to the east of Morne Bois Pin in the basin of the Riviere Trois Sources should be designated as part of the Core Area. All of this land appears to be state land. The Sondeo team also recommended that Morne Bois Pin and the steep slopes of Pic Formon between 1600 and 1800 meters be included in the Core Area. Since some of this land is currently owned by COSAR, an accommodation should be reached such that COSAR agrees to strict enforcement of the laws of the biosphere reserve, or agrees to sell or trade land within this zone. The ridges above 1600 meters of all of the mountains bordering the Riviere Trois Sources should be designated as part of the Core Zone.

The steep slopes of the entire watershed of the Riviere Trois Sources are vulnerable to rapid soil erosion, and should be designated as Agroforestry Zones. Appropriate uses of these lands include planting coffee, nut trees, pigeon pea, plantains, and the use of living terraces. Plantains are especially suitable as a food crop that can also be used to secure the soils in gullies and steep ravines. The planting of black beans alone should be discouraged on steep slopes.

Land along the margins of the Riviere Trois Sources is appropriate for intensive use, and should be micro-zoned as Agricultural Zones or Agroforestry Zones.

Burning of fields should be discouraged in all areas below the core area to prevent wildfires from damaging the higher core areas habitats.



Parc National Pic Macaya

Overview of the Target Zones: Demographic and Socioeconomic Factors

Due to very inclement weather and transportation problems, the Sondeo team was not able to visit the Deglaci region. The observations and conclusions stated below therefore apply to the Formon and Trois Sources areas only.

Demographic description

Formon. The population of Formon has grown largely through in-migration, most of it in the past thirty years. This conclusion is based on a census of fifty households which was conducted in June and July of 1988 and which showed that a majority of the residents interviewed were born outside of Formon. Within Formon, internal migration to the adjacent community of Durand has also occurred. Beginning in the 1960s, members of the Beljean family moved there, and most residents there today have the surname Beljean. Some out-migration is also taking place, most of it to Port-au-Prince. A few wealthier farmers have also relocated to the southern coast (Tiburon) to invest in the coffee market and to clear and farm state land.

Several separate communities were identified by the Sondeo team in the Formon zone. These are: "Kay Tilus", Sous Bois, Durand, Formon proper, Upper Cavalier (Kay Jeudi), Cavalier Rak Bwa ("Delta" / "Ma Blanch"), "Gran Fon", Trois Riviere, and "Kay Bout". The total permanent population of the target region is approximately 785 individuals. Table 1 provides a population estimate for each community.

There are many types of land holdings in the target region. Much of the area consists of state lands. They may either be farmed directly by the individual who rents the land from the state, or may be sublet to other farmers or farmed by others on a sharecropping system. There are two types of state lands. Lands which have been rented for many years by the same individual, sometimes called premier testament, are treated almost as private land. They are often sublet or sharecropped. There are also many parcels which are rented on a short-term basis from the state. Some state lands on the northern periphery of the target area have been closed to farming by MARNDR in response to the national park project.

Table 1: Population Estimates for the Communities of Formon

<u>Community</u>	<u>Number of Households</u>	<u>Number of Individuals</u>
Kay Tilus	12	60
Sous Bois	40	200
Durand	12	60
Formon (proper)	50	250
Kay Jeudi	10	50
Delta\Ma Blanche	12	60
Gran Fon	5	25
Trois Riviere	6	30
Kay Bout	10	50

There are also large holdings of private lands in the target region, which fall into two categories: inherited and purchased land. "Te eritay" (inherited land) forms a significant part of the region, and has been passed down from the original landowners (the Despagne, Beljeans, and Pierres). This land is either farmed directly by the owner, or sublet or sharecropped. "Te achete" (terre achete or purchased land) is also an important component of the local land tenure system. Much of this land, as in the Delta\Ma Blanche zone, is rented to local farmers by absentee landlords. The Sondeo team recommended that a land tenure study of the region would be beneficial.

Trois Sources. Several communities were also identified in the Trois Sources zone. These are: "Gran Plenn" (Grande Plain), "Plenn Mou" (Plaine Mou), "Gren Mouton" (Graine Mouton), "Tombe Kriye" (Tomber Crier), "Tranpe" (Tremper), Jeremie, and "Sivette". The total permanent population of the Trois Sources zone is approximately 125 individuals, but this does not reflect the large number of non-resident farmers who come up from Rendel to plant beans. Table 2 provides an estimate of the permanent population occupying each of the communities that were identified. A much better estimation of the population utilizing this area, particularly the non-resident population, is needed.

There are four types of land holdings in the target area: state land, COSAR land (596 hectares), private holdings owned by absentee owners, and private holdings owned by local residents. State and private lands are sharecropped and are rented to farmers from Rendel (and other lowland areas to the south), who travel to Trois Sources to cultivate beans and sweet potatoes. Beans are by far the most important of the two crops, however. Most private landowners in the Trois Sources area are absentee landlords.

Table 2: Population Estimates for the Communities of Trois Sources

<u>Community</u>	<u>Number of Households</u>	<u>Number of Individuals</u>
Sivette	10	60
Gren Mouton	5	20
Tombe Kriye/Tranpe	3	10
Gran Plenn/Plenn Mou	7	35

Most of the people who impact the Trois Sources area are non-residents, most of whom are members of COSAR. There are fifteen permanent residents of the Trois Sources area who are direct employees of COSAR, and they are permitted to plant beans on COSAR lands. The "food for work" program and the availability of COSAR land for planting beans are direct benefits to the residents of Rendel.

Many lands in the valley of the Riviere Trois Sources (also called Grande Riviere) are steep and unstable. In two locations (Sivette on the west slope of Pic Macaya at approximately 1500 meters elevation and Tombe Kriye in the upper river valley at approximately 1450 meters elevation), the local residents expressed grave concerns about avalanches. Residents of Sivette stated that they are unable to rent out some of their lands because of the danger of avalanches.

Because of its importance in the area, the role of COSAR deserves special attention. The main benefit of membership in COSAR at the present time appears to be access to COSAR lands where members are allowed to plant gardens in areas that have been cleared (theoretically for the eventual planting of coffee, but in most cases for immediate use as bean gardens). The Sondeo team interviewed several non-resident farmers (COSAR members) who were working in a relationship with "COSAR" similar to a sharecropping pattern, and split the bean crop on a fifty percent basis. Father Hallee has indicated that he believes that this was a temporary arrangement and is no longer COSAR policy.

Many areas where beans are being planted are ecologically inappropriate (i.e., too steep to be stable and thus resulting in severe soil erosion, or locations that were originally covered with primary forest inhabited by rare species of endemic plants and animals). While the stated goal of COSAR is to plant coffee on land owned by COSAR (596 hectares), the main activity of cooperative members appears to be planting beans on coop land (approximately 250 to 300 hectares). More land has been cleared in the area than can possibly be planted in coffee within the next decade at the current rate of planting (10 to 15 hectares per year). The planting of coffee and the maintenance of the coffee nursery appear to be poorly supervised. As a result, coffee is being planted in an inappropriate manner, and young coffee trees are dying.

Many members of COSAR who were interviewed expressed disappointment in the direct benefits they were receiving from COSAR in relation to coffee production (no financial rewards as of this date) or food for work (food is often late in delivery). As a result, many members are concentrating their activities in the area on growing beans on land provided by COSAR. This leads to increased pressure on COSAR to sanction the clearing of forested lands.

The completion of the planned road from Rendel to the cooperative headquarters at Gran Plenn will improve the level of supervision and coordination at COSAR. This same road has the potential, however, of creating a situation where the rate of exploitation of inappropriate habitats in the Macaya-Formon area will accelerate. This would result in degradation of the national park (core area of the MBR) and the watershed of the Riviere des Roseaux.

Current impact of the park

Formon. The closing of state lands to gardens has removed one of the main sources of gardens for the poorer peasants on the Plain of Durand and Plain of Formon, as well as Cavalier. The reduced availability of these lands has had two main effects:

- (1) Some poorer farmers have moved out of the region towards Port-au-Prince, Tiburon, and Jeremie; and;
- (2) Some other poorer farmers have redirected their farming activities towards the rak bwa stretching from Sous Bois to Morne Cavalier, accelerating the rate of deforestation in these areas significantly.

The loss of the upper steep slopes of the Massif Formon as a source of gardens has also had an effect on some of the wealthier local farmers, such as the Despagne family. These individuals have also migrated to Tiburon, where unoccupied state lands are available. An example of this is Robert Despagne, who has relocated to Tiburon.

The positive economic benefits of the park project have been mainly concentrated in the hands of the wealthier land-owning families of the Plain of Formon. The poorer families participated in the early phases of the park project and had anticipated more positive economic benefits from the project. Therefore,

there exists a problem of unfulfilled expectations. Since the market system of the region depends heavily on the availability of cash, this is a significant loss for the poorer families of the region.

The farmers that were interviewed did see a relationship between the deterioration of the environment of Formon and the degradation of their quality of life. They did not, however, see any clear relationship between the preservation of the nearby rak bwa on the edge of the Formon escarpment and the quality of their lives. They depend heavily on this region for firewood and poles for building.

Trois Sources. The Sondeo team did not find any indication that local residents in this area were affected by the park. One reason for this may be that the Contribution Office on the Jeremie side may not yet be restricting the renting of state lands, and the park agent assigned to the region lives far away in Rendel and has not actively enforced the principles of the national park program to the same extent that it has been enforced in the Formon/Durand region.

COSAR has stopped the cutting of large pine trees on their lands and has closed down their sawmill.

Emigrants from Cavalier (who may have left there due to loss of park lands in the Formon area) do not appear to have settled in the Trois Sources region as was suspected. It appears that they have continued to the Jeremie side, especially the La Hotte zone on the northwest flank of Pic Macaya.

Conclusions and recommendations on the impact of the park

The impact has definitely been greater on the poorest families. Tied to this, most direct benefits from the project for local residents flowed into the hands of the wealthiest families during the eighteen months when management of the area was completely in the hands of ISPAN. (This was undoubtedly done inadvertently and these comments are not meant as criticisms of ISPAN. Nonetheless the impact of the decisions that were made in this became a fact of life for the on-going project and must be recognized). Together, these two facts led to a situation in which the social disparity in the Formon area was exacerbated.

Three recommendations arose from these conclusions.

- 1). Project monies should be distributed more equitably among the various income-level groups.
- 2). Careful attention should be paid to working with the poorer families, which requires careful selection of project primary cooperators (see below) to include representatives from this group.
- 3). Further study of land tenure would be useful, concentrating on access to land among poorer families.

Community institutions

Peasant organization is crucial to the success of the Macaya project. There is no other way in which the labor mobilization needed for erosion control or the local management of a biosphere reserve can take place. The community of Formon has several informal institutional possibilities which could be used to channel technical assistance, however there is no strong local church organization, no groups or cooperative structures and little or no contact with extension from the ministry of agriculture.

The one formal institution, the Community Council, does not represent the majority of farmers in Formon, yet its presence should not be ignored. The council is dominated by the wealthier farmers and while it had done some community work in the past (such as road repairs or providing free labor to local farmers in times of crisis), the experience with building the park road seems to have been its downfall. After having worked for the high wages paid by ISPAN and with all the disputes that accompany any public works project, the council is not functioning at present.

Other community institutions include the "chefs section" representatives in the village, called "marechal". While their duties in the past were probably limited to settling disputes of livestock wandering into gardens, their services are used quite often by the park guards. Any time someone is caught clearing land or sawing boards or if their animals are confiscated, they must appear before the marechal. More often now, these violators are taken down to Chantal to appear before a magistrate. Thus, the imposition of laws resulting from the presidential decree in 1983 has brought the government into the lives of peasants, in a way that

rarely used to happen.

There are two churches in Formon, a Catholic and a Protestant. The Catholic priest for the parish conducts mass about once a month in Formon, and other times mass is conducted by a local sacristan. The church is also where public health workers provide vaccinations for children once a year. Both churches offer limited schooling for local children and have also provided a short-lived literacy campaign. Peasants who can afford education for their children must send them down to Les Cayes, so there is a large gap between the few with an education and the majority that have never gone to school and cannot afford to send their children.

The most recent community institutions were the University of Florida and ISPAN. The entire park project provided jobs for porters, guides, construction crews, masons, carpenters, and park guards. It built one house for the park headquarters and rented another one for temporary housing of visitors. At the time of the Sondeo, the project had also given away thirteen pigs with the promise of more to be delivered. In addition, ORE established a nursery in Trois Riviere and rented additional land in Portal Formon.

While the negative feeling towards the park has been described above, UF (even though not specifically recognized, but merely a generalized idea of the project plan) continues to have a very positive reputation in the area, due specifically to the American personnel it brought to Formon.

The most pervasive form of community institution are the large families. As has been mentioned, the Despagne control much of the land and resources in Formon, the Pierre's manage most of the state lands on the plain of Durand, and there are more than a dozen Beljean households in the little community of Durand. These are only three of the many family names commonly found in Formon, a number that is growing through in-migration. The programmatic implications for UF are that these families often live in close proximity, they share labor and resources and membership and they are tied together through inherited lands, debts, baptism and placage. They should be taken into account when planning for delivering extension services, mobilizing labor and monitoring project impact.

Community institutional structure in the Trois Sources region is strikingly different to that in Formon, largely because of the presence of COSAR, which is discussed in considerable detail elsewhere in this document. However, the limited size of the resident population, its dispersal over a relatively large geographic area, and the presence of non-COSAR migrants also play a role.

No formal community structures have been identified among the permanent residents of this area, although they do belong to churches located outside the immediate geographic area. Undoubtedly, however, informal structures exist. The role of the labor squads, for example, is an important one, although not one that varies in terms of overall function and structure between the two project zones. Further investigation into the roles of individuals, individual families, and the nature of informal community structures should be conducted.

Similarly, very little is known about community structures utilized by the migrants to the zone who are not associated with COSAR. As pointed out elsewhere, many of these families come from Cavalier and Formon, and it is presumed that much of the social structure and many of the community organizations of which they form a part are in fact based in their "home" communities.

For those families associated with COSAR, this organization is by far the most important institution in terms of their impact upon lands in the target zone. Interaction between COSAR and the project is of importance. Suggestions for interaction between the two are made below.

Conclusions and recommendations on community institutions

The role of the three wealthy families, and particularly the Despagne family, in Formon simply could not be ignored. The project simultaneously ensured that project benefits did not continue to flow into their hands in the way that they had in the past. These families were involved in project activities at a moderate level which ensured that other families within the area were well-represented in the project.

A very serious problem was faced in Trois Sources. The Sondeo revealed the extent to which current

land use is due to migrants in the area. This severely complicated UF's task of working with local farmers in at least two ways. First, simply identifying the impacting population and selecting cooperators among them was extremely difficult. Second, it was difficult to believe that migrants would perceive any long-term benefits to be accrued from changing land use management, particularly since they utilize the Trois Sources area to raise their major cash crop, black beans. Nonetheless, the task was attempted.

Several recommendations were made. (1) UF concentrated some effort on working with the permanent residents. (2) UF attempted to identify migrants entering the area from the Formon area, and concentrate some efforts on working with them. For those families associated with COSAR, this organization is by far the most important institution in terms of their impact upon lands in the target zone. Interaction between COSAR and the project therefore is of importance. (3) UF selected, in cooperation with COSAR, some cooperators among the migrants who enter the area from Rendel and work with them. This three-pronged approach, of course, leaves some migrants untouched by direct project interactions (those who come from neither the Formon area or from Rendel). For the time being, until the region is better known, this cannot be avoided. (4) Further study of the migrant population in Trois Sources is urgently needed.

Importance of the cash economy

Haitian farmers are highly market oriented and sell a large percentage of their production for cash. The importance of cash flow within these communities can hardly be overstated. This is a basic social and economic fact which strongly influences farmer strategies and which requires thorough understanding for successful project implementation.

Unlike many basically subsistence-oriented farmers, farmers in the Formon and Trois Sources regions do not produce sufficient quantities and/or are unable to store products considered to be staples. Large quantities of cornmeal, millet, wheat flour, rice and sugar are consumed and must be brought up from the coastal plains or imported via the marketing system.

A significant portion of the crops produced are sold at harvest to meet cash needs for labor, planting material, land rent, tools, education expenses, and consumption goods. Any significant amount of cash not required for immediate priority needs is often invested in livestock, as a form of savings which generally provides a return on investment, and which is relatively less risky than alternatives like investing in crop produce which requires storage. High humidity and pests make post-harvest storage of crop produce difficult and risky. Most farmers prefer to quickly sell cash crops and any excess of production beyond consumption needs. Many farmers do not seem to even keep their own seed, if there is more than a month between harvest and the next planting. Although buying seed is often expensive, farmer behavior indicates that the buying strategy is less risky than relying on storage. Tuber crops, like yams, "malanga", "mazonbel", manioc, and sweet potatoes have the advantage that harvest can be delayed, basically storing them in the ground until needed, avoiding the post-harvest storage constraint.

Many rural economies which operate at a near subsistence level and which do not have storage problems demonstrate a marketing strategy which gives the appearance of near exchange. A small amount of produce is removed from storage and transported to the market, sold and the money used immediately to purchase the consumer goods required. Since storage is a problem in Haiti, the pattern is more complicated. Farmers diversify production in time and space, as well as in terms of the crops planted. They attempt to continuously have something, somewhere, which is ready for harvest and sale. Since there are often gaps in the availability of saleable produce and some of the important crops are sold all at once, consumption goods must often be purchased with cash obtained in a different period of time than the market day on which the corresponding consumption goods are purchased. Having cash available for such purchases requires a different type and higher level of cash management than is often displayed in more nearly subsistence economies.

Major sources of cash income include the sale of crops and livestock, as well as the sale of labor. Important cash crops in the Formon and Trois Sources regions include (black) beans, yams, sweet potatoes and coffee. Interviews indicate that plantains is also an important cash crop, but Hurricane Gilbert seems to have seriously reduced plantain production during the period of the Sondeo. Livestock sales are also an important source of cash. Chickens, 15 sheep, 5 goats and a calf were available for sale in the Sous Bois market. Goats sell for \$15 to \$20, young sheep for \$20 and a good mature ram for \$50 - \$60.

Additional crops which are marketed in the region include: "malanga", "mazonbel", corn, castor beans, pigeon peas, sisal, pumpkin, squash (milton), watercress, hot peppers, chives and shallots, thyme, mangos, and in some cases citrus, avocado, coconut, apricot (mammy), and breadfruit.

Consumption goods available in the market of Sous Bois during the Sondeo include: rice wheat flour, white bread, sweet bread, corn meal, millet and millet meal, sugar, garlic, maggi (bouillon cubes), matches, cloth, clothes, peanuts, peanut brittle, cooked rice, cooking oil and margarine, salt, onions, meat, kerosene and gasoline, plastic ware and enamel ware for household use, silverware and glasses, numerous small trinkets sold from small table vendors.

Conclusions and recommendations on the importance of the cash economy

This economy is cash driven. It is not a "typical" peasant economy in which farmers first produce enough to eat and then sell "what is left over." Perhaps equally important, actual food supply for the family is not critically limited. While there are short periods of limited food supply, these periods are neither prolonged nor critical. The abundance of root crops in the farming systems in the area (described below) is a major factor in ensuring a fairly continuous supply of food. Therefore, increasing cash flow is the farmer's major goal, one that was clearly stated by farmers of every socioeconomic class.

Project interventions therefore concentrated on increasing cash flow. Farm families are caught in a situation in which cash is of critical importance, but where their access to cash is extremely limited. The economy could in this sense be described as a "petty economy," in which returns to labor are highly devalued by the extreme scarcity of cash. Many women reported, for example, spending entire days at the market to earn forty or fifty cents.

Current cash flow is expended entirely on meeting daily consumption needs--buying salt or sugar. With access to cash this highly limited, farmers are totally unable to invest in inputs which would increase their return to labor or the productivity of their land, both necessities for project success. Any immediate increase in cash flow undoubtedly goes to the same end, increasing consumption. Therefore, interventions must both significantly increase the amount of cash in the local economy, and allow an upward spiral of cash flow to be initiated.

Labor and employment

Agricultural labor is mobilized in three ways, through the household, exchange and daily wage work. Exchange is done through rotating labor squads, usually segregated by sex. Commonly, 3-10 peasants exchange days of work on each others' fields. Larger groups ("eskwads") may have a president or secretary to keep records, especially if the squad does a lot of wage work, but most are more informal. For each day of a work a member accumulates, the rest of the group owes him/her a day. This is literally called "loaning a day" (prete joune). The member who owns this day can sell the work to someone not in the squad, and the payment for each member goes to the owner of the day. Daily wage labor (joune) is also used in Formon; men receive 60 cents a day and women get 30-40 cents a day. Wage labor is one variable that stratifies farmers in Formon, as some sell their labor more often than they buy it. Another common method of securing labor is called "atribusyon", and is actually work done on credit. The laborer will usually be paid in December and an animal will be slaughtered for a feast for all laborers who have worked on credit during the year.

Agriculture in Formon is labor intensive, as in most of Haiti, and the labor needs increase with the planting of rocky hillsides. Most of Formon is not suitable for using a plow. Those few owners who do prepare with a plow have plots at least one carreau in size. While plows might also be suitable in the plain of Durand, most parcels are divided up into gardens measured in "ventsankiem" (.08 of a carreau). Labor becomes a constraint to agricultural production mostly during the planting periods, when the dates are critical and most households need a squad of men to start even a small garden. Depending on the size of the plot, poorer families do manage to secure labor for planting, weeding and harvest through household labor and inter-household exchange. The longer planting times for manioc and sweet potatoes also favor those who might have to wait for labor to be freed up before they can clear land.

There is some non-agricultural employment in Formon, although these jobs are usually only a supplement to agriculture. There are food vendors, professional market women, representatives of the police and the army, carpenters, board cutters, masons, school teachers and park guards. In Trois Sources, extra food can be had by working for the coffee cooperative which pays in food aid. Women also earn money by processing corn meal, manioc or coffee. They will travel to another community to buy coffee and decorticate it by hand, then sell it to a speculateur. Corn is carried down to the mill at Ducis where it is ground into meal and then carried back to Formon to be resold. The profit margin from these activities is very slim.

In the household division of labor, women and children are responsible for much of domestic production. Households secure additional labor through "restavek", which are children from other households who may live in a home of either kin or non-kin as a servant. These children do a large part of household and agricultural activities. These children may be baptized by the family that takes them in and when they are older, they may depend on this family for sharecropping relationships. Some households are augmented by the labor of older non-kin members, such as adult men who come from other communities, sometimes bringing their children with them. These older persons perform the same duties as "restavek". Many young couples have friends or siblings of their own age that live in the same household, sharing with household duties or exchanging labor in the gardens.

Socioeconomic domains

The farmers of the two regions, Formon and Trois Sources, have been grouped together using several criteria by the Sondeo team. This is because the project cannot work with each individual farmer and they are not homogeneous as an entire population. Farmers in the two regions have different resources available to them (land, labor and capital); and different environmental circumstances under which they farm. The end result is that farmers face different constraints in regards to use and acceptance of new technologies.

The Sondeo team had to define practical groupings of farmers, each group being relatively homogeneous, in order to target these farmers for interventions. The Sondeo discovered that the following criteria are useful for delimiting these groups:

- Land tenure;
- Access to labor;
- Geographic location and physical environment;
- Access to markets; and
- Livestock production.

In addition, a sixth criterion, seasonal migration, is specific to farmers in the Trois Sources region. Additional criteria reflect more clearly the unique situation in Trois Sources because of the presence of COSAR.

Finally, the farmers within the different groups are considered household units. However, gender differentiated activities are a criterion used to choose the primary project cooperators (described below). For example, vegetable and spice interventions will need to be targeted for women.

Formon. The farmers in the Formon area can be divided into three types.

Diversified Smallholders. These farmers generally work a "portfolio" of land parcels which fall under different tenure status. These include land parcels that are rented, sharecropped, or inherited. They may also be current or former users of state-owned lands. What distinguishes this group, however, is their access to lands through means other than sharecropping state lands. These households both buy and sell labor. They own animals as well as care for livestock belonging to other farmers.

Large land owners and/or farmers who control a large amount (more than 4 or 5 carreaux) of state land. This group is composed of a relatively small number of families who form a socioeconomic elite in Formon. These farmers buy most of their labor and provide animals for less well-off farmers to raise. While this

group depends on the Sous Bois marketplace, they do have the ability to market their produce directly to Chantal, Ducis, or even Port-au-Prince. Historically, this group is composed of early settlers who now claim the best agricultural lands and monopolize control over the state lands. These lands are then rented and sharecropped out to other farmers.

Farmers who do not own land (in the area where they live) and are primarily sharecroppers and renters dependent on state lands. These families access land almost entirely by sharecropping with the previous group, the controllers of state land. This group may rent a parcel of state land for the construction of a house and subsistence garden, but the most productive land utilized by these farmers is sharecropped. The lands are generally the poorest for agricultural production. Finally, this group must often gain access to livestock by tending animals belonging to other farmers.

Trois Sources. The population of Trois Sources is different from Formon due to the impact of non-resident farmers in the region. The local population consists of only 100 to possibly 200 persons, while hundreds of non-resident farmers cultivate gardens in the region. The permanent residents can be categorized in the same manner as in Formon. The team assumes that the migrants can also be placed in similar socioeconomic groupings. But migrants were interviewed along the trail in the highlands and this was not a good setting for verifying migrants land tenure and socioeconomic status.

Permanent Residents. These farmers are widely scattered throughout the zones of Gran Plenn, Gren Mouton, and Sivette. Most can be characterized as smallholders and all use lands within or adjacent to the Park. Their socioeconomic status does not vary greatly and there is no local elite as in Formon. This group depends primarily on the market in Rendel, but they also use the markets in Beaumont and Jeremie. Without the stratification found in Formon, most labor is done through exchange among squad members. There appears to be little exchange of labor with the non-resident farmers in the area. Livestock was noticeably missing from many of the households observed, especially in Sivette. Locals also said that this was due to avalanche risk on hillsides and losses due to Hurricane Gilbert. In the Gran Plenn community, more livestock were observed, some of the animals belonging to owners in Rendel.

Non-Resident Farmers Connected with COSAR. This group is almost exclusively from Rendel. They came up to the Trois Sources region following the start of coffee production by COSAR in Gran Plenn. The group is composed of cooperative members and their families. This includes the full-time employees of the cooperative and those day laborers responsible for clearing, planting, and harvesting coffee. Most, if not all, of these cooperative members are planting gardens in the region, both on COSAR-owned lands and state or private lands. The primary crop grown in the immediate target area is black beans, with little crop diversity. (They do, however, have other gardens at their homesites below in Rendel.) Labor for clearing and preparation of the soil is accessed by exchange among these migrants. The small size of most gardens in the target area means that it may take only one day to prepare for planting. The migrants come into the Trois Sources region for short periods of time, one or two days, most coming and leaving within the same day. Some of these non-resident farmers were observed to be bringing livestock up to graze, mostly sheep or goats.

Non-Resident Farmers Not Connected with COSAR. This group originates from a wider area, Cavalier, Formon, Rendel, and the lowland plains. They rent and sharecrop state lands and private lands held by absentee owners. While the main crop is beans, there is more diversity in crops planted in the target zone than among COSAR migrants. This includes corn and household subsistence crops. These migrants also include many marketers (trafik) from Cavalier who resell goods in the Trois Sources region and also buy beans and coffee to bring back to Cavalier. It is believed that many of these marketers also have gardens in the Trois Sources region.

Marketing

Description of the Formon Systems

The Formon region is served by several overlapping marketing systems. These include connections to a regional marketing system, direct marketing by large producers to urban centers, a specialized coffee marketing system and the activities of the population of Cavalier, which has specialized in marketing activities.

The Cayes Plains Regional Marketing System. This system serves the urban center of Les Cayes, the third largest city in Haiti, and also feeds the Port-au-Prince market, which dominates marketing throughout the country. The local market structure consists of the markets of Sous Bois and Cavalier, found in the Formon region; and the markets of Ducis, Chantal and Camp Perrin on the Cayes Plains at the base of the high plateau on which Formon and Trois Sources are located.

If producers have other business to transact or if there is a significant difference in prices, they will carry produce on their heads to the lowland markets, particularly the closest, Ducis. Market persons, mostly women called Madame Saras, also purchase produce in the local markets of Sous Bois and Cavalier and transport it to the lowlands for resale. Depending on the scale of activity, market women may transport produce by using pack animals, by hiring porters, or by carrying it on their heads. The literature as well as local informants indicate that there is a stratification of Madame Saras. Some with limited capital operate primarily between rural markets and larger markets like Ducis and Chantal. Others with more capital buy on a larger scale and operate primarily between larger markets like Ducis and Les Cayes. Still others with considerable capital operate out of Port-au-Prince, purchasing large quantities for sale in the capital.

Market interviews and price information from producers indicate that marketing margins between the small highland and larger lowland markets are extremely narrow. A very limited sample during the Sondeo indicates that marketing margins cover very little more than the cost of transportation. People often do not seem to consider the time spent on transportation and marketing, nor do the margins seem to provide much compensation for the time spent on these activities in addition to other expenses. An informant stated that beans and yams provided the largest marketing margins.

	<u>Formon</u>	<u>Ducis</u>
Beans	14 Gourdes/marmite *	16 Gourdes/marmite
Yams	50 Gourdes/sack	75 Gourdes/sack

*Note: this price is when beans are being bought as a seed source for planting; the regular price for beans is closer to 7 Gourdes/marmite. The price of yams also varies seasonally. A marmite is usually represented by a six-pound can of margarine.

At Ducis, produce can be sold to Madame Saras operating out of Les Cayes and Port-au-Prince within minutes of arrival. This allows time to buy items to transport back up to Formon, and still return within a single (very long) day. Going to Chantal requires 2 days, so that trip is made less often. Spices and vegetables are usually taken directly to Les Cayes, where they receive a better price. There, low volume and high value make the trip worthwhile, although the trip is often instigated by the need to buy other things in Les Cayes.

Direct Marketing by Large Producers. Some of the (wealthy) large producers in the area do their own marketing directly to Les Cayes and Port-au-Prince. These producers own pack animals and have the means to rent others to transport produce down from the highlands and stock produce at a depot near roads on the plains. When they have sufficient produce (40 to 60 sacks) they hire a truck to take them and their produce to Port-au-Prince or Les Cayes. They sell primarily to wholesalers in the urban centers. They want to sell quickly and in quantity, but are not willing to trust any one individual to handle their shipment for them. They usually sell to several wholesalers in order to receive cash for as much of their shipment as possible. When asked why an arrangement was not made with a wholesaler to avoid the need to travel with the produce, one of the responses was: "if you want something done right, you have to do it yourself".

However, the nine or ten large producers in the Formon area with the means to do direct marketing do not necessarily limit themselves to marketing only their own produce. They will buy produce locally to complete their shipments or even plan on marketing other produce as part of a revenue generating strategy. These producers may also rent out their pack animals, thus generating additional revenue for themselves as well as for local farmers who want to get their bulky products to markets but who do not currently own pack animals. One of the large producers interviewed gave the impression that they were in fact the major traders in the region and that the produce handled by the market women was not very significant. This

seems to be an attitude related to the scale of the operation. It is not clear what the relative volumes of produce handled by the market women and these large producers might be, but this needs to be investigated. Women from these families also often operate as Madame Saras in the local and regional markets.

Some of the relative prices cited by informants include:

	<u>Formon</u>	<u>Les Cayes</u>	<u>Port-au-Prince</u>
Yams	80 Gourdes/sack	115 Gourdes/sack	125 Gourdes 25/sack
Sweet Potatoes	6-7 Gourdes/sack	10 Gourdes/sack	14-15 Gourdes/sack
Beans	7 Gourdes/marmite	10-12 Gourdes/marmite	12-15 Gourdes/marmite

When sufficient quantities of produce are available, pack animals are used to transport down to the lowlands. When only small quantities are being transported, porters are used. Porters are often paid in kind, receiving about 1/5 of the quantity transported. A sack is 20 marmites, and a porter will be paid 4 marmites for each sack transported.

Marketing Activities of the Cavalier Population. Cavalier has been seriously affected by population pressure and the closing of park lands to cultivation. One of the outcomes has been out-migration, a second has been a specialization in marketing activities by people from Cavalier. Although these activities are closely tied to the regional marketing structure, they also include buying directly from farms even in remote corners of the region. We encountered traders from Cavalier buying directly from farms as far away as Sivette in the Trois Sources region. Some of the market women and large producers do purchase directly from farms in their area of operation, but seldom in isolated corners of the region. This activity by the Cavalier population seems to be at least an extension of the market structure, and may over time provide serious competition with the local Madame Saras. This group seems to operate on very little capital, and individual transactions typically appear to be very small. The role of this group also needs further study.

Coffee Marketing. A fourth overlapping marketing system is specific to coffee. With the coffee mill of Jean-Denis out of commission due to the effects of Hurricane Gilbert, there still appear to be many buyers of coffee in the Formon and Trois Sources regions. These buyers include traders working on commission for speculators in lowland/coastal cities, Madame Saras and other traders buying independently. In addition to the traditional commission and independent traders dealing in coffee, numerous persons from Cavalier now seem to be involved buying and transporting coffee between farm and market.

The marketing margins at the local level appear to be very small, typically in the range of one to three Gourdes per marmite of shelled coffee (worth 21 to 25 Gourdes). This includes responsibility for shelling/milling the cherry coffee purchased and transporting the shelled coffee down to the lowland market or depot. With typical day wages in the range of two to three Gourdes per day, this margin little more than covers the cost of transportation or the opportunity cost of labor to transport the coffee.

Description of the Trois Sources System

The Trois Sources region has no market on the highland plateau, but farmers sell much of their produce at the Rendel and Port-a-Piment markets in the lowlands. The Rendel market in turn is part of the Cayes Plains regional marketing system. Although it is less well served by traditional traders due to the small population and its isolation, the Trois Sources region does benefit from the Coffee Marketing system, the activities of the COSAR coffee cooperative, and the activities of the Cavalier population.

In addition to these, at least some of the Trois Sources communities use the markets at Beaumont and Jeremie. Interviews indicate that prices in the Beaumont/Jeremie regional marketing system are often better than those available at Rendel. Without having visited those markets, the Sondeo team hypothesizes that this is due to the fact that this region is further from the dominant national (Port-au-Prince) market and is also a relatively wealthy region.

Conclusions for marketing

It does not appear that access to marketing services per se is a severe constraint in the MBR region. What is a severe constraint is transportation down to the lowlands and the time required for products to arrive at major urban markets. Transportability (the relationship between value and volume) and perishability must be major considerations in evaluating marketability of any products proposed for (increased) production in the MBR region. Since yams and sweet potatoes are now marketed, it would appear that perishability is probably even more important than transportability. Among other things this means that spices and vegetables which have high value, low volume and are not highly perishable are good choices for interventions, as far as marketing is concerned.

An additional effort also needs to be made to better understand the role of different actors in the marketing system, particularly those selling directly, the Madame Saras, and the traders from Cavalier. It is possible that with their direct sales and numerous family members operating as Madame Saras, that the large land-owning families control as much as 1/4 to 1/3 of the agricultural produce marketed from Formon.

MAJOR FARMING SYSTEMS IN THE REGION

Overview

Haitian families in the project area typically have a garden around the house which contains small amounts of coffee, plantain, yams, sweet potatoes, "malanga" and/or "mazonbel", squash, vegetables, spices, etc. If sufficient space is available, they will establish the equivalent of a field garden where sweet potato, beans, "malanga" or "mazonbel", corn and perhaps yams or manioc are grown in larger quantities. This area around the compound seems to be cultivated continuously without fallow, apparently relying on small amounts of household refuse and the manure of household livestock to maintain fertility.

Farm families will have other fields further from the house in different micro environments which are cultivated for one to three years followed by a one to two year fallow. Fields within a reasonable distance of the compound usually reflect the same type of mixture of sweet potato, beans, "malanga"/or "mazonbel", corn, and manioc or yams. Fields far from the compound often have a higher proportion of less bulky crops like corn and beans, and it is not unusual for many of the distant fields on the high slopes to be limited to beans alone. Farmers manage a portfolio of different parcels with diverse characteristics and variable quality. These parcels may also have different land tenure status: purchased, divided or undivided inheritance, rented, sharecropped, etc.

Most farm families either own and/or are caretakers for livestock including chickens, sheep, goats and occasionally cattle or pigs (recent re-introduction). The larger animals are staked out to graze wherever the family has access to an area where they will not damage crops. There seems to be significant grass cover on some of the high slopes, and animals are often taken there for grazing, even though it may be several miles from the compound.

Although there are numerous minor alterations on this pattern, this is the dominant farming system of the region.

A second important cropping pattern was observed on the Formon plain. These fields of higher than average fertility are often planted in a combination of yams, pigeon peas and manioc. This appears to be a more intensive and highly profitable cropping pattern, but one which is mainly available to a few of the large, wealthy land owners.

Although the farming system mentioned above seems to be the dominant pattern throughout the region, there are important differences of degree related to certain variables.

Cropping calendar

The local cropping calendar begins with clearing land in November and December. The work is usually done by men working together in rotating labor squads or by daily wage labor. On the hillsides and rak bwa, weeds are cleared with a machete, then soil is piled up (ranble te) either with a pick (pikwa) or hoe (wou) or sometimes a machete. Rocks are piled up in mounds and probably serve to control erosion. Some use is made of the vegetation that is piled in contour rows along the hillside (ramp pay). Often a barrier of sticks will be erected around the garden to keep out wandering livestock. In the Trois Sources region, non-resident farmers begin planting beans in January.

On the grass-covered flat lands of Formon and in the valleys of the rak bwa, land clearing is done by two different methods. One is called taiye and it means to break up sod with a shallow stroke of the hoe. It is often practiced by a single farmer in his garden. Some informants indicate that this is done only to sod with a heavy growth of grass and that it would have to be followed by regular squad of men who would make deeper rows. Other informants say that they use the taiye method because they cannot afford to pay a squad for the other method. Thus it may be a method for reducing the days of squad labor needed by softening up the soil.

A quicker method is called "fouye" and it is done by several men in a row, swinging their hoes in deep strokes and turning over sod to make long straight rows approximately two feet high. The contours of these gardens probably follow rainfall and drainage conditions. Some gardens in Formon are tilled by plow. Only one farmer in the Formon area owns a plow and he rents it out to a few of his neighbors who have large parcels of land on the plain. No plows are available in Trois Sources.

After the soil is turned or cleared and the rocks are piled up, there may be a second step whereby mounds of dirt are scattered through the garden. This is called a "bit" or "bit jon" and it is formed by piling sod in mounds approximately three feet by three feet. While it is rare to see women swinging a hoe, they will be the ones who form the mounds by hand. One local said these mounds were like having two gardens, one on the mound and the other below it. In garden plots of rice, the "bit jon" is planted in sweet potatoes and manioc.

Planting begins in January, and women commonly participate in this task. Planting is done with a short machete or a pointed stick. The basic crops are yams, manioc and sweet potatoes with associated crops of black beans or corn. Beans are planted three to a hole, which "gives them an advantage". Root crops of "mazonbel" and "malanga" are scattered throughout the garden along with pumpkin. Planting is usually finished by February. In Trois Sources land preparation is much simpler than in Formon. Hillsides are cleared of weedy cover by non-resident farmers. Sometimes hillsides are burned beforehand. A pick may be used to form low contours before the beans are planted.

Weeding is generally done by women starting in March. The work is often done in rotating labor squads, but when it is paid by the day, women get 30 to 40 cents a day, roughly half the day wage of men. The work is usually done with a curved knife (serpet) with no handle. Tools are bought in the markets of Chantal and Ducis or sometimes from the eastern part of Formon in "Katriyem" (Quatrieme).

The first crop to mature will be the black beans in May. The beans are dried in the pod, either hung along a pole or spread out on a cement drying platform ("glasi"). Beans are usually stored in the pod and not thrashed until they will be used.

The second weeding takes place in May or June and often is done as part of the black bean harvest operation. Corn also reaches maturity in June, but will be left in the field and harvested only as needed. There is rarely enough corn to be sold in Formon and instead it is used for household consumption. Some corn is sold in the Trois Sources area.

In July, the second gardens are planted and the preparation for this occupies much of the available labor. Men clear the land while women gather sweet potato cuttings from the January garden. Women and children immediately follow the men in the new gardens, placing the vine cuttings in the broken up soil. Black beans may also be planted in July, probably depending on what type of soil is available to the farmer in the second garden and on availability of labor at planting time. Planting dates for sweet potatoes can last into September whereas beans are too late by the end of July. Planting dates are a little later as the elevation of gardens gets higher.

By the first week in July, two varieties of yams are ready for harvest. This harvest is either sold or saved for replanting the following year. The two varieties, guinee and "madigwe" (or sign) must germinate in the ground before they can be replanted and often the smaller yams will be placed together in a hole and covered with leaves and soil. The vines will be left in the field and will have regrowth by October which is reportedly larger than the yams in July. For two months during the harvest, men women and children will dig up yams. Those farmers who have fields large enough may have to hire labor to harvest and often a basketful of yams provides part of the payment to laborers.

In October, the harvest of sweet potatoes begins along with that of the beans planted in July. Throughout this time, several lesser crops are harvested such as squash, pumpkin, coffee, plantains, "malanga"/"mazon-bel", leeks and thyme. By the end of the year the yams and manioc are harvested. Many gardens are then left fallow and will be used to graze animals. Although manuring or use of any fertilizers is almost non-existent, tethering animals in fallow gardens and near household gardens does provide some nutrients.

In October and November a third planting normally occurs. (In 1988, it was delayed until November and December because of the effects of Hurricane Gilbert.) The major crops involved are sweet potatoes and black beans. It appears to be the smallest of the three gardens planted during the year, but helps reduce the "hungry season" prior to the harvest of the larger fields which are planted in February.

Crops

The major crops are directed towards the market and are also consumed by the household. These crops are supplemented by a variety of lesser crops which have significance for the local diet and the cash needs of households. One of these crops is corn which is intercropped in the January garden. Even though corn does not do very well, especially in Formon, most farmers plant some of it for household consumption and for livestock.

Most residences are also surrounded by other crops that can be sold at market as well, such as plantains, coffee, chives, shallots, leeks, thyme, castor beans and sisal (for making rope). Vetiver was planted many years ago and sold to factories in Les Cayes. Now, however, it is seldom planted except as a barrier for gardens. It is still often used to repair the roofing on homes and kitchens, but apparently must be brought in from other areas. Formon is lacking in many fruit-bearing trees that are commonly found in other parts of the watershed such as citrus, avocado, mango, and coconut and local residents claim they don't grow well because of the high winds. (In fact, the Antillean variety of avocados, mangoes and coconut will not grow at this altitude.)

Beans. Beans are the most important cash crop in the project region and their cultivation appears to have the greatest impact on MBR lands. The most common type is black beans although a few pigeon (congo) peas, red beans and lima beans are planted. Some residents have planted the red beans that they receive as food for work from the U.S., and they grow very well. Beans can be planted in the January garden or in separate gardens with only sweet potatoes. The second crop is planted in late June or early July and on the higher hillsides it is commonly planted without another crop. After men clear the land for beans, women usually contribute much of the labor, by planting, weeding and harvesting. The crop requires less labor input than the root crops and can be harvested in three months.

Black beans are probably the best cash crop for the market if the right kind of land can be had and if seed is available. During planting season, beans sell for up to 15 Gourdes a marmite, 30% higher than the normal price. The investment can pay off as evidenced by the numbers of farmers who travel to Tiburon to plant beans.

Disease is a major problem for bean production. Mosaic (probably both common and golden bean mosaic) are commonly observed in bean parcels. In addition, soil nematodes may be a limiting factor. This could explain why residents refer to the soil as becoming "accustomed to" beans and requiring fallow. A garden can only be planted once a year in beans before the soil becomes "accustomed to" beans and must be fallowed.

Farmers mentioned rats as perhaps the most important constraint to crop production. The team estimates that crop damage may run as high as 20 to 30 percent of production. One farmer stated that he had changed land clearing procedures from partial clearing of land to clear cutting in order to destroy the rat habitat and reduce rat damage. This type of change is likely to have important implications for the sustainability of the production system and for regional ecology.

Sweet Potatoes. This is the second most important crop in Formon according to local residents. However, it may be more accessible to poorer farmers and it is planted three times each year. Known as "patat", it is commonly intercropped with black beans in the July and October gardens or even planted alone if bean seed is not available. Patat is an important component in the local diet and is be a very important crop for the swine repopulation project. The most common variety is "tiliyan", with a vine that grows runners. Other varieties have local names such as "ti jorjine", "ti palmis", "jen", "koulkoma", and "cheristom".

As with yams, a farmer should ideally have a continuous supply of sweet potato planting material for his next garden. The second planting of the year (in July) comes from cuttings (called bwa patat) that were planted in January. The July garden of patat is be harvested the following March and provides cuttings in January.

The cropping calendar in Trois Sources seems to run about a month later than in Formon. With more rainfall at the higher altitude, farmers say they prefer to plant on the sunny slopes (te chaud).

Finding cuttings does not seem to be a problem, and if cuttings need to be purchased, they are much cheaper than yams. The biggest pest problems for patat are "chenyi" and "mawoka" (grubs or beetle larvae) and "pousinye" (a mite which eats the leaves). It is said that the later one plants in the season, the worse the pousinye will be. Locals also claim that after two to three years, a particular plot of land "becomes familiar with" the patat (vin abituwe) and they will no longer grow in the soil. The plot will then have to be left fallow for at least a year. Severe infestations of mealy bug were observed in many sweet potato plots.

The potential for procuring and maintaining clean seeding material should be investigated as a possible control for mealy bug. Substituting Irish potatoes for sweet potatoes in the rotation might also help control the mealy bug problem. Improved methods of storing tubers, like the CIP potato storage shed, should be tested. ORE is experimenting with tissue culture techniques which should be applicable to sweet potatoes and other tuber crops.

Yams. Yams are an important crop in Formon although their use as a cash crop seems to be dominated by a small number of the large land owners. Two varieties are most common, but there are many lesser varieties that farmers consider to be desirable. The two most important are the sigine madigwe and the guinee, both of which give two harvests per year. The July harvest is ready when all the leaves on the vine have turned a dark green. The yams are dug up by hand and cut from the vine with a pointed stick. The smaller ones may be stored in a hole in the garden and the larger ones taken to market. If the growth of the yam is stunted, the head of the vine is left exposed to the sun for two weeks and then is recovered with soil. The second harvest will be ready by November or December.

In January, after harvest, the main root should be replanted, and the "fingers" consumed or sold. However, having yams to plant in January is a problem for poorer farmers who may have had to sell all of their harvest for cash needs. In this case, they have to purchase yams from other farmers. The constraint of not having sufficient seed crop may be one indicator of the income levels of peasant households. Also, because of the labor inputs for yams, such as the extensive weeding, some households do not have the ability to plant many yams; they concentrate more on sweet potatoes, manioc or beans, all of which are less labor intensive.

Farmers try to take advantage of differences in soil types, preferring to plant yams at the foot of the rak bwa (which permits them to utilize nutrients released by weathering of limestone which drain to the foot of the rak bwa). The sigine madigwe is said to perform well in drier soils. The other varieties, such as the francais, give only one harvest and require twelve months to mature. The Spanish (panyol) and the iral varieties are ready sooner and produce the largest yams. If farmers happen to have some of these varieties, they may plant them close to the house or near a stand of plantain trees to protect against wind destruction

and thieves. Other varieties are known as "bakala", "plen bit", "perin", "sel", and "anglais". Most pest problems are attributed to the mawoka, which eats the yam near the vine and stunts the growth. Besides size and resistance to pests another desirable characteristic is how dry it is when eaten ("sech").

Three major problems in yam production were observed.

- 1). Most farmers prefer not to try to store planting material. Thus, when there is any delay between harvest and replanting, farmers consume or sell all materials and are therefore forced to purchase planting materials at a later date.
- 2). Planting material itself is scarce and expensive at planting season.
- 3). Many yam gardens illustrate moderate to severe nutrient deficiency.

Improved land management should reduce the problem of nutrient deficiency. The claims of a few large farmers that they have used fertilizer but have seen little result needs to be investigated.

Manioc. There are two varieties commonly used in Formon--"manyok ame" (bitter) and "manyok dous" (sweet)--and they seem to be planted interchangeably. The sweet manioc needs no preparation and can be grilled on a fire. The bitter manioc must be processed to remove the poison and is usually baked like a bread (kasav). Manioc is usually planted in January and will be ready for harvest in a year or even longer (up to nearly two years at high elevations in Trois Sources).

Harvesting in large fields is commonly performed by women who pay the owner of the field two dollars per sack of manioc root. Women and children pull the plant out of the ground ("rache manyok") and then chop the root from the stalk with a machete or break off the fingers by hand. The stalks are left in the field and a few livestock may be brought in to graze. The manioc root is prepared by peeling, washing and then grating. The grated root is then placed in a piece of cloth which is twisted so the juice is squeezed out. It is then cooked in either a deep cauldron ("chodiye") or a shallow pan ("platin") and it becomes "kasav". This bread can be sold at the marketplace or by the roadside for 2 cents apiece, but it seems to be consumed largely within the household. The poison in bitter manioc can also be removed by grating and then drying the manioc in the sun. It is commonly fed to chickens after being prepared in this way.

Only a limited amount of manioc was observed during the Sondeo, a small portion of which was affected by mosaic.

It may be useful to try some of the International Institute for Tropical Agriculture's (IITA) mosaic-resistant sweet manioc for disease control and to reduce the time spent processing bitter manioc. Similarly, some of the technologies developed at CIAT, centered around improved propagation techniques, should be considered. The acceptability of consuming manioc leaves to increase protein availability should also be investigated. General tuber interventions like improved storage or tissue culture work should also affect manioc.

Mazonbel/Malanga. These root crops are important for household consumption and can sometimes be sold in the market to raise small amounts of cash. The plants are found around homes, in canals and on the walls of ravines. They are always scattered around the edges of gardens and among the rocks in the rak bwa. Mazonbel (or "karaiyib") is identified by the leaf split halfway and is found in areas that provide plenty of water. Malanga has a leaf split all the way to the stem and can survive in drier conditions. Both plants are very hardy and are an essential part of any garden. Women often take young shoots from near the household and replant them in their gardens or give them to other households. They are often planted at the first of the year when "there is not so much sun" and then offshoots are cut and replanted in the summer. It is said that if one plants at the new or full moon, the plants will produce more "children" ("fe pitit"). Some households also plant another variety of taro root ("taiyo"). The leaves of these plants are commonly used to wrap fruit and eggs for transportation and to cover newly planted gardens to protect the seeds.

Farmers in Trois Sources said they planted malanga on sunny, well-drained slopes while mazonbel was planted on shaded slopes and in more humid areas. These cultures are affected by many of the problems cited for other tuber crops above, but no other problems specific to the crop were identified. These crops should also benefit from the general tuber interventions mentioned above.

Corn. Corn is planted at the first of the year with the yams, sweet potatoes and manioc. It is weeded in March and May and is ready for harvest by early June. In Formon, a second crop during the year is rare. It is harvested slowly, as households need it, and the stalks are fed to the livestock. Formon residents claim that it does better the higher up that it is planted and that corn prefers "te fret", or cold soils which could pertain to moisture retention or to higher altitude. Trois Sources residents claim that it does better on sunny slopes or "te chaud". It appears that there may be a complex interaction between sun, moisture retention, altitude and wind which needs further investigation. The most common variety is known locally as "chiken kot" (chicken corn).

Relatively little corn is planted in the lower reaches of the target area around Formon. Even less is found in the rak bwa gardens. Nevertheless, it is an important food crop because it is ready at a time when there is little else to eat. Even though it was rarely mentioned in Formon as a former crop in the park lands and there is usually not enough of it to sell, local farmers would like to be able to produce more corn.

Corn is a relatively more important crop in the Trois Sources region and is often planted with beans in the gardens on the high slopes within the park.

Nutrient deficiencies are a very severe problem in corn production throughout the area. It would also appear that the cultivars currently utilized in the higher elevations are not well adapted to growing conditions there, probably having been brought into the area from areas of different elevation. Corn in the target region is small, stunted, and usually has extensive damage from worms, rats and birds. Lodging is also a major problem.

Plantains. Known as "banan" in Haiti, plantains are another staple in the diet. They are inexpensive and provide food when nothing else is being harvested. During the summer months, many from Formon were buying banan at the marketplace (it often comes from nearby Cavalier), but informants indicate that there is a glut on the market by December in most years (although not in 1988 due to Hurricane Gilbert) and that banan often goes to waste. Every household in Formon is usually surrounded by plantain trees, they are also planted in ravines and sinkholes to take advantage of water runoff from hillsides. It is common to see locals giving or selling young plants to their neighbors and the nearby coffee cooperative sometimes buys banan to use as shade trees. The favorite varieties are "token", which is resistant to wind; "fran", which has the highest market value; and "larenn", which grows the highest. The biggest pest problem is the "mawoka" (a beetle larvae), which eat the roots.

Coffee. Coffee grows around most homes and most farmers let it grow wild. While coffee grown in nearby regions like Port-a-Piment is an important source of cash for Formon women who travel to there to buy, process and resell it, there is not a significant amount grown in Formon. There is definitely a market for coffee and many households sell small quantities of it. The crop is present around every household, yet it is not planted in the gardens. It is poorly maintained and production levels are extremely low. Some of the wealthiest farmers in Formon have moved to the coast, or to Tiburon where unrestricted government land is still available, to become involved in coffee speculation and large scale production.

Many coffee beans are propagated by the activities of rats and neither these nor suckers are removed. Most household plots are nearly solid with coffee and produce almost nothing except around the outside edge. The coffee is not fertilized nor is it pruned. COSAR has introduced the Catura variety and has attempted to introduce modern production practices such as appropriate plant spacing, pruning, and fertilization. COSAR hopes in the near future to introduce a variety resistant to coffee rust, which has been found in other areas of Haiti. Improved varieties do not seem to be available to farmers for their private coffee plots, nor does there seem to be any effort to teach farmers to improve coffee maintenance techniques.

Coffee is more important in the Trois Sources region than in Formon. The level of maintenance seems to be about the same, except that a number of farmers in the Trois Sources region do mulch underneath the coffee to help control weed growth and improve fertility.

Both the improved varieties and the improved maintenance techniques need to be passed on to farmers.

Other Crops. The remaining crops found near most households are not planted in great amounts or have only a small market value. These crops are often the responsibility of women in the household. Many women have small gardens of chives, shallots, leeks or thyme. They are often grown in mounds of soil

about 5' x 5', and women can be responsible for all stages of production and marketing of these crops. Households in the summer rarely had more than four or five plots of this type. Other crops which surround the household are pumpkin ("jiromou"), chayotte squash ("milton") and castor bean ("maskriti"). The castor beans are dried in the sun and then sold in the marketplace. The internal market system transports the castor beans to processing plants in Les Cayes and eventually castor oil makes its way back up to the marketplace in Formon. It is used on hair and as a purgative.

Rice ("diri") is a crop found in a few households in Formon, and it is a crop used mainly for household consumption. While men prepare the soil for the small gardens of rice (usually only .08 of a hectare), women are responsible for weeding and harvest. It has a three-month growing period and two crops can be planted per year, from March to June and again from June to October. Rice plots always have several mounds of soil scattered through the garden with manioc or sweet potatoes planted in them.

Livestock

Animals are extremely important and all but the poorest households have some access to either sheep, goats, cattle or poultry. While the majority of households own only one or two head of livestock, they gain access to livestock by taking care of the animals of others, often those belonging to wealthier peasants. This system is called "gadinaj" and the caretaker receives every other offspring if the animal is female or half the profits from the sale of a male animal. The caretaker ("gadien") has rights to all milk production, although only cow milk is consumed. Data from the census work in June and July, 1988, shows that many farmers from Durand rely on one or two livestock owners for gadinaj and one of them controls most of the state land in Durand. However, each individual household may be taking care of animals for several different owners, some who live in the lower regions of the watershed. Data from Jaffe shows that ownership of livestock in Dubreuille is even more unequal than the distribution of land holdings and a first impression indicates that the situation is similar here.

Animals are usually sold in the Formon market to middlemen who come up from the markets at Ducis or Chantal. Ducis is the largest livestock market in Haiti and it is an intermediate point for the final destination of most livestock: Port-au-Prince. Livestock are usually sold when there is a cash need in the household such as medical expenses and there are no crops available for harvest.

Cattle, goats and sheep are generally tethered and put out to pasture (preferably in higher lands), moved at least once a day and then brought back to the house every day by the young men and boys in the household. Cattle can be tied out in pastures overnight, but sheep and goats are always penned up or tied near the house to protect them from wild dogs. Most farmers complain that dogs commonly kill some of their animals every year. Another danger from tying animals out in the hillsides (which are considered to provide the best pasture), is that they sometimes fall and strangle themselves ("tonbe nan kod"). This was considered a more common reason for animals dying than illness, however anthrax ("chabon") is cited as a problem too. While cattle had been inoculated by the department of agriculture in the past (probably against blackleg), this practice has not been continued.

Sheep seem to be preferred over goats in Formon, probably because they find pasture more easily and can be sold for a higher price. There is also a higher penalty against the owner of a goat which wanders into a neighbors' garden. According to the law, the goat can be killed by the owner of the garden and he keeps a portion of the meat. However, it is uncertain how often this action is taken because of the potential for conflict in the community.

In Formon, sheep seem to have a serious problem with footrot, and many animals needed to have their hooves trimmed. Footrot seemed to be less of a problem in Trois Sources, even though it receives more precipitation, perhaps due to the higher altitude and/or better drained slopes.

Poultry. Chickens are found in almost all households and they are usually tended by women and children. They are fed some corn meal and manioc. Many chickens die from "lafyev poul", "pye sech" (Pied seche), or "fredi", all of which may be Newcastle disease. Other common complaints are that wild cats, mongooses, dogs and most often red tailed hawks ("malfini") kill many chickens. Some households have pens for their

chickens at night (or more often for fighting cocks) or they are placed in shallow holes and covered with straw. Chickens can be sold at market for \$2.00 to \$4.00 and sometimes eggs are sold, either at the market or by food vendors on the roadside.

Pack Animals. The absence of pack animals in most households means that Formon residents do not have access to the larger markets lower in the watershed. The whole family is required simply to move the harvest to the nearby market at Sous Bois. The few families which do own them have donkeys and mules and sometimes horses. Sometimes, pack animals will be rented out to other households. These families are able to bring their harvests down to Ducis, Chantal and Les Cayes and thus get better prices for them. Not surprisingly, women from these families may also work as professional market intermediaries, bringing goods back from the Les Cayes area for resale in Formon.

Pack animals are often affected by a disease in which the animal's body stiffens before it dies, possibly tetanus. Ticks also are said to be a problem.

Livestock Interventions. The project trained and established a vet technician in the project areas to help vaccinate animals and provide treatment for parasites and other problems. It also investigated the use of forage crops in agroforestry and live terracing activities to help improve animal nutrition and maintenance. Poultry and other animals were vaccinated. It should be possible to train local women to vaccinate chickens and pigs. Some of the problems of foot rot in sheep could be avoided by better care for the animals' hooves, a practice that may be of interest to local farmers.

Forest resources

The Formon region was a source for lumber before the park was created. Logging in the park had been controlled by elites in the government and from Port-au-Prince. There are several woodcutters who now find work sawing dry wood. The park guards have been successful at stopping much of the local lumber cutting in the Formon region, but not in the Trois Sources region. There are also two or three makers of charcoal, usually the poorest strata of farmers. The most common use of wood now is posts for housing (a small house requires at least thirty posts). In addition, new homes require quicklime for the walls and this puts a strain on wood resources. The quicklime is made by building large pyres of green wood and broken limestone, which burn for several days. In the virgin areas of rak bwa, patches of land are being cleared to build these pyres and then the area is ready for planting a garden. Wood resources are also used extensively for fencing around gardens and for fuelwood for cooking. While much of this is dry wood, one might ask how so much of it becomes available. Everyone in the household gathers firewood and most cooking is done using three stones. There is very little use of charcoal and charcoal grills are not even sold at the marketplace.

Farming systems in Formon

The primary difference in the Formon area can be expressed in terms of three farmer groupings which relate to circumstance and access to resources. These groups are: 1). Large land owners and/or farmers who control a large amount (more than 4 or 5 carreaux) of state land; 2). Small land owners, managing a portfolio of different types of land and with different land tenure status; and 3). Farmers who own little or no land (in the area where they live) and are primarily renters and sharecroppers.

These classifications are not based on land tenure alone, but are strongly correlated with other important variables. The large land owners typically have better access to marketing, often marketing direct to urban centers, tend to buy labor, own more animals and have better, more productive land as well as more of it, and are relatively wealthy.

The small land owners have less access to marketing, both buy and sell labor, own some animals, have some land security through ownership, and are intermediate in terms of wealth.

The renters/sharecroppers have less access to marketing, sell their labor to others, own few animals, have little land security which limits their interest in tree crops or other medium and long term interventions, and are relatively poor.

Clear relationships between socioeconomic status and the quality of lands available to farmers are also evident. The larger land owners virtually always have available to them a variety of land formation types and soil types for agriculture. This permits them to utilize the varied physical environments to best advantage. Typically, they will utilize lands on the nearly level plains (usually oxisols) to intercrop yams, sweet potato, and manioc in the January planting season, and will later utilize some of these same lands for additional plantings of sweet potatoes and beans. Some even plant rice on these lowland areas. They will utilize the less desirable lands available to them for other crops of lesser value, and will often utilize the very steep, high slopes for pasture.

Farmers with more limited access to land are forced to utilize more restrictive cropping systems. The very poorest will find that the only land available to them are the steep slopes occupied by poor, highly erodible soils. In many cases they are unable to plant yams on these lands, and crops such as rice are completely out of the question. Thus, their farming systems tend to rely much more heavily on sweet potatoes and manioc than do the systems employed by farmers with access to a greater portfolio of lands.

Farming systems in Trois Sources

The Trois Sources region is somewhat different from Formon, due to higher altitude and more rainfall. It is also distinguished by the influence of COSAR. Among the small number of permanent residents (100 to perhaps 200 individuals in 25 to 40 families), the team identified farm families which corresponded to each of the three categories of farmers identified in Formon.

However, their influence on the MBR is outweighed by that of the hundreds of non-resident farmers cultivating gardens in the region. Most of the farmers cultivating in the area actually live in the lowland plain and come to Trois Sources for brief periods of time to work in their gardens. Since these fields are often a three- to six-hour walk from the home, there is a strong tendency to focus on the production of high value, transportable crops, particularly beans, and in some cases beans and corn. Many of these migrants are among the 2000 members of the COSAR cooperative who plant beans on COSAR land and also find gardens outside of COSAR holdings. For this reason it is important to look at migrants who obtain access to land in the Trois Sources region via their relationship with COSAR, compared to non-COSAR migrants and the small permanent population in the region.

Plantains are much more important in the home gardens of farmers in the Trois Sources area than they are in Formon. There is also a reduced occurrence of other minor root crops such as "mazonbel" and "malanga". Coffee is much more common in this area. The overwhelmingly important crop in these farming systems, however, is black beans, often planted on very inappropriate sites.

Conclusions and recommendations for major farming systems in the region

Several characteristics of these farming systems are important from the perspective of project intervention. First, the typical farmer in the region is managing a portfolio of lands. These lands include various soil types and various land formations. These lands often fall into two or three of the land use zones described earlier in this document and it is hoped that over time several different types of interventions would occur on each farm. It is not reasonable to expect, however, that farmers will undertake interventions which can only be expected to have intermediate to long-term benefits. Therefore, initial project interventions on farms should concentrate on those lands which are appropriate for intensive agricultural use and on interventions which can bring short term benefit to the farmer. On-farm testing of longer-term, rehabilitative interventions should be postponed until greater confidence between project personnel and farmers is established, and until these interventions have been adequately tested on non-farm sites (described below).

Second, these farmers are managing diversified agricultural systems, although the importance of root crops is very great. This diversification represents an attempt both to ensure an adequate, continuous supply of food for the family and, perhaps more important, to ensure that there is a continual supply of products for sale in the market. Clearly, the poorest farmers have the least diversified systems, and cannot plant some of the higher value cash crops. They often cannot plant yams, for example, and their ability to

plant beans is often limited by the amount of seed that they can afford to buy. Both short- and long-term project interventions should concentrate on increasing the diversity of the farming system in order to ensure that these goals continue to be met. A wide variety of technologies should be tested, both on and off farm, and the individual farmer should be offered a menu of alternatives so that he/she can select those interventions which are most appropriate for his/her particular farming system.

Third, these systems are labor-intensive and, conversely, labor is the only outside input currently employed by local farmers. Increasing labor demand is probably not possible and cannot create the kind of changes in farm productivity that are needed in order to establish intensive, high value farming systems in the region. Interventions must therefore, over the intermediate to long run, concentrate on increasing cash flow significantly so that farmers can afford to utilize other inputs in crop production and therefore increase the return they gain from their labor. This can only occur, however, after farm families have secured sufficient cash flow to meet their immediate consumption needs. Increases in cash flow in the short term will go to increased consumption, not increased production. Thus, interventions must be designed to achieve sustained and upwardly-spiraling increases in cash flow.

Fourth, the role of home gardens in these systems is a critical one. It is a source of innovation in the farming system. Chives and onions, for example, both recent introductions of cash crops into the farming system, have taken their place in the home garden, not in extended fields. The home garden is primarily managed by women and is an important source of income for the household. Finally, the home garden does receive some additional inputs in terms of refuse from the household and therefore represents one of the more productive parcels managed by the family. Interventions should therefore place extensive emphasis on the home garden.

Finally, animals are an important element in these farming systems. They provide a source of cash income. They represent one relatively secure way in which farm families can invest. For the poorest families, the fact that they can secure ownership to animals by tending them represents one important way in which these families can improve their economic status. In the long term, the manure they produce could be an important component in the cropping system. The results of the depopulation of pigs is evident everywhere. No other animal currently present in the farming system can fulfill the role traditionally played by pigs because no other animal is (1) small enough for the poorest farmers to raise and at the same time (2) high enough in market value to represent a significant source of cash flow. Both sheep and goats have much lower market value than pigs.



Gardens should not be permitted on steep slopes in Parc National Pic Macaya

A STRATEGY FOR THE DEVELOPMENT OF COMMUNITY BASED SUSTAINABLE AGRICULTURE

Preservation of Parc Macaya requires that certain very fragile lands in the watershed be zoned for non-agricultural land use and that other areas which are somewhat less fragile be zoned for very restrictive land use, such as selective use of natural forest products. Most of these areas will be located in the upper regions of the watersheds of each of the major rivers of the biosphere reserve, such as the upper Ravine du Sud and upper Ravine Trois Sources. The majority of lands in the Target Areas will remain in "agricultural" land use. Agroforestry represents the most appropriate land use for some areas. Some of the land area in each Target Area can and should support intensive agricultural production.

Agriculture on these lands must be sustainable both in a biophysical and socioeconomic sense. Several factors offer an optimistic view for developing biophysically sustainable agricultural practices. Many soils in the area, while not excellent from an agricultural point of view, are quite satisfactory for sustained agricultural production. Rainfall is adequate for the production of many crops. Achieving socioeconomic sustainability will, however, be far more difficult to achieve than biophysical sustainability in many ways. We base the following strategy on two assumptions which have been shown to be functional in virtually every region of the world.

Preconditions for Long-Term Success

First, only farmers who achieve relatively high value production from their lands can afford to practice conservation agriculture. Employing labor or capital intensive strategies, such as constructing terraces, is not feasible for farmers who engage in non-intensive agricultural production. Farmers who raise only low value crops, for example, will find it economically more feasible to clear new land than to incur the labor and/or cash costs involved in building permanent terraces. Over the long term, then, agricultural production in the MBR must become both more intensive and achieve higher value production.

Second, farmers must perceive a fairly immediate return on investments in protecting the environment in order to assume the role of stewards for lands which are not in fact their own property. This means that sustainable agriculture, if it is to include voluntary protection of the upper reaches of the watershed, must be tied directly to benefits which accrue from the protected lands. Such benefits may include both transitory use of such lands (selective utilization of forest resources, for example) and other direct benefits such as improved agricultural production.

The strategy that is suggested for development of community-based sustainable agriculture in the region therefore includes three major elements. (1) The origin and dissemination of new technologies must be based on local point sources of innovation and must occur within the existent community structure both geographically and socioeconomically in order to be sustainable. (2) Only technologies that have been proven to be both biophysically and socioeconomically functional within the region itself can be disseminated on a mass basis. (3) Achieving overall project goals requires that innovation include a menu of technologies, some of which show short-term benefits to participating farmers, and some of which are necessary for intermediate- and long-term benefits.

Selection of target communities

Five communities in the Formon area and two communities in the Trois Sources area were selected for initial testing and dissemination of innovative technologies. These were Kay Tilus, Sous Bois, Durand, Formon Proper, and Upper Cavalier "Delta\Ma Blanch" in the Formon region and Sivette and Gren Mouton in the Trois Sources region. The seven communities were selected from the total group of seventeen communities (excluding Cavalier, which itself includes thirty-three subcommunities) identified during the rapid rural appraisal.

They were selected because they provided project personnel and local farmers with a wide range of biophysical, agroecological, and socioeconomic conditions for testing potential interventions. It is unlikely that any single group of technological innovations would prove equally suitable for the range of conditions encountered in the region. Therefore, ensuring that proposed interventions were tested under a representative sample of biophysical and socioeconomic conditions was of utmost importance.

Both land formation and soil type were included in selecting communities. Three major land formations are included among these communities: (1) uplands near the crest; (2) lowlands with access to rak bwa lands; and (3) rak bwa lands. Three major soil types typical of the zone are also included among the selected communities: (1) deep, well-drained oxisols; (2) deep, highly-weathered ultisols; and (3) relatively thin, more recent ultisols.

In addition to land formation and soil type, three socioeconomic domains can be identified in the five communities selected in the Formon area. These are diversified smallholders, the large land owners and/or controllers of state land, and the sharecroppers and renters dependent on state land. These three domains correlate in the communities with wealth, livestock ownership, and market and labor access. The three socioeconomic domains may not all be present in each community, but the selection of communities ensured their representation overall.

In the Trois Sources area, the two sites selected, Sivette and Gren Mouton, reflect the different strategies of permanent residents and non-resident farmers. This was straightforward for the permanent residents but was more complicated for the migrants, since the families were not present in the region.

The project asked Father Hallee and COSAR to help identify a limited number of non-resident farm families in the different socioeconomic domains who were interviewed in order to select cooperators. These cooperators served as case studies to better understand the migrant non-resident farmer phenomenon. With the limited resources available, the project had great difficulty doing a more general study of migrant circumstances. Knowledge of families in the Formon/Cavalier areas, where some of the non-COSAR non-resident farmers originated, allowed the project to pick a small number of non-COSAR migrants for cooperators and as case studies.

Durand. The majority of farmers are sharecroppers on state land. Their access to livestock is largely through gadinaj and they are more likely to sell labor than to buy it. Residents have difficulty accessing inputs such as seed or planting material and are tied to the Sous Bois market.

Formon proper. All three socioeconomic domains can be found, but primary control of the land is in the hands of those who control the state lands.

Sous Bois. This area contains both state renters and sharecroppers, as well as smallholders.

Kay Tilus. This community is located on the slopes above the Ravine Casse Cou. Although much of the land there is controlled by one of the residents of Formon, some of the land is owned by local residents who generally have small and diversified landholdings.

Upper Cavalier "Delta/Ma Blanche". Here one finds land under cultivation that is owned by absentee landowners. Some of the land is "Te Leta" or state land.

Sivette. This area represents the permanent residents of the area, most of which can be described as smallholders.

Gren Mouton. The majority of farmers in this area are non-resident farmers.

Initial project activities

The first year of project activities included two major thrusts designed to achieve point source testing and dissemination of sustainable technological innovations. Known, low risk technologies were immediately moved onto local farms in the target communities for farmer testing and screening. It was expected that some adaptation of these technologies for utilization under local conditions would be necessary. In general, however, these technologies included those which could be expected to achieve the greatest

immediate and intermediate impact both in terms of diffusion to other households and communities, and in terms of biophysical and socioeconomic stabilization of existent farming systems. Examples of such technologies include: varietal screening of existent black bean cultivars which have been developed for productivity; and inoculation of chickens for Newcastle's disease.

Higher risk, less well-tested technologies were first tested on rehabilitation test sites entirely at project cost and without farmer risk in the area. Technologies which proved appropriate in these areas would have then been moved onto farms for final adaptation to farmer conditions in future years. Examples of such technologies include: screening indigenous tree species for use in agroforestry systems; evaluation of native grasses for utilization as living barriers; screening exotic tree species for fuelwood production; and testing varieties of vegetables, beans and tubers for production potential under local conditions.

Approximately thirty primary project cooperators (PPC's) were selected from the seven communities named above. These individuals were to screen and evaluate suggested low risk technologies and assume responsibility for dissemination of those technologies within their immediate community circles in the coming years. Using a community-based approach, project technicians assumed responsibility for working very closely with these thirty individuals, rather than attempting to work directly with large numbers of farmers. A multiplier effect of five to seven was anticipated in succeeding years. That is, each PPC was expected to assume responsibility for disseminating appropriate technologies to five to seven other individuals within his/her community.

The primary project cooperators were selected based on the socioeconomic criteria described above. Not all of the cooperative relationships that were established in the first year proved fruitful, of course, and it was expected that as many as one-third of the initial cooperators might elect not to continue as primary project cooperators in succeeding years. Further selection was therefore expected to occur in the future, based both on expansion of point source diffusion into other communities and on selection of other individuals as primary cooperators within the original group of seven target communities.

Seven rehabilitation test sites were also established during the first year of project life. These sites were selected to represent a wide range of biophysical conditions, including the three land formations and soil types described earlier. Unlike technologies tested on farms, it was expected that a relatively high number of technologies tested on these sites would fail. The purpose of these test sites was to screen unknown technologies for application on farmer-managed test sites. However, these technologies were also expected to have a high long-term return, thus justifying both the time and cost involved in their evaluation.

Unlike the primary project cooperators, the seven rehabilitation test sites did not have to represent the socioeconomic distribution in the communities. Nevertheless, considerations of the feasibility of incorporating given interventions into existent farming systems, including socioeconomic constraints, were considered in testing at these sites. It was envisioned that the sites would be chosen from locations representative of the range of biophysical conditions in the region. The locations were selected by mid-1989, with immediate start-up of field work.

Because land availability was already a constraint, some areas required that lease or rental agreements be made with local landlords. State lands were used in several cases. A minimum of one hectare of land was needed for such sites and, in some cases, up to five hectares were useful. Local labor was hired to work under the supervision of project field technicians.

Direct project impact

Rehabilitation test sites were established and the initial thirty PPC's selected by June, 1989. On-farm screening occurred in the period June, 1989, to September, 1991. The permanent population of the entire MBR region is estimated at 900 to 1000 individuals. Migratory populations form an important part of the total population which impacts the MBR in the Trois Sources area, but the actual number of migrants impacting the area cannot be accurately measured.

Expectations of direct project impact for the permanent population were established. Minimal levels of direct impact were expected to include one hundred farm households by January, 1991. This represents

approximately five hundred individuals, or about fifty percent of the total permanent population. This assumed that one-third of the primary project cooperators would elect not to maintain a working relationship with the project and that each PPC would be able to work with only five other individuals during the period January, 1990 to January, 1991.

Project emphasis was placed on offering to farm households a menu of tested and adapted technologies. The menu was designed to offer farm households an opportunity to select those items which they could most effectively utilize from both a biophysical and socioeconomic point of view for intervention in their own farming system. Thus, direct project impact would not be measured by the level of dissemination and adoption of any given technology, but rather by the number of households which elected to adopt one or more technological adaptations from the proffered menu. Some households found that only one of the proffered items was suitable for adoption within their farming system.

It must be stressed here that the extremely short proposed lifespan of the project seriously handicapped adoption (or even testing) of many project activities. Given a two-year planning framework, it was clear that the highest levels of adoption must be expected with those technological innovations which were (1) easiest to test and adapt, (2) easiest to adopt, and (3) highest in short-term return to farmer investment. Unfortunately, these technologies did not include many interventions which could be expected to have the greatest potential impact on total MBR development over the long term.

In five years, for example, it was scarcely possible to test local tree species for suitability in agroforestry systems, much less expect to see adoption of their use on a widespread basis. Yet, over the long term, the incorporation of native species into the farming system is one of the most important potential outcomes of the MBR concept. This both ensures survival of those species within the MBR, and lends stability to the overall agroecosystem.

Similarly, five years was a very short period of time in which to achieve a sound working relationship between outside agents and the local population. As experience the world over has shown, development of fruitful, long term relationships is a stepwise process. Farmers cannot be expected to accept the word of outside agents that basically unknown and relatively exotic practices, such as terracing, are beneficial and necessary without some proof that those agents do in fact represent a valid pool of knowledge. This again points to the importance of bringing successful short-term interventions to the area.

Perhaps more important, the full benefit of even short-term interventions can only be achieved over a longer period of interaction. They provide the basis for trust and mutual respect which will allow less obviously beneficial innovation to occur. The project specifically rejected the opportunistic approach of concentrating all resources on short-term interventions, but stated clearly that a five-year planning horizon would not permit adoption of many needed technologies and would not realize full returns from the development of working relationships.

Technologies suggested for short term investigation

Activities undertaken during the first two years of project life focussed on four areas: (1) improving production of existent crops; (2) improving livestock production; (3) introducing new crops; and (4) infrastructural improvements. These four areas, taken together, provided a sound basis for providing immediate improvement in cash flow in the local economy, although some of the suggested technologies required longer than two years to reach fruition (improved cattle breeds, for example). Prior to June, 1989, on-site team members and the support team resident at the University of Florida determined which of the technologies listed below should be tested by primary project cooperators during the first two years of project life.

Four criteria are used to determine whether the project should invest time and money in testing any given short term intervention suggested below.

- 1). Does the intervention offer a means of increasing cash flow? This includes interventions which prevent outflow of cash for farm families. Buying corn meal, for example, represents a significant cash flow for some families. Therefore, improved corn production could be a priority for these farming systems, even though corn production is not highly important today.

- 2). Does the intervention improve or maintain the system's diversity? Diversity is important to the farmer because it ensures his/her food supply and a steady inflow of cash. Most farmers interviewed indicate that their farming system would be significantly improved if they could spread their sales of crops across more calendar months. Diversity is important to the project because it enhances the overall stability, and therefore sustainability, of the farming system.
- 3). Is the intervention sustainable without on-going project inputs? The University of Florida specifically rejects on-going give away programs or continual subsidization of inputs. Interventions which require these kinds of on-going input from the project make farmers dependent upon continuance of the project for the health and welfare of their own farms.
- 4). Does the intervention provide the farmer with increased return to labor? These systems are labor intensive. Most farmers state that labor is a major constraint that they face. Labor is currently the only outside input into the farming system. Therefore, increasing the productivity of labor must be a guiding principle for intervention.

Improvement of existent cropping systems

Beans. Improving bean production was a major objective of first-year project interventions. Beans are a very important cash crop. They provide farmers with a high return to labor. Their production can be improved without considerable outside inputs.

Improved Varieties. The major problem confronted today is disease. Mosaic (probably both common and golden) infects a large portion of the bean crop in the region. This is one instance where a simple change in variety could prove to be successful, one of the least expensive, most easily adopted, and easiest to maintain interventions possible in a farming system. The International Center for Tropical Agriculture (CIAT) should be asked to provide seed for testing cultivars that are resistant to these major bean diseases and that illustrate appropriate biological potential for production at elevations exceeding 850 meters and under rainfall conditions of over 3000 mm of precipitation. Screening these varieties on farms was a major activity for the first year of project life.

Seed Availability. Getting bean seed is also a problem for the poorest farmers who cannot afford to buy many seeds. One way to overcome this is for farmers to save their own seed. This is probably not feasible, given the extreme pest problem that exists in the zone, which prevents farmers from storing any grains for any considerable length of time. Improved storage facilities (recommended below as a project intervention) could play an important role here. Storage of beans should be the first major focus of efforts to improve grain storage.

If farmers do in fact move toward seed production and storage on farms, they will need to learn how to maintain disease free seed and how to select seed well. These interventions should be considered for the future. Some limited testing of response to P fertilizer was useful.

Yams. Yams are another very important element in the farming system. They are a good cash crop, produce relatively well without outside inputs, and provide farmers with a high return to labor. **Availability of Planting Material.** Obtaining yam planting material is a major problem for many farmers, particularly those of lower socioeconomic status who indicate that they cannot obtain material. ORE examined this problem in greater detail as a major priority since ORE's tissue culture laboratory could provide a possible source of inexpensive seed material for farmers.

Sweet Potatoes. Sweet potatoes are an extremely important element in the farming system. Virtually all farmers raise some sweet potatoes and they are both consumed and sold. They, like yams, grow relatively well without fertilizer inputs. Their market value is lower than that of yams, which made intervention in their production a lower priority than yams if interventions to increase the availability of yam planting material, especially for the poorest farmers, were possible. Interventions in sweet potato production received a high priority.

Pests. Pests are the major problem in sweet potato production. Mealy bug infestations are easily observable in many sweet potato plantings and probably represent the major pest problem. The occurrence of this pest can be alleviated by increased use of fallow, chemicals, or use of uninfected planting material. The latter was the most promising approach for the short term in the region given that land

availability was already a major constraint and that cash flow was not sufficient to permit the use of chemicals. Again, ORE's tissue culture facility could provide one avenue of producing clean planting material.

Grubs were also mentioned by some farmers as a pest in sweet potatoes. It is unclear that the damage caused by this pest is extensive, and the team was not able to accurately identify the grub. The extent of this problem and possible interventions should be investigated.

Manioc. Manioc is an important element in some farming systems. It is produced both in home gardens and as an intercrop in other parcels. Its market value is unclear, but farmers do sell it. The first recommendation was that the project team gain more information about the role that this crop plays in the farming systems, its marketability, and problems in its production.

Improved Cultivars and Planting Techniques. Some improvements in production were possible through the introduction of improved cultivars and/or planting techniques. Manioc in the area produces only a very small tuber, which may be due to disease, cultivar, planting type, or nutritional status. CIAT has available both improved manioc cultivars and, perhaps more important, has developed a number of tested technologies for improving planting techniques with this crop. Sufficient interest was not exhibited by PPCs to improve cultivars and improve planting techniques. In general, because interest on the part of PPCs was not high, work with manioc was regarded as a much lower priority than work with beans, yams, and sweet potatoes.

Summary of Improvements in Cropping Systems

Beans, because of their high value and the relative ease of improving production, received first priority during year one of project life. Emphasis was placed on screening new varieties for resistance to disease and improved seed storage. Yams were a second major priority. Increased seed availability is the major problem faced by farmers. Sweet potatoes were a third priority. Manioc represented a significant opportunity for intervention, but requires further investigation prior to attempting interventions. Improved cultivars and/or planting techniques may represent avenues for increased production.

Activities suggested for year one of project life were:

1. Screening of improved bean varieties;
2. Improved grain storage, with emphasis given to bean storage;
3. Production of yam planting material; and
4. Production of pest-free sweet potato planting material.

Livestock production

Livestock play an important role in the farming systems found in this region. They provide a major source of cash income for farmers. For the poorest farmers, gaining access to animals through taking care of them for other farmers is one major way of improving economic status. The project therefore included interventions with livestock as a major thrust in its first year activities.

Swine. No other animal can replace the hog in the local farming system. This animal has an extremely high value. It provides very high returns to labor, since the labor demand involved in caring for hogs is relatively low, especially compared to crop production. The hog introduced by the project is very similar to the traditional Creole hog, and those introduced in the region to date have exhibited 100% survival. They are foraging and growing relatively well, even under limited food supply conditions. Farmers are delighted with them.

Swine Repopulation. Loss of the swine population clearly represents a disastrous blow to the cash economy of farmers in the region. The swine repopulation effort which had already been initiated was continued with a more effective distribution system and with further investigation into feeding practices and potential. It was strongly suggested that all PPC's become focal points of the repopulation effort and that they receive swine at the earliest possible opportunity, designed to coincide with maximum availability

of food resources for the animal. Swine were further reintroduced through distribution to those farmers who work with the PPC's during year two of the project. A monitoring system for performance of the animals was instituted. It was also suggested that the project develop its own swine reproduction facility, preferably in the Formon area, in order to alleviate supply problems and to assist in acclimatization of the animals to the zone.

Chickens. Chickens are present in virtually every household. They are rarely consumed, but many families do consume eggs. Both chickens and eggs represent very good market items. Returns to labor are extremely high since chickens receive virtually no care. Most forage for food, although they may also receive corn meal when it is available in abundance.

Poultry Health. Project personnel worked with IRD to provide improved poultry care in the area, including providing inoculations. An inoculation program was instituted. It was recommended that the female PPCs be trained to inoculate chickens and be provided with initial inputs by the project.

Cattle. Many farmers have access to cattle, which are highly prized. The wealthier farmers own cattle. Poorer farmers gain access to them through caring for others' animals. The caretaker receives the milk produced. In addition, he/she is entitled to keep every other offspring, which becomes an important mechanism for increasing wealth.

Nutritional Status. Nutritional status of the animals observed in the area is good, but could probably be improved. The damage caused by parasites would undoubtedly also be reduced by improved nutritional status. Further, introduction of tall grasses, grasses as retaining walls or living barriers on terraces, and improved growth of local grasses were all project priorities for the intermediate and long term because of their role in conservation and rehabilitation practices. Improving nutritional status is probably both the easiest intervention, the lowest cost intervention, and the intervention that achieves the highest returns to the project because of the importance of forages in conservation and rehabilitation.

It was therefore suggested that limited project efforts to improve livestock (including sheep and goats, discussed below) nutritional status be undertaken in year one of the project. This was limited, in the first year, to planting test plots of tall grasses on a few selected farms in the region. Introduction of improved short grass pastures (such as African star grass or kikuyu grass) was not envisioned because these species could easily become severe pests in the park reserve area.

Improved Breeds. In the longer run, dairying could be an ideal enterprise for this region. In the short run, improving milk production could both increase the protein intake of the family and, over an intermediate time frame, provide a source of cash income. Introduction of some improved breeds could provide an initial step toward some longer term goals.

Importing animals is expensive and extremely risky since it is common for the majority of imported purebred animals to die in settings like those found in this region. Therefore, artificial insemination is suggested. Holstein lines should not be utilized. Brown Swiss or Jersey would probably make appropriate crosses. The Sondeo team suggested that initial efforts at artificial insemination be made in year one of the project, and the health and performance of the first generation cattle be carefully monitored.

Disease and Pests. Some farmers report disease problems among cattle, probably blackleg. There has existed a government inoculation program in the past, but it is unclear that this program is now operative. It is also unclear that this is a service that farmers are willing and/or able to support financially. IRD cooperated with UF in the inoculation of cattle.

Parasites are also a problem. However, it was clear that farmers were unable to purchase medicines. There were many higher priorities for their cash use. Although no action was recommended in the Sondeo, the UF discovered that a subsidized program would be effective.

Sheep and Goats. Both sheep and goats are present on many farms. They are normally tethered and are often brought to the homesite at night. Both are lower value animals than swine, and farmers exhibit relatively little interest in improving their production. They also require more labor than chickens. Therefore, it was suggested that both sheep and goats rank below swine and poultry in terms of interventions in livestock production.

Nutritional Status. Like cattle, sheep and goats appear to have relatively good nutritional status. Nonetheless, improvement is also probably possible here. It was suggested that some farms where tall grasses are tested include sheep and goat producers.

Diseases and Pests. Foot rot is a fairly common problem for sheep producers, particularly those who tether their animals on the higher clay content soils since these soils retain more moisture. Some efforts to partially control this disease could be made, particularly introducing the practice of trimming their hooves. However, it was unclear that this was a significant priority at all for farmers and, it did not constitute a major project priority because farmers did not exhibit much interest during the Sondeo. This problem is not common to all farmers and would, therefore, be an intervention limited to specific zones.

Parasites, particularly internal parasites, are undoubtedly a health problem for both sheep and goats. Even more so than with cattle, use of commercial products is not feasible because that there are other, much higher, priorities for cash use by farmers. Some success in parasite control on sheep using tobacco and wild artichoke has been achieved in Peru. However, it is unclear that the time and effort that would be involved in finding local plants which could serve as a deparasitant are justified by either the seriousness of the problem or the importance of these animals in the farming system. No action was therefore recommended.

Summary for livestock production

The role traditionally played by swine in the local farming system simply has not been, and probably cannot be, fulfilled by any other element. Therefore, major project attention was focused on improving the swine repopulation effort. This occurred through the primary project cooperators in year one, and through their cooperators in succeeding years. Poultry, particularly chickens, are also a critical element in virtually every household. Disease (probably Newcastle's) is the single most serious problem faced by women raising chickens. The project therefore initiated a program of inoculation, carried out by IRD.

Of other livestock, cattle are the most important and should receive the greatest attention. Improving animal nutritional status was recommended. Tall grasses were tested on a few local farms in year one of the project. Ultimately, this region would be ideal for dairy production, and some artificial insemination to improve the local breeds' dairy production should be attempted. Control of parasites is recommended as a short term intervention, and would clearly become more important if dairy production does in fact increase in importance.

Activities suggested for year one of the project included:

1. Strengthening of the swine repopulation effort;
2. Development of a poultry inoculation program;
3. Testing of tall grass pastures on a few local farms, including some which produce sheep and goats; and
4. Artificial insemination of cattle on a few farms to increase milk production.

Introductions

Biophysical conditions in much of the area are very promising for the introduction of several high value vegetable crops. Transportation and marketing considerations necessitate that new crops exhibit low perishability and relatively high unit value. The following crops were suggested as potential test items for new crop introductions. Prior to June, 1989, a short list of five to six potential introductions was developed for testing during the first year. The home garden mentioned above served as the focal point for testing these alternatives; however, UF felt it should bear the cost of field testing.

Garlic was given receive top priority. This is a relatively high value item in Haiti. It is easily transported and exhibits very low perishability. The national market is very large and Haiti currently imports large amounts of garlic.

Leeks, chives, and onions all were investigated since these crops have already been incorporated by some households but have clearly not been disseminated as widely as possible.

Spices also received fairly high priority. Thyme is already produced on some farms for sale (again, in the home garden). These crops provide high value per unit weight and exhibit extremely low perishability. A market for them definitely exists.

Pigeon pea is raised on many area farms. Farmers indicate that this crop brings a good market price, although not as high as black beans. Farmers also indicated that they would be interested in producing more of this crop.

Potatoes and carrots have already been tried by some farmers and found to grow well. While their value per unit weight is lower than that of some of the crops indicated above, farmers have indicated a willingness to carry them to market and were apparently satisfied with the market value for them. Both could be considered for planting outside the home garden, and represent possible alternating crops for sweet potatoes in some fields. This could alleviate the pest problems currently associated with the continual, widespread production of sweet potatoes.

Some farmers also currently raise **amaranth**. Further investigation is needed to determine the degree to which a market for this crop exists.

Markets for other root crops, such as **beets, turnips and cabbage** definitely exist. Further investigation of the market potential for such crops is needed, however, since it is unclear that their market price will be high enough to offset the problems associated with their transportation.

Lima beans, lentils and kidney beans could represent alternatives to black beans for some farmers. Again, further investigation into the market value of these crops is needed before their introduction is attempted.

Both the production potential and the marketability of crops such as **spinach and other greens** are questionable. Further investigation is needed before these crops are tested on farms.

Summary for Introductions

Introduction which have the highest market potential should be tested first on site. These include garlic, spices, and pigeon peas. Other crops which show high potential include carrots and potatoes. Of lower priority are crops for which little marketing information exists, such as turnips, lentils, and spinach. The team developed a short list of five crops for testing in home gardens in the first year of project life and testing was initiated in June, 1989.

Infrastructure development

Haitian rural society demonstrates a serious lack of infrastructure development. In fact, some people would argue that Haitian rural society has undergone a process of decapitalization since 1804. The project considered undertaking certain activities which would contribute to the development of infrastructure within the MBR communities. Some of the types of infrastructure which the project considered supporting included: grain mills, crop storage facilities, oxen plows, bee hives, replacing traditional home construction with the use of concrete blocks, drying floors and improving the health of pack animals as a means to improve transportation and access to markets.

Improved crop storage

Storage of crops is a very serious constraint in the region. Losses to pests are so high that farmers sell virtually everything they produce at harvest, often rebuying the same product at a higher cost later in the year. It is virtually impossible for farmers to take advantage of any seasonal variation in market value of crops without storage. Further, inability to store crops further limits the farmers' ability to adequately

manipulate cash flow to meet family needs. This problem is serious enough when most cash flow goes to consumption needs. As purchase of outside inputs becomes more important, the need to be able to manipulate cash flow adequately becomes even more important. All of these methods would help improve the quality of the product available for market and reduce storage risk by reducing the loss of crops due to spoilage and/or physical damage. Therefore, the development of improved post harvest technologies, particularly crop storage technologies, was the highest priority of the interventions in suggested infrastructural. Several interventions are possible.

Storage of grains is a very serious problem. Tubers can, to some degree, be "stored in the ground." This option is not possible for either beans or corn, forcing farmers to sell their entire production at one single time during the year. This contributes to the problem of seed availability for planting beans, which is a constraint particularly important for the smallest farmers. Therefore, improved grain storage was a very high priority activity.

The simple metal storage bins produced in Camp Perrin were considered as one alternative. However, the potential for introducing other technologies which do not require this level of investment should also be examined.

Storage of tubers could also be useful to farmers. Currently, tubers are basically "stored in the ground." They are immediately consumed at harvest, and replanting occurs immediately. This contributes to the problem in securing yam planting material, since farmers who cannot maintain continuous production are forced to buy planting material at relatively high cost. Introduction of CIP potato storage technology could contribute to the preservation of many of the tuber crops produced in the region.

Drying floors would help reduce humidity and improve the storability of produce.

Honey production

Honey is a high value product and honey production seems to be a possible activity throughout the region. If vegetable production increases, bees can be expected to become an even more important element in the farming system. The introduction of more fruit trees (suggested under intermediate and long term interventions) would increase their value even further.

Traditional techniques of raising bees and removing the honey produced often require the destruction of much of the hive. This problem is a relatively easy one to overcome. Simple improved bee hives would prevent this destruction and allow increased and more efficient honey production. Several models used in Africa and Latin America can be made almost completely of local materials, thus avoiding the high cost of using improved hives like those employed in the United States. The Sondeo team recommended giving this activity relatively high priority; however, the Haitian specialists contacted by the project could not dedicate sufficient time for this activity to be carried out.

Pack animals

Transporting produce down to the lowlands is one of the important constraints in the marketing system. Although pack animals may be borrowed or rented, the number of pack animals available does seem to limit access. The importance of pack animals can be expected to grow if agricultural production is in fact intensified in the area. Producing a relatively high number of pack animals is clearly a relatively long term program. For these reasons, the team investigated the potential for interventions in this area very early in project life; however, at that time, cattle were far more important for the farmers.

Animal health. Several farmers interviewed indicated that animal health is a major problem with pack animals. A disease, possibly tetanus, appears to be relatively important. It was recommended that the team investigate this further. However, pack animals were not of interest to the major target group. Nevertheless, pack animals were inoculated at the same time the other animals were inoculated.

Increased availability. Donkeys probably represent the most appropriate animal for most farmers in the region. While both mules and horses are more desirable animals from some points of view, their size

precludes their maintenance by some of the poorest farmers. The project team investigated the potential for increased production of donkeys in the region. However, primary project cooperators are unable to afford the costs associated with such animals.

Grain mills

Grain mills would reduce the need to take produce, particularly maize, down to lowland markets to be ground into meal. The labor saved could be used in other productive activities. The project team therefore investigated the possibility of introducing small grain mills into one or two communities in the region on a trial basis. While local farmers were interested in this technology, the amount of grain to be milled was not sufficient to justify costs associated with the construction of mills.

Mechanical land preparation

Mechanical land preparation could reduce the labor and time necessary for land preparation for some agricultural activities. This is admittedly limited to some of the plains and areas of less severe slope. Use of animal traction for plowing would also facilitate some of the terracing interventions that are likely to be tried in some of these same areas. The potential for utilizing mechanical land preparation was investigated. This was a relatively low priority activity. In addition, no one was available to train others in the use of plows or other mechanical devices, and no animals had been trained to pull plows.

Home construction and water facilities

Homes and the resources used in their construction are probably the primary infrastructure consideration throughout the region. The use of wood for building houses is one of the major sources of consumption of forest resources. The introduction of concrete block technology could replace the traditional construction techniques and reduce a major source of demand for wood--a large portion of which is used to produce the lime used in traditional mortar. This was a relatively low priority activity.

Although water cisterns might help improve the quality of life and reduce labor demands somewhat, they are not essential to vegetable production in these high rainfall areas, like they are in some other regions of Haiti. There appears to be relatively little economic return on an investment in cisterns in this region. The improvement in the quality of life is a worthwhile objective, but due to the low economic return this project placed a low priority on such investments.

Intermediate and Long-Term Interventions

Interventions which can be expected to produce positive benefits in an intermediate time frame should be tested either on farms (when risk is low) or on rehabilitation test sites (when risk is relatively high). Larger scale dissemination of these interventions should not generally be expected to occur during immediate project life. Interventions which can be expected to produce positive benefits only in a long term time frame were tested first at rehabilitation test sites. Dissemination of these technologies is unfeasible if total project life fails to exceed ten years.

While the original two-year time frame of project funding seemed to preclude any intermediate and long term intervention possibilities, the Sondeo team felt strongly that it was necessary to start testing potentially useful ideas that would take longer to evaluate but potentially have significant longer term impact on the residents of the area and on the integrity of the Park area. The ideas presented by the team fell into three general categories: Agroforestry, forestry and infrastructure development.

Agroforestry

Introductions. Agroforestry techniques were developed to supply an intermediate type of intervention providing more protection to the environment than agriculture alone. For this, introduction of exotic fruit and nut species were suggested in the Sondeo.

Utilization of native species. Windbreaks, living terraces and land rehabilitation projects using native species formed an important part of the research of native tree species in the project area. In addition, native species of trees were reintroduced in cleared *rak bwa* areas to reestablish them. It was estimated that windbreaks alone might be capable of enhancing present agricultural production.

Utilization of Exotic Species. While much of the research effort was oriented towards improvements of existing crops, some effort was placed on testing exotic crops or tree crops which appeared promising in certain biophysical environments. Some of the exotic species originally suggested by different members of the Sondeo team include: cardamom, apples (Israeli, Florida and other varieties), grapes, black pepper, lentils, radish (regular and Japanese type), beet root, turnip, cabbage and other cole crops, spinach/greens, many types of spices and herbs, grain amaranth, garlic, carrot, ginger, etc. In addition, various edible fungi (mushrooms) could be investigated in the native forest and/or introduced as a specialized crop.

Terracing and land rehabilitation. Living terraces of forage and woody plant species were attempted on some of the severe erosion areas in order to stabilize the soil. The trees were leguminous trees capable of coppicing or re-sprouting after lopping. Forage can be cut for sale or used to enrich the terrace soils. Several exotic tree species and forage species are available for this. (NOTE: Care has been taken not to introduce species of known or potential explosive growth that could endanger the native species of the Park area such as *Leucaena* and kudzu!).

Forestry

Sylvicultural treatment of at least some of the remaining forest areas outside of the core area of the Park is appropriate. Predominantly pine areas were designated as selective harvest and pole production areas through zoning. Thinning operations could be a productive intermediate program enhancing the longer term development of harvest of selected larger trees for lumber production. Utility pole production using pressure or otherwise treated poles may be a viable alternative to presently available poles for rural electrification.

Smaller poles presently available from thinnings were found to be appropriate for construction forms and scaffolding. As the road is improved to the areas surrounding the Park, these production areas could prove to be a supplemental source of jobs and income for inevitably increasing populations of residents.

Hardwood species for special use may also become available for limited harvest in areas outside the core, but would require additional study to ascertain feasibility and impact on the Park.

Infrastructure development

Storage of crops is a serious constraint in the region. Introduction of tuber storage technology could contribute to the preservation of many of the tuber crops produced in the region. Simple metal storage bins produced in Camp Perrin could be useful in reducing the some of the important post harvest losses to rats, mice and other pests. Drying floors would help reduce humidity and improve the storability of produce. All of these methods would help improve the quality of the product available for market and reduce storage risk by reducing the loss of crops due to spoilage and/or physical damage.

Honey production seems to be a possible activity throughout the region. Traditional techniques of raising bees and removing the honey produced often require the destruction of much of the hive. Simple improved bee hives would prevent this destruction and allow increased and more efficient honey production.

Homes and the resources used in their construction are probably the primary infrastructure consideration throughout the region. The use of wood for building houses is one of the major sources of wood consumption. Programs should be developed to produce lumber for home construction. Programs should also be developed to produce fuel wood required for production of the lime used in traditional mortar.

Transporting produce down to the lowlands is one of the important constraints in the marketing system. Although pack animals may be borrowed or rented, the number of pack animals available does seem to limit access. Animal health interventions, particularly with regard to what appears to be tetanus would likely increase the availability of pack animals and access to transportation for both merchants and farmers.

Water systems, including cisterns, would help improve the quality of life and reduce labor demands. For these reasons, water systems must be developed.

Proposed interactions with NGOs

ORE's Activities in the Formon Area. The Sondeo team was able to establish the immediate goals and activities of ORE in the Plain of Formon target area of the Macaya Biosphere Reserve as follows:

The demonstration of hedgerows using *Calliandra*, Guinea grass, Elephant grass, and a hybrid *Pen-nesitum* RD 22 (which does better at higher altitudes). ORE had requested that the University of Florida provide recommendations concerning the use of indigenous rather than imported plants, which was done.

The nursery capacity of ORE in the Formon area is currently about 50,000 trees a year. ORE is concentrating on local coffee and shade trees, and the production of various indigenous forestry trees, and fruit trees such as avocado and citrus. They are making a special effort to look for a rust resistant coffee tree and are searching for indications of the presence of rust in southern Haiti. ORE believes that searching for different varieties of coffee and seeking to improve culture techniques are of high priority. ORE initially steered away from the question of growing apples in Formon because they decided that only the richer farmers would be able to make the investment necessary to grow apples. At the time apples were initially considered, ORE had made the decision that priority should be given to the introduction of citrus and avocado trees. Since this activity has been completed, ORE has indicated that they might now be interested in reconsidering apples as a high revenue crop for the area.

The question of yam improvement and the scarcity of plants is being addressed by ORE as follows: a) ORE is mini-setting yams in Camp Perrin for planting in Formon and plans in the very near future to start mini-setting in Formon (this may have already begun); and b) ORE is developing techniques for the tissue culture of yams. This would be primarily for the introduction of new varieties and the maintenance of high-yield disease-free plant material from which mini-setting would subsequently be done.

Two varieties of sweet potatoes - Tapato & Viola - provided by Dr Frank Martin as the most suitable of those he had selected at Mayaguez, Puerto Rico have been multiplied and distributed to Formon. Tests showed that these varieties gave higher yields than the highest local varieties.

ORE is starting up a seed improvement program under the supervision of Agr. Eliassaint Magloire (who majored at the University of Florida in plant genetics and has 8 years experience in this field in Haiti, Mexico, and the US). They planned to address the question of high altitude corn for the Formon area in 1989. ORE has already produced seed of the lowland chicken corn, which is the farmers' first choice for lowland and accounts for 80% of the corn used in Haiti. They plan to use this seed in other project areas. Their seed improvement program also involves the selecting and breeding of black bean seeds.

In November Agr. Maurois Francois conducted a vegetable demonstration activity on ORE's demonstration sites, and with 5 participating farmers. Cabbage, leeks, onions, carrots and beets were chosen as the most transportable, relatively high revenue crops.

ORE has signed up about 20 farmers in the area (both the flat lands and the hilly slope) to participate in some or all of the above activities. They intentionally kept the number of participants low. They were concerned that UF not start a program in the same area which would involve the same farmers before a protocol with ORE had been worked out, and areas of mutual benefit established.

ORE's resources in the Plain of Formon area. Thirteen permanent employees who act as animator/extension agents, nurserymen and the other activities mentioned above. Supervision by Agr Eliassaint Magloire, Francois Alexis and the inspection team is divided between this area and the four other areas where ORE works. ORE has stated that they realize that a full time agronomist level supervisor should ideally be in charge of these activities.

Anticipated Interactions with ORE. Project Macaya anticipated that a close working relationship with ORE would be possible because of the close proximity of project headquarters. In addition, Project Macaya shared the same concerns and responsibilities for many activities on the Plain of Formon. The anticipated close interactions with ORE were in the following areas:

- 1). Project Macaya and ORE worked together to develop a fully coordinated action plan for dealing with farmers on the Plain of Formon.
- 2). Project Macaya and ORE shared ideas, plant materials, and planting technologies in order to maximize the quantity and quality of the agricultural efforts of both projects on the Plain of Formon.
- 3). UF and Project Macaya provided ORE with research and development experiences, data, and personnel in areas applicable to the stated goals of ORE's projects in the Macaya Biosphere Reserve region.
- 4). UF and Project Macaya provided ORE with the names and biological data on appropriate endemic plant species that are most appropriate for specific locations within the Macaya Biosphere Reserve.
- 5). UF and Project Macaya provided ORE with technical information on the state of the ecosystem within the Macaya Biosphere Reserve and surrounding areas (gathered via such data sources as aerial photography and satellite imagery).
- 6). The University of Florida made available to ORE the Macaya Biosphere Reserve Natural Resource Data Base being compiled as part of this project.

UNICOR's activities in the Trois Sources area. The purpose of the Cooperative Agreement between USAID/Haiti and UNICORS was to establish soil-conserving and watershed-regenerating technical interventions in targeted watersheds in the southwestern peninsula of Haiti. UNICORS was to provide material and technical assistance to upland peasant farmers so that they could increase the productive potential of their hillside farms and increase on-farm income, while at the same time contributing to the stabilization and improvement of the targeted watersheds.

UNICORS and its cooperating organization COSAR occupy a very important place in the management of fragile lands within the Macaya Biosphere Reserve. The land now owned by COSAR and managed for coffee is in the Trois Sources Target Area on the west side of Pic Formon, and the Pic Cavalier region of the Plain of Formon. Much of this land is steep. Its soil characteristics and vegetative cover are essential to the stability of the upper watersheds of the Riviere Port-a-Piment and the Riviere des Roseaux. We were very impressed with the stated goals of UNICORS to stabilize steep slopes through wise land use activities, and with their proposed activities. We believe that many of these activities have been slow to be implemented in the Trois Sources region because of three important factors: 1) the Trois Sources area is very remote (although a road is currently under construction that will have a dramatic impact on this feature); 2) there have been problems in identifying proper land use practices on steep slopes; and 3) the Riviere Trois Sources is a tributary of the Riviere des Roseaux, which is technically not part of the Targeted Watershed Management Project (note: this area is south of Pic Macaya, and therefore should be considered part of the TWM project). The Sondeo team recommended that the University of Florida Project Macaya interact with UNICORS in the western region of Pic Formon and the Riviere Trois Sources basin in the following ways.

- 1). Project Macaya and UNICORS worked together to insure that the upper watershed of the Riviere Trois Sources (technically south of Pic Macaya, and therefore part of the TWM project guidelines) was included in the UNICORS plan of action for their cooperative agreement with USAID.
- 2). Project Macaya provided technical input to UNICORS on current land use practices in the Target Area.
- 3). Project Macaya provided detailed recommendations on proper land use activities in the Target Area, and produced a detailed map of proposed Land Use Zones.
- 4). Project Macaya provided technical assistance to UNICORS by sharing research data, the Macaya Biosphere Reserve Data Base, aerial photographs, satellite image data, and other technical services that could assist UNICORS in its project goals.
- 5). Designated representatives of Project Macaya and UNICORS met to discuss the mutual needs and goals, and to plan for the implementation of the activities of each group in the Target Area.
- 6). Designated representatives of Project Macaya and UNICORS should meet to discuss the possibility of trading very fragile COSAR lands for other less fragile habitats, and to seek other ways of

permanently protecting the most fragile zones in the Trois Sources basin.

- 7). UNICORS supported the activities of Project Macaya by providing lodging facilities.

Anticipated Interactions with DCCH and IRD. Both DCCH and IRD are important NGGs to the Macaya Biosphere Reserve and the Targeted Watershed Management projects. Their specific areas of geographic concentration, however, are outside of the current Target Areas for Project Macaya. Therefore, during this phase of Project Macaya there were no direct on-site interaction. However, both DCCH and IRD are working on the boundaries of the proposed biosphere reserve, and have a major interest in the plan for the Macaya Biosphere Reserve. DCCH is concerned with the stabilization of steep slopes in the Ravine du Sud, and IRD is concerned with livestock diseases in the Macaya region. The University of Florida via the Macaya Biosphere Reserve project interacted with DCCH and IRD in the following ways:

- 1). Planned together for the design and implementation of the Macaya Biosphere Reserve.
- 2). Met together to evaluate the progress of the biosphere reserve activities, and to share data and experiences.
- 3). Shared technical data.
- 4). Undertook long-range planning together for the entire biosphere reserve region.
- 5). Interacted on specific sub-projects, such as livestock disease control (IRD) and steep slope stabilization via agroforestry (DCCH).
- 6). Project Macaya personnel served as ecological consultants for DCCH and IRD.

Specific conclusions from the Formon sondeo

This listing of conclusions concentrates on the area of the Plain of Formon. The overall area extends from Sous Bois in the southeast, Cavalier in the valley south of the escarpment of Formon/Durand, Morne Cavalier to the west, the steep slopes of the Massif Formon to the north, and the ravines and rolling hills of the Plain of Formon to the east in the region of the Ravine Casse Cou.

Region

- 1) The zones of the target area are: "Kay Tilus", Sous Bois, Durand (pronounced "Diran", Formon proper, Upper Cavalier ("Kay Jeudi"), Cavalier Rak Bwa ("Delta"\Ma Blanch"), Grande Fond, Trois Riviere, and "Kay Bout".
- 2) The permanent population of the target region is divided as follows: Kay Tilus (12 houses with 60 people); Sous Bois (40 houses with 200 people), Durand (12 houses with 60 people); Formon proper (50 houses with 250 people), Kay Jeudi (nearly continuous with lower Sous Bois, but add another 10 houses with 50 people), Delta\Ma Blanch (12 houses with 60 people), Gran Fond (5 houses with 25 people); Trois Riviere (6 houses with 30 people); Kay Bout (10 houses with 50 people).
- 3) The total permanent population of the target region is approximately 785 individuals.
- 4) There are many types of land holdings in the target area. Much of the area is state lands that are either used directly or are rented out or sharecropped. There are also large holdings of private lands in the target area, which fall into two main categories: inherited and purchased land. Much of this land, as in the Delta\Ma Blanch zone, is rented to local farmers by absentee landlords. The Sondeo team recommended that a land tenure study of the region would be beneficial.
- 5) The inhabitants of the region fall into three main groups. 1) Well-to-do local farmers who control most of the capital of the region, and strongly influence the lives of less well-to-do farmers because they buy labor and control access to land. 2) Diversified small landholders who work portfolios of land parcels of diversified land tenure status. 3) Sharecroppers and farmers who rent small parcels of either state or private land (often of poor soil quality or in remote locations).

Crops

- 1) On the Plain of Durand the major crops are yams, sweet potato, cane, plantains, milton, manioc, "malanga", okra, and coffee. The crops are planted in small gardens (1/25 carreau) on well-drained oxisols.
- 2) The large open areas of the Plain of Durand were planted with sweet potatoes. These "patats" were

growing poorly, showed signs of phosphorous deficiency (purple colored leaves) and were infested with mealy bugs.

- 3) There are five varieties of yams grown on the Plain of Durand and Plain of Formon: Yam Guinee, Yam Miral (also known as Yam Rural), Yam Francais, Yam Anglais, Yam Espagnol.
- 4) The yam miral (rural) is a perennial plant. It therefore is available throughout the year, and can continually be harvested. It has the potential of being a crop that can help farmers through the "hungry months" of May and June, which precede the harvest of July.
- 5) The yam rural puts out prolific vegetative growth, and can fill a garden with vines and leaves, or even kill trees. People who plant coffee were especially concerned about planting it. So, there are some special concerns about yam mural that need to be addressed.
- 6) In the region of Sous Bois the household gardens have planted thyme and onions. Small patches of these crops were found in each garden, and the farmers indicated that they were trying them on an experimental basis. These crops also showed up in the gardens of wealthier peasants in Formon, but were much less common in the gardens of poorer peasants on the Plain of Durand.
- 7) Corn was not growing well on the Plain of Durand or Plain of Formon. The plants are small in stature. The Sondeo team believed that the problem was with the variety of corn being planted rather than with the soil conditions.
- 8) Crops are restricted (or negatively impacted) by the high winds that frequently blow across the Plain of Formon from the east.

Animal resources

There appear to be significant problems with the health of domestic animals in the region. An immediate positive impact was made on the local farming system by providing the assistance of trained veterinary technicians to address the specific problems identified by the Sondeo.

- 1) Cattle are abundant on the Plain of Durand, and appear to be in excellent condition.
- 2) Goats are not abundant, but are in "ok" condition.
- 3) Sheep are not abundant, and are in moderate to poor condition. The main problem is foot rot.
- 4) Chickens are in poor condition. They are subject to "pie sech", which is either Newcastle disease or fowl cholera.
- 5) Pigs are doing well. Thirteen individuals (3 males and 10 females) were initially distributed in the region in May, 1988, and allocated to individual peasants by a lottery system. Five more pig distributions were made between December, 1988 and June, 1991; 16 males and 55 females were allocated. The pigs have grown well, and many exceed 140 lbs. They are being well taken care of and are valued. The pigs are produced by the French pig project, and have been specifically bred to resemble the traditional Creole pigs of the country.
- 6) Horses are more uncommon in the region, especially at Cavalier, than the team anticipated. We commonly received complaints about a disease in horses where they became stiff and tense, usually ending in death. We believe that this disease is likely to be tetanus.

Plant problems

- 1) Sweet potatoes are suffering from mealy bugs, and phosphorous deficiency.
- 2) Corn is stunted, probably the result of a lowland variety being planted at too high an elevation.
- 3) Coffee is suffering from several problems. The leaves are showing signs of mosaic. Plants are often infected with a viral disease. Plants also often show signs of nitrogen deficiency.
- 4) "Mawoka", a beetle larvae, is destroying the roots of root crop plants such as "malanga" and "mazonbel".

Market orientation

- 1) The primary orientation of the region is towards the well-established market at Sous Bois. Participants in this market need cash to purchase household items.
- 2) There is a market at Cavalier.
- 3) The main market for the region is at Ducis, which is one of the largest livestock markets in Haiti.

Current impact of the park

- 1) The closing of the state lands to gardens has removed one of the main sources of gardens for the poorer peasants on the Plain of Durand and Plain of Formon, as well as Cavalier.
- 2) The absence of the availability of these lands has had two main effects.
 - a) Some poorer farmers have moved out of the region towards Tiburon and Jeremie.
 - b) Some other poorer farmers have redirected their farming activities towards the rak bwa stretching from Sous Bois to Morne Cavalier. The rate of deforestation in these regions has accelerated significantly.
- 3) The loss of the upper steep slopes of the Massif Formon as a source of gardens has also had an effect on some of the richer local farmers. These individuals have also migrated to Tiburon, where unoccupied state lands are available.
- 4) The positive economic benefits of the park project have been mainly concentrated in the hands of the richer land-owning families of the Plain of Formon. The poorer families participated in the early phases of the park project, and had anticipated more positive economic benefits from the project. So, there has been a problem of unfulfilled expectations.
- 5) The farmers we interviewed did see a relationship between the deterioration of the environment of Formon, and the degradation of their quality of life.
- 6) The farmers interviewed do not see any clear relationship between the preservation of the nearby rak bwa on the edge of the Formon escarpment and the quality of their lives. However, they depend heavily on this region for firewood and poles for building.

Recommended land use activities

- 1) The ban on garden activities and the cutting of trees above 1600 meters on the steep hills of Formon should continue.
- 2) The rolling hillsides between 1000-1600 meters should be zoned Agroforestry. An active program to utilize this zone for firewood-producing trees, local endemics of economic importance, coffee, plantains, shrubby crops such as Congo peas, and living terraces should be initiated.
- 3) There is a significant problem of soil erosion in the Ravine Casse Cou and the Ravine Dalest. These ravines, and their associated connecting ravines, should be reforested (Agroforestry and Special Use Zones).
- 4) The Plain of Formon and Plain of Durand are appropriate locations for intensive agriculture with some attention being paid to soil stabilization in the ravines of the area.
- 5) There are several areas of unique value on the Plain of Formon and Plain of Durand, such as the small pond near Madam Robert's house and several karst domes scattered throughout the plains. These should be designated as Special Use Zones. The margins of the small pond, for example, should be reforested with appropriate endemic trees and shrubs (as well as some food trees such as plantains). Continued access to the small pond for washing and watering animals would be available at one location near the main trail.
- 6) The rak bwa at Sous Bois, Formon, Durand, and Cavalier are of great significance to the natural patrimony of the country. These areas are also of significance to the daily activities of the local residents of the region. Poles and firewood are collected there. Some pockets are suitable for agriculture.

The rak bwa should be maintained as a continuous unit from the Ravine Dalest and Ravine Casse Cou to the Rak Bois Sous Bois, Bwa Formon, Bwa Durand, and upper ridges of Morne Cavalier. This will provide a continuous buffer zone for the area of the biosphere reserve. Careful zoning will allow pockets in this continuous rak bwa (once reconstructed) to be used for agricultural purposes, and should allow limited exploitation of the forest for poles and firewood, since these resources are renewable.

Specific conclusions from Trois Sources sondeo

This listing of conclusions is organized into sections, and covers the region surrounding the COSAR building on the Gran Plén. A road is currently being built into the region, but at the moment access is via a rough trail from the end of the road. The area is here designated as the Trois Sources region, based on

the central feature of the zone, the Riviere Trois Sources. This river flows west from the saddle (ridge) connecting Pic Formon and Pic Macaya. The steep slopes of each mountain dominate the region. The local communities and population characteristics of the region are described below.

Region

- 1) The zones of the target area are: "Gran Plenn", "Plenn Mou", "Gren Mouton", "Tombe Kriye", "Tranpe", Jeremie, and "Sivette".
- 2) The permanent population of the Trois Sources region is divided as follows: Sivette has 10 houses with 60 people; Gren Mouton has 5 houses with 20 people; Tombe Kriye & Tranpe have 3 houses with 10 people; and Gran Plenn & Plenn Mou have 7 houses with 35 people.
- 3) The total permanent population of the Trois Sources region is approximately 125 individuals.
- 4) There are four types of land holdings in the target area: state land, COSAR land (596 carreaux), private holdings owned by absentee owners, private holdings owned by local residents.
- 5) Most private land owners in the Trois Sources area are absentee.
- 6) Most of the people impacting the Trois Sources area are daily migrants, most of whom are members of COSAR.
- 7) There are 15 permanent residents of the Trois Sources area who are direct employees of COSAR and are allowed to plant beans on COSAR lands.
- 8) State and private lands sharecropped are being rented to farmers from Rendel (and other lowland areas to the south) who travel to Trois Sources to cultivate beans and sweet potatoes.
- 9) "Food for work" and the availability of COSAR land for the planting of beans are direct benefits to the residents of Rendel.
- 10) Many lands in the valley of the Riviere Trois Sources are steep and unstable. In Sivette and "Tombe Kriye", the local residents expressed grave concerns about avalanches. Residents of Sivette stated that they were unable to rent out some of their lands because of the danger of avalanches.

COSAR

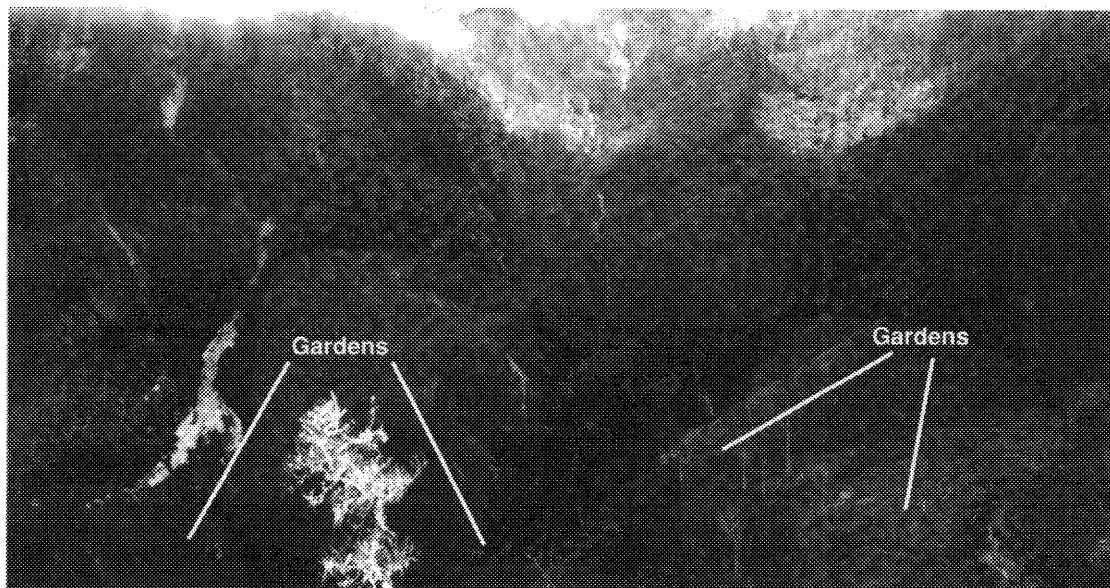
- 1) While the stated goal of COSAR is to plant coffee on land owned by COSAR (596 carreaux), the main activity of cooperative members appears to be planting beans on coop land.
- 2) Many areas where beans are being planted are ecologically inappropriate (i.e., too steep to be stable and thus resulting in severe soil erosion, or locations that were originally covered with primary forest inhabited by rare species of endemic plants and animals).
- 3) The main benefit of membership in COSAR at the present time appears to be access to COSAR lands where members are allowed to plant gardens in areas that have been cleared (theoretically for the eventual planting of coffee, but in most cases for immediate use as bean gardens).
- 4) More land has been cleared in the area of COSAR than can possibly be planted in coffee within the next decade at the current rate of planting 10-15 carreaux per year.
- 5) The completion of the planned road from Rendel to the cooperative headquarters at Gran Plenn will improve the level of supervision and coordination at COSAR.
- 6) This same road has the potential of creating a situation where the rate of exploitation of inappropriate habitats in the Macaya-Formon area will accelerate, with a resulting degradation of the national park (core area of the biosphere reserve) and the watershed of the Riviere des Roseaux.
- 7) The planting of coffee and the maintenance of the coffee nursery appears to be poorly supervised. As a result coffee is being planted in an inappropriate manner, and young coffee trees are dying.
- 8) Many members of COSAR that were interviewed expressed disappointment in the direct benefits they were receiving from COSAR in relation to coffee production or food for work, and as a result many members are concentrating their activities in the area on growing beans on land provided by COSAR (leading to increased pressure on COSAR to sanction the clearing of forested lands).

Crops

- 1) The primary crop grown in the Riviere Trois Sources watershed is black beans.
- 2) A very small percent of the total cleared area in the Trois Sources area is planted in coffee. Of the 596 carreaux owned by COSAR, approximately 56 were estimated by COSAR to be currently planted in coffee (9%). The Sondeo team did not observe 56 carreaux planted in coffee, and believes that the 56 carreaux figure is a significant overestimation of the amount of land COSAR has planted in

coffee.

- 3) There is no sign of coffee disease on the plants observed by the Sondeo team.
- 4) Sweet potatoes are grown as an alternate crop.
- 5) Lima beans are grown in large numbers in the Sivette area.
- 6) Plantains appear to grow better in the Trois Sources region than in the Formon/Durand region.
- 7) Corn appears to grow better in the Trois Sources region than the Formon/Durand region.
- 8) Permanent inhabitants of the Trois Sources area have gardens of "malanga", "mazonbel", castor beans, pois Congo, plantains, sweet potatoes, okra, cane, and corn. Malanga is sometimes sold at markets, while "mazonbel" is eaten locally.
- 9) None of the "new" vegetables observed in the Formon/Durand region, such as thyme, leeks, and onions, were observed in the Trois Sources region. The one exception to this was a garden in Gren Mouton, where a small herb garden was maintained for an absentee landlord.



Animal Resources

- 1) Sheep are the most common domestic animal, and are widespread. All appeared to be in good condition. The sheep population in the Sivette area was very reduced by Hurricane Gilbert.
- 2) There are a few cattle. Most farmers interviewed were tending cows.
- 3) Chickens were present at most permanent settlements. There were many chicks. Chickens appear to be vulnerable to Newcastle's disease in the Trois Sources area, just as they are in the Formon/Durand/Cavalier areas.
- 4) Knowledge of wild animals in the region is nonexistent, indicating that the hunting tradition for bushmeat has disappeared. Nobody recognized the Zagouti (*Plagiodontia aedium*), which is known to be present in low numbers in the area. Local residents do not see any commercial or food value in the wild animals, including birds. The mongoose is present in the area, and is known to kill some chickens. Rats (both black and Norway) are present, and represent a significant agricultural pest.

Problems

In general, disease problems for plants and animals in the Trois Sources region are less severe than for the same organisms in the Formon/Durand area. This is probably because so much of the land has been cleared within the past ten years.

- 1) Newcastle's disease is a serious problem for chickens.
- 2) No foot rot was observed on sheep, probably because of the steep slopes and well drained soils.
- 3) No mealy bugs were observed on sweet potatoes.
- 4) No mosaic was observed on coffee leaves.

Forest resources

- 1) There are a large number of large pine trees in the region. Many of these have commercial value as a source of planks, and a harvest strategy could be developed.
- 2) These pines are reproducing well, and the pine forest appears to be capable of replacing itself if wise forestry practices are followed. There is no need to establish a nursery to reproduce pines. The important older seed trees should be carefully marked to prevent their being harvested for short-term gain.
- 3) The harvesting of pines on a commercial basis in designated Agroforestry areas should be possible. Mature harvestable pines would be 35-40 years old. There is a reasonable mix of different aged pines in the region, so a limited market could be developed in the near term.
- 4) The large pines in many areas of the watershed are suitable as a source of naval stores (turpentine, gum).
- 5) Fuelwood in the form of fallen branches and dead trees appears to be sufficient to support the daily needs of the limited permanent population.

Market orientation

- 1) The most important market for inhabitants of the Trois Sources area is Rendel.
- 2) Permanent residents in the area, especially Sivette and Tomber Crier, are less market orientated than inhabitants of Formon/Durand, probably because they have less cash.

Current impact of the park

- 1) The Sondeo team did not find any indication that local residents were affected by the park. One reason for this is that the Contribution Office on the Jeremie side is not yet restricting the renting of state lands, and the park agent assigned to the region lives far away in Rendel and has not actively enforced the principles of the national park program to the same extent that it has been enforced in the Formon/Durand region.
- 2) COSAR has stopped the cutting of large pine trees and closed down their sawmill in the area.
- 3) Emigrants from the Cavalier area do not appear to have settled in the Trois Sources region, but rather to have gone on around to the Jeremie side, especially the La Hotte zone on the northwest flank of Pic Macaya.

Recommended land use activities in the Trois Sources region

- 1) All land to the east of Morne Bois Pin in the basin of the Riviere Trois Sources should be designated as part of the Core Area. All of this land appears to be state land.
- 2) Sondeo team recommended that Morne Bois Pin and the steep slopes of Pic Formon between 1600-1800 meters be included in the Core Area. Since some of this land is currently owned by COSAR, an accommodation should be reached such that COSAR agrees to strict enforcement of the laws of the biosphere reserve, or agrees to sell or trade land within this zone.
- 3) The ridges above 1,600 meters of all of the mountains bordering the Riviere Trois Sources should be designated as part of the Core Zone.
- 4) The steep slopes of the entire watershed of the Riviere Trois Sources are vulnerable to rapid soil erosion, and should be designated as Agroforestry Zones. Appropriate uses of these lands are planting coffee, Congo beans, plantains, and living terraces. Plantains are especially suitable as a food crop that can be used to secure the soils in gullies and steep ravines. The planting of black beans alone should be discouraged on steep slopes.
- 5) Land along the margins of the Riviere Trois Sources is appropriate for intensive use, and should be micro-zoned as Agricultural Zones or Agroforestry Zones.

General Conclusions and Recommendations

The Sondeo team strongly recommended that the current two-year contract be viewed as the first phase of a much longer effort to establish the Macaya Biosphere Reserve. Short-term interventions were needed both to stabilize existent farming systems and to gain the confidence and establish fruitful working relationships with local farmers. Some of these interventions were successful within the time frame envisioned in current project planning.

However, establishment of the biosphere reserve itself is not a two-year project. While plans for its development were instituted in that time period, the reserve concept is not one which can be made functional in such a short time frame. Left without further technical assistance, many of the most important concepts of the biosphere reserve will not be in place.

Further, long-term applied testing is needed to bring to full fruition the concept of a biosphere reserve. Learning how to utilize native species, how to encourage their regeneration, and how to provide a stable environment for their continued reproduction, for example, are not goals that can be achieved over the short term. Achieving these goals requires longer-term interaction between trained scientists and the local stewards of the land.

This project therefore adopted the two-pronged thrust described in this document. Evaluation of adoption of short-term interventions at the end of two years was recommended. It was further recommended that an evaluation of feasibility of intermediate and long term interventions be made at that time, utilizing that opportunity to correct and further focus and narrow emphases for continuing intermediate and long term interventions.

Specific recommended activities for the first year of project life included:

1. Screening of improved bean varieties;
2. Improved grain storage, with emphasis given to bean storage;
3. Production of yam planting material;
4. Production of pest-free sweet potato planting material;
5. Strengthening of the swine repopulation effort;
6. Development of a poultry inoculation program;
7. Testing of tall grass pastures on a few local farms, including some which produce sheep and goats;
8. Artificial insemination of cattle on a few farms to increase milk production;
9. Testing of several new crop introductions, with highest priority given to garlic, spices, pigeon peas, carrots, and potatoes;
10. Introduction of improved tuber storage facilities;
11. Introduction of improved beehives;
12. Establishment of rehabilitation test sites for field tests of agroforestry introductions, utilization of native species, and terracing and land rehabilitation techniques;
13. Development of an initial study of silvicultural treatment of some remaining forested areas; and
14. Establishment of test plots of some introduced species for agroforestry and forestry interventions.

Many other interventions are possible. This list of fourteen includes only those which were estimated to have the highest possible return to investment of project time and money.

Sondeo to Pic Macaya, Pa Lan Kont, Deglakis and Bwa Formon

Introduction

This section was originally an appendix to the report of the Sondeo conducted by a team from the University of Florida in November and December 1988, and submitted to USAID/Haiti on January 25, 1989. At the time of the Sondeo, the team was unable to make the long and arduous journey by foot to the top of Pic Macaya, and to locations in the Ravine du Sud known as Pa Lan Kont and Deglakis.

Between January 28 and February 5, a second team spent nine days in the region of Pic Macaya conducting a Sondeo. The members of this team were: Jenness McBride (UFMBRP), Paul Paryski (UFMBRP), Charles Emile Philoctete (USAID), Denique Philoctete (UFMBRP), Felix Saint Pierre (MARNDR), Roy Voss (UFMBRP), Charles A. Woods (UFMBRP), and Ellen S. Woods (Volunteer).

Itinerary

The team assembled at Formon on January 28, 1989 at the MBR field headquarters on the Plain of Formon adjacent to the house of Madam Robert (Tila Despagne). The weather was too rainy to climb on January 29, and so the team worked in the Plain of Formon region. On January 30, the team departed Formon (957 meters) and hiked west to the Plain of Durand (1200 meters) and then up the west trail over the ridge Morne Formon (2000 meters) and down into the most western end of the Ravine du Sud (1400 meters). The journey took 8 hours because of rain and very slippery trails. Camp was made in an open rocky "galet" in a region known by the guides as "Pa Lan Kont". On January 31, the team hiked up the east wall of the saddle connecting Pic Formon with Pic Macaya to the camp ground in the saddle at 1870 meters ("Fraz"), and then north up the shoulder of Pic Macaya to the top of the mountain at 2347 meters. The name "Fraz" was applied to the resting place in the saddle by several people, especially Redicule Beljean, who knows the area well. We are not sure of the meaning of the local name, which may come from alternative to the Creole word for "chitchat". The journey from Pa Lan Kont to the summit of Pic Macaya took 7 hours. The trail along the way had to be cleared by guides with machettes because there had been so much regeneration of vegetation since our last visit to the top of Macaya in 1986. The Sondeo team split into two sections, with McBride, Paryski, C.E. Philoctete, and D. Philoctete camping at "Fraz" in the saddle, and F. Saint-Pierre, R. Voss, C. Woods, and E. Woods camping at the summit of Pic Macaya. All members of the team visited the top of Pic Macaya. Data on the status of the forest, flora, and fauna were gathered for two days along the transect from the saddle to Pic Macaya .

On February second, C.E. Philoctete, D. Philoctete, and F. Saint-Pierre began the journey back to Formon, while Paryski, McBride, Voss, C. Woods, and E. Woods journeyed down the Ravine du Sud via Pa Lan Kont to Deglakis. The journey from Pic Macaya to Formon took 10 hours, while the journey to Deglakis took 5 hours. The route to Deglakis is along a poorly-defined trail to Pa Lan Kont. It then follows the river bed of the Riviere de la Ravine du Sud, which is difficult to traverse. Much of the water in the river flows underground, and it is necessary to slowly climb around and sometimes over huge boulders. Occasional springs enter the river from the steep walls of the ravine, and two major streams join the ravine. The team spent the night of February 2 at the place where the trail to Formon via "Kay Ogil" joins the

ravine, and near where a waterfall enters the ravine from the Macaya side. The region was referred to by the guides as the beginning of Deglacis. They stated that the Deglacis region continued on downstream for another three hours walk. The area above the ravine to the north and west of our camp, which is on a high shoulder of Macaya where the slope is not too steep to plant, is called "Kay Ropano" by the guides in reference to the house with a tin roof that Ropano Despaigne built there. The team interviewed inhabitants of Deglacis the evening of February 2, and the morning of February 3. The team then hiked out of the Ravine via "Kay Ogil" to the headquarters at Formon on February 3. The journey took 7 hours. McBride and Paryski departed for Camp Perrin on February 3. Voss, C. Woods, and E. Woods worked in the Plain of Formon and Rak Bwa Formon on February 4 and 5 before departing for Camp Perrin on February 5.

Current Status and Recommendations

Trails

The team discussed each trail in the region, and made the following recommendations concerning the use of each existing trail.

Pa Lan Kont Trail. This trail connects the western region of the Plain of Durand near Morne Cavalier with Pa Lan Kont in the Ravine du Sud. This trail should remain open to provide access to Pic Macaya and the side trail to Pic Le Ciel. There should be specific guidelines (restrictions) for the use of this trail. They are:

- 1) No domestic animals should be allowed on the trail.
- 2) No garden crops should be transported on this trail.
- 3) No pine planks should be transported on this trail.
- 4) A graphic sign (or several) should be posted to indicate that the use of the trail is restricted.
- 5) The trail should be open for all other uses, including tourism, recreation, scientific, reforestation, and selected resource exploitation as determined in the MBR zoning plan.

Pic Le Ciel Trail. This trail departs from the Pa Lan Kont trail at the top of the Formon ridge at 1950 meters and climbs up the shoulder of Pic Formon to Pic Le Ciel. We recommended that a permanent weather station be established at Pic Le Ciel, and that the clearing there become an observation spot for the Formon/Macaya region. Therefore, the section of the trail from the main Pa Lan Kont trail to Pic Le Ciel should be well marked and well maintained. There should be a sign at the intersection of the two trails that indicates the position of the trail, and states that access is for research purposes only.

Pic Formon Trail. This trail continues on from Pic Le Ciel to the top of Pic Formon, and then descends the north slope of Pic Formon to the saddle. This trail is very important for gaining access to Pic Formon, and traverses some of the wettest and most significant habitats in the biosphere reserve. It could expose these areas to exploitation, so great care must be taken in controlling access to Pic Formon along this trail. We recommended that this trail be maintained at a very low level to restrict access to persons needing to get into the Pic Formon region for research purposes. The trail should not be clearly marked. A sign at Pic Le Ciel and the saddle should indicate that access to Pic Formon is by permit only.

Pic Macaya Trail. This trail ascends the south slope of Pic Macaya from the shoulder connecting Pic Formon and Pic Macaya. Access to the trail is via a short connecting trail up the east face of the saddle from Pa Lan Kont. The trail is very steep, and there are several dangerous sections. The trail is also vulnerable to damage from erosion if too much traffic exposes the vegetative cover to the effects of heavy rainfall. This trail should be kept open and well maintained. There should be signs at the saddle indicating that access to Pic Macaya is by permit only. The camp grounds at the saddle, and at the top of Pic Macaya should be well maintained, and stocked with a supply of fire wood (so nearby forest trees can be protected). There should be an information sign at the top of Pic Macaya indicating the summit elevation and a bit of the lore of Macaya. No permanent building or ajoupa should be constructed at the top of the mountain.

Macaya Ridge Trail. The trail along the east-west running ridge of Pic Macaya should be kept open, but not well maintained. This trail will allow researchers to work on the top of Pic Macaya, but should not be so well maintained that vulnerable areas of the peak are exposed to exploitation. This ridge trail could eventually be connected with the trail up the west slope of Pic Macaya from Sivette to make it possible for hikers to travel from the south side of the biosphere reserve to the north. However, this should only be done after the administrative structure of the biosphere reserve is well established, and the need for this trail clearly established.

Pa Lan Kont-Deglacis Trail. This trail connects Pa Lan Kont in the upper Ravine du Sud with Deglacis at 1000 meters in the mid-Ravine du Sud. The trail is through rugged terrain, much of it along the stream bed of the Riviere de la Ravine du Sud. It is dangerous to traverse. No clear trail exists in many sections. We recommend that a survey team explore the region carefully to determine if a good route exists for a trail. However, since environmental damage has been so extensive in the Deglacis area, no clear connecting trail should be constructed that will expose the upper Ravine du Sud to the well-populated Deglacis area.

Upper Formon - Deglacis Trail (via Kay Ogil). This trail connects the upper Deglacis area (1020 meters) with Formon. Exploitation of the habitat along this trail has been very extensive. On the Deglacis side there are many pastures and gardens, and the site of an ajoupa (currently destroyed) maintained by Ropano Despagne. There are three areas on the top of the Formon ridge (1925 meters) where trees have been cut and the forest cover burned in an apparent attempt to establish gardens. Farther down on the south slope of Formon the trail passes through a small plateau and clearing (called Kay Ogil) where there has been a severe landslide and overgrazing. Farther down the Formon ridge the area adjacent to the trail shows extremely severe impact from overgrazing. Redicule Beljean (our guide, and an ex park guard) informed us that local peasants feel that the area should be planted in gardens once again, and that the pressure to continue to graze animals and to clear the soil cover is mounting.

We recommended that the use of the Formon-Kay Ogil-Deglacis Trail be severely restricted. People should be allowed to use this trail only to go back and forth to Deglacis. The main use of this trail should be by personnel associated with the biosphere reserve. Signs should be posted at the beginning and end of the trail, and at Kay Ogil indicating that access to this trail is "restricted". No domestic animals or farm crops should be transported on this trail. In this way the extreme habitat destruction in the Ravine du Sud at 1000 meters, and on the adjacent slopes of Formon and Macaya can be reduced. This is the area where the rehabilitation zones of the biosphere reserve will be concentrated. There is another trail that provides access to Deglacis (see following) which should become the main commercial route between Deglacis and Formon.

Lower Deglacis Trail. The trail that connects the lower part of the Deglacis district with Formon was not seen by the Sondeo team. This trail needs to be surveyed, and its ecological impact analyzed. However, from the description of the trail, which is east of the Kay Ogil trail, it appears as if this route is the most suitable for gaining access to Deglacis for the following reasons:

- 1) It traverses the Formon ridge at a lower elevation.
- 2) It connects Formon with the center of the Deglacis region.
- 3) It is farther away from the ecologically sensitive areas on the slopes of Pic Macaya and the ridge of Formon.
- 4) It is on the eastern margin of the biosphere reserve.
- 5) It does not pass close to the environmentally sensitive (and threatened) basin just east of Kay Ogil.

Plain of Durand to COSAR Trail. There is a need for a good route by foot or mule between the center of the biosphere reserve project on the Plain of Formon and the COSAR/western Formon-Macaya area. This route already exists. The Sondeo team recommended that the project develop and maintain this important route.

Regions

Pa Lan Kont. The Pa Lan Kont region is located at the head of the Ravine du Sud. Most ajoupas in the basin have been destroyed (probably by Hurricane Gilbert). The main damage at the moment is the result

of 36 sheep that are grazing at 1365 meters on the slope of Macaya near a new ajoupa. There is enormous damage to the area from overgrazing. These sheep should be immediately removed from the area. They are preventing the regeneration of trees and grass, and the loss of vegetation of the steep hillsides has resulted in several landslides. There is a second ajoupa lower down in the ravine. There was much bean frass ("pay pwa") around this ajoupa, and a cooking fire was burning when we passed the area. There was no sign of recent garden activities.

These ajoupas should be removed, and all agricultural activities in the Pa Lan Kont region should be curtailed. It is the opinion of all members of the Sondeo team that there is already too much damage in the Pa Lan Kont basin, which is just below Pic Macaya. Any agricultural activities in the basin should be forbidden for the following reasons:

- 1) Fires escaping from gardens in this area will sweep up to Pic Macaya (as they have from Tombe Kriye on the west side of the saddle below Pic Macaya).
- 2) There are already severe landslides in the basin and along the margins of the Riviere de la Ravine du Sud that threaten the watershed.
- 3) Pa Lan Kont is in the "heart" of the most forested region of the biosphere reserve in the center of the main "Core Area", and therefore only conservation activities should be permitted in the region.
- 4) The slopes on both sides of the Ravine du Sud in this area (1150-1400 meters) are too steep to be able to hold their soil cover if the vegetation is cleared.

The Creole name "Pa Lan Kont" means a place that "is no tea party."

Deglacis Region. This area spreads along the lower Ravine du Sud between 1000 meters (where the Kay Ogil trail joins the ravine) and the confluence of the two main arms of the ravine that come together to form the Riviere de la Ravine du Sud at 800 meters. The area was settled in about 1965 by Ti Yoyo Valerius, who named it "Deglacis." The Deglacis area stretches for 6 km down the Ravine du Sud from 1000 meters to 600 meters, with the center of the district being at the confluence of the two arms of the ravines where there is the largest amount of flat ground (800 meters). There is reported to be a small store ("boutik") at that location. The Sondeo team was informed that the lower trail from Deglacis to Formon connects near that location.

The Sondeo team interviewed over 30 people on the morning of February 3, 1989. The group ranged in age from eight to fifty, and included eight women and 2 children. The pioneer (Ti Yoyo) was a member of the group, but did not speak often. The main spokesman for the group was Espano Blaise, a young man of about 25. There were several very dignified looking members of the group, who we believe represented a very good cross section of the inhabitants of Deglacis. All of the people said that they were permanent residents of Deglacis. They seemed to have close ties with Formon, and most frequently used the market at Sous Bois. However, there were also ties with Camp Perrin, where some of the people had come from, and where some went to market.

The women all reported having children ($n = 8$, range 3-13, average 7.5). We did not ask all of the women how many of their children survived. In order to estimate the number of people in the area we used the standard Haitian measure of 6 people per household. We estimated that there were fifty houses in the district. By this measure the population of the Deglacis region would be 300 people.

It appears from all of the reports the Sondeo team received that Ropano Despagne is one of the major "developers" of the Deglacis area. He rents large blocks of land, and subleases blocks of this land to poor farmers. He built two buildings in the region, one ajoupa along the Kay Ogil trail on the north slope of the Formon ridge (recently destroyed by Hurricane Gilbert), and a tin-roofed ajoupa on a slight plateau on the south slope of Macaya above Deglacis. Large gardens of beans were being planted at the time of the Sondeo near Ropano's "kay tol".

The environmental damage in the Deglacis area is extensive. There is much erosion on the steep slopes of Macaya. Landslides are common, and there are areas where entire hillsides have slipped ("the mountain is throwing up"; in Creole, "mon lan ap vonmi te"). The waterfall at 1000 meters near the main stream bed of the Riviere de la Ravine du Sud, had been extensively rearranged since our visit to Deglacis in 1986 (a sign of the force of the run off from the deforested slopes above the waterfall). No water is running in most sections of the river bed. The residents of Deglacis are aware of the problem. Many peasants

indicated that they knew that they should not be cutting the trees or planting gardens on the steep hillsides, especially on Macaya, but they wanted to be paid to stop making gardens. More accurately, they stated that they heard that the project was helping to make jobs for people in the Formon area, and that they felt they should be able to get jobs to help replant trees and rehabilitate the environment of the ravine. Espano Blaise, the spokesman for the group, was very emphatic and vocal about the justification for providing opportunity for the local residents of Deglacis. They want to be a part of the "rebwasman" of the ravine.

The concentration of endemic species of plants and animals in the area of the Macaya Biosphere Reserve is one of the most important features of the region. In an effort to evaluate the knowledge of local residents about some of the more dramatic of these species, questions were asked about species that members of the Sondeo team knew well. Many peasants interviewed knew about the "zagouti", an endangered species of endemic rodent which they said lived in the big forest up on the high slopes of Macaya. More men than women knew of the animal. They did not know about the *Solenodon* (*Solenodon paradoxus*), although we know from previous work that peasants often confuse the "zagouti" and "nez long", and call them both "zagouti". We have collected remains of *Solenodons* just west of the eastern boundary of the biosphere reserve on the east slope of Macaya, so we are confident that the species occurs in the area. Peasants did not know of the presence of Black-capped Petrels on the ridge of Macaya. They were very aware of rats, "rat kay", mongooses, and "chat maron". The peasants of the Ravine du Sud eat cats ("chat maron"), which they catch with dogs. They also eat "zagouti" when they can catch them, but they are harder to catch than are cats.

Our economic questions received the same pattern of answers which we received on the Plain of Formon. Most peasants sold their labor, and worked in "esquads". They earned (or paid) 2-3 gourdes per day. They sell beans for 7 gourdes per marmite if they have many, and buy them for 12 gourdes when beans are scarce and they need them for planting. Coffee sells for 15 gourdes per marmite. There are no horses or mules in the area, because the terrain is too steep. There are some cows. There are no problems with their cows, sheep or goats. There are problems with their chickens ("pye sech"). Many animals were killed by Hurricane Gilbert.

Regional recommendations

- 1). Ropano's "kay tol" was not used as suggested in the Sondeo. Instead, a small two-room house was built for the technicians who worked in this area. In addition, a nursery was established in the area. The Sondeo team recommended that the region be renamed "Belvi" in order to emphasize the beautiful vistas and better life in the region as a result of the project.
- 2). All areas of the ravine west of the 800 meter contour line as it crosses the bed of the Riviere de la Ravine du Sud should become part of the "Core Area". No agriculture or agroforestry should be allowed in this zone, and all ajoupas and houses should be removed except for MBR structures. The boundary of the "Core Area" should be micro-zoned along the eastern margin so that it is easy to follow, and easily identified by everyone working or living in the area. The boundary should follow easily observed contours and landmarks, and should reflect the major concern of the MBR project to protect the watershed.
- 3). Some areas east of the 800 meters line and boundary of the Core Area (running approximately north-south) were to be the sites of major extension efforts by the MBR project. Agriculture, agroforestry, and forestry activities were promoted. A careful plan for zoning the region so that activities were focused on suitable habitats was developed. Some areas still need to be micro-zoned as "Outlying Core Areas" when the slopes are very steep, or the danger of severe environmental degradation indicates reason for concern.

Pic Macaya. The top of Pic Macaya was damaged by the high winds and rain of Hurricane Gilbert. There was a bad landslide just east of the trail from "Fraz" to the top of Macaya, and all trees and soil have washed away, leaving a white gash. There are many fallen trees on the top of the mountain, most of them old pines that had previously died. Some of these trees may have been in excess of 250 years old. The trail along the east-west ridge of Macaya is impassable because of all of the fallen trees, and will have to be reestablished. There is evidence of regeneration of pines in the form of seedlings in sunny patches resulting from clearings in the forest canopy caused by the fallen giant pines. The sphagnum moss which covers

many areas on the summit of the ridge of Macaya was saturated with water, and the dense vegetation of the climax forest on the summit appeared to be in good shape.

The Black-capped Petrels were still present on the south slope of Macaya at 2000-2340 meters west of the trail from "Fraz" to the summit. The terrible landslide to the east of the trail may have eliminated the nesting habitat for petrels in that region, and no petrels were heard east of the trail during the nightly census periods. Other birds known to occur on the summit of Pic Macaya were observed, including the White-winged Crossbills. Therefore, we concluded that the hurricane did not disrupt the avifauna of Pic Macaya. We did not observe any White-winged Warblers, which may be the rarest bird in Haiti, and which is known to breed on Pic Macaya. A careful study of the status of this species is needed.

The mongoose is now common on the top of Pic Macaya. Many signs of mongooses were observed (feces, feeding areas), and there are indications that the mongoose is preying heavily on rats living on the summit. The population levels of both Black and Norway rats were considerably reduced since our last inventory of small mammals on Pic Macaya in 1986. The mongoose could also be having a very negative effect on the status of ground-nesting birds, and any endemic mammals that might survive on the peak. The potential hazard of the mongoose on the colony of Black-capped Petrels is unknown, but the members of the Sondeo team were concerned. The team recommended that immediate efforts be undertaken to trap mongooses off of the summit of Pic Macaya. The team also recommended that a census of small mammals be made on the top of Pic Macaya to determine if any *Nesophontes hypomictus* still occur there, and to evaluate the impact of mongooses and rats on the status of the White-winged Warbler and Black-capped Petrel. This survey is important for two reasons:

- 1) If *Nesophontes* is found to still occur on Pic Macaya it would focus international attention on the Macaya Biosphere Reserve.
- 2) If mongooses and rats are killing White-winged Warblers, Black-capped Petrels, and *Nesophontes hypomictus*, it is possible that these endemic animals will become extinct before a plan to control rats and mongooses in the biosphere reserve is developed.

Bwa Formon. The "rak bwa" forest is the major vegetative cover along the edge of the escarpment marking the southern and western boundaries of the Formon and Durand plains. "Rak Bwa" is a Creole term that is widely used and universally recognized in Haiti as a description of a forest (usually in a semi-natural state, but sometimes quite fragmented). This vegetative zone is the remnant of the broadleaved forest that once covered the entire plain and karst hills between Les Platons and Morne Cavalier. In most areas this forest has been cut for gardens. The forest remained intact along the region of the escarpment and on the karst hills because the presence of huge blocks and jagged broken fragments of karst ("kase dan") made the areas unsuitable for agriculture. In some areas rocks were more diffuse, and these were used as pastures (cleared or over grazed). In many areas, however, the landscape is so rugged that there is very little soil and it is necessary to walk along the top of the tall sharp rocks (called dog-tooth limestone). These areas were covered with a mature climax forest where many important endemic plants and animals occurred. Without a doubt, the rak bwa forest between Sous Bois and Morne Cavalier is the most significant area of biological diversity left in Haiti. It is also one of the most threatened.

The Sondeo team surveyed all of the areas of the rak bwa where "zagoutis" were observed by Charles Woods in 1986, and where Donald Dod collected endemic orchids. Most of these areas have been destroyed. All of the big trees have been cut in most places, and many areas have been cleared and burned for gardens. The gardens are in areas of enormous rocks, and are planted mainly in pumpkins ("jouwoumou") and taro ("mazonbel" or "karaiyib"). The newly cleared gardens are hazardous to work in, since the farmer must slowly walk over the sharp tops of jagged rocks. Many crevices are over five feet deep. The loss of the remaining forest along the edge of the biosphere reserve is a tragic consequence of the establishment of Parc National Pic Macaya, and the closing of the park to activities such as making gardens. Since the rak bwa forest was outside the park boundary in most areas, peasants turned all of their agricultural activities to these zones. Many of the peasants we interviewed were from as far away as Les Platons. Few were from the immediate area. The majority of the destruction of the rak bwa habitat has been since 1986. The rate of destruction is accelerating, and much new forest is being cut.

The Sondeo team was alarmed at the rate of habitat loss. The rak bwa area is generally of poor quality for agriculture, but is of great significance as a biological reserve for endemic plants and animals for the

biosphere reserve. It is the most biologically significant habitat in the biosphere reserve. The Sondeo team recommended an immediate emergency effort to find a solution to the problem of the rak bwa. The area needs to be carefully "micro-zoned", with major areas being included as part of the Core Zone, and a few smaller areas being designated as "Special Use Zones." An overall management plan for the rak bwa was developed. The cutting of the forest in many areas must be stopped. Areas where the forest has been cut since 1986 can still regenerate because of the coppicing nature of many of the trees. In areas that the MBR teams identified as having high ecological importance for endemic plants and animals, gardens should be removed (or at the very least, no new gardens should be planted and old garden should be allowed to return to secondary forest). These areas should be marked with spray paint, or orange marking tape.

As an educational process for the local residents of the Sous Bois/Bwa Formon/Bwa Durand areas, a video tape could be made to show to local peasants the importance of the rak bwa to the way of life of the region. The video could document all of the ways the presence of the rak bwa improves the quality of life, and provides them with renewable natural resources such as: 1) poles for construction; 2) wood for fuel; and 3) water. In this way people may be more easily convinced of the importance of abandoning their gardens, and focusing their agricultural activities in more appropriate areas. In the case of non-resident farmers, local community action could be taken to exclude them from planting in inappropriate areas. We believe that it is important to establish a tradition whereby the local residents actively protect their own "community" from exploitation by non-resident farmers from such remote districts as Les Platons and Cavalier. These non-resident farmers are responsible for much of the habitat destruction in the rak bwa.

General Conclusions and Recommendations

The Sondeo team presented the following conclusions based on the survey of Pa Lan Kont, Pic Macaya, Deglaci, and the Rak Bwa Formon during the period 28 January- 5 February, 1989.

- 1). The presence of existing trails in the biosphere reserve influence land use practices in the area. The Sondeo team recommends developing a strict plan for the use of the trails, and a way to enforce the recommended guidelines.
- 2). The upper Ravine du Sud is being severely damaged by overgrazing in the Pa Lan Kont area. All sheep and cattle should be removed from this area. The presence of mature pine in the vicinity will allow for the rapid recovery of forest cover on the degraded and steep mountain slopes as soon as the source of the overgrazing is removed. All grazing in the area should be forbidden.
- 3). There were many recent landslides in the Ravine du Sud. Some of these landslides were very severe, such as one on the Formon side of Pa Lan Kont near where the sheep are grazing, and another near the summit of Pic Macaya west of the "Fraz" - Pic Macaya Trail.
- 4). The soil cover on the steep slopes of the Ravine du Sud, are severely damaged.
- 5). While many old large pine trees were blown down by Hurricane Gilbert on the summit of Pic Macaya, there was little damage to the forest on the peak. The lower story vegetation is in good shape, and abundant water is stored in the sphagnum moss ground cover. The avifauna of the peak is in good shape. Mongooses appeared to be much more abundant than they were in 1986, and are a threat to the endemic vertebrates of the Core Area.
- 6). The loss of forest cover in the Deglaci area is especially severe. The population of the area is at least 300, with many of these people being permanent residents. The watershed of the Riviere de la Ravine du Sud is threatened by the accelerating rate of deforestation. The local residents were involved with the rehabilitation efforts in the zone.
- 7). The Deglaci area is so important to the watershed that the Sondeo team recommended enlarging the Core Area to include all areas of the Ravine du Sud west of 800 meters (in the base of the ravine) and the saddle, as well as the areas west of the saddle indicated in the previous Sondeo report.
- 8). The Sondeo team recommended that the MBR project work in the Deglaci region as one of its Target Zones, which was done.
- 9). The rate of loss of the forest in the rak bwa zones outside of the national park is accelerating, and a crisis situation is rapidly developing. A plan was developed for managing the rak bwa resource. Emergency action should be taken to prevent any more rak bwa forest from being cut. The future biological diversity of the biosphere reserve is threatened by the present trend. Therefore, it is our

recommendation that the entire rak bwa between Sous Bois and Morne Cavalier be designated as part of the Core Area, with small parcels being selected out as special use zones.

- 10). The research recommended by the Sondeo team which was undertaken in the Core Areas of the park included:
 - a) Weather stations at Pic Le Ciel, and the headquarters at Plain of Durand. A special effort was made to document rainfall and temperature at Formon and Deglacié.
 - b) Plant propagation trials were undertaken to develop a plan for propagating endemic forest trees and shrubs.
 - c) Soil stabilization trials on steep slopes were undertaken.
 - d) Investigations on the status of small mammals were undertaken within the Core Area.
11. The Sondeo team recommended that a Pic Macaya Guides Association be developed. This association would keep a record of all guides and porters who have worked on projects, and each member would be provided with a certificate. The MBR team pursued this matter, and made final plans for the establishment of the Pic Macaya Guides Association.



Pic Macaya Guides Association

SUMMARY OF RECOMMENDATIONS

Land Use

1. The team recommended that a general land use plan for the MBR be established. Five land use categories were recommended: (1) Core Zone; (2) Special Use Zones; (3) Forestry Zone; (4) Agroforestry Zone; and (5) Agricultural Zone.
2. In Formon, the ban on garden activities and the cutting of trees above 1600 meters on the steep hills of Formon should continue.
3. The rolling hillsides between 1000 and 1600 meters should continue to be zoned for agroforestry use.
4. The Ravine Casse Cou and the Ravine Dalest and their associated connecting ravines, should be reforested (Agroforestry and Special Use Zones).
5. The Plain of Formon and Plain of Durand are appropriate locations for intensive agriculture. Attention should continue to be paid to soil stabilization in the ravines of the area.
6. There are several areas of unique value on the Plain of Formon and Plain of Durand which should be designated as Special Use Zones.
7. The rak bwa should be maintained as a continuous unit from the Ravine Dalest and Ravine Casse Cou to the Rak Bois Sous Bois, Bwa Formon, Bwa Durand, and the upper ridges of Morne Cavalier. Careful zoning will allow pockets in this continuous rak bwa to be used for agricultural purposes, and should allow limited exploitation of the forest for poles and firewood.
8. All land to the east of Morne Bois Pin in the basin of the Riviere Trois Sources should be designated as part of the Core Area.
9. Morne Bois Pin and the steep slopes of Pic Formon between 1600 and 1800 meters should also be included in the Core Area. Since some of this land is currently owned by COSAR, an accommodation should be reached such that COSAR agrees to strict enforcement of the laws of the biosphere reserve, or agrees to sell or trade land within this zone.
10. The ridges above 1600 meters of all of the mountains bordering the Riviere Trois Sources should be designated as part of the Core Zone.
11. Burning of fields should be discouraged in all areas below the core area to prevent wildfires from damaging the higher core area habitats.
12. The steep slopes of the entire watershed of the Riviere Trois Sources should be designated as Agroforestry Zones.
13. The planting of black beans alone should be discouraged on steep slopes.
14. Land along the margins of the Riviere Trois Sources is appropriate for intensive use, and should be micro-zoned as Agricultural Zones or Agroforestry Zones.

Project Strategy

1. Project interventions must continue to concentrate on increasing cash flow, particularly on interventions which both significantly increase the amount of cash in the local economy which allow an upward spiral of cash flow to be initiated.
2. Careful attention must continue to be paid to working with the poorer families, which requires careful selection of project primary cooperators to include representatives from this group.
3. Further study of land tenure would be useful, concentrating on access to land among poorer families.
4. In Trois Sources, efforts must continue to concentrate on working with the permanent residents.
5. Efforts should continue to be made to work with migrants in the Trois Sources area.
6. Further study of the migrant population in Trois Sources is urgently needed.
7. Transportability (the relationship between value and volume) and perishability must continue to be major considerations in evaluating marketability of any products proposed for (increased) production in the MBR region.
8. Interventions on farms should continue to concentrate on those lands which are appropriate for

- intensive agricultural use and on interventions which can bring short term benefits to the farmers.
9. On-farm testing of longer term, rehabilitative interventions should be postponed until greater confidence between project personnel and farmers is established.
 10. Both short and long term interventions should continue to concentrate on increasing the diversity of the farming system. A wide variety of technologies, both on and off farm, should continue to be offered as a menu of alternatives so that the farmer can select those interventions which are most appropriate for his/her particular farming system.
 11. Project interventions should continue to place emphasis on the home garden as a source of innovation in the system, and at least fifty percent of project cooperators should be those who manage home gardens.

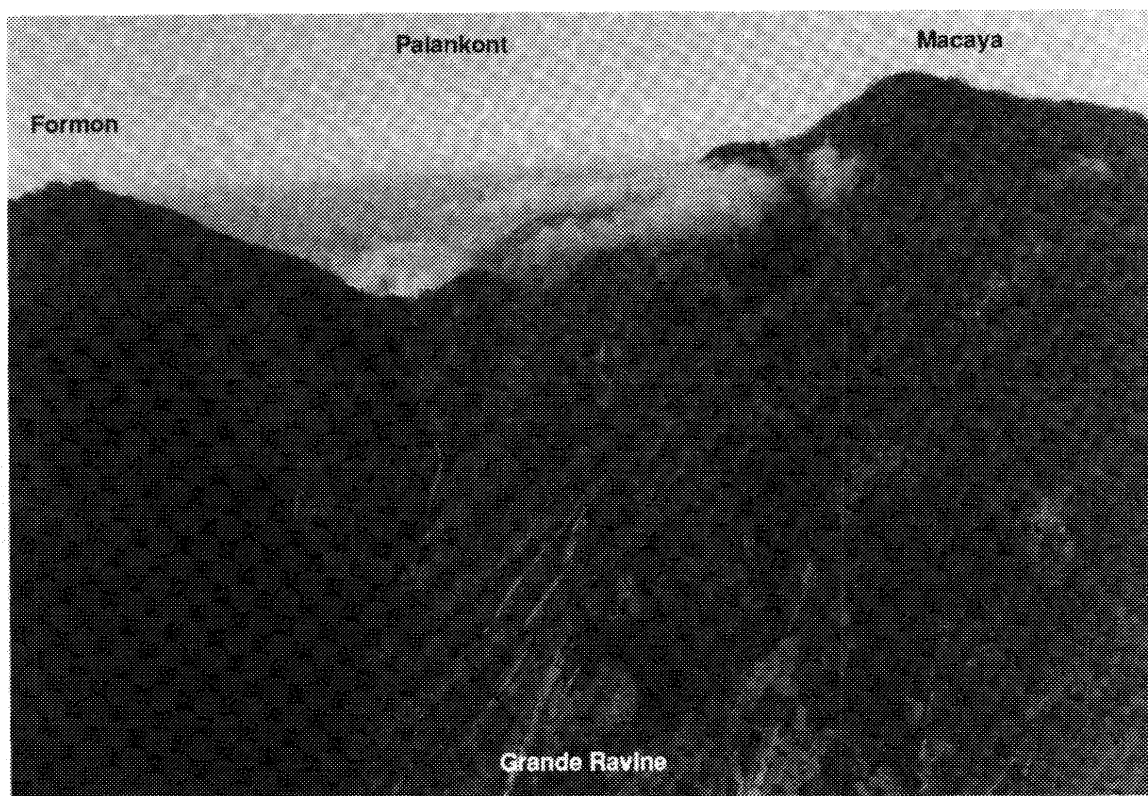
Short-Term Interventions

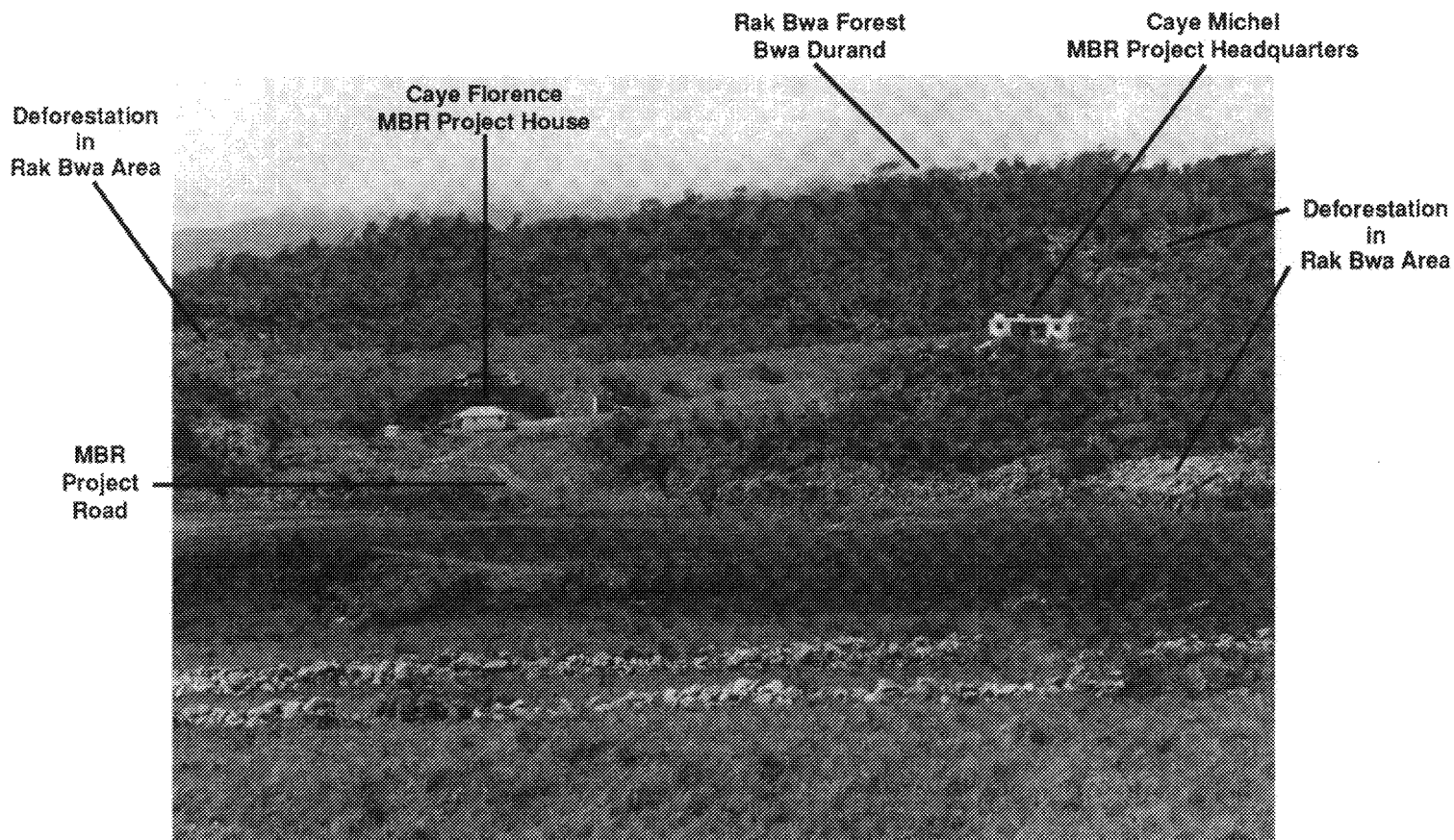
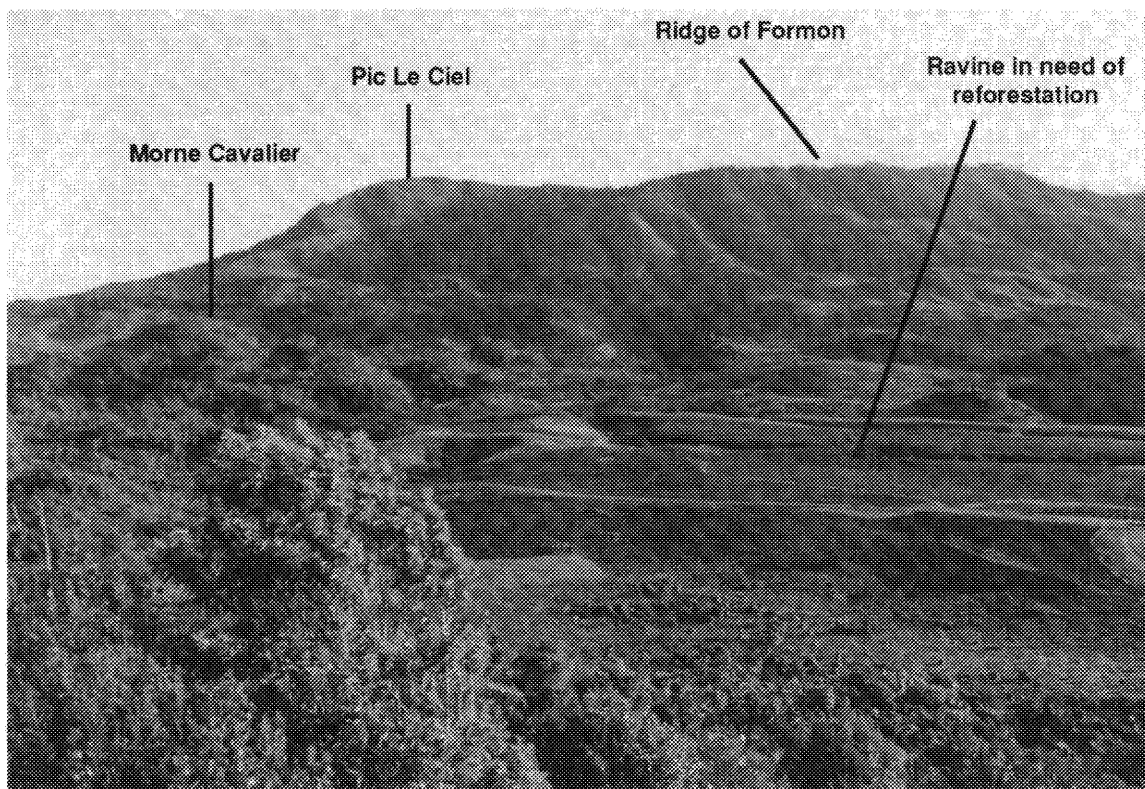
1. Short term interventions should continue to focus on four areas: (1) improving production of existent crops; (2) improving livestock production; (3) introducing new crops; and (4) infrastructural improvements.
2. Beans received first priority during the early period of project live. However, this program should not be encouraged because significant environmental damage is done as farmers clear areas for planting beans.
3. Yams were a second major priority. Increased seed availability is the major problem faced by farmers. Therefore, a program to alleviate this problem should be developed.
4. Sweet potatoes were a third priority. Controlling pests and providing planting material are major opportunities for intervention.
5. Manioc requires further investigation prior to attempting interventions. Improved cultivars and/or planting techniques may represent avenues for increased production.
6. Poultry, particularly chickens, are also a critical element in virtually every household. Disease (probably Newcastle's) is the single most serious problem faced by people raising chickens. Therefore initiate the program of inoculation should continue to be carried out.
7. Of other livestock, cattle are the most important and should receive the greatest attention. Improving animal nutritional status is recommended. Tall grasses should be tested on a few local farms.
8. Some artificial insemination to improve local cattle production should be attempted. Control of parasites is recommended as a short-term intervention.
9. Those crop introductions which have the highest market potential should continue to be tested first.
10. Investments in improved infrastructure can, in some cases, significantly improve the productivity and/or profitability of the farming system. Interventions which enhance the farming system itself should be given the highest priority, with other interventions receiving relatively less attention.
11. By far the most important of these is the introduction of improved post harvest technologies.
12. Honey production is an activity that could assume greatly increased importance, and the major limitation is the type of hive used. Therefore, introduction of improved hives should have high priority.
13. A much lower priority involves improved access to pack animals. Improved animal health and introduction of increased numbers of donkeys are potential interventions.

Intermediate and Long-Term Interventions

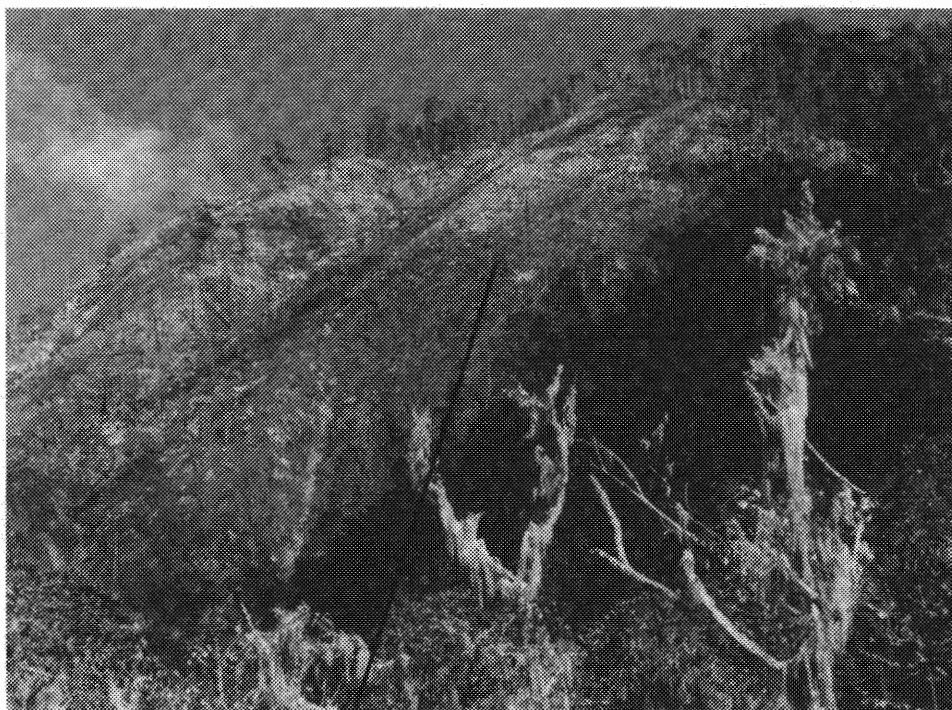
1. While the brief time frame of project funding would seem to preclude any intermediate and long term intervention possibilities, the Sondeo team felt strongly that it was necessary to start testing potentially useful ideas that would take longer to evaluate but potentially would have significant longer term impact on the residents of the area and on the integrity of the Park area.
2. The ideas presented by the team fell into three general categories: Agroforestry, forestry and infrastructure.
3. Agroforestry techniques should continue to be developed.
4. Windbreaks, living terraces, and other land rehabilitation techniques utilizing native species should form an important research effort.
5. Native species should continue to be reintroduced in degraded or cleared rak bwa areas.

6. Living terraces of forage and woody plant species should continue to be established on some of the areas of severe erosion.
7. Silvicultural treatment of at least some of the remaining forested areas outside of the core area of the park continues to be appropriate.
8. Predominantly pine areas may be designated as selective harvest areas, particularly for pole production.
9. Hardwood species for special uses may also become available for limited harvest in areas outside the core, but would require additional study to ascertain feasibility and impact on the park.
10. Introduction of selected exotic species should be carefully evaluated.



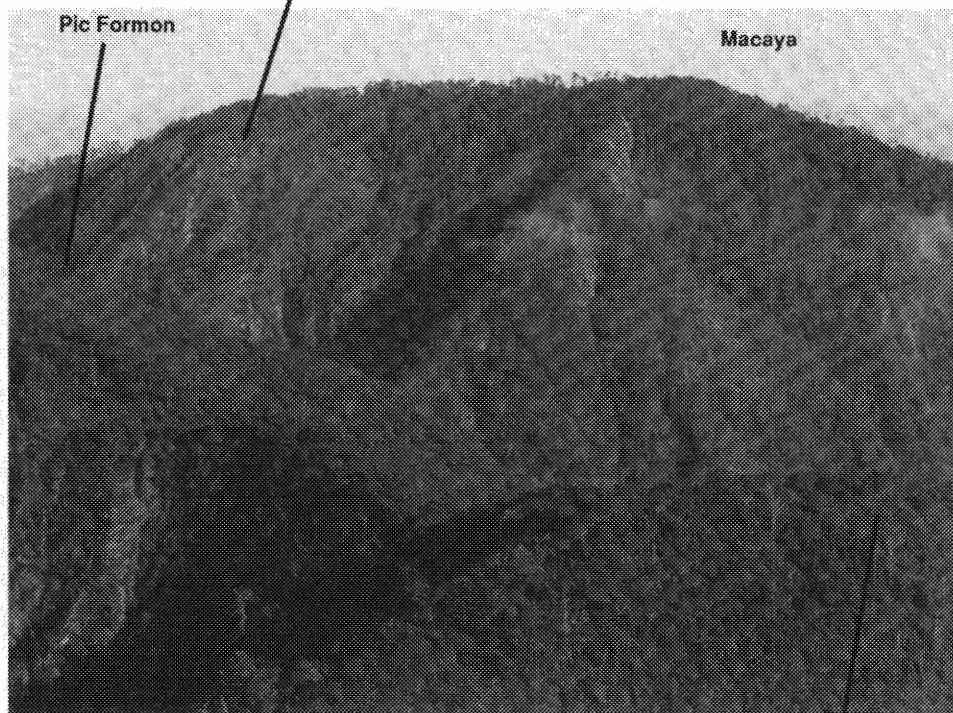


View of MBR Project Area
April 1992



1986

View of an extensive area of burned habitat on the southwest shoulder of Pic Macaya taken in 1986. Black-capped Petrels breed on the southern cliffs.



1977

Le Ciel

Central core of Parc National Pic Macaya taken from a small airplane in 1977. Note the differences in vegetative cover between 1977 and 1986.

Chapter II

The 1988-1992 Macaya Biosphere Reserve Project

DESCRIPTION OF PROJECT

The goal of this project was to preserve and rehabilitate the unique ecosystems and manage the natural resources of the area in and around Pic Macaya which contain Haiti's last relictual mountane cloud forests, thus assuring sustainable economic development of the region.

PROJECT RATIONALE

The University of Florida Macaya Biosphere Reserve Project proposed to complete the development of a well-planned and sustainable program in natural resource management in the area of the Parc National Pic Macaya and the populated areas that surround the park. This is an area of very rugged terrain where communication and land use planing are very difficult. The area is of enormous economic importance because it is the watershed for five major rivers of southern Haiti (Grande Ravine du Sud, l'Acul, Port-a-Piment, Guinaudee, Glace), all of which provide household and irrigation water for agricultural activities on the plains surrounding the peaks that form the heart of Parc Macaya. The forest cover on these peaks also causes water to condense out of the clouds that blow across the high ridges, thus multiplying the water recharge capacity of the watersheds, and providing water via "cloud precipitation" even during times when rainfall is limited. Severe damage to the forest cover on the ridges of Pic Macaya and Pic Formon, which would occur very rapidly if the area were left unprotected (Cohen, 1984), would have devastating consequences on the agricultural productivity of lowland areas, especially the Plaine des Cayes. We estimate that the loss would be as high as 19 million US dollars per year, not including tragic loss of life and property.

In addition to the economic importance of the core areas of Parc Macaya and the surrounding buffer zone, the area also serves as a refuge for several species that are on the verge of extinction in Haiti. These are not obscure species, but rather include one of only two endemic warblers found in Haiti and one of only two native land mammals still surviving. If they become extinct, two of the most significant vertebrates of the island will be lost. The presence of these two unique species in the area, as well as the presence of one of the last confirmed breeding colonies of Black-capped Petrels in the West Indies, is significant enough that it opens the possibility for ecotourism in the Macaya area. The Macaya area is also an important winter haven for migratory song birds from North America. Such habitats are under great stress throughout the Neo-Tropics--a source of great concern for many conservation organizations. So, there is much to gain from preserving the Macaya region--and much to lose if it is lost.

Because of the importance of the Macaya region, the University of Florida proposed that Parc Macaya and the surrounding buffer zone be managed as the Macaya Biosphere Reserve. USAID has supported these efforts during the past five years as the Macaya Biosphere Reserve Project (MBRP). The project is at a critical phase. The UF/MBRP attempted to build a fully operational program in national parks management and biological conservation within the "Service for the Protection of the Environment" under MARNDR. This was an entirely new concept for MARNDR. Since MARNDR had almost no experience in this area, and no staff with training or project experience in high-elevation agroforestry, steep slope and avalanche stabilization, watershed management utilizing and protecting native species, and biological conservation, the task for MARNDR was daunting. All of the necessary programs were built from the base up. Therefore it was extremely important for the University of Florida to serve as the model and source of training for the emerging program at MARNDR.

The UF/MBR Project was necessary to assist the Direction des Ressources Naturelles of the Ministry of Agriculture to achieve its ambitious goals: 1) making the Service for the Protection of the Environment at MARNDR capable of managing Parc National Pic Macaya and reserved areas on a long-term basis by training Haitian agronomists and agroforestry technicians; and 2) developing a solid, sustainable infrastructure for the Macaya Biosphere Reserve that would promote improved and profitable agricultural development without destroying the environmental conditions necessary for the long-term stability of southwestern Haiti. We estimated that the SPE and the Service des Forêts of MARNDR would be capable of assuming these responsibilities by 1994.

PERSONNEL

Principal Investigator:	Dr. Charles A. Woods, Florida Museum of Natural History
In-Country Coordinators:	Paul E. Paryski (MBR Administrative Coordinator) Florence Sergile (MBR Technical Coordinator)
Present Support Staff:	1 Administrative Assistant 2 Agronomists 3 Agroforestry technicians 3 Agricultural technicians 2 Animators 2 Chauffeurs 4 "Guardians" 2 Nurserymen 2 Housekeepers 1 Water carrier 1 Peace Corps Volunteer

TOTAL PRESENT FUNDING:

USAID/Port-au-Prince
Cooperative Agreement 521-091-A-00-7170-00, \$1,737,396

TOTAL LIFE OF PRESENT PROJECT: 1 October 1988 to 31 May 1992

OTHER COLLABORATING ORGANIZATIONS:

MARNDR Ministère de l'Agriculture des Ressources Naturelles et du Développement Rural and
SPE Service de Protection de l'Environnement, District Agricole du Sud; Service des Forêts
ISPAN Institut de Sauvegarde du Patrimoine National
ORE Organization for the Rehabilitation of the Environment
UNICORS Union des Cooperatives de la Région du Sud
Peace Corps

INSTITUTIONAL CAPABILITY AND EXPERTISE

The University of Florida has an exceptionally large range of expertise in the natural and environmental sciences, agricultural development and research, and the conservation of neo-tropical wildlife and natural resources. No other university can match this multi-disciplinary expertise and its depth of experience. The University of Florida has been carrying out research and managing projects in Haiti for the last two decades. It completed the biophysical inventories and preliminary management plans for Haiti's three National Parks (La Visite, Macaya and the Citadelle) (Dod, 1984, 198 ; Judd, 1986; Judd et al., 1990; Woods, 1986, Woods & Ottenwalder, 1986, Woods et al 1988). Dr. Charles A. Woods is an expert on Hispaniolan natural history, especially its fauna. Paul Paryski has lived and worked in Haiti for the past 16 years and initiated Haiti's national parks system; he has had training in national park development and management. Florence Sergile is an Haitian agronomist who has been trained at the University of Florida in conservation and management of natural resources and biosphere reserves.

AREA OF INTERVENTION

The project focussed on the area around Pic Macaya (2,347 meters) and Macaya National Park (approx. 5,000 hectares) in Haiti's southwest peninsula, and includes the land immediately south of it, roughly 50,000 hectares. The upper areas of the national park contain the last relictual vestiges of natural montane cloud forest in Haiti and the greatest biological diversity (more than 100 endemic plant species, 20 endemic bird species, 2 endemic mammals--mostly endangered) in the Caribbean basin. The Pic Formon area receives up to 6,000 mm (240 inches) of rain per year. The slopes of Pic Formon and Pic Macaya in the center of Parc Macaya are the water catchment areas for seven major rivers, several of which supply water for the most agriculturally-productive area in Haiti, la Plaine des Cayes. Near Pic Formon, the mid-elevation (950-1200 meters) "Rak Bwa" forest along the edge of the steep escarpment of the Plains of Formond and Durand contains the highest levels of biological diversity in all of Hispaniola, and should serve as an extension of the core area of the higher slopes of Parc Macaya. Intense deforestation during the last 30 years threatens the entire ecosystem, as well as the watershed for all seven rivers. One endemic bird species, and one endemic mammal may be extinct everywhere else in Haiti, and the MBR project has documented that they are on the verge of extinction in the Macaya area. These unique characteristics and natural features qualify the area for designation as a World Heritage area, and a National Biosphere Reserve. If properly developed, the area also has the potential of developing into a center for ecotourism, since it is an area of unique natural beauty.

PRINCIPAL OBJECTIVE

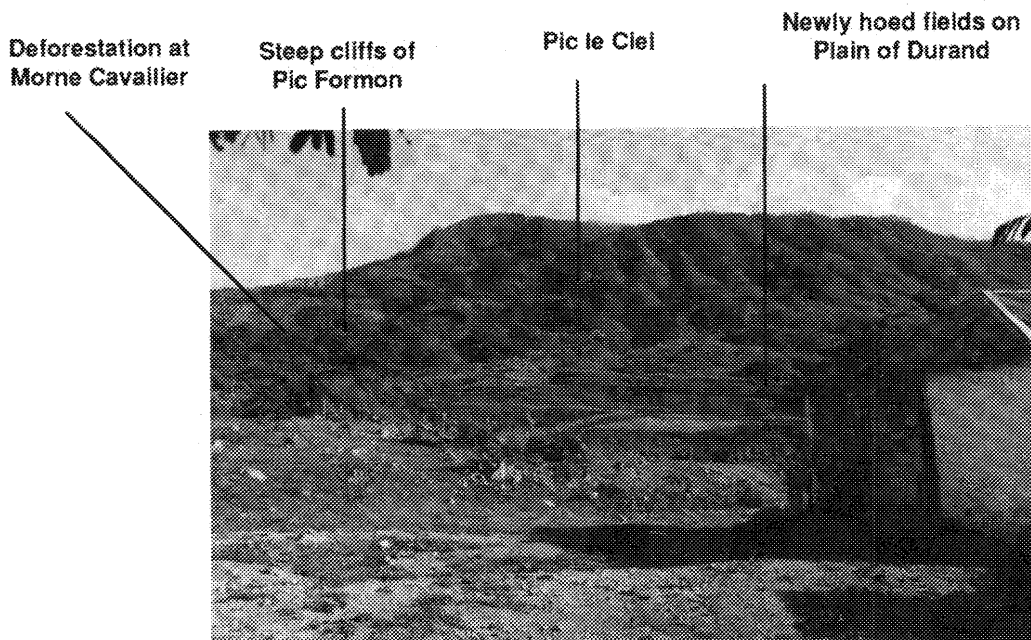
The principal objective of the project is the protection of the natural ecosystems (Pine forest, hardwood forest, Rak Bwa), biological diversity and natural resources of Parc Macaya and the surrounding areas thus assuring the sustainable development of the watershed area.

BACKGROUND

Although the start-up date for the USAID UF/MBRP was October 1, 1987, major project activities did not effectively commence until November of 1988. The political and social instability in Haiti and the November 29th election massacre and its concomitant effects on U.S. policy towards Haiti necessitated a change in project strategy and, in effect, an amendment to the project Cooperative Agreement (C.A.) Despite considerable adversities caused by civil unrest, difficult and unsafe site access, and severe civil budget constraints, we feel that during the first three years of on-site project activities significant progress has been made towards achieving project goals. We completed a detailed "Sondeo" (rapid reconnaissance) of the three target zones (Formon, Deglacié, and Rivière Trois Sources) to analyze their population, ecology, and agricultural systems. A critical part of the core zone has been protected by a barbed wire fence. Conservation and rehabilitation measures have been undertaken in three major environmentally degraded and eroded ravines. Trial-selected new crops have been introduced. PPCs (primary peasant cooperators) have been trained in soil conservation methods and new planting techniques. Rustic pigs which are highly-adapted to peasant conditions have been distributed and have been propagated very successfully; these pigs are increasing the income of the peasant farmers considerably in the target zones. The Formon-Dubreuil road has been improved to provide safer, all-season access for project personnel, and market access for peasant farmers, as well as to aid in community development. The upper Ravine du Sud is being protected and soil conservation methods are being used by the peasants living in this area. A floristic study of landslides has been completed to enable us to use ecologically-appropriate and low-cost methods to stabilize very steep slopes and avalanches. A local guides organization has been established as has served as a regional organization devoted to the protection of Macaya. A model program has begun to rehabilitate critically degraded mini-ecosystems by recreating previous plant communities. A video documentary about the project has been prepared and was widely distributed.

We have experienced great frustration and passed through many crises, but feel proud to have accomplished so much. We are particularly satisfied that in 1990 and 1991 after exceptionally heavy rains, there has been no flooding or heavy soil run-off in the Acul and Ravine du Sud river basins. The vegetative regeneration of the Formon core zone is nothing short of remarkable. And the best development of all is the acceptance of the MBR philosophy and strategy by the World Bank, and the creation of an expanded Service for the Protection of the Environment at MARNDR.

The final achievement of project objectives and goals will require a long-term commitment, perhaps ten or more years, and adequate human and financial resources. The World Bank may be able to provide this long-term stability, and to build on the solid base created by the University of Florida. The University of Florida is prepared to make such a commitment. We are confident that with "transitional funding" from USAID, there is a good chance for success. Without this funding, and if the World Bank and MacArthur projects are not implemented, then an ecological and economic disaster in the area is certain to occur before the end of the end of the century.



View of Formon – April 1992

IMPLEMENTATION STRATEGY AND PRINCIPAL ACTIVITIES:

The project has developed activities that will permit the MBR to be sustainable after the termination of the project (25 years). We worked closely with the GOH (ISPAN and SPE) and local peasant groups, and the Projet Sove Te. Almost all of the project personnel was Haitian. They form a qualified and well-trained team capable of managing the MBR for the GOH in the future. The project provides a functional model for conservation and rehabilitation programs in other critical areas of Haiti.

OBJECTIVES AND STRATEGIES

1. Planning the management and development of the Park and surrounding land as biosphere reserve (MBR) including: functional zoning for appropriate land use, developing "species recovery plans" for the most critically endangered species found in the Macaya area (White winged Warbler, Haitian Solenodon, and Black-capped Petrel), participatory integration of resident peasant population, creation of basic infrastructure and establishment of adequate institutional supervision.

Sondeo of MBR Target Zones: A professional team visited the target zones (Formon, Durand, Cavalier, Trois Sources from 27 November to 12 December, 1988; and Deglacis from 28 January to 4 February, 1989) to analyze current status and recent changes in the biogeophysical and social environment in the MBR area and to produce a strategy for project activities. The Sondeo is spelled out in Chapter I of this volume.

Aerial Photographic Surveys: An air survey was done on 8 July 1988 as a basis for analysis of land use patterns, habitat and vegetative land cover. Carlos Cosio, a graduate student in aerial reconnaissance techniques and analysis of the University of Florida, was in charge of the study. The flight occurred in a chartered Cessna Stationaire, flying on an east-west flight grid with a determined altitude of 9,000 ft. Five hundred seventy six (576) semi-systematic vertical color transparencies of the MBR area and oblique color transparencies of the MBR area and southern watershed area were obtained.

Between April 7-16, 1992 a team from the UF/MBRP surveyed the project area between Camp Perrin, Formon on the slopes of Pic Formon, and the east and north sides of Pic Macaya in the region of Duchity. The team was composed of Charles Woods, Jose Ottenwalder, Donald Dod, Paul Paryski and Florence Sergile. The techniques used to evaluate ecological and environmental conditions included traveling into the region by jeep, walking into more remote areas on foot, and surveying the entire region by small plane (on April 14th). The general observations of this survey are summarized below.

- 1). The core areas of Parc National Pic Macaya (the high ridges of Pic Macaya and Pic Formond, and most of the upper watershed basins between the peaks) remain intact, and steeper and more remote regions of the park are still covered with a variety of forest types (most natural, and serving as important reservoirs of biodiversity as well as critical components of downslope watersheds).
- 2). The most eastern extension of the ridge of the Formon Mountain Range is almost completely deforested, especially in the regions of Bellefontaine and Lahatte. This deforestation is resulting in an expanding and massive landslide near the entrance of the Grande Ravine (the headwaters of the Riviere Grande Ravine de Sud), and to a significant degradation of the Ravine de Sud watershed.
- 3). There has been a great deal of deforestation of the "Rak Bwa" forests in the area of Formon and Durand during the last 6 months. This forest (old trees in areas of rugged exposed karst limestone) is the most important remaining forest in Haiti in terms of biodiversity, and it is now very threatened.
- 4). For the first time in the history of our work in the Formon/Durand area of Macaya, no parrots were heard or seen. The answers to our inquiries indicate it is likely that parrots are being collected by some people at their nest sites and eaten or sold for food.
- 5). The entire area of the plains of Formon and Durand have been cultivated for the planting of "potats". This is part of the regular two-year cycle of land use in the area, but this year the local farmers are

planting far into the boundaries of Parc Macaya, often on steep slopes, and along the edges of steep ravines and water courses. As we walked out of the interior of the park, we passed six farmers at different locations walking into the park with hoes and machetes.

- 6). These unwise and desperate agricultural activities make all of the watersheds of the region, and particularly the Riviere l'Acul, vulnerable to heavy rains. On the night of April 13th, such heavy rains fell in the region. During our flight over the area on the morning of April 14th there was evidence of soil erosion and damage to the l'Acul watershed. The Riviere l'Acul was clouded with mud, and the water level was very high, a major change from the afternoon before when we came out of the area by jeep and crossed the low and clear-flowing river.
- 7). The water catchment area of the Riviere Cavailon/Saut Mathurine hydroelectric plant is almost completely deforested. Much of this deforestation is very recent.
- 8). The area to the north, northeast and east of Pic Macaya (including Catiche, Duchity, and Beaumont) has been severely deforested. A few years ago this area was 40% forested. Most of the deforestation is very recent, and seems to be due to charcoal production. We saw many bags of charcoal being brought down from the area. There is a new road being built by the Cheron family of Duchity, with local assistance from people living in the region. The new road will pass from Duchity westward towards Lacadonie. This road was begun two months previously, and will be completed by July.
- 9). The national park is now surrounded by roads (the project road to the south, the UNICORS road to the west, the Cayes-Jeremie road to the east, and the new Cheron road to the north).
- 10). There is little or no forest cover east of Camp Perrin all the way to the Ridge of Pic La Selle in eastern Haiti. The areas to the north of the coast of the Cote de Fer region seem to be undergoing desertification. There are some patches of forest in the regions north of l'Asile, and a large area of important scrub forest on the peninsula west of Duverger projecting into the Baie d'Aquin.
- 11). The rainfall that occurred the night of April 13th (the night before our air survey), produced flooding and heavy silt content in almost all of the rivers of the area. The Riviere l'Acul below Le Prete was brown, and there was a great plume of mud at the mouth of this river west of Les Cayes. The Riviere Cavailon was so high that muddy water was spilling in a huge rapid over the hydro-electric dam. Saut Mathurine, which had been completely dry for almost a year, was spectacular, as were the lesser waterfalls along the Riviere Cavailon. Huge amounts of silt were being deposited in the sea on the morning of April 14th. This flooding and heavy erosion after a single rainfall is a consequence of the severe deforestation of the past few months. During the past few years such flooding and erosion were observed only after weeks of heavy rain.

Summary: The forest cover in the core of the Macaya Biosphere Reserve is still intact. This forest is the last remaining upland forest in Haiti, and is rich in endemic plants and animals (we found two new species of orchids on this trip, and are currently describing a new species of mammal that is a "living fossil"). The forest protects the upper watersheds of the six major rivers of the region. The area is still worth protecting because of its importance in water generation, watershed stability, and soil protection. However, the area is under siege. The rate of deforestation accelerated during the last six months. New roads are going to make it possible to rapidly deforest the area in the next two years. The deforestation has not yet reached the irreversible stage, however, and several projects are possible that could slow the rate of deforestation, protect the critical core areas of the region, and provide economic alternatives to local residents who are desperate for food and sources of cash. Reforestation activities, especially in already badly degraded steep areas of the watersheds, can help stabilize the watersheds, and provide work and opportunities to educate local people.

We feel that the MBR project has worked to the degree that the amount of deforestation in the Formon/Durand area could have been much worse, and certainly the amount of environmental damage in the Durand/Formon region is much less than that in the Duchity area on the northeast side of Macaya where the project did not undertake any work. In general, the survey indicates heavy and recent deforestation in most areas we investigated except for the central area of Parc Macaya, and some Rak Bwa forest areas near the project headquarters at Formon/Durand.

Revision of work plan to place more emphasis on natural resources component: The 1990-1991 work plan was revised, at the request of USAID, to place greater (70 + %) emphasis on the preservation of natural resources and less on direct assistance to the peasant farmers. The new work plan concentrated on such activities as: consulting with MARNDR on park boundaries, appropriate legislation, park guards and the World Bank Environmental project, completion of the MBR plan, interpretive and identification

signs, species recovery plans, expansion of links to international conservation organizations, and increased rehabilitation and avalanche stabilization activity.

The plan was accepted by USAID, and rehabilitation activities significantly increased in Dupuy/Malfini, Source Cresson, Fond Bleu, Chambellan, Kay Tilus, Kay Delome, Durand.

Mapping and zoning

A preliminary map of functional zoning was produced as part of the management plan (Appendix I. Map of the Macaya Biosphere Reserve Project). The blue-print of this map has been widely distributed to organizations and institutions in Haiti for suggestions and corrections.

As a part of the program of cooperation between Parks Canada and the UF/MBRP, Parks Canada furnished the UF/MBRP with detailed and professional quality multi-purpose maps of the MBR zone. These maps are synthesized from the 4 original maps: 5371 II Pic Macaya, 5471 III Beaumont, 5370 I Coteaux, 5470 IV Camp-Perrin at the scale of 1:50,000 and contour lines at 40 meters. We have used the maps for such activities as planning and zoning.

Fencing critical zones of Park : We have consulted with MARNDR to assist them with the fencing of the critical areas of the Park (core zone). In 1989, about 4,000 meters of fencing were obtained from MARNDR/SUD and placed between the forestry and agro-forestry zones north of the Plains of Durand and Formon at the 1,300-meter altitude contour line and extending in the east to the Ravine Sec, across the Ravine Fond Bleu and down hill of Source Cresson and morne Malfini in order to protect the upper slopes of the Formon ridge and to limit access to the core zone. This effort was successful until July, 1991, when the fence broke in many area and people entered the core zone for planting beans. In August, 1991, 500,000 gourdes of PL480 funds were available for physical protection of the national park. Unfortunately, due to the September coup, no action has been taken.

Maps of the proposed boundaries were given to the Service de Protection de l'Environnement (SPE) of the Ministry of Agriculture (MARNDR). A meeting was conducted in September, 1991, with Michele Oriol, UF/MBRP, SPE and the Direction Generale of MARNDR to form a committee (Comite Elargi de Bornage) to establish park boundaries, plan the training of the park agents and make arrangements with the Direction Generale des Impots (DGI) to limit the core zones. Some of the land identified as private land will be dropped and probably bought in the future. The Comite Elargi de Bornage would have been formed by UF/MBRP technical coordinator and assistant, director of SPE, Agronome de District, representative of the Forces Armees d'Haiti for the Departement du Sud, and three private individuals. The meeting and the park agent training planned for beginning of October, 1991, were cancelled.

Park agents should be hired and trained to understand the ecological patterns which form the base line of the zoning and fencing and to reinforce the law beyond the fence line.

Production of signs: Contact was made with a Port-au-Prince graphics studio in August, 1991, and sign construction company, MultiAd, which prepared a plan for twenty-five interpretive and announcement signs to be placed in the MBR area to delineate the functional zones and to identify the MBR and certain of its unusual features. Positive signs are highly recommended and force us to develop special programs such as forage (cut and carry), pasture, and back yard vegetable production as alternatives to the peasants. Three hundred stickers were also produced for wide distribution in Haiti and specially in Port-au-Prince and les Cayes.

SPOT satellite imagery: SPOT satellite imagery was ordered to be shot in January, 1989, December, 1990, January, 1991, and July, 1991. Because of cloud cover during the 1990 and 1991 imaging process, only the 1989 shots are available. In December, 1990, land classification and ground proofing was done for 1989, it is in data base form which will be effectively used in comparing land utilisation over other time periods. Copies of the images were widely distributed to the interacting institutions.

Construction and refurbishing of shelters for project staff: A headquarters at Kay Michel comprised of eight small rooms, two bathrooms, two lofts, and a large dining/conference room, and two peasant houses were built for technicians and visitors. They were furnish them with beds and were supplied with

kitchen and household utensils. Construction began in 1990; numerous improvements are still necessary. Bathrooms are semi-functional, the kitchen needs to be outside, the walls are too porous for such a humid climate, etc.

A small warehouse was built to store materials and supplies necessary for extension and production on the farm. This frees the loft at Kay Michel and the small room of the white house, and allows easier access to tools and supplies.

Construction of shelter for TWMP radio repeater: a small shelter (four square meters) was constructed in 1991 to house the TWMP radio repeater which will allow those organizations in the TWMP to communicate with each other and with USAID. The repeater enables all Targeted Watershed Projects to communicate with one another and with USAID Port-au-Prince. Communications are difficult: very often telephone lines are out of order; the headquarters in Durand/Formon and UNICORS have no telephone lines.

Access Road: Since 1989, the project has improved the access road from le Pretre to Durand/Formon (approximately 20 km) with the collaboration of the MARNDR/District agricole des Cayes. The portion from Platons to Durand was open to good four-wheel drive vehicles. This road is still very dangerous and a crossing needs to be constructed in the l'Acul near the village of Canot.

Organization of NGO/MBR Committee (*Groupe d'Interet pour Macaya-GIM*): The first meeting for the GIM was held 15 December 1988 and two others were held on January 19 and March 22, 1989, to create a structure for the GIM and to organize GIM activities to promote UF/MBRP objectives. An executive GIM committee composed of Jean-Yves Banatte (DCCCH), Sebastien Hilaire (IRD), Michele Oriole (Cooperative des Artisans du Sud), and Paul Paryski (UF/MBRP) was selected; a presentation of slides of the MBR and a discussion of its current status was given by Paul Paryski; June 15, was set as the date for a Pic Macaya expedition by GIM members; plans were made for creating an MBR educational program with MARNDR/SUD; GIM decided to start a data base on the status of the MBR. GIM was being developed, and positive activities undertaken. A group of six people of les Cayes went to climb Macaya. However, since 1990, due to political unrest, it became impossible to have meetings and or to take any action.

2. Assisting the 1,750 peasants in the MBR to increase agricultural production and their household income using ecologically appropriate means so they will not be forced by economic pressures to continue the deforestation and destruction of ecosystems.

Primary Project Cooperators: The selection of Primary Project Cooperators (PPCs) is described in chapter one. From May 22 to June 11, 1989, peasants were interviewed by UF/MBRP sociologist Paul Monaghan in all three target zones. He was assisted by Roy Voss, Janness McBride, Felix St. Pierre, the GOH technicians and Paul Paryski. Thirty-one farmers, representing different socio-economic classes and possessing lands with different types of ecological/biological conditions, were chosen to screen and evaluate technologies outlined in the MBRP Sondeo Report. These farmers would assume responsibility for later dissemination of these technologies within their communities. Project technicians worked closely with the PPCs during the first year on technologies such as vegetable crops, new varieties of existing crops, soil conservation and livestock improvement. The cooperators themselves would replicate successful technologies on the farms of five other individuals within their communities. Some of the PPCs in the Riviere Trois Sources area were chosen by COSAR, the local coffee cooperative.

We worked directly with all the PPCs chosen for the first round of assistance. All thirty-one (31) PPCs selected have been provided training in improved production of beans, carrots, beets, onions, and cabbage. PPCs in Durand (4), Formond (7), Kay Shale (4), Sous Bois (4), Cavalier (3), and Deglacis (5) have conducted on-farm trials of vegetable and bean varieties (52 tests completed) and of soil improvements techniques (44 tests in progress). Four PPCs in Trois Sources (Rendel) started testing vegetable production techniques in March.

This strategy has proved to be very dynamic, but requires time and training to be fully functional and well-accepted. People who were not PPCs or members had a negative attitude toward the project. They felt left out. In the Deglacis area, the team has a high turnover, and our two technicians, Eugene Remy

and Dorvil Jean Baptiste, worked well with this situation. "It's like a club," they said. In the Formon area, which has the wealthier people, it is a significant socioeconomic issue.

Farming systems trials on farms: Experiments were performed on vegetables on the 3.1 hectares area of the Durand Farm to develop improved intervention techniques for intensification of peasant agriculture on appropriate lands within the project area, to determine the productivity of various intervention strategies on a land area of known quality and under direct control, and to provide a demonstration area for introduction of new crop varieties and new cropping techniques.

Approximately 100 plots with over 30 varieties of vegetables have been established. Legume cover crops have been tested in four plots and bean varieties (27 varieties tested) have been tested in over 40 plots. Collard greens and cabbage are grown on the farm in transplanting beds for distribution to PPC farmers as transplants as a method of preventing plant loss by insect pests in the farmers' gardens. In 1991, some PPCs adopted this method and sold their seedlings to other farmers.

Extension with the Primary Project Cooperators (PPC): The establishment of 7,500 m² farming systems varietal trials (17 varieties of beans, four varieties of carrots, two varieties of onions, four varieties of beets, three varieties of corn) at the UF/MBRP experimental farm and 350 m² of FSR trials in the gardens of the 27 peasant farmer primary project cooperators; Ms. Denni Espinal, UF Farming Systems Technician, assisted in the planing and establishment of the farming system plots at the demonstration farm and on the cooperator plots during the first two weeks of July, 1989.

Extreme degradation of the soil on the demonstration farm and on many of the cooperator's farms and the unseasonable lack of rain have resulted in variable results, ranging from complete failure to rather good results for some of the improved varieties of beans and vegetables. Pest control is a major problem and will necessitate the intensification of control using integrated pest management techniques.

Three bean varieties (Salignac 86, Mersant and Black Turtle) have proved to be much more productive than the local varieties or any other tested varieties. Carrots, especially Chantenay and Chantenay Andina, proved to be very successful. Other successful varieties included: KK Cross (cabbage), Granex (onions), and Detroit Dark Red (beets).

Distribution of vegetable seeds and black beans to project cooperators: In February, 1991, 137 marmites of selected bean seed, nine one-kg cans of carrot seed, four one-kg of cabbage seed and 17 cans of beet seed was sold at cost (roughly 50% below the market price) to those PPCs who had undertaken soil conservation measures. Some seed was distributed free of cost to the extremely poor PPCs in the Trois Rivières zone.

The demand for the seed far exceeded the supply, indicating that the peasant farmers have decided that improved bean seeds and vegetables can significantly improve their household incomes. However, some major problems appeared in this program and the distribution of black beans to local farmers is suspended because: 1) UF/MBRP distributed beans which were planted in the core zones. 2) The protocol between UF/MBRP and ORE which provides that ORE sell seeds to the local farmers in Formon is inadequate: people have to walk more than eight hours round-trip from Deglacié to Formon, or Civette to Formon, to buy five grams of carrots and cabbage. 3) The conservation measures were too recent to guarantee protection of the soil and to insure us that these programs will last. The PPCs are very often the poorest among the cooperative team. They cannot afford to test crops or farming systems. It will take at least five years to demonstrate, assist and promote new technology.

The project made an effort to sell at low cost a wide variety of vegetable seeds and to have a demonstration and extension program to develop the "jardin lacou" with vegetables in order to improve peasant diet and income. In the extension program, the economic value of each crop was emphasized. Again, the program is too recent to be accepted by the peasants.

Table 3. Fertiliser trials and value of crop

Treatment	P	P + K	K	Temoïn
Growth*	30	25	15	12.5 cm
# of pods/plant	27	27	5	5 pods
# of seeds/gousse	5	6	5	5 pods
Yield kg/ha	400	333	127	133 Kg
Production cost	1499	1499	1499	582 gourdes
Production value	2400	2000	760	800 gourdes

* Growth quantified 46 days after seeding

Table 4 Vegetables: trial and yield

Species	Variety	Yield/ha
Beet	Detroit Dark Red	34 Tonnes
Carrot	Chantnay/Andina	05 Tonnes
Cabbage	Copenhagen Market	16 Tonnes
Cabbage	KK-Cross	38 Tonnes
Onions	Granex	39 Tonnes

Field research on population dynamics: Dr. Suzanna Smith, a UF anthropologist, visited the UF/MBRP from April 3-16 and from August 17-September 6, 1991, in the Formon and Cavalier target areas to conduct field research on population changes, household size and composition based on the paper *"Three communities in the MBR: a report of fieldwork in the Southern mountains of Haiti"* (Monaghan, 1988). This research in *"Population changes in the Pic Macaya Biosphere Reserve, Haiti: Implications for conservation and development project"* identifies no significant changes in household composition, but indicates that women are more likely to migrate towards urban areas. There was no difference between men's and women's education attainment, except that the proportion going to school was higher due to the fact that wage labor provided by the project gave villagers the economic means to send their children to school. The majority of the sample was from the MBR area, with a majority of the population represented by a younger age group; therefore household sizes are expected to increase.

Distribution of rustic pigs: The distribution of rustic pigs was one of the most successful activities. Pigs received from the PPCs have been redistributed among the members of the PPC. Although people complained about receiving male pigs, they made compensations. They developed a strategy on their own: a bigger male was traded for a smaller female, one got an animal for a fast cash profit, the other for reproduction. Each litter produces an average of seven piglets. We estimate that the total number of pigs that be in the region to be 5,706. If we consider that all males (26 + 2,840) and half of the female population

(1,420) is sold during pregnancy or at birth for 150 gourdes, our assistance to local farmers amounted to 644,100 gourdes.

Table 5. Distribution and estimated population of swine

Distribution date	# males	# females	# known	# estimated
18/5/88	10	3	84	997
31/12/88	2	9	203	2971
4/90	1	17	113	1148
7/90	1	6	7	367
11/90	3	5	8	64
2/6/91	9	18	27	159
Total	26	58	442	5706

Breeding life 24 month; 2 litter/year; waiting period for reproduction maturity = 8 months: average 2 males and 2 females. Maturity from day of distribution to day of birth = 4 months.

Meetings with PPCs: Meetings were held every 15 days with the PPCs in the project area. There was good dialog between the technical team and the cooperators and team members. The local people could express their needs; the technical team made no promises that the project could not afford to keep. The PPC asked for a better planning and vegetable seed purchases, more demonstrations and a wider range of vegetables, animal health programs, and in the Cavalier area people asked for seminars on reforestation. We understood also from these meetings that: 1) the peasants of the area need to grow cash crops that can be stored and marketed--even in small quantities--but all year round. 2) The PPC is a very dynamic component of the project. Several PPCs and members dropped the program due to emigration and peer pressure. In other areas such as Deglacié although some PPCs dropped the program, their numbers increased from 14 to 27 teams. 3) Raising cattle, sheep and pigs is the major activity for the wealthier, who also grow beans and yams. The poorest are used as day laborer for 2 gourdes a day. People planting sweet potatoes on the Plain of Durand are not involved in the project; they are from Platons and la Source. 4) The impact of the project is that all the people that we employed bought from 1/4 of a cow to a cow and one to five sheep. 5) Convincing poor people not to plant on critical area is a challenge; demonstration and implementation of programs which require changes in habits requires more time than was available.

Soil conservation: The PPCs of Deglacié seemed to be willing to use soil conservation practices. About one hectare of forage species (Napier, Wynn grass and Guatemala) was planted for reproduction and distribution to the cooperators. One thousand meters of hedge rows were planted on their plots. People in the Kay Tilus region are overusing the landslides and are not yet willing to improve their cultural practices. Indeed, they had asked our technicians to plant grass on their lands four weeks after the seeding of black beans. This was a little trick to remain cooperators, and to fool us since by working in their plot after germination we would be responsible for any damage which occurred in the plot.

Our technicians were not able to assist the cooperators of Cavalier in 1991. Instead, focus was placed on rehabilitation programs in the Kay Tilus, Malfin/Dupouy and Deglacié areas. However, in this region there is a tremendous sense of community and some leaders came very often to be taught techniques in horticulture, reforestation and soil conservation.

Formation of the *Association des Guides de Macaya* (AGM): We have developed an association of guides and porters to assist interested parties wishing to undertake expeditions to the core zone of the MBR (e.g., Pic Macaya); to provide non-destructive employment and revenue to the poorest sector of the local community; to bond the poorest peasants to the goal of protecting the MBR which will become both a source of pride and income for them. In March, 1989, 61 members, primarily persons who have been porters or guides on previous expeditions, were enrolled. A list of necessary equipment and suggested training was compiled. Some equipment was distributed and the members of the AGM were taught how to erect and dismantle tents as well as to recognize bird species. Members decided to help protect the MBR core zone. However, during the month of July, 1991, lots of these guides were clearing and burning plots to grow black beans. The members complained about not receiving enough from the project. However, by our records, we noted that they have more money than ever (they had received pigs and worked for more than 30 days during 1991 for the project) and as a result, they had more money to plant more beans.

3. Rehabilitate critical zones that have been environmentally degraded, e.g. steep slopes, springs, endangered habitats, and thereby improving the quality of the watersheds, and water generating capacities of the associated water catchment areas, and protecting the biological diversity of the core zones of the Macaya Biosphere Reserve.

In 1991, we concentrated our efforts on natural resource and ecosystem preservation and rehabilitation. We are pioneering techniques of rehabilitating montane ecosystems especially natural plant communities and avalanche stabilization.

Model ecosystem rehabilitation

Our goal is the rehabilitation of critically-degraded areas by reproducing the plant communities and ecosystems that were once there. Plant communities were mapped and species were chosen according 1991 observations and the floristic study of Judd, 1986. The rationale in collecting reproductive material was heliophile and useful species for the people in the area. Seeds and wildlings were collected from such native endemic trees as pine, bwa tremble, koma, sapotelle and calebassier. We applied the following methods: direct seedlings in the Kay Michel/Morne Pipirite area, seedlings on beds and in bags. The Service des Forêts of MARNDR has contributed 10,000 pine seedlings (root trainers) to the project, an indication of their support. The best method is seedlings in beds, and transplanting on site during the rainy season with a mud coating when necessary. Studies are being conducted to determine when these trees flower and seed and to determine how to best reproduce them in our nurseries. This is the first study of the reproduction of montane cloud forest species in Haiti. The results are published in "Arbres et Arbustes de Macaya." To date, we are rehabilitating 100 hectares in the Dupuy/Malfini area which contains the headwaters of the Rivières L'Acul, 100 hectares in the Chambellan/Deglacis area, 50 hectares in the Kay Tilus/Kay Delorme steep slopes and five hectares in the Source Cresson area.

Table 6

Activities	Kay Tilus	Deglacis	Dupouy	Total
Seedling	5000	10700	73000	88700
Wildling	1000	-	1000	2000
Direct seedling	2000	16700	20000	38700
Plantlets	3400	10600	14000	28000

N.B. A total of 2,600 plantlets have been planted in the degraded area near Durand.

We faced several problems: 1) Many endemic species are not useful and do not have a creole name; 2) Degraded area where the program was planned was pasture (overgrazed areas); 3) Pre-germination of direct seedlings does not work. Bare root seedlings should be adopted since they are more economical. Germs are too fragile and data on weather is not reliable yet. 4) In the Kay Tilus area, the program was not a success since people own the land there and do not accept any conservation measures on their land. 5) Free grazing is a limiting factor. 6) And last but not least, for five months, the nurseries had no supervision or maintenance. 68.5% of our production was lost; we were expecting an 80% rate of success in the nurseries. These seedlings should have been planted in October with the rainy season; a second season (October-April) should have started. The seedlings we were able to plant this season cost \$2.75 instead of the planned \$0.50 because we had to pay all the technicians (\$6000/month x 12) between March, 1991, to March, 1992, and day labor during (\$750 x 6 month) between September, 1991 and May, 1992.

Stabilization of avalanches: The preliminary survey of landslide sites in the lower agricultural zones of Kay Tilus (1,000-1,300 meters), located approximately 60 landslides in a one km² area, which vary greatly in size (10-500 square meters), geomorphology (debris avalanches, rotational slope), habitat (crops, short grass pasture), and slope (average 60%; range 30-100%). Landslides apparently provide the establishment sites available for indigenous plant species in this agriculturally degraded zone. An abstract of "Landslide Stabilization and Rehabilitation in the MBR Haiti, an Ecological Approach" was submitted to the International Symposium on the Research Needs and Applications to Reduce Erosion and Sedimentation in Tropical Steeplands (scheduled for June, 1990). Four of the technicians were trained by Ms McBride in methods of landslide protection and economics. The technical team worked on five avalanches an area of approximately seven hectares. In May, 1991, they were sown with direct seedlings of 2,000 *Prunus* sp. and koma, 500 *Pinus occidentalis*, and bamboo cuttings, and in May, 1992, with 3,000 plantings of sapotella, koma and *Prunus* sp. The stabilisation of these avalanches depends on 1) Land utilisation: these steep slopes are considered private land and are cultivated twice a year. The soil preparation still involves slash and burn; and 2) The seeding is too recent to insure that they will be stabilized.

Correction of Ravines: This involves the construction of stone check dams in the Ravine Cresson and Fond Bleu. A total of 169 check dams were erected totalling about 450 m³ of stones and costing \$40/m³. In degraded areas, bamboo cuttings are planted behind the dams and cost \$.20 per cutting.

4. Create a data base containing information on the history of Macaya, studies of the region, its natural history, and past and present interventions for use by all interested institutions and individuals.

Bibliography: During the past four years, we built an important data base counting 3,500 entries with an index of more than 5,000 words on publications on the natural history of the region, and extended it to the whole Caribbean area for the sake of biogeography and species dispersion. This has already been distributed to scientists interested in the area. It is in press. However, the computer program used for the data base, "Procite," is much more powerful on soft copy than hard copy due to the program's ability to search for specific terms among the entries.

Survey on land tenure for park boundary: Michele Oriole, a Haitian sociologist, had been contracted. The objective of the study is to eliminate confusion about the actual boundaries of Parc National Pic Macaya and to provide information about land tenure in the Kay Tilus and Quatrieme sections (southeast of Crete Formond) so as to permit an acceptable boundary for the Park (core zone) in these areas. She produced a report on the land tenure of Plain of Durand and a complete document for the entire watershed management project ("Situation Fonciere et degradation ecologique a Durand"). This document contains good recommendations for the PST.

Update of floristic Survey of 1983: Dr. Walter Judd of the University of Florida visited the Rak Bwa Durand/Formon, 9-17 November 1989. Eighty-one additional taxa are reported from the region with ten species endemic to the Massif de la Hotte. Important and/or useful trees and shrubs on the ridge of Morne Formon and Ravine Fond Bleu and Rak Bwa Formon were identified and collected.

Native and endemic species: We also collected data on each species tried in the region. A handy booklet, "Arbres et Arbustes de Macaya," is now published to help the forestry technicians propagate native and endemic species of the montane ecosystems. The data on each species includes the plant community, a

concise botanical description, time of seed availability, flowering season, length of time for germination, weight of seeds or fruits according to the easiest way to purchase them.

Soil fertility: Three experiments have been conducted to determine the fertility and soil nutrient status of the soils of the Durand area, and soil sample were collected for analysis. Soil conditions in the project area are much more variable than was initially thought. Acidity (pH) of the soils ranges from very acid (pH 4.5) to non-acid (pH 7.0). Phosphorus appears to be the most limiting nutrient and may be improved by green manure crops, composting and additions of phosphorus. The lack of organic matter in the soil is also a major problem and affects both the physical and chemical characteristics of the soil. They are predominantly ultisols with slightly acid soil reaction in the Formon and Cavalier areas. Oxisols were only found in significant areas in the mid-les Platons area. Erosion hazard was found to be significant even at relatively low slopes of 12% to 15% at many locations. We have been able to deduce qualitative results of the soil fertility tests undertaken to determine the relative importance of pH, phosphate, potassium and organic matter for soil fertility in the Durand plain. This test enables us to help the peasants increase productivity most cost-effectively in the relatively flat plains in the Formon area. Phosphates are the most limiting factor in soil fertility.

Table 7. Fertiliser trials

Treatment	P	P + K	K	Temoin
Growth*	30	25	15	12.5 cm
# of pods/plant	27	27	5	5 pods
# of seeds/gousse	5	6	5	5 pods
Yield kg/ha	400	333	127	133 Kg

* Growth quantified 46 days after seeding

Forms for follow up: Forms for follow-up and collecting data on activities were produced for check dams, pig reproduction and distribution, PPC teams, and nursery activities.

The final product will be a booklet (series of bound printouts) listing all of the available information on the natural resources of Haiti, and well as a set of computer diskettes. These data sets will be available to all organizations working on natural resources in Haiti, but their primary distribution will be to the SPE of MARNDR and the MBR steering committee.

5. Develop an educational and interpretive program that will: i) train personnel and local people; ii) produce educational booklets on the parks and birds that can be used in training/general education/public relations work; iii) create booklets on the conservation of water, soil, trees, and wildlife for distribution to local schools (pilot project); iv) place interpretive and interdictory signs in appropriate places.

We developed a series of seminars for technicians and villagers on the following subjects: definition, importance and cost of a project; importance of reforestation; damage of slash and burn; natural resources: definition, type, importance, and management; erosion and its consequences.

1) Park agent training: In collaboration with MARNDR/SUD, the UF/MBRP organized a two-day seminar (24-25 January 1990) for the 15 *Agents de Parc* assigned to Macaya on such topics as erosion

control, the ecology of the MBR, Haitian law and the MBR, administrative procedures presented by representatives of the GOH such as the *Prefet des Cayes* and the *Commandant* of the *Département du Sud* and the UF/MBRP MBR Coordinator. As a result, the Agents du Parc are much better equipped to protect the MBR and assist in the development of the area. Further training for the Agents is certainly indicated and desired.

2) Farming Systems Trials and Diagnostics: This four-day seminar was offered to technicians and agronomists of PST and the GOH (31 July to 4 August 1989) to train the agronomists and technicians working in the TWM Project in the techniques of farming system trials and diagnostics. Twenty-seven participants were selected and the seminar was presented at ORE headquarters, graciously offered by Mousson and Sean Finnigan. Drs. M. Swisher and P. Bergeret conducted the seminar with assistance from P. Paryski and R. Voss. Each participant received a full package of pedagogic materials. Actual field trials were conducted. The seminar was videotaped for future use. The seminar was an outstanding success. The 27 participants were trained in farming systems research techniques and received diplomas from the University of Florida. All of the participants expressed their satisfaction with the seminar and requested that follow-up seminars be planned. The Departmental Director of MARNDR, who participated in the seminar, proposed that MARNDR establish formal ties with the UF for a training program.

3) Integrated Pest Management and Farming Systems Analysis: The presentation of the Second Farming Systems Seminar was highly effective and greatly appreciated. Statistical methods of easy analysis were demonstrated for decision making in pest control management. A three-day training course was given on June 4, 5 and 6, 1990, for 25 technicians and agronomists from TWMP and MARNDR using the facilities of ORE. It was conducted by M.E. Swisher and K. Barfield.

4) Journée de formation des techniciens de l'UF/MBRP: The technicians of the project organized this workshop on 21 March 1991 in les Cayes. This was a one-day seminar led by the Technical Coordinator and Natural Resource Agronomist. The technicians presented their accomplishments during the period they had been working for the project. They were familiarized with the logical organization of projects in general and specifically with the UF/MBRP, with costing activities, and with the preparation of quantified activity reports. TWMP Coordinator Philoctete participated in the training session.

5) Atelier de Durand/Formon: A one-day workshop was organized by the technical coordinator and her assistant for the technicians at Formon on July 23, 1991. They were to perform some project activities in order to understand how to plan for their programs. This is a good exercise for technicians; they learn to plan and look for information necessary to succeed in their activities.

These kinds of seminars are highly recommended in countries like Haiti, where the word "project" implies the availability of a lot of money to do nothing. Since the seminar, the UF/MBRP technicians have understood more fully the organization of a project and of the UF/MBRP, and their responsibility toward their employer.

6) Importance of reforestation and the damage of slash and burn: A one-day seminar was organized by the technical staff for people of Cavalier on July 9, 1991. The subjects were presented by Jean-Ronny Merisier and Jean-Claude Piverger, agroforestry technician. Practical exercises were carried out. Twenty people actively attended this seminar. The following week, a group of five people organized a meeting with the technical staff to begin a nursery in Cavalier.

4) Importance of the Natural resources and consequences of erosion: A two-week seminar was performed by the assistant technical coordinator and one technician from 4 to 16 May, 1992, for the villagers of Formon, Durand and Cavalier. We had a very heterogeneous group, chosen by a steering committee (Groupe pour la défense de Macaya). People of different ages and different levels of education (illiterate, 4th grade, and higher levels of education) wanted to learn from this seminar. Time was a constraint, therefore we worked (with great difficulty) with all the groups. An evaluation was distributed which confirmed our concern about the different target groups. Some people attended this seminar to be technicians and be employed by the project, others could not follow the course. We recommend giving much shorter seminars to specific target groups.

Outside Seminars:

1) **Presentation of UF/MBRP Papers at International Farming Systems Symposium University of Arkansas 9-11 October 1989:** Two papers on the project were prepared by UF team members: "A Biosphere Approach to Developing Sustainable Agriculture and Protecting an Upper Watershed of Haiti" by Voss, Paryski & Swisher and "Farming Systems and Sustainable Agriculture: the Haiti Model" by Swisher, Woods & Monaghan. Information about new and successful farming systems interventions was exchanged.

2) **Land tenure seminar:** The administrative coordinator attended this session from 3 to 6 October 1989. Land tenure is considered to play a major role in the project. This training was helpful especially during the Sondeo, since land tenure is a criteria for choosing peasant farmers participants in the PPC program.

3) **35th Workshop of the International Union for the Conservation on Nature(IUCN)/Commission on Parks and Protected Areas(CNPPA), Santo Domingo, D.R. 29 April to 3 May 1991:** The director of the project attended this seminar to address problems of conservation strategies for the Caribbean. Among the issues discussed were: tourism and parks, the foundation of a Caribbean Heritage Park System and Trust Fund, parks and NGOS, parks and economics, parks and local communities, revenue generation, parks and culture, Caribbean data bases, biosphere reserves, training and education, improving management, and participation in the IV IUCN World Congress in Caracas in February, 1992. Other organizations were informed about the UF/MBRP, its goals and activities; working contacts in particular with Caribbean groups IUCN, CANARI, Parques Nacionales and international organizations such as ICBP, IUCN, IUCN/CMU, OAS, Nature Conservancy, Audubon, WILD were established; an article on UF/MBRP for *Parks* magazine was planned; pledges of support for UF/MBRP from the Nature Conservancy under their Parks in Peril Program, and the Audubon Society were obtained; an invitation to participate in the IVth World Congress on National Parks and Protected Areas in Caracas in February, 1991, was received; the establishment of an international biosphere reserve including l'Etang Saumatre, Trou Caiman, Lago Enriquillo National Park (Dominican Republic) and the Baboruco National Park (Dominican Republic) was discussed.

4) The director of the project was also invited to Aspen Colorado in August, 1991, to present a panel on the project. This seminar was under the Global Warming Program. He met with experts in environmental issues.

5) The director and the technical coordinator attended *IUCN IVth World Congress on National Parks and Protected Areas* in Caracas from 10 to 21 February 1992. Due to the political situation of Haiti, funding was provided for by the Caribbean Natural Resources Institute (CANARI). Subjects discussed included the strategy for conservation in Haiti with emphasis on the Macaya Biosphere Reserve Project presented by Paul Paryski, and a draft of an educational poster drawn by Laura Waltz for critique was presented by Florence Sergile to a group of specialists in environmental communication in the Antilles.

6) Agronomist Jean-Ronny Merisier, the project's assistant technical coordinator, attended a seminar May 20 to June 8, 1991, on *"Diagnosis of soil erosion causes & observations on improvement measures"* (*Diagnostic des facteurs de l'erosion et raisonnement des aménagements*). The course was organized by the National Association of Haitian Agronomists (ANDAH) and the French National Tropical Agricultural School (Montpellier, France). The subjects discussed included: the causes and types of soil erosion, study of an experimental device for observation of erosion phenomena on a watershed, photo-interpretation, cartography, analysis of soil rehabilitation in Haiti, forest management, eroded areas and ravine improvement, the role of the agronomist and the engineer in soil conservation. Merisier has not only improved his own knowledge of soil conservation measures, he also organized a series of mini-seminars on this subject for the rest of the project technical staff and with the local people of Formon and Cavalier.

7) The UF/MBRP Peace Corps Volunteer, Sharon Hart, and Jean-Denis Chery, UF/MBRP technician, attended a Peace Corps seminar at Ibo Beach on 29-30 January 1991 on the *"Development of small community projects,"* projects that could be funded by the Peace Corps. Both the PCV and Technician Chery learned how to formulate, present and organize a small community-based project. Together they formulated a small project--the tapping of a Formon spring to feed local public fountains--and each one was promised \$5,000 to carry out the project. Unfortunately, there were a lot of misunderstandings among the two villages and ORE in collaborating on the project. However, the technicians formulated a new

ecotourism project just before the coup.

8) **Training of veterinary auxiliaries:** Two young villagers of Formon were selected by the community to receive training on primary health care for animals by DCCH in August, 1989. Such training will provide them with alternate employment and will help the population to raise better cattle.

9) **Seminar on animal husbandry and care:** From August, 1991, to May, 1992, three villagers and one technicians of the project attended a series of workshops organized by the veterinary service of IRD/PST. The villager of Deglaci dropped the course. The three others are now very useful in their village. They can now organize sessions with their people and give the first aid to the cattle and poultry of the area.

Other Training

Short-term training: We were delighted that Jericca St. Vil, project technician was selected by USAID to participate in a short-term training program in the United States under the Caribbean and Latin American Short Term Training program. She was supposed to attend her session by October, 1991; unfortunately the political situation of the country canceled this activity.

Presentations

The technical direction presented five slide-shows to show people the MBR area, its importance and what the project was doing.

Production video documentary on UF/MBRP: A 59-minute in-house video was produce in September, 1989, to show the Formon target area and project activities for interested organizations and individuals. The videotape was shown during the TWM evaluation meeting on September 26 and at the UF in Gainesville. Based on a review of the videotape, in May, 1990, the UF produced a documentary, "The Forest in the Clouds," on the UF/MBRP for broadcasting on PBS in the USA.



CONCLUSION AND RECOMMENDATIONS

Time is the most limiting factor in a project as unusual as the Macaya Biosphere Reserve Project. In working with nature and with socio-economic aspects, time is required in order to implement and evaluate each program element.

RECOMMENDATIONS

General

Long-term activities and funding must be assured since success cannot be achieved in the short-term. We cannot expect to rehabilitate ecosystems and eroded landslide sites, change technologies, and educate people in three years.

Technicians taking jobs in this area should be on probation for a three-month period and be evaluated by the Technical coordinator, the director of the project and the team. The purpose of such an evaluation is to insure that the project has better-trained technicians, to insure that they are committed to the project and to avoid rapid turn over.

NGOs of different fields should collaborate their efforts rather than competing with each other.

Efforts should be made by the government (Ministry of Agriculture) to apply law enforcement. As a typical example, during the month of June, a number of peasants in the Formon area planted gardens in the park (core zone) north of the park boundary fence. Many of them had come from other areas such as La Source and Troisieme seeking labor. In planting the gardens, they destroyed more than 100 hectares of regenerating secondary forest. The peasants justified their invasion by saying that: 1. the "Agents de Parc" are no longer functioning; 2. President Aristide stated that he would give peasants any state land they were exploiting and that all land should be planted to provide food for the country; 3. The University of Florida has no authority to interdict planting in any land.

Agriculture

Vegetable crops should be planted preferably in household gardens because they require intensive care; they can be planted in small areas by adolescents, and women.

Technical packages should be delivered with seed distribution.

Wide ranges of species, seasonal and/or contra-season varieties should be studied and disseminated to allow for year-round production and marketable production. Peasants in Durand/Formon, Duglacs cannot afford industrial production with the available limited market network.

Storage should be included as part of agricultural production to allow farmers to store and sell their products (Yam, beans, potatoes) for better prices over intermediate terms.

Marketing systems and networks should be evaluated before starting any production program. Transportation habits and demand are limiting factors of production.

Black bean production and distribution should be eliminated in any program in the buffer zone because villagers are planting black beans in the core zones.

Animal Husbandry

A forage and animal health program should be included in all households which are increasing income-producing activities. Villagers of Formon, Durand and Duglacs are cattle raisers.

Socio-economics

Wage and prices of labor should follow those of the region to avoid economical disbalance.

Sociologists and/or anthropologists should be studying and evaluating population trends of the region. Forms and short-term training should be developed by these scholars so that they may be used by local technicians to update regional data.

Family planning is being requested by the women villagers. It should be seriously considered because the explosion in demographics explosion will exacerbate problems associated with scarcity of the resources.

Regional

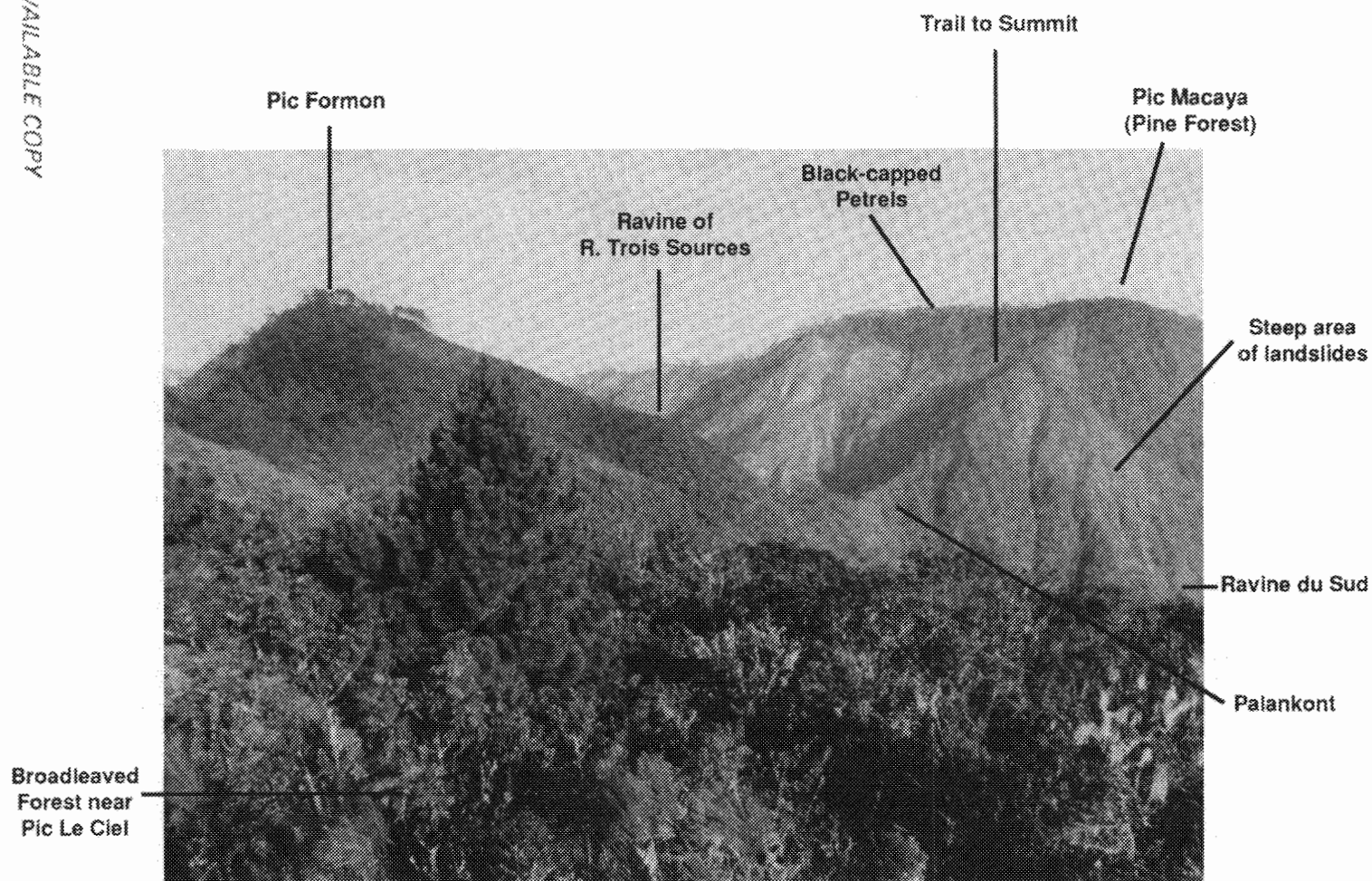
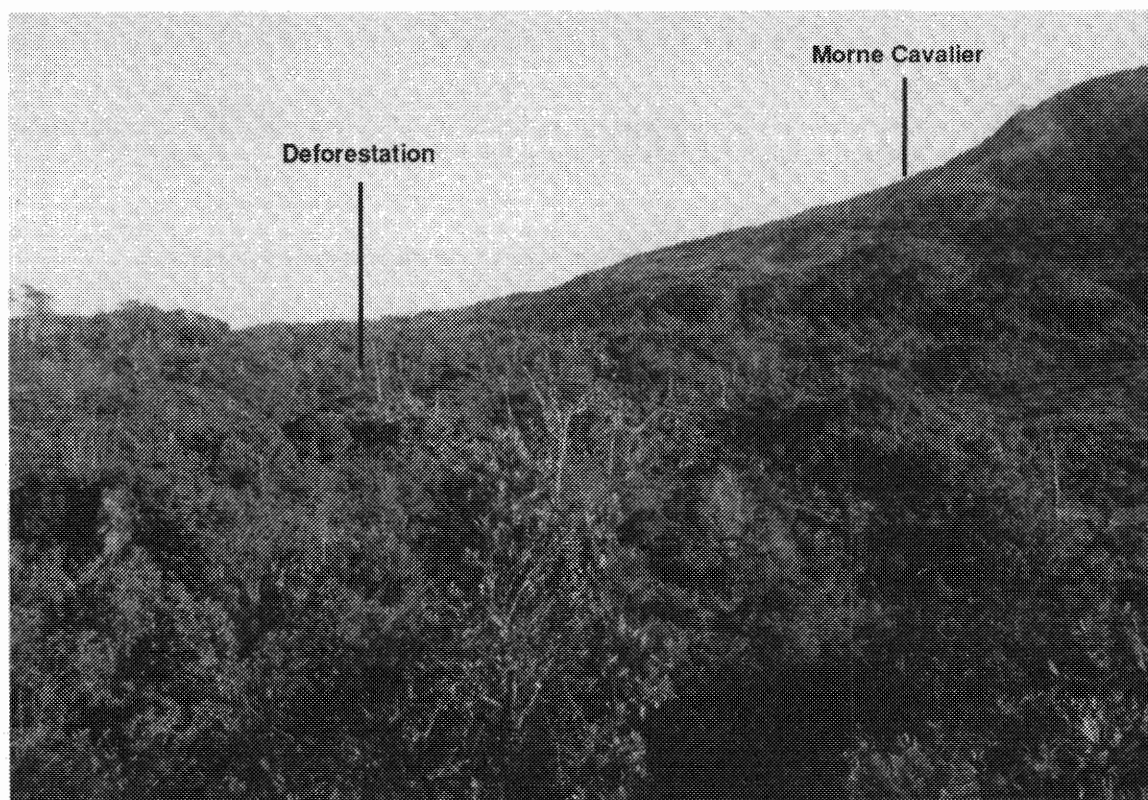
The Trois Sources area should have its own program directed by an agronomist specializing in rural development and hillside production. A furnished headquarters should be provided to this person.

Training and Extension for local people should be done by project technicians, not from cooperator to cooperator. A program should be developed for Cavalier.

Matters of land distribution should be left in the hands of the Haitian government. "Invisible" sociological elements should be well-studied before any action is taken.

Raising cattle, goats and sheep is one of the major economic activities in the MBR. But free-range grazing has been one of the major causes of deforestation and soil erosion in the MBR. Increasing the production of high-yield forage crops, training people to cut and carry forage crops and the delimitation of pasture zones are possible solutions to this problem.





Interior View of Parc National Pic Macaya

Chapter III

The Future of National Parks and Biosphere Reserves in Haiti

The concept of protecting the natural areas of Haiti goes back at least to the law of 17 August 1955. The official creation of "Parcs Nationaux Naturels" was a decree published on 23 June 1983, which set aside "Morne La Visite du Massif de la Selle" (2000 hectares) and "Morne Macaya du Massif de La Hotte" (2000 hectares) in addition to eight sites selected as "Parcs Nationaux" and "Sites Naturels" in the decree of 18 March 1968. These ten sites, but most especially Parc National La Visite and Parc National Pic Macaya, represent the components of the national parks of Haiti. Since 1983, the Florida State Museum has worked on a contract with USAID to complete an inventory of Parc National La Visite and Parc National Pic Macaya, and to develop a management plan for each park, as well as for the national parks program. Between 1983 and 1988 the parks were under the joint administration of ISPAN (then under the Institut National Haitien de la Culture et des Arts or INAHCA) and the Ministere de l'Agriculture, des Ressources Naturelles et du Developpement Rural (MARNDR).

Location

Parc National Pic Macaya. The area now designated as Parc National Pic Macaya is located 36 km NW of Les Cayes and 195 km W of Port-au-Prince at latitude 18 21' N and 74 01' W. The park is situated around the two dominant mountains of the region, Pic Formon (2219 meters) and Pic Macaya (2347 meters) and includes the large and deep ravine between these two mountains that serves as the headwaters of the Riviere Ravine du Sud. The plains south of Pic Formon (Plaine de Formon and Plaine de Durand) are also included in the park as are the rocky (karst) hills along the southern boundaries of these plains.

The two high peaks tower above the surrounding plains (which are between 1000 and 1500 meters in elevation) and a series of mountain ridges and receive moist air blowing inland from the Golfe de la Gonave to the north (via the northeast trade winds) or the Caribbean to the south (via sea breezes). The result is an area of extremely high rainfall (in excess of 3000 mm/year). Parc National Pic Macaya is the source of five major rivers of southern Haiti (Rivieres Port-a-Piment, Roseaux, Ravine du Sud, l'Acul, and Glace). Extreme deforestation in the area jeopardizes the stability of these rivers, and the rich agricultural lands

below (Cohen, 1984; Lowenstein, 1984). Parc National Pic Macaya serves the dual role of conservation of the national patrimony in protecting numerous endemic species of plants and animals and protecting the watersheds of the most important rivers of Haiti that spread outwards from the park like the spokes of a wheel.

Access to Parc National Pic Macaya is difficult. The journey is via the route to the south from Port-au-Prince passing through Les Cayes, 196 kms west. Beyond Les Cayes the route is via a gravel road to Le Duc and then a rough dirt road to Le Pretre that crosses the Riviere l'Acul. Beyond Le Pretre the route ascends to the plateau of the Plaine de Formon by a series of sharp switchbacks to the town of Les Platons. The entire trip from Port-au-Prince is 129 kms, although the 33 kms from Les Cayes to Les Platons are the most difficult. Beyond Les Platons a new and very rough road has been constructed by the MBR project, ten kilometers across the southern edge of the Plaine de Formon to the location of the Park Headquarters at Caye Michel.

Purpose

The purpose of the national parks has never been clearly stated. The law of 17 August 1955 regulated cutting, transporting and selling wood, and the Rural Code of Francois Duvalier (28 May 1962) strictly controlled forest resources and activities in forest reserves. The decree of 23 June 1983 creating "Parcs Nationaux Naturels" lists eight park-related responsibilities for MARNDR that can be summarized as: 1) protecting ecological conditions; 2) undertaking an inventory of plants and animals; 3) studying the characteristics of endemic species in relation to geology, soils, climate, etc.; 4) identifying areas having important ecological characteristics; 5) preserving national parks from physical deterioration; 6) supervising and working with the scientific community in studies in the parks and natural sites; 7) disseminating information concerning the parks and sites; 8) making the facilities of the parks available to visitors.

We have synthesized these statements, as well as our many conversations with personnel from the governmental and private sectors, into the following list which we believe represents the purpose of the national parks program in Haiti.

1. **The protection of natural ecological conditions and processes.** The two most important consequences of these actions are: A) the preservation of watersheds, thereby improving the quality of life for all inhabitants of Haiti in areas adjacent to or under the influence of national parks; B) the preservation of natural species diversity and therefore the national natural patrimony.
2. **The promotion of the national natural patrimony.** The two most important consequences of this activity are: A) the education of the citizens of Haiti about the unique features of their country that make Haiti special; B) the increased possibility that wise decisions of long-range importance can be made concerning the utilization and development of the natural resources of Haiti.
3. **The development of a recreation and tourism program** that will take advantage of the unique physical location and beauty of the parks as well as the special features of the flora, fauna or geology. We believe that it is possible for the citizens of Haiti to benefit from the parks at both the local and national levels without damaging the quality of the parks if a careful management plan is developed and implemented.

Resources

The geological and biological resources have been surveyed and are discussed in detail in the original biogeophysical reports funded by USAID in 1986, and in the second volume of this series of final MBR Project Reports entitled *The Natural History of Southern Haiti*. These reports indicate that Parc National Pic Macaya and the surrounding MBR area are of great importance because they have so many unique features. The geology reveals details about the past of Haiti when it was more than one island and when the tops of the mountains were shallow marine environments. The great ravine of the Ravine du Sud between Pic Formon and Pic Macaya is part of an enormous fault that cuts across the southern peninsula of Haiti from Tiburon to the Cul-de-Sac plain. The floristic features indicate that Parc Macaya has 69 vascular plants that are endemic. The total vascular plant flora of Macaya includes 130 species that are endemic to Hispaniola, which is 28% of the flora of the park. Among flowering plants, the degree of endemism of Parc Macaya is even greater, with 124 species endemic to Hispaniola (34% of the total) and 68 endemic to the park itself, which is 19% of the flowering plants of the park. The importance of Macaya is even more dramatically pointed out when just orchids are analyzed. Of the 133 species of orchids known

to occur in Parc Macaya, 38 are endemic to the Massif de La Hotte itself.

There are 65 species of birds recorded from Parc Macaya. The most endangered species is the White-winged Warbler, *Xenoligea montana*, but the Hispaniolan White-winged Crossbill and Black-capped Petrel are also very endangered and important.

In Parc Macaya, 17 species of endemic terrestrial mammals are known, but 15 have become extinct (88%). Fourteen bat species still occur in the park. Five species of bats of the Macaya fauna are vulnerable to extinction if caves are disturbed and habitat is destroyed (Morgan and Woods, 1986). The loss of endemic mammals is dramatic, but not unique. The average loss of endemic species throughout the Caribbean during the past 3000 years has been 88% (Woods, et al., 1986). One species of mammal, the La Hotte Hutia (*Rhizoplagiodontia lemkei*) is restricted to the MBR area. An intense effort to find additional new species of mammals in the Macaya area was unsuccessful, but it is likely that five species became extinct in the last 30 years (Woods, et al., 1986), which is the period of greatest habitat loss in the Macaya area (Cohen, 1984).

Management

We have presented a detailed outline of our recommendations for the management of the MBR area and Parc National Pic Macaya in the volume of the series of final reports entitled *A Stewardship Plan for the National Parks and Natural Areas of Haiti*. We prefer the term stewardship rather than management since we believe the latter concept implies an active state of manipulation and alteration. Stewardship can be passive and can allow the ecosystem to recover and maintain itself without extensive manipulation. The Macaya area is highly disturbed, however, and during the initial stages of creating the park and MBR, an active management program will be necessary.

Zones

The zones of Park Macaya fall into two categories, each of which is divided into three areas. One zone is associated with specific activities, and requires constant attention and an active role by national park personnel and visitors alike. This zone is called the "Designated Use Zone", and it is subdivided into the following areas.

- 1). "Recreation Areas" are where visitors can camp, hike, observe scenic vistas and enjoy special features of the parks.
- 2). "Education Areas" are where national parks personnel have created a special enriched environment to educate visitors about the special features of the parks. These areas include nature trails, special signs at designated locations, site exhibits and the public areas of the Park Headquarters.
- 3). "Maintenance and Service Areas" are where national parks personnel work and store equipment and supplies that are necessary to improve the quality of the park. These areas include the work areas near the Park Headquarters, the depots where equipment is stored, the security areas where guards work, the living quarters of the Park Headquarters, the stables and facilities for horses, mules and donkeys used in the functions of the park (including tourism), and the garages and shops associated with park vehicles and machines.

The second zone of the parks is associated with restricted activities, and is called the "Limited Visitation Zone". The primary goal of activities in this zone is conservation of the soil, water, flora and fauna. The largest and most important region of this zone is the "Biological Preserve Area". No exploitation of any kind should be allowed in areas so designated. When areas are of potential importance to the conservation of specific organisms, soils or watersheds, but are currently degraded, an active role is required to restore the habitat to a condition where it can be regarded as a Biological Preserve Area. These areas, each of which will be designated as a "Restoration Area", are where active management is necessary with the long-range goal of improving their quality so that future management will not be necessary. The last area of the Limited Visitation Zone is where research is permitted on a limited and carefully-controlled basis. These areas should be distinct from Biological Preserve Areas, and are designated as "Research Areas".

All of these zones and areas are discussed in the text of the Stewardship Plan.

Construction Controls

The construction of all structures, trails and signs should be supervised by a central office of the national parks program where a record is kept of all construction activities. A central file in the Park Headquarters should contain work plans, information on costs and photographs of completed projects. A routine inspection of all projects in the parks should be completed by the Director at least once a year, and on a regular basis by each Park Supervisor.

Roads

Clearly, access to the park and MBR area is an important feature, and these roads must be maintained and improved before an active tourism program is possible. Great care must be taken in the improvement of the Parc Macaya access road, however, since a good road would open the region up and remove the last barrier to the exploitation of the interior of the Massif de La Hotte--its isolated location. We do not recommend the improvement of this road until an active national parks program is in place with personnel committed to the supervision of all activities in Parc National Pic Macaya.

Administration

Suggested Annual Budget

The total budget for the national parks program over a five-year period includes: 1) set-up costs for the Central Office in Port-au-Prince; 2) set-up costs for each park; 3) signs; 4) exhibits; 5) publications; 6) training personnel; 7) research; 8) operating the Central Office in Port-au-Prince; 9) operating parks; 10) operating vehicles; 11) salaries in the Central Office; 12) salaries in parks. The total for these items over a five-year period is \$2,514,790. We believe that research is a very important component of the implementation of a national parks program. The cost of the proposed research plan is \$165,000 over a five-year period, or 6.6 % of the total budget. Research projects also serve as effective ways to train the staff of Parcs Haiti. Personnel from the staff of the national parks program should work closely with all researchers, and this should serve as an important part of the training process for all national parks personnel during the first five years. The staff should also participate in the publication process of the results of the research.

Routine Maintenance Schedule

A regular schedule should be established to insure communication between the parks and the central office in Port-au-Prince. The Park Supervisors should spend three weeks in the parks and one week in the central office. Within the parks a regular schedule should be established for the following components.

1. **Access Roads.** A work crew should be assigned to improve the quality of the access road to each park. After the roads are in suitable condition to allow safe and routine access to the parks, they should be "maintained" on a monthly basis to remove fallen rocks and to insure proper drainage.
2. **Trails.** The trails through the areas of each park where vegetation is dense, and especially in Parc Macaya where trails become overgrown with cutting bamboo (*Arthrostyidium haitiense*) and sharp spined blackberries (*Rubus* spp.), should be cleared as necessary. In steep areas, care should be taken to landscape the trails at the same time to prevent erosion.
3. **Park Headquarters.** The Park Supervisor should develop a checklist for regular maintenance of the toilets, sinks, septic tanks, cisterns, stoves, refrigerators, propane gas tanks, and public quarters.
4. **Central Office.** The Director should develop a checklist for regular maintenance of all equipment, supplies and buildings. The Director should delegate this authority to the Assistant Director for Administration.
5. **Vehicles.** The Director should develop a schedule where all vehicles are maintained on a regular basis. Each vehicle should receive a thorough inspection every three months.

Infrastructure

The national parks program should be organized as a discrete unit with the ability to make decisions on policy, management and budget. All aspects of the park should be under the direct control of the Director.

The location of the national parks program within the structure of the GOH has never been clearly designated. We recommend that the best solution is to name the program **Parcs Haiti** and make no reference to a department, service or institute. **Parcs Haiti** should be a free-standing unit of the GOH under the supervision of a board of trustees designated as the **National Parks Advisory Council (PANAC)**. The organization of the program is diagramed below.

National Parks Advisory Council (PANAC)

Director of Parcs Haiti (Chairman)
 Director of ISPAN
 Director of Direction des Ressources Naturelles (MARNDR)
 Director of Office of Tourism
 Director Societe Audubon d'Haiti (SAHPE)
 Prominent Private Citizen
 International Representative

"Parcs Haiti"

Central Office
 Component Parks and Natural Sites

The National Parks Advisory Council (PANAC) would be responsible for meeting with the Director of Parcs Haiti once a year, and assisting the Director to improve funding, develop long-range goals, and resolve points of conflict.

The actual administration of all aspects of national parks in Haiti should be the responsibility of the Director of Parcs Haiti.

The organization of Parcs Haiti can be accomplished in any one of three ways.

- 1). We recommend creating a new program within Haiti, which should be named **Parcs Haiti**. During the first year, a Director, as well as a complete staff for each park (Park Supervisor, seven park guards, 15 park workers), and a complete office staff should be hired. The additional two positions (Assistant Director for Education and Recreation; Assistant Director for Conservation and Research) are important to the development of the parks and parks program, and require some technical abilities. If individuals with the interest and training for these positions do not exist in Haiti, then international personnel could be used to fill the positions on an interim basis (two to five years). The University of Florida, IUCN, Parks Canada, U.S. National Parks Service and World Wildlife Fund should be consulted for advice as to whom to hire. The University of Florida-Florida Museum of Natural History would be willing to coordinate this process.
- 2). The second possible way to create Parcs Haiti would be to contract out the complete operation of Parcs Haiti to an international organization concerned with conservation and national parks. This concept has the advantage of being able to draw upon the experiences of a group specializing in national parks work, and it would allow a strong national parks program to quickly begin protecting the national parks of Haiti (which is very important). This group could also serve a valuable teaching function. The technique has the potential disadvantage of bringing foreigners into conflict with the GOH on sensitive environmental matters, and being perceived as imperialistic. On balance, we feel that this is a good idea and should be implemented if our primary recommendation is not possible. This organization should be under the supervision of the National Parks Advisory Council (PANAC) described above. The IUCN, Parks Canada, or the University of Florida-Florida Museum of Natural History are logical choices for this group. The group contracted to run the national parks program for a period of five or ten years could work closely with SPE.
- 3). The third possibility for creating Parcs Haiti is to allow the SPE to run the national parks program of Haiti under the World Bank Environmental and Forestry Project. If the WBP project starts up, then the funding and administrative structure are already in place. However, it will take a long time

for SPE to achieve the technical skills necessary to run the national parks in Haiti, and we recommend that an outside NGO or international organization be employed to assist SPE. We also recommend that SPE use the name Parcs Haiti if it does assume complete responsibility of the national parks of Haiti.

Our primary recommendation for the creation of a national parks program in Haiti is the creation of an entity called Parcs Haiti with the authority and commitment to protect the natural patrimony of the country. Any of the three alternative ways of creating Parcs Haiti would accomplish these goals, and be able to implement a national parks program. The decision on which program to follow should be made as soon as possible. The decision should be made by a majority vote of the seven individuals discussed as the National Parks Advisory Council (PANAC) which can meet on a one-time basis to make this decision or become a permanent group advising the national parks program (Parcs Haiti).

Personnel

The organization of Parcs Haiti is diagramed below.

PARCS HAITI

Director

Assistant Director for Education and Recreation

Assistant Director for Conservation and Research

Central Office

Secretary/Bookkeeper

Librarian-Secretary

Chauffeur

Commissar

Office Guardian

Parc National La Visite

Park Supervisor

Guardian Headquarters

Cook

Park Guards (7)

Park Workers (15)

Parc National Pic Macaya

Park Supervisor

Guardian Headquarters

Cook

Park Guards (7)

Park Workers (15)

Logistics

The Central Office in Port-au-Prince will coordinate all aspects of the program in national parks. All files, plans, publications, correspondence and budget information should be maintained there. The Director will interact upwardly with the National Parks Advisory Council, (of which he would be Chairman), laterally with other programs, and will supervise all activities of Parcs Haiti. The Assistant Directors will work with the Director and regularly tour the parks and make reports. The Park Supervisors will implement the directions from the Central Office and supervise the personnel in each park. The Park Supervisors should spend one week each month in Port-au-Prince planning and working with the Central Office staff. A radio communications network should exist between the Central Office and each park, and there should be regular discussions between the Park Supervisors and a designated person in the Central Office.

The Director of Parcs Haiti should meet with the National Parks Advisory Council (PANAC) at least once a year.

Interpretation and Research

The programs in interpretation and research are two of the most important elements of the national parks program. Interpretation is the responsibility of the Assistant Director for Education and Research. Research is the responsibility of the Assistant Director for Conservation and Research.

Interpretation

We recommend the following as part of a program in interpretation.

1. A series of temporary exhibits in the Central Office.
 2. Informative signs at entrances to each park.
 3. A brochure on the parks that promotes the most important features, provides a species list of important organisms, discusses the rules of the parks and the purposes and concepts of the parks.
 4. Four exhibits in the Park Headquarters at Morne La Visite.
 5. Two exhibits in the Park Headquarters at Macaya.
 6. Two permanent exhibits in the Central Office Park Headquarters.
 7. Nature trails with educational signs in each park.
 8. Information signs at designated locations off the road beyond Furcy and along the road to Jacmel.
 9. Training the Park Supervisors to be interpretive naturalists.
 10. Developing a slide-tape presentation (French, Creole, and English versions) for use in informing the public of the national natural patrimony and the value of the parks.
 11. Designating national species, and promoting a better understanding of these species. We recommend the following.
 - a). The Hispaniolan Hutia = "Zagouti" (*Plagiodontia aedium*) - National Mammal
 - b). The Grey-crowned Palm Tanager = "Oiseau Quatre Yeux du Sud" (*Phaenicophilus poliocephalus*), or the Hispaniolan Trogon = "Calecon Rouge" (*Prioteles roseigaster*) - National Bird
 - c). The "Bois Tramble" (*Didymopanax tremulum*) or the Hispaniolan Pine = "Bwapen" (*Pinus occidentalis*) - National Tree
 - d). "Ti Crapo Morne" (*Eleutherodactylus wetmorei*) - National Frog
 - e). "Couleuvre Bois" or "La Madeleine" (*Uromacer frenatus*) - National Snake
 - f). "Tortue l'Etang" (*Trachemys decorata*) - National Turtle
 - g). Iguana (*Cyclura cornuta*) - National Lizard
 - d). Fuchsia or "Belle Fleur Morne" (*Fuchsia pringsheimii*) - National Flower
 - e). Karst topography = "Casse Dent" - National Geological Landscape
- These important elements of the natural patrimony of Haiti can be promoted on postage stamps, in newspaper articles, and by conservation posters. They can also be topics of discussion in schools. Some are already portrayed on the "Connaitre et Proteger la Richesse Naturelle d'Haiti" conservation poster done by the University of Florida MBR project.
12. There should be a regular weekly newspaper feature on the national parks written by the Director of Parcs Haiti or by special contributors. This feature should be in a prominent place in at least one newspaper, and should also be featured in the English language "Haiti News".
 13. Writing booklets on the special features of each park based on scientific results from inventories and ongoing research projects.

Research

Research is one of the most important elements of the development process for the national parks program. An active research program will generate data on which management decisions must be based. The research program will also provide data on the national natural patrimony (endemic species, many of which will be new). A third and very important aspect of the research program is that by actively involving the staff of Parcs Haiti in the research activities (field work, analysis, publications) one of the most important and difficult aspects of the training program is completed (i.e., training in specific topics) at the same time that data and publications are being generated. Research personnel can also be required to lead training sessions while they are in Haiti. The most important research goals are outlined below. The costs of these programs are listed in the budget sheet and in the Stewardship Plan.

1. Publication of previous data from inventory.
2. Research on the composition and requirements of the major plant communities (five year study).
3. Research on the basic biology of the endemic mammals and endangered species (five year study).
4. Biology of the Black-capped Petrel (three year study).
5. Regular inventories of avifauna (five year study).
6. Meteorological studies (five year study).
7. Invertebrate faunal studies (five year study)
8. Habitat requirements of herpetofauna (three year study)
9. Publication of the above data (year five)

Park Headquarters and Stations

The facilities of the parks program are very important because they not only provide a working environment for Parcs Haiti personnel and colleagues, but also demonstrate the existence of a viable parks program. This is especially important within the boundaries of Parc National Pic Macaya where the Parc Headquarters buildings, depots and outposts demonstrate that the park is official and that authority exists in the region.

Location

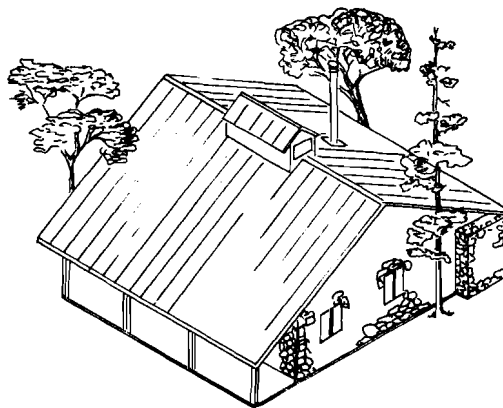
1). **Park Headquarters at Parc National Pic Macaya.** This structure has been completed as part of the MBR project. It is located at the edge of the Rak Bwa forest on the upper Plain of Durand. The name of this facility is Caye Michel.

2). **Depots and maintenance facilities** have been completed at Parc National Pic Macaya in the area of the Park Headquarters at Caye Michel.

3). An inexpensive caye was constructed at 1000 meters elevation in the ravine of the Riviere Ravine du Sud as an outpost.

4). An inexpensive caye should be located on the north side of the park near 700 meters beside the Riviere La Guinaudee. An existing caye could be rented. This caye would serve as an outpost.

5). An inexpensive caye should be located on the west side of the park near the end of the UNICORS/COSAR road in the valley of the Riviere Trois Sources. This caye would serve as an outpost for park guards, and would help protect the very vulnerable and important western side of Park Macaya from exploitation.



Integration of the Parks with Region

Parcs Haiti should make a special effort to improve the quality of life in the region around the parks in the Massif de la Selle and Massif de la Hotte. If the parks are going to be successful they must be viewed

as being of local importance as well as of national interest.

Jobs

The personnel for the parks should be hired from the region. The available jobs are listed below. Regular jobs = R; Occasional jobs = I.

Number positions

1. Park Guards	7 (R)
2. Park Workers	15 (R)
3. Headquarters Cook	1 (R)
4. Headquarters Guardian	1 (R)
5. Other building guardians	3 (R)
6. Guides (estimate)	10 (I)
7. Special project workers (estimate)	10 (I)

Additional jobs will be available as the Park Guardposts are being constructed, and during phases of reforestation. Jobs will also be generated as recreation and tourism increases. Some of the jobs will be as guides, while others will be associated with renting horses and donkeys to visitors and selling supplies. The amount of money in the region will increase because of regular salaries being paid to Parcs Haiti personnel, and as income increases, the demand for supplies will also increase. This will provide more jobs in the service sector of the region (store owners, farmers).

Watershed Management

One of the main goals of the park is to improve the quality of the environment and of the associated watersheds. This will have some impact on residents on the plains and along the rivers below the parks. More water will be available for human consumption and irrigation. The quality of the water will improve. The rate of soil erosion will decrease. The danger of flooding to inhabitants living near rivers will decrease. All of these will be possible because major reforestation projects will be undertaken.

Model Programs

The following model programs are proposed for the region of each park in association with Parcs Haiti.

1. Communal Firewood Forest Project.
2. Reforestation and Fruit Tree Propagation Program in cooperation with MARNDR and PVOs in region.
3. Technical Assistance Program coordinated by the Park Supervisors.
4. Local artisan, handicraft and recreation program coordinated by the Assistant Director for Education and Recreation.

Recommended Training Programs

Since there has not been a tradition of natural science, conservation or national parks in Haiti it is necessary and important to implement an active training program for all national parks personnel. After careful reflection, we believe that the most cost-effective and appropriate training program is for seminars and workshops to be conducted in Haiti. Each workshop or training session should last about two weeks and be conducted by a carefully-chosen international individual or group. There should be four training sessions per year (every three months). These training sessions should take place in the Central Office and within the parks. There could be more frequent training sessions during the first year of the program, and fewer as the program develops.

Individuals

Everybody associated with the parks should receive training of some kind. The nature of the training is listed below.

Location and Training

With the exception of the Director, who should be encouraged to participate in the international parks seminar, all training should be in Haiti. In this way specific training is possible and all staff of the Parcs Haiti program can receive some benefit from the instructors and training sessions. In addition, research biologists can be involved in the training sessions. This is economical, and encourages individuals to focus their training on the Parcs Haiti program rather than use the training as a stepping stone to a higher position in Haiti or to leave Haiti. It is important to create as much long-term stability in the Parcs Haiti program as possible.

Specific Courses and Topics

The need for specific topics will be identified with greater precision as the staff is hired and the backgrounds of the Parcs Haiti staff become known. At the initial stage, we can identify the following topics which are of major concern.

National Parks Topics

1. History of National Parks and National Parks Policy
2. Biosphere Reserves
3. Administration of National Parks
4. Personnel Management and Decision Making
5. Financial Accounting and Data Management
6. Public Relations and Advertising
7. Museum Studies and Interpretation
8. Principles of Recreation and Tourism



Research and Conservation Topics

1. Biological and Geological History of Haiti
2. Collecting Techniques - Botanical
3. Collecting Techniques - Zoological
4. Ecological Methods
5. Cartography, Photo Interpretation and Remote Sensing
6. Major Features of Haiti - Botanical
7. Major Features of Haiti - Zoological
8. Major Features of Haiti - Geological, Hydrological, Meteorological
9. Analysis of Data, Statistical Techniques and Publication Techniques

Some of these 16 topics can be combined into one training session, while others (such as "major features of Haiti") represent more than one training session. The training sessions can be organized as distinct one- to two-week workshops and seminars. All research staff should be required to present training sessions as part of their research activities, and to include selected Parcs Haiti personnel in specific aspects of their research activities.

Concluding Comments on Parcs Haiti and Biosphere Preserves

The concept of a "Biosphere Reserve" has been approved by UNESCO under the Man and the Biosphere Program (MAB). Biosphere reserves are examples of the major landscapes of the world complete with

characteristic landforms, flora and fauna as well as the various patterns of human use and adaptation of the region. The concept of a biosphere reserve is intended to promote a balanced relationship between people and the natural environment. The emphasis in biosphere reserves is demonstrating the value and need for conservation and the link between wise land use policies and conservation and sustainable development. We believe that the concept of a biosphere reserve is very well-suited to the nature of the national parks in Haiti, and we encourage Parcs Haiti to proceed with the implementation of a biosphere reserve program.

Biosphere reserves consist of core areas, which are lands already under some form of protection. In the case of Parcs Haiti, the core areas are Parc National Pic Macaya, Parc National La Visite (and the various sites listed in the decree of 23 June 1983). In biosphere reserves core areas are surrounded by zones of cooperation, in which a variety of resource uses may take place, such as forestry, ranching, farming or tourism. In these zones, solutions to man-environment problems are sought by involving local people. This is often done by including selected local residents in the discussion process, often via a committee made up of Parks personnel and local residents. The zone of cooperation is analogous to the "Buffer Zone".

The concept of a national park surrounded by a buffer zone, as discussed in the Stewardship Plan lends itself well to the principles of a biosphere reserve, especially considering the complicated land use practices in and around the national parks and the need to resolve the problems of land use and land tenure in the region. This was the plan followed by the MBR project. The habitats of Parc National La Visite and Parc National Pic Macaya are unique and worthy of inclusion in the biosphere reserve program of the world's major landscapes. In 1985, there were 243 biosphere reserves in 65 countries.

Parcs Haiti should begin from its conception to organize the national parks program, and especially the two great national parks of Haiti as biosphere reserves. The idea could be to link them all into one big, multi-part biosphere reserve that would include both national parks and the freshwater lakes of the Cul-de-Sac Plaine. This biosphere reserve, which could be called the La Selle Biosphere Reserve, has biological importance because this entire ecosystem was part of the old "South Island". Or Macaya Biosphere Reserve and La Selle Biosphere Reserve could be separate entities.

Biologically and historically significant areas in northern Haiti could be grouped together in a second biosphere reserve, which would be named the Henry Christophe Biosphere Reserve. This distinction makes sense, since all of northern Haiti had a very different geological and biological history from southern Haiti.



View of the Future: The Proposed World Bank Project

In 1990, a plan was developed by USAID, UFMBRP, and the World Bank Environmental and Forestry Project to work together to assist Haiti in developing a first-class environmental program. This plan was scheduled to begin in 1991. The World Bank Project was designed to protect the natural resources of Haiti. This project was planned to draw on the experiences of the University of Florida activities in Haiti, and to build on the programs developed in the national parks at Parc National Pic Macaya and Parc National La Visite as well as the planned Macaya Biosphere Reserve. The World Bank Project is still pending. It has been suspended, we hope temporarily, because of the unsettled political situation in Haiti. USAID, the University of Florida and MARNDR have worked together to try to "Bridge" between the present transitional situation, and a time in the future when programs such as the World Bank Project will make it possible to have a strong program in land stewardship and conservation in Haiti. USAID is working to design an ongoing project in soil and water conservation in the Macaya area, and the University of Florida is working on a project in environmental education and wildlife conservation via a grant from the MacArthur Foundation. These activities are described in the next section of this chapter. The present section describes the planned World Bank Environmental Project in detail. We hope that the project outline presented here will provide a clear picture of the possible future direction of programs in natural resources and environmental protection in Haiti.

The University of Florida Macaya Biosphere Reserve Project (UF-MBRP) has worked as an NGO contractor of USAID in the buffer zone south and west of Pic Macaya National Park, and in the southern half of the park itself. This project was initiated on October 1, 1987, and was originally expected to continue receiving funding from USAID through August, 1993. However, political guidelines of USAID have prevented the UF-MBRP from working in association with the Government of Haiti (GOH) for most of the life of the project, and the MBR project was not continued beyond May 31, 1992. The activities of the UF-MBRP include: (i) sociocultural inventories of the southern district of the park and adjacent districts (buffer zone); (ii) zoning the southern regions of the park and buffer zone; (iii) analysis of land use trends; and (iv) development of an agricultural and agroforestry extension program in the southern Macaya region. Since the GOH has the administrative responsibility for all state lands, and since Parc National Pic Macaya is on public domain state lands under the administrative control of MARNDR, the management of the Pic Macaya Project has been complex (and flawed). The situation is further complicated by the mission of the USAID project, which defines the geographical limits of its Targeted Watershed Management Project as south of Pic Macaya. The existing UF-MBRP, therefore, is technically only concerned with half of the existing national park and associated buffer zone.

The Proposed World Bank Project and the The Environmental Protection Service

The Service de la Protection de l'Environnement (SPE) was first created by Administrative order in 1984, and legally recognized by the "Conseil National de Gouvernement" Decree of 1987. The intention of Division of Natural Resources (DRN) of MARNDR in creating the SPE was to place the supervision of all national parks in Haiti under the SPE, but this was not done at an earlier time because of a lack of resources and funding. For example, it was the Forest Resource Service (SRF) instead of SPE that assumed responsibility for the park at La Visite in 1988 because SRF had funds available and SPE did not.

In 1990, at the time the World Bank Team was designing the project, the SPE had three professional staff (including the Chief). These three people were assigned to two sections:

- a). Section on Research and Inventories;
- b). Section on Preservation and National Parks.

The World Bank Project was designed to enhance these two existing sections, and to add one additional section, the:

c). **Section on Cartography and Information Services.**

The new section would be responsible for centralizing the data collected by the inventories, and the information collected by the other two sections, and making it available in the form of data bases and Geographic Information Systems (GIS).

One additional section of the SPE existed on paper, the Section on the Prevention of Pollution, but it did not have any staff members assigned to it in 1990. This section would not be enhanced as part of the World Bank Project.

Although two of SPE's three professional staff members have training in conservation and management of natural resources, the SPE does not have the financial resources and staff numbers necessary to initiate and complete an inventory of Haiti's fragile ecosystems, to develop the decision making skills necessary to manage these fragile ecosystems, and to supervise staff that would be assigned to remote locations in the existing national parks. As a result, SPE has not yet been able to assume full responsibility for the management of the existing national parks. It is important, therefore, that the SPE be strengthened as a Service so that it has the resources to assume all of its assigned duties in managing parks and protecting the environment.

Objectives and Strategy of the Proposed World Bank Project

Objectives

The objectives of the environmental protection component of the World Bank Forestry and Environmental Protection Project are to strengthen the Service de la Protection de l'Environnement (SPE) to enable it to:

- a). **Generate Data on National Land Use Patterns.** Data collected by aerial photography, satellite imagery, and ground survey teams would be used to evaluate land use trends, and to identify areas of special concern.
- b). **Inventory Forest Resources on a National Scale.** Forest areas would be surveyed throughout the country to compile data on the forest resources, remaining areas of forest cover, total available forest biomass, and the rate of deforestation.
- c). **Identify New National Parks and Forest Reserves.** Data gathered by the land use survey and the inventory of forest resources would be used to identify candidate ecosystems for a new national park, as well as to identify forests which should be designated as forest reserves to be managed by the SRF.
- d). **Manage the Two Existing National Parks and Create One New National Park.** The SPE would assume immediate responsibility for the management of the existing national parks at Morne La Visite and Pic Macaya, and would complete the activities necessary to add one additional national park to the system by year four of the project.
- e). **Preserve Endangered Plant and Animal Species.** The SPE would compile data on threatened, rare, and endangered species, and establish a program to protect these species.
- f). **Establish and Manage Databases on Renewable Natural Resources.** The SPE would enter the data from baseline sociocultural and environmental profiles of candidate ecosystems, as well as other data on the environment and the status of forest resources gathered in the course of the project, into various data bases for general access.
- g). **Monitor Long-term Land Use Trends Throughout the Country.** The SPE would use its role as the information center on renewable natural resources, and its trained staff and available data bases to constantly monitor and evaluate land use trends throughout the country.

Strategy

The above objectives would be achieved through the following project actions.

- a). The SPE would be provided with offices and buildings (central office, Macaya field office, and la Visite field office), transportation, equipment, training, operating funds, and salaries (on a declining basis).
- b). An internationally recruited "Resident Ecologist" (as full-time consultant for 5 years) would assist the Chief of Service of SPE and other SPE staff members in the implementation of all programs assigned to the SPE. Short-term consultants would be financed under the project for up to a total of 15 man-months in the areas of: (i) anthropology/sociology; (ii) conservation biology; and (iii) remote sensing (aerial photography and satellite imagery) to assist the SPE carry out various special survey and management tasks.
- c). The SPE would conduct inventories of renewable natural resources to identify candidate areas for the establishment of one additional national park and of future forest reserves to be managed by the SRF.
- d). The SPE would immediately assume administrative responsibility for the national parks at Pic Macaya and Morne la Visite, and implement the management plans by: (i) assigning supervisory staff and establishing local facilities; (ii) employing park rangers to prevent illegal exploitation; and (iii) creating land use zones.
- e). For the management of the park areas and surrounding buffer zones, the concept of "biosphere reserves" would be implemented.
- f). Endangered species of plants and animals would be protected by: (i) better administration of current laws requiring permits to collect and/or export plants and animals; (ii) better cooperation and communication between the SPE and enforcement agencies; (iii) establishment of nursery programs to propagate and grow endangered plant species from both lowland and upland habitats; and (iv) provision of habitat for threatened animals in the national parks.
- g). Access by the public to the parks would initially be discouraged by not improving roads linking the parks to major population areas during the life of the project.
- h). The SPE would develop a data base on the natural resources of Haiti that would allow the Service to document the status of present environmental conditions, and track future changes. Some of the activities that would allow SPE to fulfil this mission would be: (i) inventories of ecosystems and forest resources of Haiti via aerial photography, satellite imagery, and ground surveys; (ii) sociocultural inventories and environmental profiles of specially designated areas; and (iii) inventories of forest biomass and potential productivity.

Major Project Components of the Proposed World Bank Project

The activities that would be undertaken by the SPE are as follows:

National Land Use Assessment

Satellite imagery will be utilized to periodically monitor country-wide land use trends, and the distribution of ecosystems. The Landsat (EOSAT) Thematic Mapper (TM) as well as the SPOT system will be used. This will provide a fairly detailed inventory of land use patterns and land resources. Manual interpretation of these images will include "geocoding" the images, and entering data on urban areas, barren ground, brush lands, major rivers, coastal wetlands, roads, forest cover (3 crown density measures), sand, agricultural areas, water, wetlands, and rural villages. Because the topography in Haiti is so steep, and the area of the country is relatively small, resolution becomes an important consideration. The SPOT system provides higher resolution, and it therefore would be used (along with aerial photography) for more detailed analyses. The ground truthing for the analysis of the satellite images will be conducted by the SPE inventory team. The local SPE staff will be trained in interpretation and validation with the assistance from a short-term consultant.

Flight surveys (aerial photography) would be utilized to determine local land use patterns, the locations of specific renewable natural resources, and to document activities in areas of special concern such as

core zones in national parks, and inventory areas. Some aerial photographs exist, such as the outdated photographs in Geodetic Institute, and the photographs of the Pine Forest Reserve obtained by the SRF which include a small section of the national park at la Visite. These existing photographs would not be recent enough, nor cover areas of primary concern by the SPE (la Visite, Macaya, Environmental Profile Areas), and so additional aerial photographs and flight surveys will be contracted by the SPE to fill in the gaps in the available data, and to provide additional data needed to identify candidate areas, study forest resources, and document land use trends. These aerial photographs would be interpreted initially by the firm as part of the contract, but later by the staff of the SPE once they have the proper training to do so.

Forest Resources Inventory

Forest Attributes. The forest types delineated on the satellite images would be sub-sampled to produce more specific information on the forest resources (crown cover, species, density, and biomass volumes). An aerial survey and ground inventory of the chosen forest types would provide data on forest cover and density attributes. The SPE would produce biomass tables for every forest class. Biomass and area totals for all classes would be summed to represent total biomass availability.

Patterns and Nature of Wood Harvesting and Utilization. An essential question for forestry planners in Haiti for each forest type is: at what level and for what products is current wood harvesting sustainable? Data generated by aerial photographs and satellite images (land use and forest type) provide valuable information on proper forest classifications. Biomass tables for each forest sub-stratum provide information on potential wood availability. This information alone, however, presents an inaccurate picture of actual wood use, wood production, and true trends in the availability of forest resources, and additional sociocultural inventories would therefore be carried out under the project.

Sociocultural Inventories of Forest Resource Utilization. Sociocultural surveys in selected regions would be used to inform planners of the level at which wood harvesting is sustainable by documenting rates of wood production, patterns of harvesting and utilization, all of which are geographically and forest type specific. Sociocultural surveys of wood use would be carried out with the assistance of short term consultants for each forest stratum to describe local wood consumption patterns; specifying the types, amounts, and purposes of wood utilized; and the use of other tree products and utilizations (shade, forage, medicinal, etc.). The sociocultural information would help planners document the causes and rates of forest depletion and degradation. Upon the completion of the inventory by the SPE, the SRF staff would continue a more detailed analysis of national wood production and consumption patterns, and initiate planning activities for future forestry investments.

Long-Term Forest Inventories on Observation Plots (Ecosystem Monitoring). Permanent plots would be installed in ecologically representative sites in each of the forest reserves and national parks. These would be designated as "research zones". The goal is to establish at least one research zone of 100 hectares in each national park and forest reserve. The research zones will be fenced and/or guarded to protect them from illegal exploitation. Data on species composition, growth, natural succession, and the propagation of indigenous species would be gathered and monitored.

Creation of a New National Park and Additional Forest Reserves

Sociocultural Inventory of Candidate Areas. The SPE staff would carry out the sociocultural inventory of each candidate area using the Rapid Rural Assessment method. The information to be gathered would include: (i) demographics of the population living within and adjacent to the proposed reserve or preserve; (ii) the nature of the use of natural resources; (iii) patterns of access and authority on the site; (iv) land tenure arrangements; (v) ownership of the resources; (vi) the economic benefits accrued by user populations; and (vii) descriptions of local agricultural strategies. The inventory would also identify existing or potential conflicts that might result from management interventions, and would recommend alternative methods that should be used to mitigate or avoid conflict, maintain user welfare, and integrate users and local residents into the plan for the protection of the ecosystem. A Social Science Specialist would assist in these aspects of the sociocultural inventory as a short-term consultant.

Environmental Profiles of Candidate Areas. The purpose of the environmental profile would be to establish baseline environmental information on each area that is being considered as the future national park, or as a forest reserve. Each profile would describe the major biotic and abiotic features, their status, and would recommend measures to preserve biological diversity.

New National Parks. The procedure for establishing the new national park and future forest reserves would be as follows. (i.) The SPE would conduct intensive sociocultural surveys and environmental profiles to determine the suitability of each candidate area. (ii.) The SPE would prepare a document describing and summarizing the information on forest cover, forestry potential, natural resource components, status of important endemic flora and fauna, existing patterns of human occupancy and land use, and land tenure characteristics for each candidate area. The document would include a recommendation as to the best land use strategy for each candidate area (i.e., national park or forest reserve). (iii.) The Director of the DRN would review the recommendations with the SRF and SPE and decide which recommendations would be implemented under the project. The SRF would manage areas designated as forest reserves. The SPE would manage all national parks. (iv.) The designation of the area as an official national park would occur after the MARNDR prepares a draft Decree for consideration by the Council of Ministers. The Decree would identify which sites would be transferred into the public domain as national parks or forest reserves under the MARNDR. Following the approval of the Decree by the Council, it would be published in *Le Moniteur* and become official. (v.) The SPE would arrange for the delimitation of the designated national park area, and the legal transfer of the surveyed area under the jurisdiction of MARNDR.

Management of National Parks

Haiti's existing natural national parks and forest reserves are: Parc National Pic Macaya (5,500 hectares), Parc National la Visite (3,000 hectares) and the Pine Forest Reserve (27,871 hectares). The SPE would manage the two national parks (which are "preserves"), while the management of the Pine Forest Reserve (which is a multi-use "reserve") would remain under the SRF. The following activities would be implemented by the SPE under the project for each of the two existing national parks, and for the one additional national park that would be established.

Delimitation of National Parks. The SPE would contract the Direction Generale des Impots (DGI) for a survey to delimit each national park with funding provided under the project. A technician from the SPE would accompany the DGI survey team to advise the DGI as to where the boundary should be located relative to existing natural and man made features of the area (i.e., avoid problems of including existing farms or excluding ecologically important features). The SPE would offer to provide the DGI survey team with tents, camping equipment, and logistical support while the team is in the field. The SRF would be responsible for making arrangements to survey and delineate forest reserves.

Demarcation of National Parks. SPE would work in teams, under the supervision of the park supervisor, to place concrete monuments and metallic signs at regular intervals along the perimeter of each national park. Signs would be posted at the junctions of major trails and access roads. Three guard houses would be built in each national park to shelter the park rangers, and to demonstrate the presence and authority of the SPE in the management of the parks. A total of 18 km of fences would be erected in vulnerable or exposed areas to protect core and research zones. Part of the process of demarcation would also be the establishment of clearly marked land use zones where management activities will be undertaken by the SPE. The boundaries of zones would be marked by paint on trees or wooden posts.

Supervision of the National Parks. For each park, an agronomist would be assigned as the Park Supervisor. The Park supervisor would be responsible for all activities in the park, and would also supervise the activities of the NGO Agroforestry Extension project in the area. A forest technician would be appointed as the supervisor of the park rangers. This technician would have received training in ecology, flora and fauna, and park management in order to qualify him to supervise the activities of the rangers.

Protection of the Resources of the National Parks. The SPE would employ park rangers who would be carefully selected, appropriately trained by EMAF, regularly paid, and consistently supervised. These conditions would be similar to those of the forest rangers of SRF. The park rangers would patrol the lands

within the park boundaries, staff check points to control the entrance and exit of vehicles and pedestrian traffic, and encourage compliance with park regulations, laws, and zoning guidelines. Park rangers would undertake other protective services as designated by the supervisors of each national park, and would stay in regular contact with their supervisors via radio-communications. At the completion of the project, a total of 32 park rangers would be employed.

Limitation of Access Roads as a Management Tool. Access to the parks via well-developed and direct roads would be discouraged during the project in order to provide an extra measure of protection for the parks until an adequate protective infrastructure is established. The national parks, especially La Visite, have potential as tourist attractions. However, access roads are necessary before this potential can be realized. The existing road from Port-au-Prince to Morne la Visite provides the most direct access to the park; however, it has been impassible to vehicles since 1988. The SPE has determined that this road should not be rebuilt during the period of the project, and that it is necessary to keep this road closed to vehicles until the park infrastructure is well established. Otherwise, irreparable harm could be done by illegal wood harvesters and transient cultivators who make seasonal gardens. The SPE would rebuild the access road from the south of the park at La Visite because access via the town of Seguin can be controlled. It would rebuild the service roads through the park and to Unit II of the Pine Forest Reserve at Morne Rouge. The SPE would not build a major access road into the north side of Parc National Pic Macaya because of the same concerns. The strategy is to develop the infrastructure of each national park before developing easy access to the parks.

Management of "Occupied" Lands within Park Boundaries. One of the major problems faced by the SPE is the presence of farms within the boundaries of the national parks. Some of the problems created by the presence of farms in the parks are: (i) farmers expand existing gardens into new areas; (ii) livestock are grazed in inappropriate areas; (iii) slash and burn garden plots developed in remote areas; and (iv) the introduction of dogs, cats, rats, and mongooses, all of which follow humans into a region and do great harm to endemic animals. These problems are most significant at the Parc National La Visite, where farmers living within the park were given a letter from MARNDR in 1983 notifying them that they were to leave the park by 1985. Farmers in the park have not paid leases on the land they occupy since that date. The relocation of these farmers was not carried out because they were notified that they would receive land outside the park in exchange for the land they currently occupied. Under the project, the SPE would resolve this problem as follows. Leases would not be collected from any farmers who have already received a relocation notice. The land within the boundaries of each park would be zoned for proper land use at the earliest possible date. Some lands within the boundaries of the parks would be determined to be suitable for agriculture, and would be zoned as "agricultural zones" or "multiple use zones". Peasants who live and farm in these areas would be allowed to remain there. In areas within the national parks which are zoned for preservation (i.e., "core zones" and "research zones"), peasants would receive a letter similar to the one that was used by MARNDR at Parc National la Visite notifying them that they would be relocated. In the special case where a farmer living within the boundaries of Parc National La Visite received an eviction notice in 1983, but the land he occupies is zoned for agriculture by the SPE under this project, then the eviction notice would be voided and a new lease agreement would be written using the above guidelines.

Relocation of Peasants from Core Zones. It is important that the letter of eviction be accompanied by a detailed proposal for compensation for the loss of the land that is currently occupied. The SPE favors paying the peasants a one-time payment for "damages", thus providing them with the means to lease lands at another location. In order to prevent the public from perceiving that they can gain land and money by occupying national park lands, payments will only be made to farmers who can show documentation that they were holders of leases for the land, or occupied the site for at least five years.

Management of Leases. In areas of the national parks zoned for agriculture, agroforestry, or multiple use, land leases would be used by the SPE as a management tool. Leases would be written by the SPE with the assistance of legal counsel, and would define which land use activities are allowed and which not. Leases would initially be for periods of one year, and would not commit the SPE to allowing the peasants to lease the land indefinitely. If a farmer is found to occupy land in an area that is zoned for agriculture, agroforestry, or multiple use activities, and if the Park Supervisor feels that the presence of the farmer does not compromise the management plan for the park, a long-term lease would be established. Long-term leases would encourage farmers to invest in their holdings and take advantage of the extension

services provided by the project. These long-term leases would be for ten years, but would give the SPE the authority to cancel the lease annually if the land use conditions of the lease were violated by the farmer. The supervisor of each park would be responsible for the administration of the lease program in that park.

Lease Fees. There are no legal guidelines within MARNDR as to where money collected from leases on MARNDR land should be deposited. Revenues from fines and the sale of forest contraband are deposited in the Fund for Reforestation (1972 law) that is being administered by the SRF. The fees collected from leases and from fines in the national parks would also be deposited in the Reforestation Fund. These funds would be used in accordance with existing law, and would be handled in the same way that leases are to be handled in the Pine Forest Reserve.

Supervision of the Agroforestry Projects Surrounding Each National Park. The DRN would contract with NGOs to provide agroforestry extension in the national parks and in the buffer zones surrounding the national parks at La Visite and Pic Macaya. In the case of Pic Macaya, extension activities would have been initiated in the area north of the park, since the UF-MBR Project was already working in a similar capacity in the buffer zone south of the park. This part of the World Bank Plan would have to be amended since the University of Florida USAID-sponsored project ended on May 31, 1992. The objective of the Agroforestry Extension contract is to provide agricultural extension services to peasants living near the park to reduce the danger of environmental degradation within the park, and to increase local standards of living. The NGOs would also operate agricultural extension services to design recommendations for agroforestry and soil conservation techniques that would be used to establish conditions for leases, and to verify that conservation treatments on leased lands have taken place. The SPE would supervise the agroforestry extension activities through the Supervisor of each national park (resident agronomist).

Biosphere Reserve Plan. The SPE would develop an extended management plan for both the park and buffer zone as the first step in expanding the UF-MBR Management Plan to the north slope of Pic Macaya. In developing this plan, the SPE would modify the management plan developed by the University of Florida under a contract with USAID for the national park at Pic Macaya to include: (i) information on biosphere reserve land use zones; (ii) activities that would be appropriate for the surrounding buffer zone; and (iii) the recommendations contained in the existing UF-MBR Management Plan.

Identification of the New National Park and Forest Reserves

The SPE would identify approximately 30,000 hectares that would be suitable as candidate areas for the new national park to be established in the fourth year of the project, and as candidate areas for future forest reserves. The method of selection and establishment of these areas is described in detail above. The major ecosystems to be investigated are as follows.

- a). **Mangroves:** Because of their inter-tidal location, mangrove ecosystems play an essential role in the reproductive cycles of numerous aquatic species, and provide habitat for economically important species of crustaceans. In addition, threatened species such as the Greater Flamingo and West Indian manatee occur in bays and lagoons associated with mangrove habitats. Although a majority of the original mangrove stands have been eliminated, forests remain in the bays of Acul and Caracol (near Cap-Haitian), the Cayes region (Aquin Bay, Ile a Vache), the north coast of the southern peninsula (Baraderes and Corail). The most substantial remaining forest is in the Artibonite River Delta (south of Gonaives and west of l'Estere). Though mangrove forests are within the public domain, rights of access and harvest are often allocated among private individuals. Preliminary studies indicate that the Caracol and Artibonite areas are the best candidates for protection because they have the largest concentrations of threatened and endangered species. Preliminary plans by the SPE indicate that this ecosystem has the highest priority as a candidate for the new national park.
- b). **Broadleaved Forest:** Though much of Haiti was originally covered with this forest type, most of the Broadleaved stands have been converted to coffee plantations and multi-purpose agricultural plots. Important Broadleaved forests remain south of the town of Borgne, on the Morne Brigand and Boucan Michel and west to the headwaters of the Anse-a-Foleur watershed and in the Pic Macaya area. Some of these are on state land, especially west of Pic Macaya, and could be attached to the Macaya Biosphere Reserve as part of Parc National Pic Macaya.

- c). **Semi-arid Forest:** This forest community, which is characterized by the dominance of *Prosopis*, has historically been the source of most of the charcoal produced in Haiti, and vast areas of semi-arid forest have become degraded through over-harvesting. Remnants of natural stands remain on the edges of the Etang Saumtre (in the Cul-de-Sac plain), Grand Gosier (southeast Haiti) and near Cotes de Fer (southern coast of Haiti), and on the western end of Ile de la Gonave. All of these lands are within the public domain.
- d). **Arid-land scrub:** This ecosystem is characterized by cactus, succulents and thorny shrubs. The zone which most characterizes this vegetation type is found along the coast between the towns of Anse Rouge and Gonaives (Savane Desolee). The cactus and scrub resources are rapidly being depleted for charcoal and fuel uses. This land is apparently within the public domain.

Preservation of Endangered Species

The preservation of biological diversity is difficult in Haiti because of the extent of deforestation that has already occurred, and special efforts would need to be undertaken to preserve some of the country's most endangered plant and animal species. SPE staff would propagate and grow endangered species of the lowland endemic plants such as *Attalea crassipatha*, *Geonoma oxicarpa*, and *Zombia antillarum* at the MARNDR owned Levy Farm near Camp Perrin. A similar program for upland species such as *Juniperus eckmanii* would be undertaken in conjunction with SRF at the nursery in the Pine Forest Reserve. The SPE would develop special programs to improve the habitat for threatened species by including these activities in the management plan for the existing national parks (for example providing nest boxes for birds who normally nest in cavities in mature trees).

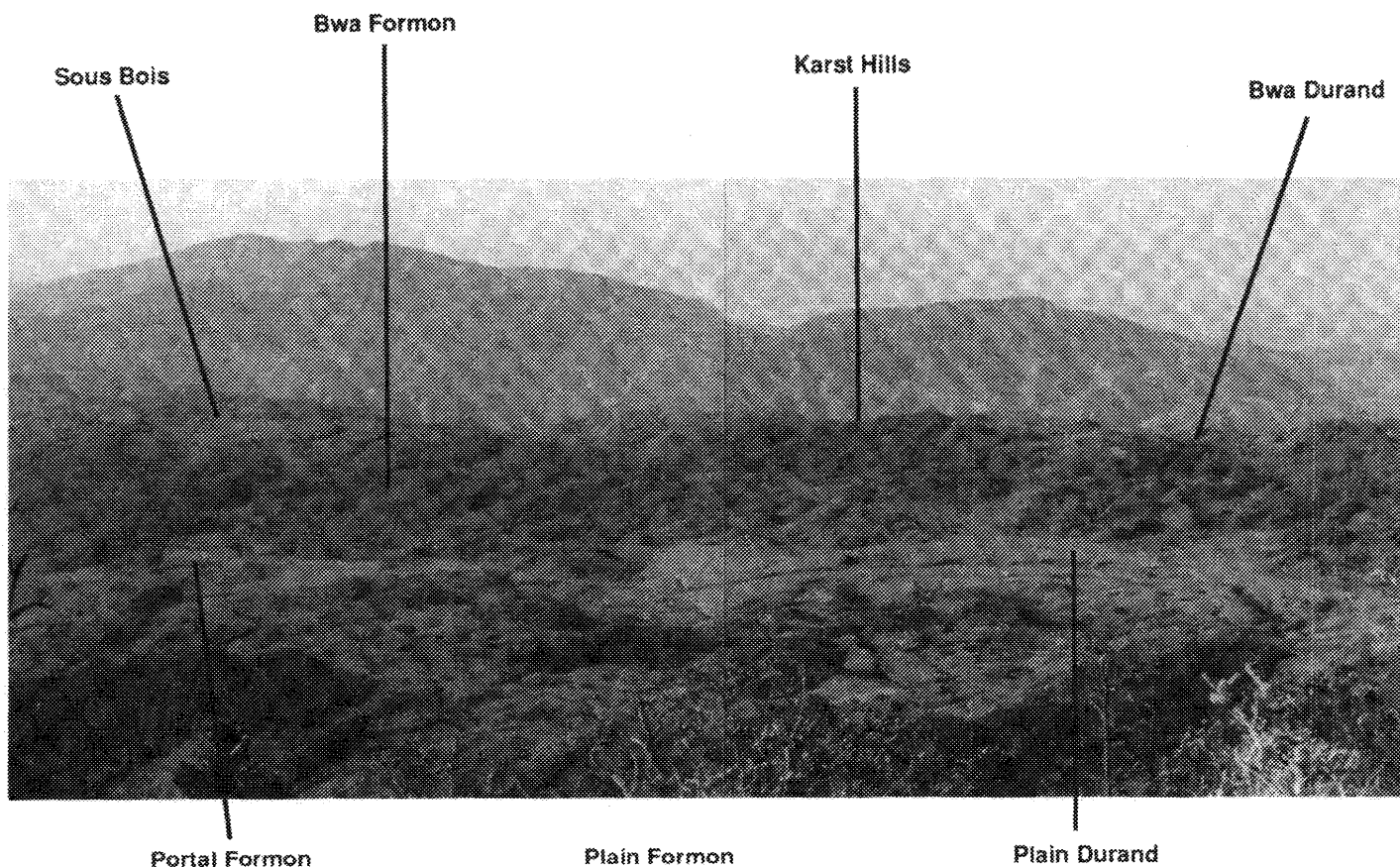
The SPE would also protect endangered species and preserve biological diversity by: (i) developing "recovery plans" for endangered species and species of special concern, and include the activities and recommendations contained in these recovery plans in the management plans for each national park and forest reserve; (ii) administering current wildlife regulations and laws by requiring permits and licenses to hunt, collect plants and animals, and export any native species of plant or animal; (iii) drafting additional legislation to protect the native flora, fauna, and threatened habitats of the country; and (iv) seeking international assistance from organizations such as the International Office of the Nature Conservancy, or Wildlife International, to purchase special parcels of threatened ecosystems which serve as habitat for rare, threatened, or endangered plants and animals.

Phasing of the SPE Project

One of the requirements of the project is to have a sufficient number of agroforestry technicians on the staff of the SPE to carry out objectives of the project. Since there are not enough agroforestry technicians currently available in Haiti to staff the project, agroforestry technicians will need to be trained at the EMAF. The SPE immediately need to hire two agroforestry technicians during the first year of the project to work on the inventory team. These technicians are currently available. There are not enough agroforestry technicians available to be assigned to the existing national parks and to Levy Farm, since the first graduating class from EMAF will not be available until the third year of the project. At that time, the SPE will hire eight agroforestry technicians (La Visite = 2; Macaya = 4; Levy Farm = 1; Inventory Team = 1). In the fourth year of the project, an additional national park will be established by the SPE, and two additional agroforestry technicians will be hired (for a total of 12 for the SPE in the project). During the first two years of the project, while the agroforestry technicians are being trained by EMAF, the agronomists assigned to each national park and to Levy Farm will serve in an expanded role that will include supervising the park rangers and plant propagators. The expanded duties of the supervisors would also provide the opportunity to increase their experience at all levels in the operation of national parks, and increase their abilities to supervise and manage their respective parks.

Other phasing that will occur in the project are as follows. All activities at Parc National La Visite will begin during the first year of the project, but only half the guards will be hired at Parc Macaya. In year two of the project, all of the guards will be hired at Parc Macaya. In year three of the project, all of the agroforestry technicians will be hired for the existing parks, and the staff of each park and Levy Farm will be operating at full staff levels. In year four of the project, one additional national park will be added to the SPE system of protected areas, and a full staff will be hired. The agronomist that will supervise this

park will have been employed by the SPE since the first year of the project, and will have previously been assigned to the Levy Farm. This person will begin duties as supervisor of the new national park with three years of SPE experience, and a knowledge of management techniques based on previous experience with the national parks program.



MBR Project Area

MacArthur Foundation Grant "Bridge"

As part of the original "View of the Future" of Haiti, another grant was put in place to help make the World Bank Plan work more effectively. This grant was awarded to Charles Woods to help support conservation activities in Haiti. The original goal of the MacArthur Foundation Haiti Conservation Project was to help "Bridge" the ongoing USAID, UFMBR, and MARNDR programs. However, all of these programs have now been suspended or terminated. Therefore, the MacArthur Foundation Conservation Project has become even more of a "Bridge". It is a bridge across the environmental disaster that is occurring in Haiti as a result of the current international embargo and difficult economic conditions that now face the country.

The current goal of the MacArthur Foundation Haiti Conservation Project is to help "Bridge" the gap between the existing UFMBR Project, which closed down on May 31, 1992, and a time in the future when new conservation projects emerge--when the current political and economic difficulties in Haiti are resolved, and the international embargo is lifted. The activities of the UFMBR project are summarized in the three volumes of the MBR Project Final Report: *Final Project Report of the University of Florida Macaya Biosphere Reserve Project*; *The Natural History of Southern Haiti*; and *Stewardship Plan for National Parks and Natural Areas of Haiti*.

The recent political events in Haiti have thrown efforts to protect the natural patrimony and natural resources of the country into turmoil. The infrastructure that supported the National Parks of Haiti has collapsed. The World Bank Environmental and Forestry Project has been suspended. Local farmers in the countryside are degrading the natural resources of the region at an increased (almost frantic) rate. Because the World Bank Project has been put on hold, MARNDR does not have the resources to continue to protect Parc National Pic Macaya. A "Bridge" is desperately needed that will help preserve what remains of Haiti's natural patrimony in the Macaya area, and that will make it possible for the pieces to be picked up when conditions in Haiti become normal once again. The major activities of the MacArthur Foundation Project are summarized below. These activities will change as conditions in Haiti change. The project is planned to run from August, 1991 through August, 1994.

Summary of Proposed MacArthur Foundation Activities

- 1). Continue to support the Parc National Pic Macaya and the Planned Macaya Biosphere Reserve "infrastructure" (buildings/nursery) at Formon/Durand until there is a long-term GOH or PVO program in the area. Jean-Rony Merisier will visit Formon on a weekly basis to continue to demonstrate that an official GOH project is in place, and to carry out educational and administrative activities for the MacArthur Conservation Project.
 - a). Guardians will continue to be paid (PADF will assume responsibility for half of the Guardians for mutually shared facilities);
 - b). A nurseryman will continue to be employed (PVO will assume full responsibility for this position and for all reforestation activities when it becomes active in the area);
 - c). Some trees will continue to be propagated (PVO will take over this responsibility as part of its project);
 - d). Signs will be checked and secured. The signs will be the responsibility of UF/MacArthur.
- 2). Continue some conservation activities in the Formon area, and along the south side of the Pic Macaya region (old PST area). PVO will carry out these activities in the MBR area, and UF/MacArthur will have the responsibility for the area of Parc National Pic Macaya. Jean-Rony Merisier will visit the Formon region on a regular basis, and will coordinate all activities there between Charles Woods, Florence Sergile, PADF, and MARNDR.
 - a). Some reforestation activities will continue to be carried out in critical areas where erosion is a major factor (to be assigned to PVO).

- b). Check dams will be maintained, and additional check dams will be built at critical areas (PVO in MBR, UF/MacArthur in Parc National Pic Macaya).
 - c). Additional signs will be placed in areas of special concern (UF/MacArthur).
- 3). Biological Conservation activities will continue in the area of Pic Macaya, the Rak Bwa Broadleaved Forest near Formon/Durand, and the hill country between Catiche and Pic Macaya. Charles Woods will design all of these activities, and will train Florence Sergile and Jean-Rony Merisier to help carry them out. Some of these activities will include the assistance of Dr. Jose Ottenwalder, Caribbean conservation specialist. These activities include:
- a). Writing Species Recovery Plans for all living species of Solenodon ("living fossils"). This activity is very important since these creatures are the rarest native mammals of Haiti, and perhaps the entire West Indies;
 - b). Writing a Species Recovery Plan for the White-winged Warbler, a very rare ground-nesting warbler that is endemic to Hispaniola, and may survive only in the Pic Macaya area;
 - c). Writing a Species Recovery Plan for the Black-capped Petrel. One of the last surviving colonies of this nearly extinct bird is on the south face of Pic Macaya near the summit.
 - d). Writing a Species Recovery Plan for the Zagouti. This species is now very endangered in the area of MBR, and is extinct in northern Haiti;
 - e). Beginning a long term bird-banding project in the Rak Bwa Broadleaved Forest near the project headquarters at Formon. This sub-project is seen as a way of involving local Haitian volunteers and PVO groups in a plan for natural resource monitoring, and is a vehicle for training Haitians in the principles of wildlife conservation. It is also a way of establishing a program that could attract North American volunteers, and attract the attention of North American organizations such as the Audubon Society. A similar plan has been very successful in Belize, where the Massachusetts Audubon Society has adopted Belize as a major focus of interest and fund raising.
This sub-project will operate under the official Bird-Banding Permit of Charles Woods, and will build on his past bird-banding work in Haiti, so that it is starting from a point of strength and will be using an existing data base on migratory birds.
- 4). Florence Sergile, Charles Woods will be in Haiti on a regular basis to "lobby" for environmental concerns in the Macaya area, and in the areas of the other national parks of Haiti. Jean-Rony Merisier will be in Haiti on a permanent basis, and will design and implement educational programs, and also lobby for environmental issues. The targets of these constant lobbying efforts will be:
- a). MARNDR
 - b). FAN
 - c). Audubon Society of Haiti
 - d). Private individuals with an interest in the environment
 - e). International Agencies working in Haiti (AID, OAS, EEC, UN, World Bank, Canadian Govt., etc.)
 - f). Tourist organizations
 - g). Educational groups, especially the University of Cayes.
- 5). Charles Woods will work with the existing data generated by the project over the last decade to publish a body of literature and educational materials that can be used in Haiti to promote conservation and the wise use of natural resources. These materials will be used in the MacArthur Foundation Environmental Education activities, and will be available to the next generation of individuals, GOH programs, or international organizations that will be working to save the remaining natural resource base of Haiti (and to prevent the total economic and ecological collapse of the country). Some of these activities include:

Posters

- a). Distributing the Conservation Poster on Haiti, "Connaitre et Proteger la Richesse Naturelle d'Haiti," that was produced as part of the present USAID-sponsored project (MBRP) with financial assistance from the MacArthur Foundation;

- b). Creating a second Conservation Poster that will be a "companion" poster to the above two posters, and which can be used in schools and workshops for educational purposes. This poster will include text and clearly labeled messages that will allow for discussions and learning sessions (now being planned and scripted);
- c). Creating a third Conservation Poster that will concentrate on only the Macaya Area, and rely on photographs and text (now underway);
- d). Working with well-known Haitian artist Andre Normil to create a "Haitian Painting" that will be turned into an attractive poster. This poster will feature a mixture of real Haitian plants and animals mixed together with the traditional non-Haitian plants and animals featured in most Haitian art. This poster will be widely distributed in Haiti, especially to tourist facilities. We hope that it will create an interest in native plants and animals. If we are lucky, it will also start a movement in Haiti where local artists will begin to increasingly include local plants and animals in their paintings.

Educational Books and Booklets

- a). A booklet with color photos and important environmental information on the National Parks of Haiti (in French).
- b). A book on propagating native trees and shrubs of MBR area in Haiti. This is a follow-up to the book, *Arbres et Arbustes de Macaya*.
- c). A book on the *Mammals of Haiti* (living and extinct), that will include substantial information on wildlife conservation (English, but with a long French abstract, and French and Creole names)
- d). A book on the *Birds of Haiti*, with many photos and color plates. This book will be suitable as a field guide by local people and tourists who want to visit the parks. It will include a substantial section on conservation of birds and habitats;
- e). A book on the *Natural History of Haiti*. This book will be an expanded version of the book that is being published as part of the final report of the MBR project (*The Natural History of Southern Haiti*). Both books can serve as source books on all aspects of the "Natural History" of Haiti;
- f). Other books and booklets that are determined to be appropriate and important, including a possible book on the *Useful Trees of Haiti*.
- g). An educational coloring book in conjunction with the "Connaitre et Proteger la Richesse Naturelle d'Haiti" conservation poster.

Conservation Video

- a). A conservation video, using VCR technology, will be created that will show scenes of natural areas in Haiti, including views of the natural forest in the most remote sections of the parks, and scenes of the parks from the air. We will have copies of this video with French, Creole, and English sound tracks. We hope that it will be shown on Haitian TV, and other places where attitudes towards the natural environments and natural resources of Haiti can be improved.

The production of all of these posters, books, and booklets will be done out of the office of Charles Woods, and using the facilities of the Florida Museum of Natural History in Gainesville. The computers and existing MBR data bases at the Museum/University will be used in the preparation of these materials.

- 6). The project will work with MARNDR to try to protect the park. The project does not have enough resources to support activities that are not very specific to the goals of the MacArthur Project, which are to
 - a). Protect the existing national parks.
 - b). Educate people about the importance of natural resources in Haiti.

- c). Create species recovery plans, and a useful published literature on natural resources. These will improve the chances of success for future programs in natural resource management in Haiti.
- d). "Bridge" the gap between the existing USAID/MBR program, and possible future programs (including the important World Bank Environmental and Forestry Project).

Personnel Supported by the MacArthur Foundation Grant

Charles Woods

Project Director (at no cost to the project, since all salaries and benefits are paid by the University of Florida). Most of his time will be spent at the Museum coordinating the scientific, educational, administrative parts of the project and planning and executing the conservation posters and environmental books, but he will work in Haiti on a regular basis, and will conduct all conservation biology activities.

Florence Sergile

Technical/Administrative Coordinator. Beginning June 1, 1992, her salary and benefits will be paid by the MacArthur Foundation Grant (until another outside source of funds is found). This additional cost was unplanned for, and will stretch the activities of the project to the limit. However, her position is an important one, and her training in Agriculture (BS at Damien), Biosphere Reserves (MS at UF), and Natural Resource Management (on the job training with the UFMBR project for five years) make her invaluable. She is a native of Haiti, speaks French, Creole, English, and Spanish perfectly, and knows the infrastructure of Haiti. During the course of this project, Florence will be trained in the principles of Conservation Biology and Environmental Education. Florence will spend half her time in Haiti, and half her time at the Florida Museum of Natural History. It will be Florence's responsibility to seek additional funding to support activities of the project. She will work with a variety of funding agencies, PVOs, and private individuals in Haiti.

Jean-Rony Merisier

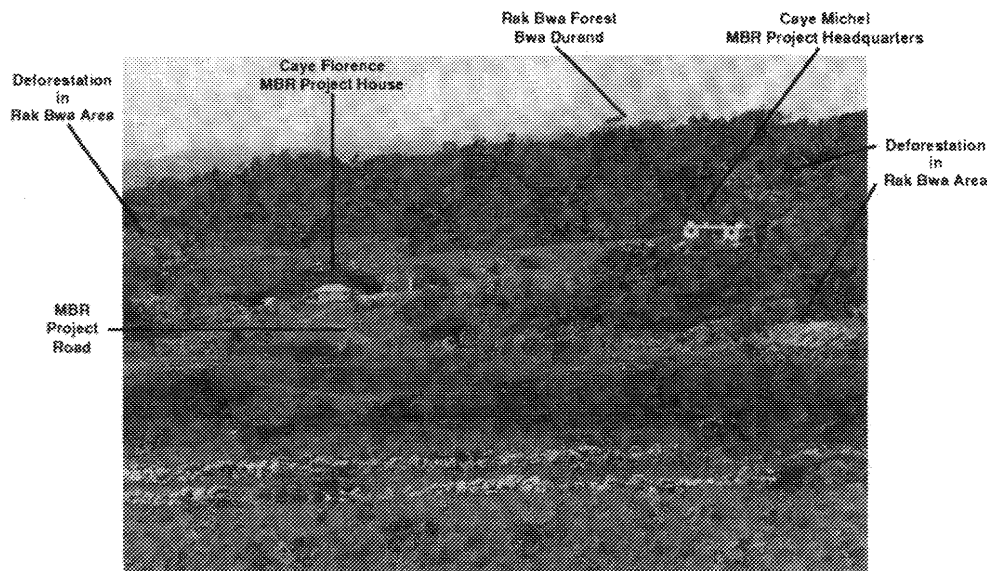
Education and Conservation Specialist. Beginning June first, his salary and benefits will be paid by the MacArthur Foundation Project. Mr. Merisier is also a native of Haiti, speaks French, Creole, and English well, and knows the infrastructure of Haiti. He has served as a teacher, project administrator, and "Agroformateur". For the past two years, he served as an Agronomist for the MBR project. Mr. Merisier's duties will be to implement the educational program, and to work with GOH and private organizations in Haiti to help bridge between the present poorly-defined program and the future programs in conservation and natural resource management (as well as the emerging program in national parks). One of our goals in the MacArthur Foundation Project are to train Mr. Merisier to take a major role in future work in conservation in Haiti. Mr. Merisier visit the area of the project in Formon on a regular basis. During the important transitional period through the summer of 1992, he will be in Formon each week.

Local Guardians and a Nurseryman at Formon. These individuals will be paid for initially by the MacArthur Foundation Grant, and then shared jointly and paid jointly by PADF after their project is activated.

Guardian at the Project Headquarters at Les Cayes (Paid for by MacArthur Foundation Grant).

Chauffeur to be used as needed (Paid for by MacArthur Foundation Grant).

Volunteer Field Biologist to work on site at Formon/Macaya (basic living expenses to be paid for by MacArthur Foundation Grant).



View of MBR Project Area
April 1992

Facilities to be utilized by MacArthur Foundation Grant

- 1). The Project Headquarters in Les Cayes. This facility will serve as the:
 - Project Office
 - Project Depot
 - Project Conference Room
 - Project Residence Facility
 - Vehicle Maintenance and Storage Facility
- 2). The Small White House at Formon/Durand. This project house will serve as the field residence for Merisier, Sergile, Woods, etc. when they are in the field at Formon. It will also serve as the field depot for signs, equipment, materials and literature for the MacArthur Foundation Conservation Project.
- 3). The Large House at Formon/Durand (Kay Michel). This building will be shared with PVO. It was designed by the UFMBR Project as the Headquarters Building for the Parc National Pic Macaya. If the World Bank Project becomes active, we will coordinate activities in the building so that there is a place for SPE/World Bank/MARNDR/Parcs Haiti or whomever else is taking a major role in the administration and supervision of Parc Macaya. There is sufficient space for PVO to use it as an office and depot. In general, the facility at Kay Michel can serve as:
 - Office Space
 - Field Depot
 - Center for Field Research (Species Recovery Plans)
 - Field Residence and Meeting Facility for Volunteers and Project Related Staff, as well as PVO staff.
- 4). Office, publication, educational, and research facilities of Charles Woods at the Florida Museum of Natural History/University of Florida, Gainesville, Florida. This facility will be the overall center of administration/coordination for the MacArthur Foundation Conservation Project.

Phone: 904-392-1721;
Fax: 904-392-8508.

Financial

The funds for the grant are managed by the University of Florida Research Foundation (Private). Funds for the grant are disbursed via checks from this Foundation account. There is also a bank account in Haiti in Gords at the Bank of Boston in Port-au-Prince in the name of "Dr. Charles A. Woods Conservation Fund". This bank account is used only for financial matters relating to the MacArthur Foundation Conservation Project.

Address of Project

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Literature Produced in Association with, or as a Result of Data Collected During the Course of the University of Florida Work in the MBR Project Area

Anonymous. 1990. Balancing Agriculture and conservation: the Macaya Biosphere Reserve. *Focus*, (January/February): 1-2, 4-5.

Dod, D.D. 1984. "Massif de la Hotte, Isla Peculiar: orquideas nuevas iluminan su historia.," *Moscoso* 3:91-99.

Dod, D.D. 1984. "Orquideas nuevas para la Espanola y otras notas. V." *Moscoso* 3:100-120.

Dod, D.D. 1986. "Orquideas (Orchidaceae) nuevas para la Espanola y otras notas. VI." *Moscoso* 4:188-199.

Dod, D. D. and W. Judd. 1986. Orchidaceae of the La Visite and Macaya National Parks, Haiti. *Grant report prepared for USAID/Haiti under contract no. 521-0169-C-00-3083-00*. Gainesville, FL.

Franz, R. and D. Cordier 1986. The herpetofaunas of the proposed National Parks in southern Haiti. *Unpublished report*. U.S. Agency for International Development, Port-au-Prince, Haiti.

Gali, F. and A. Schwartz 1986. The butterflies (Lepidoptera: Rhopalocera) of Morne La Visite and Pic Macaya. *Unpublished report U.S. agency for International Development*. Unpublished report, Port-au-Prince, Haiti.

Higuera-Gundy, A. 1989. Recent vegetation changes in Southern Haiti. In *Biogeography of the West Indies: Past, Present and Future*. Edited By Charles A. Woods, Acquisitions and Production Editor J. Ross H. Arnett, 191-200. 896 pp. Gainesville, Florida: Sandhill Crane Press, Inc.

Abstract: Pollen and chemical evidence from analysis of sediments in the lac Miragoane indicates minimal disturbance of local vegetation at pre-Columbian period but deforestation and erosion since European arrival.

Judd, W. S. 1986. Floristic study of la Visite and Macaya National Parks, Haiti. U.S. Agency for International Development/Haiti. Port- au-Prince, Haiti.

Judd, W. S. 1986. "First report on *Micropholis polita* (Sapotaceae) and *Hamelia ventricosa* (Rubiaceae) from Hispaniola." *Moscoso* 4:222-225.

- Judd, W. S. 1986. "A new species of *Wallenia* (Myrsinaceae) from Haiti." *Sida* 11(3):329-333.
- Judd, W.S. and J.D. Skee, Jr. 1987. "Two new species of *Meriania* from the Massif de la Hotte, Haiti." *Systematic Botany* 12:374-380.
- Judd, W.S., J.D. Skee, Jr, and C.K. McMullen. 1990. The flora of Macaya Biosphere Reserve: Additional taxa, taxonomic and nomenclatural changes. *Moscossa* 6:124-133.
- Abstract:** Floristic and vegetational survey documents high degree of endemism. Description of the plants communities. Discovery of several new angiosperms. A total of 470 species are listed by families.
- Judd, W. S. 1987. Floristic Study of Morne la Visite and Pic Macaya National Parks, Haiti. *Bulletin of the Florida State Museum Biological Sciences* 32 (1): 1-136.
- Judd, W. S. and J. Skee, Jr. 1987. Three new angiosperms from Parc National Pic Macaya, Massif de la Hotte, Haiti. *Bulletin of the Florida State Museum, Biological Sciences* 32 (1): 137-150.
- Monaghan, P. September 1990. Macaya Biosphere Reserve Project. *An evaluation of the UF/MBR Primary Project Cooperators' program: a report on fieldwork with program proposals*. International Programs, or 258 Museum, University of Florida.
- McFadden, B. J. 1986. Geological setting of Macaya and La Visite National Parks, southern peninsula of Haiti. *Unpublished report*. U.S. Agency for International Development, Port-au-Prince, Haiti.
- Paryski, P., C. A. Woods, F. Sergile 1989. Conservation Strategies and Preservation of Biological Diversity in Haiti. In *Biogeography of the West Indies: Past, Present and Future*. Edited By Charles A. Woods, Acquisitions and Production Editor J. Ross H. Arnett, 855-878. 878 pp. Gainesville, Florida: Sandhill Crane Press, Inc.

Abstract: Haiti is one of the most environmentally degraded countries in the world, and faces serious economic and social problems. Haiti is also one of the most biologically significant countries of the West Indies. Many endemic plants and animals, some of which are confined to Haiti, occur on the island of Hispaniola. Developing conservation strategies in Haiti, therefore, is both challenging and important. The most effective conservation strategy in a country with so many socio-economic problems, and with so much habitat already seriously degraded, is to link conservation with issues of high national priority and sustainable economic development.

In Haiti conservation has advanced in the past decade in spite of many other problems by associating conservation with major environmental issues such as watershed management and soil conservation. Two significant national parks have been established with the continued support of the government partly because increasing numbers of people in Haiti have become aware of the relationship of forested mountain regions to water resources in surrounding areas. National parks and conservation have also been justified in Haiti because of the tourist value of its flora and fauna, and an effort is now underway to link historical patrimony with natural patrimony. The proof that the association of conservation with watershed management, sustainable economic development, and historical patrimony is effective is demonstrated by the establishment of new national parks at a time when economic and political problems have dominated the country. As environmental degradation and species extinction continue at alarming rates throughout the world, Haiti can serve as an example of the possibility for the development of effective conservation strategies in the face of difficult socio-economic hardships.

Skean, J. D. Jr. 1989. Rediscovery of *Ekmaniocharis* (Melastomataceae). *Moscovia* 5: 194-201.

Abstract: *Ekmaniocharis crassinervis* Urban, a rare species in a monotypic genus endemic to the Massif de la Hotte of Haiti, has been collected for the first time since the two collections of Erik Ekman in 1917 and 1926. A detailed description of the species and the localities of collection are presented.

Skean, J. D., Jr. and W.S. Judd. 1986. "A new *Mecranium* (Melastomataceae) from Hispaniola." *Brittonia* 38:230-237.

Smith, S., P. Monaghan and F. Sergile 1991. Haitian improvement. *Focus*. (November/December):5.

Swisher, M. E. 1990. Viewpoint. *Focus*. (January/February): 1, 3.

Thompson, F. G. 1986. Land mollusks of the proposed National Parks of Haiti. *Unpublished report*. U.S. Agency for International Development, Port-au-Prince, Haiti.

Williams, E. E. 1989. Old problems and new opportunities in West Indian Biogeography. In *Biogeography of the West Indies: Past, Present and Future*. Edited by Charles A. Woods, 1-46. Gainesville, Florida: Sand Hill Crane Press, Inc.

Woods, C. A. 1986. Executive summary of Stewardship Plan for the National Parks of Haiti. *Unpublished report*. USAID/Haiti, Port-au-Prince.

Abstract: Management and conservation plan is proposed for the two official national Parks of Haiti, La Visite and Macaya. Lists of the fauna and flora species are included.

Woods, C. A. 1986. *The mammals of the national Parks of Haiti*. Port-au-Prince. University of Florida, USAID/Haiti.

Abstract: Inventory of the endemic species as well as the introduced species. Proposes recommendations for conservation of endemic species.

Woods, C. A. 1989. The biogeography of the West Indian rodents. In *Biogeography of the West Indies: Past, Present and Future*. Edited By Charles A. Woods, Acquisitions and Production Editor J. Ross H. Arnett, 741-798. 896 pp. Gainesville, Florida: Sandhill Crane Press, Inc.

Abstract: Four families, 18 genera, and 61 species of endemic rodents were widely distributed in the West Indies before the arrival of Amerindians approximately 7,000 yBP. Most forms are now extinct. Except for *Geocapromys ingrahami ingrahami* on East Plana Cay in the Bahamas, at least one species has become extinct on each island where rodents originally occurred. Seventy nine percent of recognized rodent taxa have become extinct since 39,300 yBP. On Hispaniola, 93 percent of the known endemic rodents are extinct, and 100 percent are extinct in the Lesser Antilles. The oldest known endemic rodents of the West Indies are echimyid rodents of the Subfamily Heteropsomyinae, all of which are now extinct. This group appears to have dispersed directly from South America to Puerto Rico or Hispaniola in the late Oligocene or early Miocene, perhaps following the Aves Ridge or Lesser Antilles. Capromyids evolved from echimyids on Hispaniola. Three of the four capromyid subfamilies are confined to Hispaniola. After the differentiation of the capromyid subfamilies on Hispaniola, the Subfamily Caprominae originated on Cuba, where 87 percent of capromyines still

occur. While only two capromine genera are recognized, 75 percent of all capromyid species are in the Subfamily Capromyinae. The species richness of capromyines on Cuba is likely the result of Cuba's geography. It is a large island with isolated mountain areas surrounded by extensive low lying archipelagos. This geography led to a proliferation of species during Pleistocene climate cycles, and has insulated recent capromyines from the pattern of extinction found on other islands. Seventy seven percent of surviving West Indian capromyids are restricted to the Cuban Archipelago. Giant hutias of Jamaica, Hispaniola, Puerto Rico and the St. Martin Bank are here split into two subfamilies, but not assigned to a family. One subfamily (Heptaxodontinae) is convergent with the capromyid Subfamily Plagiodontinae, which may explain the strong resemblance between *Quemisia* and *Plagiodontia*.

Woods, C. A., and L. Harris. 1986. *A stewardship plan for the national parks of Haiti*. USAID/Haiti, Port-au-Prince.

Woods, C. A. and J. Ottenwalder and S.D. [1986]. The birds of Parc National La Visite and Parc National Pic Macaya Haiti. *Birds of the National Parks of Haiti*. Unpublished report USAID/Haiti under Contract Number 521- 0169-C-00-3083-00.

Abstract: Inventory and status of the avifauna of the Haitian national parks. Accounts on earlier explorers and ornithologists and on the different habitats occurring in the regions. Endangered species are annotated with recommendations.

Woods, C. A., and F. Sergile. 1990. Natural Sciences. In *Haiti: A Research Handbook*. Editor R. Lawless, 297-330. 354 pp. New York: Garland Publishing, Inc. Natural Sciences Bibliographic Reference for Haiti.

Abstract: The history of biological investigation of Haiti dates to 1492, and the first voyage of Christopher Columbus to the New World. Many of the early accounts are not in English but in Spanish or in French since Spain and France had established two colonies on the island of Hispaniola. The revolution of Haiti began in 1791, and the subsequent insurrections and the battle that led to the independence of the country in 1804 slowed down the flow of natural history information from the country. From 1917 to 1934 many field studies were carried out by scientists from the United States, especially investigations from the Smithsonian Institution. After the occupation of Haiti by the U.S. Marine few studies by biologists were carried out, especially during the 1960 due to the unrest and political repression of the Duvalier era.

Field research in Haiti during the 1970s and 1980s increased, especially by biologists from the University of Vermont and University of Florida. This era initiated a period of alarm on the part of the Haitian government as the realization of the rate and amount of environmental destruction became documented. Increasing numbers of projects and publications began to focus on the conservation of natural resources. The projects associated with the planning and establishment of national parks revealed the presence of a surprising number of new species of plants and animals, and ushered in a new era of scientific investigation in Haiti.

Woods, C. A., M.E. Swisher, J. Lichte, R. Voss, P. Monaghan, P. Paryski, S. Mathieu, M. Aubry 1989. *The Macaya Biosphere Reserve project: report of observations, conclusions and recommendations*. University of Florida, USAID/Haiti. 104 pp.

Abstract: Contains an overview of the Macaya and adjacent regions Formon, Rivière Trois Sources with illustrations, maps and a page of important dates of the Haitian History. Describes the current status and recent changes in the biogeophysical environment, the physical, socio-economic and demographic factors, the major farming systems in the region. It proposes specific recommendations for land use, short, immediate and long term interventions and a suggestion for the first year activities for each zone and a strategy for the development of community based on sustainable agriculture.

Consultations and reports

Cohen, W. B. 1984. "Environmental degradation in Haiti: and analysis of aerial photography." U.S. Agency for International Development. Unpublished report, , Port-au-Prince, Haiti. University of Florida.

Cloutier, A. A. Dufrense, and P. Warnock 1990. Evaluation of UF/MBRP conducted June 12-22, 1990. Memorandum to P. Paryski, H. Popenoe, M. Swisher, R. Voss, and C. Woods. 20 June 1990.

Abstract: Document the human resources, the successes and immediate needs. Cloutier and Dufrense identify park planning difficulties and make recommendations. The whole team evaluate the institutional and socio-economic aspect. An appendix of evaluation forms and method is included.

Espinal, D. 1989. Crop establishment and management. Data collection and evaluation. Report.

Abstract: Establishment of plantation critical dates data to be collected during plantation and at harvest time are spell out for each crop: carrot, beet, cabbage, oignons in Spanish and in English. A good appendix is also given with forms and drawings.

Florida Museum of Natural History. 1992. Connaitre et Proteger la Richesse Naturelle d'Haiti. Color poster.

Kubitz, Linda, Producer; D. Poucher, Executive Producer. 1990. Video: *Forest in the Clouds*, Institute of Food and Agricultural Sciences, University of Florida.

Abstract: A 27-minute video documenting the successes of the UFMGRP, and demonstrating the need to continue efforts to save Haiti's last remaining rain forest. The program has been aired on Florida Public Television.

Monaghan, P. 5 November 1988. *Three communities in the Macaya Biosphere Reserve: A report on fieldwork in the Southern Mountains of Haiti*. International Programs, or 258 Museum, University of Florida.

Abstract: Three communities, Formon, Cavalier and Trois Sources were visited in summer of 1988. Each was found to be uniquely different in its relationship to the lands within Pic Macaya National Parc. Farmers from Formon (est. pop 1000) and Cavalier (Pop. 2700), were frequent users of the state-owned lands in the Macaya watershed. These two communities have been most affected by the closing of park lands to farming in 1985. They have responded to this with seasonal out- migration and increased pressure on surrounding hillsides. The third community, loosely centered around the Riviere Trois Source, has a permanent population of roughly 500 but the region is also subjected to a large number of seasonal migrants from lower in the watershed. Many of these have only recently been farming in the area and have been brought into the watershed through the COSAR (Cooperative St. Andre de Rendel) coffee cooperative. Recommendations are made to direct further research on the communities and offer several program suggestions.

Monaghan, P. September 1990. *An evaluation of the UF/MBR Primary Project Cooperators' program: a report on fieldwork with program proposals*. International Programs, or 258 Museum, University of Florida.

Abstract: This evaluation provides documentation of UF/MBRP activities that can be used for their effectiveness and used for future project planing. The report proposes new program and recommend correction for new ones. The Primary Project Cooperator (PPC) program should continue with different activities such as vegetable production, pig farming, use of hedgerows to protect eroding hillsides. The size of the population that farms in or near the Macaya Biosphere Reserve is currently less than 500 households, but there are a number of migrants who continue to farm in the Macaya

region and in their home communities in the lower watersheds. Because the population growth trends in the Macaya watershed, the population living closest to the reserve will be rapidly changing in the next decade. For probably the first time, the majority of the current population was born in the highland Macaya region, in contrast to many of the established farmers who migrated from communities lower in the Macaya watershed during the past 30 years. This burgeoning population of young people (in early twenties and younger) is now starting families, building houses, and working lands still owned by their parents. At the same time, the established patterns of region in-migration continue. Inside the boundaries of the Parc National, above the Ravine Kaskou, farmers from the distant communities of La Source, and Troisieme come to plant gardens. DeGlasi, in the isolated Ravine du Sud, is settled by farmers who come from Formon, Camp Perrin and Duchitty. In the western region of the MBR, the greatest population movement originates in the lower Port-!-Piment watershed. Each of these population movements have different characteristics and shape the patterns of exploitation of the region's natural resources. The improvements in transportation into the MBR in the past two years, and declining agricultural production in the lower watershed communities, mean that in-migration and its associated negative environmental effects will increase.

Monaghan, P. 1990. Development policy and environmental degradation in Haiti. Unpublished paper presented at Club Creole Black History Month Talk Series.

McBride, J. 1989. A proposal to study landslide ecology and restoration in Macaya Biosphere Reserve, Haiti: Patterns of Vegetations and soil nutrient recovery. 23 pp.

Abstract: Ecological restoration of vegetation of landslides requires both substrate stabilization and accumulation of soil nutrient pools. Floristic study, comparison of succession on agriculturally induced and naturally occurring landslides, experimental analyses of vegetation and experimental manipulations of successional patterns of vegetation on recent landslide are proposed.

McBride, J. 1990. Draft proposal (Abstract): Soil nutrient influences on acceleration of succession on Landslides in Macaya Biosphere Reserve, Haiti. 25pp

Parafina, S. January 1991. Macaya Biosphere Reserve Project. *Final report: analysis of the project interventions in the Riviere Trois Sources zone and vegetable marketing in the l'Acul riviere*. International Programs, University of Florida, Gainesville, Florida. Available at 3028 McCarty Hall. International Programs, 258 Museum, University of Florida.

Abstract: The majority of the farmers are migrant. The major crop are black beans, maize and sweet potatoes. Trial with vegetables (carrots, beets, potatoes) were made. Farmers of the Trois Sources region have little interest in vegetables which are labor and time intensive system of production. Focus of short term interventions should be concentrated in household gardens. Market price was analyzed according to demand, season to plan horticulture projects.

Paryski, P.E. 1992. Parc National Macaya. Brochure. Sous Presse.

Smith, S. 1992. *Population changes in the Pic Macaya Biosphere Reserve, Haiti: implications for conservation and development projects*. Available at Home Economics, Mc Carthy Hall or 258 Museum, University of Florida.

Abstract: Population growth in Haiti poses a significant problem for the conservation and sustainability of the country's remaining natural resources and for the economic survival of its people. In the Macaya Biosphere Reserve (MBR) area, increasing population pressure could have a devastating affect on the long-term conservation and restoration of fragile forest ecosystems. Fifty households in the MBR villages of Formond and Durand were surveyed in 1988 and 1991 to determine increases or decreases in household size due to reproduction and migration among household member, along with information about their reasons for entering or leaving the residence.

The major findings are 1) the majority of the sample was born in the area; 2) there were no significant changes in household size between 1988 and 1991; 3) men appeared to be less likely than women to have migrated from the MBR area; 4) the majority of the population was concentrated in the younger age groups; 5) proportion of the people that received an education increased between 1988 and 1991; 6) the difference between men or women on educational attainment was negligible in 1991. The findings suggest that the MBR project is faced with very difficult questions about how to conserve remaining resources in an intensively used agricultural area. The results indicate that population changes associated with life cycle factors, gender, and economic opportunities play an important role in natural resource use.

Swisher, M. E., P. Monaghan and C. A. Woods. 1989. *Farming systems and sustainable agriculture in Haiti*. Paper presented at the Farming System Research and Extension annual Symposium.

Abstract: The Macaya Biosphere Reserve must come under the stewardship of the people themselves by improving their life with farming systems methodology. This approach provides farmers with technology that they can utilize in the short to intermediate term, in order to allow them the luxury of becoming long term planners and stewards.

Swisher, M. E., P. Monaghan, and C.A. Woods. 1990. *Farming systems and sustainable agriculture: the Haiti model*. Unpublished report. International Programs, or 258 Museum, @TERMS2 =

Abstract: The Macaya Biosphere Reserve project in its attempt to preserve the national park Macaya proposes sustainable agriculture to improve the standard of living of farmers in the area, who are using the park land to practice slash and burn agriculture. The document analyses the constraints in the social, economic and bio-physical environment, and identifies that achieving sustainable agriculture requires long term investment, changes in agricultural practice, and in the nature of the constraints faced by the farmer rather than concentrating on solving problems within the farm gate.

Woods, C. A. 1989. *Report of Sondeo to Pic Macaya, Pa Lan Kont, Du Glasi, and Bois Formon, Macaya Biosphere Reserve project, Haiti*. University of Florida, USAID/Haiti.

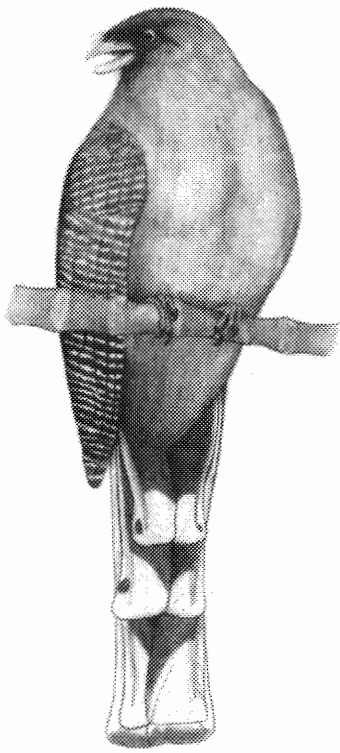
Abstract: This document is an appendix to *The Macaya Biosphere Reserve project: report of observations, conclusions and recommendations*. It traced the itinerary of the team and describes the methodology and the environmental degradation of each region, and proposes a local Guides Association.

Woods, C. A. 1989. Introduction. In *Biogeography of the West Indies: Past, Present and Future*. Edited By Charles A. Woods, Acquisitions and Production Editor J. Ross H. Arnett, i-xvi. 896 pp. Gainesville, Florida: Sandhill Crane Press, Inc.

Abstract: History of biogeography in the West Indies since 1911.

Woods, C. A. and J. Eisemberg 1989. The land mammals of Madagascar and the Greater Antilles: Comparison and analysis. In *Biogeography of the West Indies: Past, Present and Future*. Edited By Charles A. Woods, Acquisitions and Production Editor J. Ross H. Arnett, 799-826. 896 pp. Gainesville, Florida: Sandhill Crane Press, Inc.

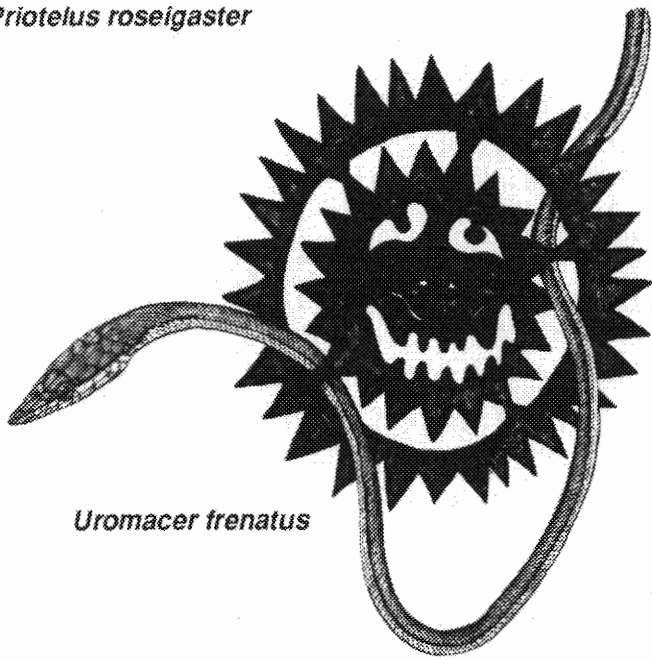
Abstract: Comparison between extinction and remaining mammals of Madagascar and the Greater Antilles. Biogeographical discussion.



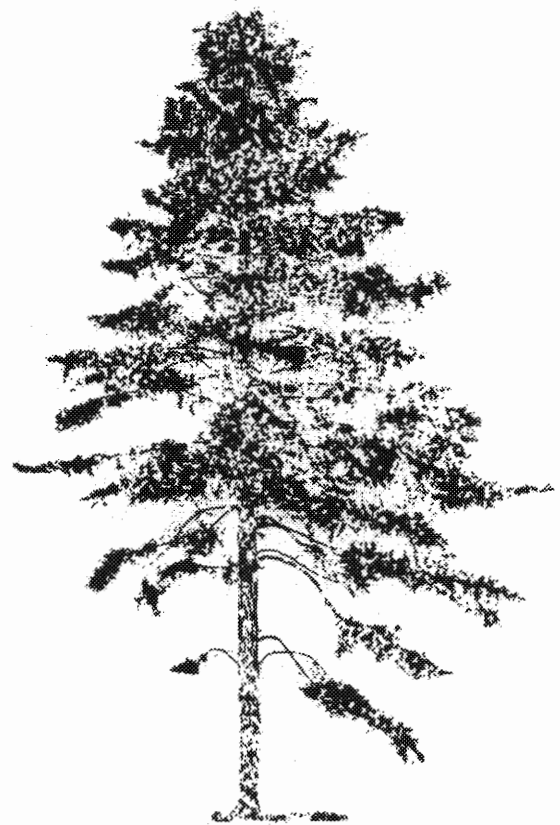
Priotelus roseigaster



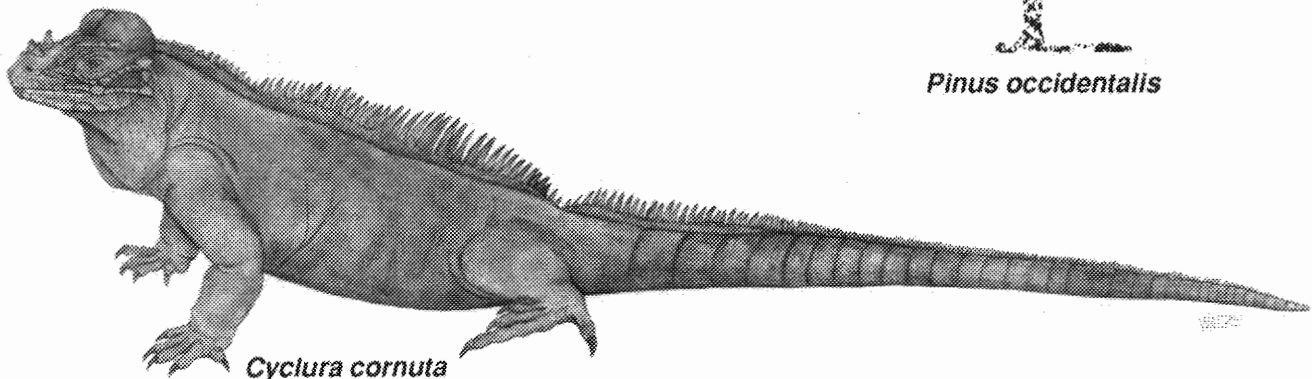
Plagiodontia aedium



Uromacer frenatus

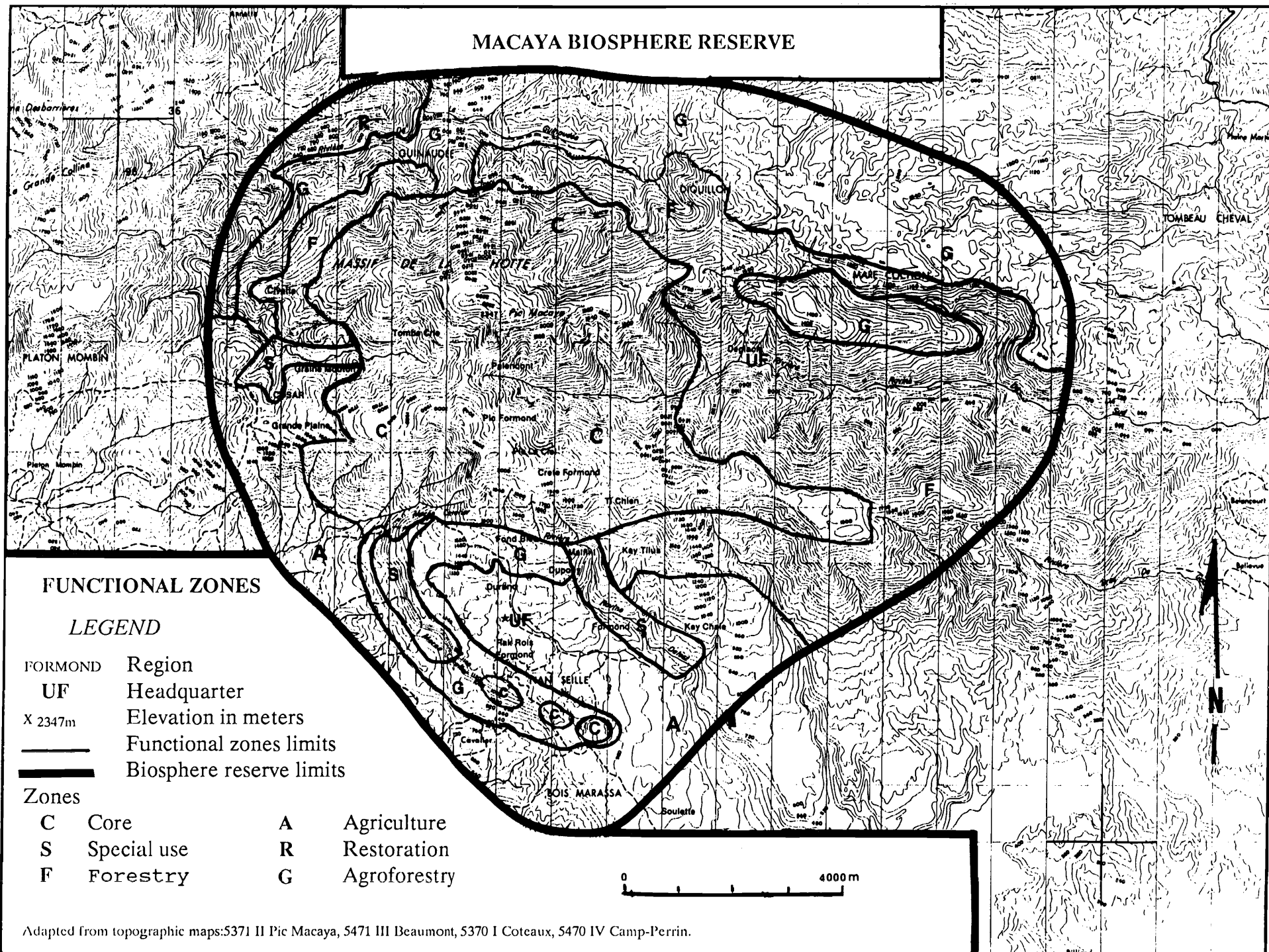


Pinus occidentalis



Cyclura cornuta

MACAYA BIOSPHERE RESERVE



FUNCTIONAL ZONES

LEGEND

- FORMOND Region
 UF Headquarter
 X 2347m Elevation in meters
 — Functional zones limits
 — Biosphere reserve limits

Zones

- | | | | |
|---|-------------|---|--------------|
| C | Core | A | Agriculture |
| S | Special use | R | Restoration |
| F | Forestry | G | Agroforestry |

0 4000 m

Adapted from topographic maps: 5371 II Pic Macaya, 5471 III Beaumont, 5370 I Coteaux, 5470 IV Camp-Perrin.

APPENDIX II

RESUME DES SEMINAIRES

LE REBOISEMENT

I. INTRODUCTION

Historique de la culture traditionnelle. Autrefois le système était basé sur les éclaicies de forêts et le brûlis avec une jachère longue, donc le sol restait fertile. A présent, la tendance est de déboiser, et de brûler pour cultiver, avec une jachère très courte. D'années en années le sol s'appauvrit davantage. Les forêts disparaissent et la production agricole devient de plus en plus faible. S'il faut continuer à déboiser et à brûler le sol pour le cultiver, en peu de temps le pays sera un désert et la vie de tous sera menacée. Faut-il accepter un pareille situation? N'est-il pas urgent d'entreprendre certaines actions pour changer cette situation?

II. IMPORTANCE D'UN ARBRE

Définition. Une plante est un être vivant qui respire, grandit, se développe, se reproduit, vieillit et meurt.

Rôle de la plante. Les plantes sont l'essence de la vie. S'il n'y a pas d'arbres, il n'y a pas de vie. Elles sont importantes dans la production agricole puisqu'elles fournissent des produits de consommation aux hommes et animaux. Les arbres contribuent à la production de bois de chauffe, de bois d'oeuvre et à la protection des sols et des bassins versants.

Les sols et leur importance. Les sols constituent le milieu où poussent les plantes. Les sols riches sont ceux qui peuvent fournir aux plantes l'eau et les éléments nutritifs dont elles ont besoin pour assurer une production et leur perpétuité. Les sols non protégés sont voués à la dégradation par l'érosion. Peu à peu, ils s'ameunissent et s'appauvrissent au détriment de leur propriétaires. Les sols de montagne sont très pauvres parce que les sols de plaines sont très riches. Le dialogue est recommandé pour que les présentateurs perçoivent les impressions de l'audience sur les sols de sa région.

III. REBOISEMENT

Importance du reboisement. Des programmes de reboisement sont établis pour lutter contre le déboisement, l'érosion et permettre la continuité de la vie et des espèces végétales et animales. Ils peuvent être effectués au moyen d'essences forestières ou fruitières. Les programmes de mises en terre des plantules ont lieu pendant la saison pluvieuse si l'arrosage n'est pas possible.

Contraintes au reboisement. Elles sont nombreuses et comprennent une relation plaine montagne-montagne plaine non existante, l'élevage libre, les systèmes culturels, l'accès difficile aux zones dégradées et le manque de moyens financiers.

Pépinière. Les pépinières sont placées à proximité d'un point d'eau. Il faut y inclure un abri et un dépôt, des réserves d'eau, des outils (pelles, pioches, tamis, arrosoirs, sécateurs, machettes, hoes, transplantoirs etc.).

Les semences sont choisies à partir du climat et des besoins d'une zone. Elles sont semées sur plates-bandes, en pots ou en sachets. Une terre de bonne qualité est nécessaire pour assurer un maximum de réussite. Le temps entre la germination et la transplantation varie selon les espèces. L'arrosage et le désherbage sont les principaux soins à apporter aux plantules.

Plantation. La plantation ou mise en terre des plantules peut s'effectuer selon les critères d'utilisation des sols et des besoins de la zone avec des essences forestières et/ou fruitières, ou une association de ces espèces avec des cultures annuelles.

La mise en terre s'effectue au début de la saison pluvieuse. Les plantules seront placées dans des trous un tiers plus grand que la motte de terre enrobant leurs racines. A la trouaison la terre de surface est déposée d'un côté et la terre du fond de l'autre côté du trou. Très souvent il est recommandé d'ajouter un peu de fumier ou de compost bien décomposé au fond du trou. La terre de surface y est remise d'abord et celle du fond sur les côtés de la motte et en surface ensuite. Il est nécessaire de bien fouler le sol pour assurer les contacts entre la nouvelle motte de terre et le sol.

IV. ORGANISATION LOCALE

Les programmes de reboisement peuvent s'effectuer au niveau local par des comités. La vulgarisation joue le rôle de porte-parole entre le comité et les populations de la région.

L'exécution du programme comprend les propositions de projet, le budget et l'identification des agences de d'aide qui le financeraient.

CONSERVATION ET GESTION DES RESSOURCES NATURELLES

I. RESSOURCES NATURELLES

Définition, importance et types de ressources. On entend par ressources naturelles tout ce qui existe dans la nature et qui n'est pas dû aux actions de l'homme. Parmi les ressources naturelles, nous citons l'homme, les animaux, les végétaux, le sol, l'air, l'eau, le pétrole, le soleil. Personne ne peut ignorer l'importance de ces ressources, quand on sait que tout être humain les utilise pendant toute la durée de sa vie.

Les ressources naturelles se divisent en:

- ressources naturelles renouvelables: les animaux, les végétaux, l'homme.
- ressources naturelles non-renouvelables: le pétrole.
- ressources naturelles épuisables: le sol, le pétrole.
- ressources naturelles non-épuisables: l'air, l'eau.

L'emploi abusif d'une ressource renouvelable ou non-renouvelable conduit d'abord à sa rareté et ensuite à sa disparition.

1. L'EAU

Importance de l'eau. L'eau est le constituant fondamental de la matière vivante. La teneur en eau des tissus animaux et végétaux varie entre 70-90% suivant le tissu et l'âge de l'organisme.

Cycle hydrologique. Près des 3/4 de la surface terrestre est recouverte par les eaux (océans, lacs, cours d'eau). Sous l'action des rayons du soleil, les surfaces liquides s'échauffent et l'eau s'évapore. La vapeur d'eau est dispersée dans l'atmosphère par le vent. En s'élevant dans l'atmosphère elle trouve des conditions de température permettant sa condensation. La vapeur d'eau condensée et saturée donne naissance aux précipitations qui réalimentent le sol, les océans, les cours d'eau. C'est le cycle de l'eau.

Répartition des eaux précipitées. Les eaux précipitées empruntent deux voies principales: l'infiltration et le ruissellement.

a) Infiltration. Une partie des eaux précipitées pénètre dans le sol à travers ses pores. C'est l'eau d'infiltration. Par réchauffement du sol, une fraction de l'eau infiltrée retourne dans l'atmosphère par évaporation. La quantité d'eau évaporée dépend de la saison, du climat et de la couverture végétale.

Une autre fraction de l'eau d'infiltration est utilisée par les végétaux pour satisfaire leur besoin en eau. Ils en rejettent une partie sous forme de vapeur d'eau dans l'atmosphère par la transpiration. Enfin, une dernière portion de l'eau d'infiltration descend dans les couches profondes du sol et alimente la nappe souterraine. L'eau de cette nappe revient à la surface en donnant naissance aux sources et aux rivières.

b) Ruissellement. L'eau de ruissellement est celle qui coule sur la surface du sol sans y pénétrer. Le ruissellement dépend de l'intensité de la pluie, de la nature du sol, de la pente, du terrain, du couvert végétal. Les eaux de ruissellement alimentent les ravines et les rivières et de là se jettent à la mer. Elles entraînent les particules de sol, engendrent l'érosion, provoquent les crues des rivières et les inondations qui peuvent en découler.

Gestion de l'eau dans les mornes. L'agriculteur qui se soucie de la protection de ses terres doit adopter des techniques qui permettent d'augmenter l'infiltration et de diminuer le ruissellement. En le faisant, il protège sa terre contre l'érosion et profite avantageusement de la pluie. Il peut faire appel aux cultures en terrasse, aux haies vives, rampes de pailles ou aux cordons de pierre. Dans beaucoup de cas, il devrait abandonner les cultures annuelles en faveur de cultures perennes.

2. LE SOL

Définition et composition. Le sol est le résultat de la décomposition de la roche mère sous l'action du climat et des organismes vivants. Les sols diffèrent selon la nature de la roche mère et selon le degré d'évolution qu'ils ont atteint.

Le sol est surtout formé par une succession de couches différenciées entre elles par leur texture, leur structure et leur couleur. La perte d'une ou des couches du sol due à l'érosion entraîne son appauvrissement, donc la capacité du sol à donner de bons rendements diminue d'années en années si des mesures de protection ne sont pas prises à temps.

Importance du sol. Le sol abrite de nombreuses espèces animales et végétales généralement microscopiques, qui par leurs activités font du sol un milieu vivant. On peut citer les vers de terre, les algues, les champignons et les bactéries. On peut bien vite comprendre l'effet néfaste que produit la pratique du brûlis entrepris généralement par la grande majorité de nos agriculteurs.

3. LES PLANTES

Définition et importance. Les végétaux sont des êtres vivants. Ils naissent, se nourrissent, grandissent, se reproduisent et meurent. Toute plante provient d'une semence. Celle-ci peut être une graine, une bouture ou tout autre morceau de végétal.

Les végétaux comme tout être vivant, se nourrissent. Contrairement aux animaux, ils fabriquent eux-mêmes leurs propres substances organiques à partir des composés minéraux. Les plantes tirent leur nourriture du sol et de l'air. A la lumière du jour, grâce à la chlorophylle des feuilles, les plantes absorbent le gaz carbonique de l'air et rejette l'oxygène. Le gaz carbonique fixé sert à la formation de la matière organique végétale. Ce phénomène s'appelle assimilation chlorophyllienne ou photosynthèse. Les plantes puisent dans le sol l'eau et les éléments minéraux qui leur sont nécessaires par les poils absorbants de leurs racines.

4. RELATIONS: EAU - SOL - PLANTES

L'eau, le sol et les plantes forment un système dans lequel les éléments sont unis entre eux par des relations intéressantes. Dans les conditions naturelles (forêt, terre en friche) il s'établit un équilibre entre le sol, l'eau et les autres organismes vivants. Les éléments suivent un cycle naturel au bénéfice des êtres vivants.

L'eau de pluie en particulier est un facteur physique important dans la décomposition de la roche mère, la formation du sol et son évolution. L'eau est nécessaire au déroulement des réactions chimiques dans le sol et au maintien des activités biologiques. L'eau est un facteur indispensable au développement des plantes. Elle est le constituant fondamental de tissus végétaux et en représente en moyenne 75% du poids frais.

Le sol est non seulement le support physique de la végétation mais aussi le pourvoyeur de la plante en eau et éléments nutritifs. En tant que support, le sol doit réunir certaines conditions physiques telles que profondeur et porosité permettant un bon développement du système racinaire.

Les végétaux protègent le sol contre l'érosion. Par leurs feuilles, les arbres arrêtent les gouttes de pluie qui arrivent au sol avec moins de force. L'eau, le sol, les végétaux sont des ressources naturelles que la nature met à la disposition de l'homme et qu'il peut mobiliser en vue de satisfaire ses besoins.

"Dlo, tè ak plant se twa roch dife ki kimbe chodiè lavi nou. Si youn pa la ou sinon si yo souike nou nan gro problem" (L'eau, le sol et les plantes constituent les trois roches de notre foyer)

sur lesquelles repose la chaudière de notre vie. Si l'une d'elles manque ou se dégrade, la chaudière ne peut se tenir et la vie devient impossible) (J.R. Mérésier).

Bref, ce sont les interventions inadéquates de l'homme qui viennent perturber l'ordre des choses, rompre l'équilibre naturel et causer des dommages parfois irréparables au sol et aux organismes vivants qui ne trouvent plus les conditions optimales à leur épanouissement.

II. LES RELATIONS DE L'HOMME AVEC SON MILIEU

En observant notre milieu, nous voyons des maisons, des plantes, des animaux, etc.. Tous ces éléments constituent notre environnement.

Notions d'environnement. On appelle environnement, l'ensemble de tous les éléments vivants ou non vivants qui entourent les hommes et autres êtres vivants.

Les relations de l'homme avec les êtres vivants. Un des besoins vitaux de l'homme et des animaux est de se nourrir. Ils doivent manger pour compenser les pertes d'énergie résultant de leurs activités. Les humains se procurent en abondance d'une grande diversité de ressources animales et végétales par la pêche, la chasse, l'agriculture ou l'élevage.

Les hommes entretiennent des relations avec les espèces aquatiques (consommation de fruits de mer), avec les essences forestières (utilisation du papier, du bois) avec les espèces cultivées ou élevées (produits alimentaires et pharmaceutiques, bétail).

Les relations de l'homme avec les non-vivants. Les hommes entretiennent également des relations avec les non-vivants, c'est-à-dire avec l'air, l'eau, le sol et le climat. Ces relations agissent dans les deux sens. Les humains influencent ces éléments mais ceux-ci sont aussi influencés par ceux-là. En ne prenant pas les précautions nécessaires pour respecter leur environnement, les hommes déséquilibrent le milieu conduisant à sa dégradation, sa pollution et des catastrophes naturelles.

III. AMENAGEMENT DU MILIEU

Concept d'aménagement et ses implications. Aménager un milieu c'est établir des structures qui le rendent fonctionnel. Donc, aménager une zone ou un bassin versant c'est y installer suivant les besoins des populations et avec leur participation des écoles (Ministère de l'Éducation nationale), des routes (Ministère des Travaux Publics, Transport et Communication), des dispensaires (Ministère de la Santé publique), des centres de productions agricoles (Ministère de l'Agriculture), des marchés et des centres de loisirs (Ministère des Affaires Sociales), des institutions de sécurité sociale (agents de police), des centres d'information et de documentation générale (Ministère de l'Information); bref, tout ce qui dans la réalité constitue un besoin pour les utilisateurs préposés.

IV. EROSION DES SOLS

L'eau et le sol constituent sans aucun doute le capital le plus précieux sur terre. L'homme demandant au sol de produire plus et plus vite, a ouvert la porte à l'érosion en créant un déséquilibre qui souvent dégénère parfois en catastrophe. Il est d'autant plus rapide que les conditions sont plus précaires (sols à réserves faibles et climat agressif).

L'érosion est la dégradation des sols ou de l'environnement sous l'influence de certains facteurs (eau, vent, climat, homme etc.) conduisant à une situation irréversible en absence de toute mesure de protection ou de conservation. L'érosion est un processus lent qui agit en trois temps d'amont en aval: 1) détachement des particules du sol dans les montagnes; 2) transport de ces particules vers les piémonts; 3) dépôt des sédiments dans les plaines puis à la mer.

On en distingue trois types: 1) l'érosion géologique ou naturelle qui se produit sans l'intervention de l'homme, mais par l'action de l'eau, du vent et des plantes sur les sols; 2) l'érosion accélérée qui découle des mauvaises interventions de l'homme; 3) l'érosion catastrophique qui est

enregistrée après les cas d'orage ou de tempête et qui laisse d'énormes dégâts au niveau des ressources naturelles.

On distingue 2 formes d'érosion. L'érosion éolienne quoique peu courante en Haïti peut être observée dans la région des Gonaïves et l'érosion hydrique très courante dans le pays et qui se divise en érosion chimique et érosion mécanique. Les types d'érosion mécaniques sont énumérés comme suit: érosion en splash, en nappe, en rigole, en ravine, en torrent, en masse, les éboulements et les glissements de terrains.

On distingue 2 grands facteurs d'érosion. Les facteurs physiques ou naturels causés par les événements géologiques, la topographie (pente et altitude) et le climat (pluie), et les facteurs humains causés par la destruction des forêts au profit des cultures, par l'abus des pâturages et par les mauvaises pratiques culturelles.

V. EXERCICE PRATIQUE. INSTRUMENTS UTILISES EN CONSERVATION DE SOL ET D'EAU

- a) Le niveau A. Construction - Importance - Manipulation.
- b) Les outillages simples pour les mesures des pentes.

VII. LES MESURES DE CONSERVATION DE SOL ET D'EAU

A. Mesures Mécaniques

Les mesures mécaniques les mieux adaptées en Haïti sont: les labours en courbes de niveau, les rampes de pailles ou fascinage, les murets et cordons de pierres et les seuils en pierres sèches.

B. Mesures Biologiques et Biomécaniques

Les mesures biologiques et biomécaniques comprennent les couvertures végétales (forêts, pâturage), les associations de cultures, la jachère, les bandes enherbées, les haies vives, le clayonnage, le compostage, les engrais verts et le paillis

C. Agrosylviculture (Agroforesterie)

L'agriculture des pays tropicaux confronte de grands problèmes, particulièrement liés à la pression démographique et à l'état érosif des terres exploitées. La culture itinérante, pratique fondée sur la mise en jachère des terres exploitées durant plusieurs années et la déforestation de nouvelles surfaces est restée pendant longtemps le système de production vivrière prédominant des régions tropicales. On constate actuellement qu'il est difficile de nourrir les hommes avec ce système de production à cause de la diminution des surfaces cultivables (perte de sol par érosion, développement et création de nouvelles villes et désertification). Il s'avère donc nécessaire de se pencher sur l'amélioration de la production agricole. C'est dans cette optique que depuis plusieurs années, l'agrosylviculture a été conçue en vue d'apporter des solutions de remplacement aux systèmes séculaires de la culture itinérante.

Définition. L'agrosylviculture est définie comme le système de production ou la méthode de mise en valeur des terres où des arbres sont associés aux cultures, à l'élevage ou aux deux à la fois.

But. L'agrosylviculture a pour but d'exploiter ou optimiser les interactions positives (arbres, arbustes avec les plantes cultivées) de façon à obtenir une production totale, élevée, diversifiée et soutenue par unité de surface, à partir des ressources disponibles et/ou limitées, dans les conditions écologiques, technologiques et socio-économiques existantes.

Principe. Il est important de savoir qu'il ne suffit pas d'être versé dans les trois disciplines individuelles qui sont l'agriculture, l'élevage et la sylviculture, mais que l'association de celles-ci

soulève des aspects techniques spécifiques dont on doit tenir compte si l'on veut que la méthode conduise aux résultats attendus. Ainsi on devra considérer la notion de *vocation des terres* pour évaluer les types de terre qui conviennent mieux à tel système agroforestier ou les meilleures combinaisons conduisant à la plus forte productivité.

L'agriculteur devra répondre à plusieurs questions.

- Quelle essence arborée est la plus intéressante et la plus avantageuse dans les conditions données?

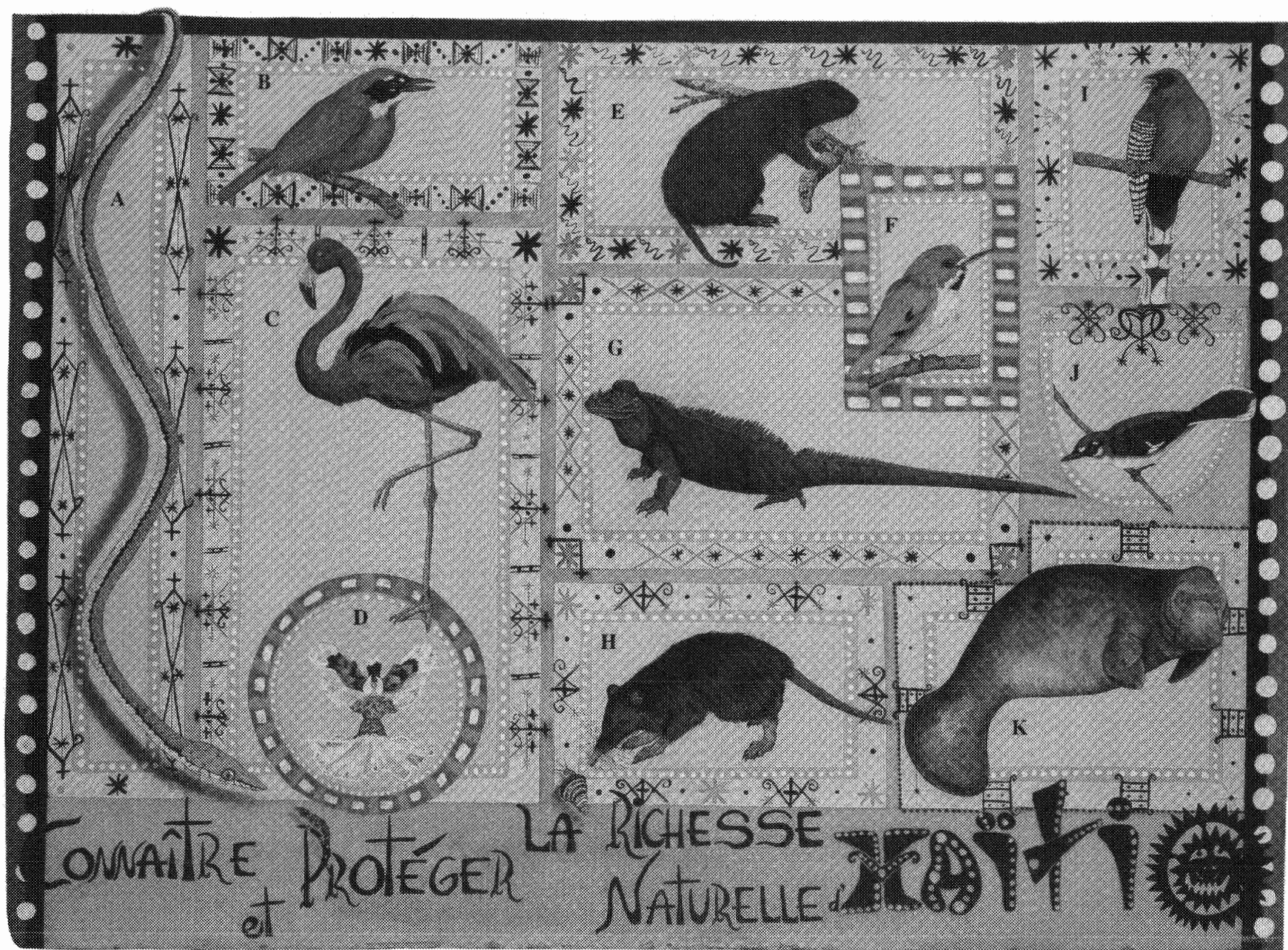
- Quelle est la meilleure distribution spatiale des arbres dans les champs?

- Quelle est la meilleure méthode d'établissement et d'organisation des différentes cultures sous-étagères?

Il est nécessaire de réfléchir au rôle et à l'utilisation des arbres dans l'exploitation et des essences fourragères importantes pour l'élevage.

Les systèmes agrosylvicoles les plus connus sont le sylvopastoralisme, l'agrosylvopastoralisme, l'aquisylviculture et l'apisylviculture.

L'agroforesterie est d'une importance impérative dans les conditions socioéconomiques et climatiques d'Haïti.



A. Couleuvre madeleine

Uromacer frenatus

Ce reptile est diurne et arboricole. Il se nourrit d'anolis et d'autres petits lézards. Il est inoffensif.

B. Quatre je sid

Phoenicophilus poliocephalus

On le rencontre que dans la presqu'île du sud à partir de Jacmel et Fauché. C'est un oiseau de plaine et de montagne qui se nourrit de baies sauvages.

C. Flamant rose

Phoenicopterus ruber

Les flamants roses sont de gros oiseaux qui vivent en bande au bord de l'Etang Saumâtre, de Trou Caïman, dans la baie de Fort-Liberté et le delta de l'Artibonite. Ils construisent leur nid par terre avec de la boue et des branchages pour y pondre un oeuf.

D. Orchidée sauvage

Tolumnia variegata = *Oncidium variegatum*

L'orchidée est une fleur remarquable. Cette espèce pousse dans les montagnes accrochée aux arbustes. Elle fleurit en avril. Les fleurs sont disposées en grappe et ressemblent à de petites danseuses.

E. Zagouti

Plagiodontia aedium

Il est un rongeur qui vit dans les forêts sur formations de roches karstiques de la presqu'île du Sud. Il se nourrit d'écorce de jeunes branches, de feuilles, de fruits et de tubercules. Il est menacé d'extinction à cause de la destruction des aires boisées.

F. Colibri ou chicorettes

Todus angustirostris

Les colibris sont de petits oiseaux que l'on ne trouve que dans les Grandes Antilles. Le mâle et la femelle fouillent à tour de rôle un petit tunnel de 30 cm où seront pondus trois ou quatre oeufs. Ils sont très utiles car ils se nourrissent d'insectes.

G. Iguane

Cyclura cornuta

L'iguane est un reptile herbivore très pacifique malgré son apparence. Sa queue lui sert de réserve alimentaire et de moyen de défense. Il vit en colonie dans les terres chaudes et arides.

H. Nez long ou Nin long

Solenodon paradoxus

Le nez long est un insectivore nocturne, timide et peureux qui se terre sous les roches de formation karstique. Il se déplace lentement et fouit pour trouver sa nourriture. Il a déjà disparu dans plusieurs localités. Il en existe très peu encore dans la région de Duchity. C'est l'animal le plus menacé d'extinction en Haïti.

I. Caleçon rouge

Priotelus roseigaster

Le caleçon rouge est l'un des plus beaux oiseaux d'Haïti. Il se nourrit de fruits et d'insectes. Ils voyagent par paire et nichent dans des cavités d'arbres de montagnes où la femelle pond deux oeufs vert pâle. Ils sont familiers, ce qui explique leur rareté car les enfants les chassent facilement avec des lance-pierres.

J. Ti chit kat je

Xenoligea montana

C'est un oiseau très rare. Il vit caché dans les bosquets de feuillus des montagnes d'Haïti. Il est menacé parce qu'il pond ses oeufs par terre ou sur les branches au raz du sol.

K. Lamantin

Trichechus manatus

Le lamantin est un mammifère herbivore aquatique. Il n'a aucun prédateur naturel, l'homme est son plus grand ennemi. Il est menacé d'extinction. On en rencontre actuellement que très peu à l'embouchure du fleuve de l'Artibonite et dans la région de Montruis.